UGC NET Paper 1

General Paper on Teaching and Research Aptitude

Unit I: Teaching Aptitude

Study Guide

इकाई I : शिक्षण अभिवृत्ति

A guide to study alone, with an index to the 1,000-question MCQ booklet

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How to use this guide

Dear student, this guide teaches Unit I of Paper 1, Teaching Aptitude (शिक्षण अभिवृत्ति). It is written for you if you are studying alone, without coaching, and perhaps in a second language. Read it slowly. Every paragraph is here because a question can be asked on it.

This guide has a partner: the **MCQ booklet** for Unit I, with 1,000 questions in 20 sets. The guide teaches; the booklet tests. You will move between the two many times.

How each section is built

Every section of the guide has the same parts, always in the same order, so that you always know where to look.

- **Exam footprint.** One line on how often the topic is asked and in what form. It tells you how much effort the topic deserves.
- **Core.** The explanation in plain words, with a table where things are compared, and an example from an Indian college or university classroom. A key term is followed, the first time it appears, by its Hindi equivalent in brackets, so that you can connect it with Hindi books and with the Hindi version of the question paper.
- **Often confused.** The pairs that examiners use to trap candidates, with the one difference that decides between them.
- **Fact watch.** Dates, figures and official positions that change with time. Each one says when it was checked. Not every section has one.
- **Going Further.** The finer points, the less familiar names and the calculations that the harder booklet sets (11 to 20) need. Leave this layer for your second reading.
- **Check yourself.** Five to eight new questions in the formats of the real paper. The answers, with a short reason, are at the end of each chapter.

When you get a booklet question wrong

At the back of this guide is the **Booklet Index**. For each of the 20 sets it gives, against every question number, the section of this guide that teaches the answer. If you get Question 23 of Set 6 wrong, look up Set 6, find 23, and read that section again. Do this every time. It is the fastest way to close a gap.

A word about facts

Students memorise what a guide says, so every name, date and figure in this guide has been checked against official or standard sources. Where a fact has changed, the guide gives the present position, says when it was checked, and names the older figure that you may still find in other books. If an older book disagrees with this guide on such a point, this guide is the more recent. Before your examination, look at the page “Facts to recheck before your examination” at the end.

None of the questions in this guide is copied from a question paper, a book or coaching material. All are written afresh for practice, in the same formats as the real paper.

Your study route

Follow these seven steps in order. Tick each one when it is done.

1. Read the Core layer of every chapter, answering each “Check yourself” as you go.
2. Attempt booklet Sets 1 to 6. After each set, use the booklet index to reread the section behind every wrong answer.
3. Attempt Sets 7 to 10. These sit at the level of the real paper.
4. Read the Going Further layer of every chapter.
5. Attempt Sets 11 to 14, rereading by the index as before.
6. Revise from the confusable-pairs table and the revision sheets, then attempt Sets 15 to 17.
7. Treat Sets 18 to 20 as stretch practice. A modest score there is no sign of being unready.

This unit in the examination

How many questions, and how they are marked

Paper 1 has 50 compulsory questions. Each carries 2 marks, so the paper is out of 100. There is no negative marking, so never leave a question blank. Paper 1 and Paper 2 are taken together in one sitting of three hours, as a computer-based test.

The syllabus has ten units, and the paper is designed to take **five questions from each unit**. So Teaching Aptitude is worth about 10 marks. Sometimes a sixth question touches this unit too, because a topic such as SWAYAM or classroom communication also belongs to the units on ICT, Communication or Higher Education.

The formats you will meet

The paper no longer asks only plain one-answer questions. In recent papers a large share of questions use these formats:

- **Multiple-select.** Five numbered statements; you choose the option that lists the correct ones, such as “1, 3 and 4 only”.
- **Match the list.** Four items in List I to be matched with four in List II; you choose the option with the right set of pairs.
- **Sequencing.** Four or five steps or stages to be put in order, such as the levels of Bloom’s taxonomy or Piaget’s stages.
- **Two statements.** Statement I and Statement II; you decide whether each is true or false.
- **Assertion and Reason.** You decide whether each statement is true, and then whether the Reason explains the Assertion.
- **Classroom scenarios.** A short situation, and the question asks what the teacher should do or what kind of method or evaluation is being used.

Every “Check yourself” in this guide uses these formats, with more room given to the harder ones.

The topics that recur most

A survey of what candidates have recalled from the papers set by NTA since December 2018, and of what the standard Paper 1 books treat as core, shows these topics coming back again and again, roughly in this order:

1. Types of evaluation: formative and summative, norm-referenced and criterion-referenced, often as a classroom scenario (Section 6.3).
2. Bloom’s taxonomy and the domains of objectives (Section 1.4).
3. SWAYAM, SWAYAM PRABHA and MOOCs (Section 4.6).
4. Teacher-centred and learner-centred methods (Section 4.1).
5. The levels of teaching: memory, understanding and reflective (Section 1.5).
6. Learning theories and the learning process (Section 2.3).
7. Effective teaching and “what should the teacher do” scenarios (Section 1.7).
8. Group and cooperative methods such as seminar, symposium, panel and workshop (Section 4.3).
9. Piaget, Vygotsky and other thinkers on cognitive development (Section 2.2).
10. Micro-teaching and Flanders’ Interaction Analysis (Section 1.8).
11. The Choice Based Credit System and grading (Section 6.7).
12. The nature of teaching and models of teaching (Sections 1.1 and 1.9).

Each section of the guide opens with an **Exam footprint** line that tells you how often its topic is asked and in what form. Spend your time accordingly, but do not skip the rarer sections: the booklet, and sometimes the examination, will ask about them.

Chapter 1
Teaching: Concept, Objectives, Levels, Characteristics and Basic Requirements

This chapter begins with the question every other chapter depends on: what is teaching, and how is it different from training, instruction, conditioning and indoctrination? It then follows teaching as a process in three phases, shows how aims become precise objectives and learning outcomes, and sets out the taxonomies used to classify objectives. It ends its first half with the three levels of teaching – memory, understanding and reflective – which the examination asks about more often than almost any other topic in Unit I.

1.1 The concept of teaching

Syllabus: Teaching: Concept

> **EXAM FOOTPRINT**
>
> Asked often; usually a multiple-select item on what teaching is or aims at, or a statement contrasting teaching with indoctrination, conditioning or training.

Core

What teaching is

Teaching (शिक्षण) is a purposeful, planned activity carried out by one person to bring about **learning** (अधिगम) in another person. Three words in that sentence matter. “Purposeful” means it has an aim; an act done without any aim of producing learning is not teaching. “Planned” means it is deliberate, not accidental. “In another person” means there is always a learner.

So teaching is not defined by finishing the syllabus, by keeping the class quiet, or by speaking for fifty minutes. A teacher may do all three and still not be teaching, if nothing is done to help anyone learn.

Teaching is also **goal-directed**. It aims at desired changes in the learner: in knowledge, in skills, and in attitudes and values. If there are no objectives at all, the activity is not teaching in the proper sense.

Three things meet in teaching

Philosophers say teaching is **triadic**: someone (the teacher) teaches something (the content) to someone (the learner). Take away any one of the three and the idea of teaching falls apart. For this reason books call the teacher, the learner and the subject matter the three focal points of teaching. Good teaching attends to all three: a competent teacher, a ready learner, and content suited to that learner.

Do not mix this up with a second “tripolar” idea found in Indian textbooks. There, the three poles of the educative process are **educational objectives, learning experiences and evaluation**; Indian books usually link this triad with B. S. Bloom. The same three elements are drawn in Section 6.1 as the evaluation triangle, where the idea is traced to Ralph W. Tyler; examiners may use either name, so recognise both. The first triad names the persons and the content; the second names the stages of planned education. If a question asks for the “three focal points of teaching”, the answer is teacher, learner and subject matter.

Note also: the fact that teaching is a profession is a true social fact, but it does not explain why teaching is triadic. The triadic structure comes from the meaning of the word, not from teaching being a job.

Famous definitions

The examination often gives a definition and asks who wrote it. The wordings below are the forms in which Indian textbooks and examination papers quote these definitions; learn the key phrase in each.

Thinker	Definition (key words)	How to remember
H. C. Morrison (1934)	"An intimate contact between a more mature personality and a less mature one, designed to further the education of the latter."	Two personalities: mature and less mature
N. L. Gage	"A form of interpersonal influence aimed at changing the behaviour potential of another person."	Influence and behaviour potential
B. O. Smith	"A system of actions intended to induce learning."	System of actions
Edmund Amidon (with Hunter)	"An interactive process, primarily involving classroom talk, between teacher and pupils."	Interaction and classroom talk

Gage's definition has two parts worth noticing: it involves two persons (one influences the other), and it involves intention (it is "aimed at" change). Morrison's stresses the personal relationship. Smith's stresses a planned set of actions. Amidon's stresses talk in the classroom, which is why his name is also linked with the analysis of classroom interaction (see 1.8).

Teaching and learning

Teaching and learning are related, but they are two different processes. **Learning** is a relatively permanent change in behaviour that comes from experience or practice. Learning is both a process (what goes on while one learns) and a product (what one has learnt at the end).

- Teaching is the means; learning is the end. Teaching is judged by the learning it produces.
- Teaching always involves the intention to bring about learning. Learning does not need teaching: much is learnt from experience, observation, play and self-study, often by accident. So learning can occur without teaching.
- Nobody can learn on another person's behalf. The learner must give attention and effort. That is why arousing and keeping up the learner's **motivation** (अभिप्रेरणा) is a necessary part of teaching.

The teaching continuum

Several activities look like teaching. Books arrange them on a line, or **continuum**, according to how much they use the learner's understanding. From the lowest to the highest:

1. **Conditioning** (अनुबंधन). A fixed link is built between a stimulus and a response, by pairing (Pavlov's classical conditioning) or by reinforcement (Skinner's operant conditioning). The learner need not understand anything. A dog that salivates at a bell has been conditioned. It sits at the bottom because it by-passes understanding.
2. **Training** (प्रशिक्षण). A specific skill is built by practice and drill until it reaches a set standard: typing at forty words a minute, handling a microscope, marching in step. Training shapes behaviour and skill; it does not need to ask why.
3. **Instruction** (अनुदेशन). Organised knowledge is passed on in a systematic way. Instruction can be given without a teacher present: by a programmed text, a computer, a radio or television broadcast, or a well-written book.
4. **Teaching**. The widest of the four. It includes skill and knowledge, but it also aims at understanding, reasoning, attitudes and values, and it engages the learner's judgement. In the strict sense it involves a person interacting with the learner.

Each later term is wider than the one before. All teaching includes instruction, but not all instruction is teaching: a computer can instruct, but it does not teach in the full sense. Training is narrower than teaching because it builds a particular skill, while teaching also seeks understanding and critical judgement.

Indoctrination

Indoctrination (मतारोपण) means implanting beliefs so that they are held as beyond question, whatever the evidence. The learner is discouraged from examining them. Propaganda works in a similar way on large groups.

What makes something indoctrination is its manner and its aim, not its subject matter. Any content – history, science, religion, politics – can be taught with reasons, evidence and freedom to object, or it can be indoctrinated without them. Teaching transmits knowledge, develops skills and builds values while allowing the learner to reason. Imposing an ideology that may not be questioned is indoctrination, not teaching.

Tutoring (individual help to one learner) and counselling (help with personal or academic problems) are also different from indoctrination: both leave the learner free to question.

The nature of teaching

The following features of teaching are often asked as true/false statements.

- **Teaching is interactive.** It is a two-way process. The teacher acts, learners respond, and the teacher adjusts to their response. It depends on feedback.
- **Teaching can be observed, analysed and improved.** Classroom talk can be recorded and coded (interaction analysis), and single skills can be practised in short lessons (micro-teaching). Both are in 1.8.
- **Teaching can be formal or informal.** A lecture in a hall is formal; a word of advice in the corridor can also be teaching.
- **Teaching is both a science and an art.** It is a science because it rests on principles of learning, motivation and evaluation that can be observed, tested and taught. It is an art because each teacher applies these principles creatively, in a personal style, to a particular class. N. L. Gage wrote a book on this theme, *The Scientific Basis of the Art of Teaching* (1978).
- **Evaluation is part of teaching.** The teacher must know whether the objectives are being reached, so as to adjust the next step. This is a reason of its own; it does not follow from teaching being communication.
- **Teaching is communication** (संप्रेषण). It fits the communication model: a sender (teacher) sends a message (content) through a channel (speech, board, slides) to a receiver (learner), who sends back **feedback** (प्रतिपुष्टि). **Noise** is anything that distorts the message: physical noise (traffic, a broken fan), semantic noise (unfamiliar words, jargon), and psychological noise (anxiety, prejudice, fatigue). Communication is treated fully in Unit IV.

CLASSROOM EXAMPLE

In a B.A. political science class at a government college in Hisar, Dr Meena explains the Preamble. She first makes students repeat the words "sovereign, socialist, secular, democratic, republic" until they can say them without a slip – this is closer to training. Next she hands out a printed note on the meaning of each word – this is instruction, and a book could have done it. Then she asks: "Why was 'socialist' added in 1976? Do you agree with the change?" She lets students argue and asks each for reasons. Now she is teaching in the full sense. Had she told them "never question the Preamble; just accept it", that would have been indoctrination, even though the subject is the same.

OFTEN CONFUSED

- **Training and teaching.** Training builds one skill to a standard by practice; teaching also seeks understanding and judgement.
- **Instruction and teaching.** Instruction is organised passing on of knowledge and can be done by a machine or book; teaching involves a person and aims at understanding. All teaching includes instruction, not the reverse.
- **Indoctrination and teaching.** Decided by manner and aim, not by subject: indoctrination wants belief without question; teaching gives reasons and welcomes objections.
- **Conditioning and training.** Conditioning links a stimulus to a response with no understanding at all (lowest); training builds a skill by practice (one step higher).
- **Gage and Smith.** Gage: "interpersonal influence … behaviour potential". Smith: "system of actions intended to induce learning".
- **Morrison and Amidon.** Morrison: "more mature and less mature personality". Amidon: "interactive process … classroom talk".
- **Teacher-learner-content and objectives-learning experiences-evaluation.** The first is the triad of teaching (three focal points); the second is the tripolar view of the educative process.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Green's topology of teaching

Indian textbooks present a map of activities related to teaching which they credit to the American philosopher Thomas F. Green (author of *The Activities of Teaching*, 1971). It puts teaching in the middle and two families of activities on either side.

- On one side are activities that shape **behaviour**: training, and further away, conditioning.
- On the other side are activities that shape **knowledge and belief**: instructing, and further away, indoctrinating.

On each side the first activity (training; instructing) is closer to teaching because it still engages the learner's intelligence. The second (conditioning; indoctrinating) lies farthest from teaching. A trap: conditioning and indoctrinating are both farthest from teaching, but they are on different sides; one shapes behaviour and the other belief.

Task sense and achievement sense

Philosophers notice that the word “teach” is used in two ways. Philosophers of education usually trace this to Gilbert Ryle’s distinction between “task” words and “achievement” words.

- In the **task sense**, to teach is to try to bring about learning. One may teach in this sense even if nobody learns (“I taught all afternoon but they learnt nothing”).
- In the **achievement sense**, to say “I taught them the theorem” claims success: it implies that they learnt it.

A comparison usually credited to John Dewey makes the achievement point: teaching is like selling. Nobody can be said to have sold anything unless someone has bought it; in the same way, teaching in the achievement sense implies learning.

Scheffler and the learner’s judgement

Israel Scheffler, author of *The Language of Education* (1960) and *Reason and Teaching* (1973), is known for stressing critical thought and the learner’s independent judgement: teaching in the full sense respects that judgement. The teacher gives reasons and is willing to have them examined. This is the line between teaching and indoctrination: an indoctrinator wants belief without examination. So three things are true of teaching together: it is intentional, it presupposes a learner, and in the task sense it need not end in learning.

CHECK YOURSELF

1. Which of the following is the best description of teaching?

(A) Completing the prescribed syllabus within the session
(B) Maintaining silence and discipline in the classroom
(C) A purposeful activity intended to bring about learning in another person
(D) Any activity from which someone happens to learn

2. Arrange the following from the activity that relies least on the learner's understanding to the one that relies most:

1. Teaching
2. Conditioning
3. Instruction
4. Training

Choose the correct answer from the options given below:

(A) 2, 4, 3, 1
(B) 4, 2, 3, 1
(C) 2, 3, 4, 1
(D) 3, 2, 4, 1

3. Match List I with List II.

List I (Thinker)	**List II (Key phrase in definition)**
A. H. C. Morrison	I. System of actions intended to induce learning
B. N. L. Gage	II. Interactive process of classroom talk
C. B. O. Smith	III. Intimate contact of a more mature and a less mature personality
D. Edmund Amidon	IV. Interpersonal influence on behaviour potential

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-I, D-II
(B) A-III, B-IV, C-II, D-I
(C) A-III, B-I, C-IV, D-II
(D) A-III, B-IV, C-I, D-II

4. Given below are two statements.

Statement I: A lesson on any subject can become indoctrination if beliefs are presented as beyond question and the learner is discouraged from examining them.

Statement II: Instruction always requires a living teacher to be present with the learner.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

5. Given below are two statements, one labelled Assertion (A) and the other Reason (R).

Assertion (A): Arousing and sustaining motivation is a necessary function of teaching.

Reason (R): No one can learn on behalf of another; learning needs the learner's own attention

and effort.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

6. Which of the following statements about teaching are correct?

1. Teaching is a two-way process that depends on feedback.
2. Classroom teaching behaviour can be observed, analysed and modified.
3. Learning can take place only when there is teaching.
4. Teaching can be formal or informal.

Choose the correct answer from the options given below:

(A) 1 and 2 only
(B) 2, 3 and 4 only
(C) 1, 2 and 3 only
(D) 1, 2 and 4 only

7. In Thomas F. Green's map of activities related to teaching, which activity lies on the "behaviour" side and is closest to teaching?

(A) Instructing
(B) Conditioning
(C) Training
(D) Indoctrinating

1.2 Teaching as a process: phases, variables and operations

Syllabus: Teaching: Concept

EXAM FOOTPRINT

Asked occasionally; usually a match or sequence item placing activities in the pre-active, interactive or post-active phase, or naming the teacher and learner as variables.

Core

Three phases of teaching

Teaching does not happen only in the classroom. It begins before the teacher enters the room and continues after the bell. Books divide it into three **phases of teaching** (शिक्षण की अवस्थाएँ). The idea comes from the American educationist Philip W. Jackson, whose book *Life in Classrooms* (1968) distinguished "pre-active" from "interactive" teaching. Later writers added the post-active phase, and Indian textbooks and the examination treat all three as the phases of teaching linked with Jackson.

1. **Pre-active phase** (पूर्व-क्रियात्मक अवस्था): planning, before teaching.
2. **Interactive phase** (अंतःक्रियात्मक अवस्था): the actual meeting of teacher and class.
3. **Post-active phase** (उत्तर-क्रियात्मक अवस्था): evaluation and reflection, after teaching.

There is no "reactive phase" or "inactive phase". If an option offers one, reject it.

What the teacher does in each phase

The tasks done in each phase are called its **operations**. Learn this table; questions often give an operation and ask for its phase.

Phase	Operations
Pre-active (planning)	Fixing the objectives of the lesson Selecting the content Arranging (sequencing) the content Deciding the teaching strategies and methods Choosing teaching aids and preparing questions
Interactive (in the classroom)	Sizing up the class: perceiving its size, mood and readiness Diagnosing the learners: finding what they know and where they struggle Action and reaction: presenting stimuli, questioning, responding to answers, giving feedback and reinforcement
Post-active (after teaching)	Defining the exact dimensions of the change in behaviour achieved Selecting suitable testing devices and techniques Evaluating the outcomes Changing strategies in the light of the evidence, and reflecting

The interactive phase covers everything from the moment the teacher enters the class to the end of the lesson. Sizing up the class is an interactive operation, not a post-active one, because it is done while facing the learners.

How decisions differ between phases

- In the pre-active phase the teacher has time. Decisions are calm, deliberate and rational.
- In the interactive phase the teacher must respond at once to what learners say and show, both in words and without words (a puzzled face, a yawn). There is little time to think, so decisions are quick and **spontaneous**.
- In the post-active phase the teacher again has time: to analyse what happened and reflect.

The phases form a cycle

What the teacher learns in the post-active phase feeds the next pre-active plan. The three phases are therefore not a straight line but a **cycle**. Post-active reflection can also grow into small classroom research, called action research (Unit II). Experience makes planning quicker, but it never removes the need to plan.

The phase of an act is decided by when it is done and why. Preparing questions at home is pre-active. Asking the same questions in class to find out what students know is interactive. Marking the answer scripts afterwards and revising one's notes is post-active.

Teaching variables

A **variable** (चर) is anything that can change. When we analyse teaching, three kinds of variable are named.

Variable	Who or what	Why so called
Independent variable (स्वतंत्र चर)	The teacher	The teacher plans, decides and manages the process; the teacher's actions are what bring change

Variable	Who or what	Why so called
Dependent variable (आश्रित चर)	The learner	The learner's behaviour changes as a result of teaching; the learner acts according to the teacher's planning
Intervening variables (मध्यवर्ती चर)	Content, methods, strategies, teaching aids, environment	They stand between teacher and learner and carry the teacher's influence to the learner

The teacher selects and organises the intervening variables. It is through them that the teacher's influence reaches the learner and changes the learner's behaviour.

A warning on a related use of the words. In an **experiment**, the factor the researcher changes on purpose (for example, the method of teaching) is the independent variable, and the outcome that is measured (for example, achievement scores) is the dependent variable. So in a study comparing two methods, "method" is the independent variable. In the analysis of teaching, the teacher is the independent variable. Read the stem to see which use is meant.

CLASSROOM EXAMPLE

Mr Rakesh teaches B.Com. accountancy at a college in Sirsa. On Sunday he fixes the objective "students will prepare a bank reconciliation statement", picks three examples and decides to begin with a short quiz – pre-active. On Monday he notices the class is tired after a test, so he drops the quiz and starts with a real passbook; a student's wrong answer shows that many confuse debit and credit, so he stops and re-explains – interactive, with spontaneous decisions. On Tuesday he marks their worksheets, finds that two-thirds made the same slip, and decides to change his method next time – post-active, feeding the next plan.

OFTEN CONFUSED

- **Pre-active and post-active.** Pre-active is planning before the lesson; post-active is evaluation and reflection after it. Selecting a testing device is post-active; fixing objectives is pre-active.
- **Sizing up the class and evaluating outcomes.** Sizing up is interactive (done while facing the class); evaluating outcomes is post-active.
- **Teacher as independent variable and method as independent variable.** In analysing teaching, the teacher is independent; in an experiment on methods, the method is independent.
- **Dependent and intervening variables.** The learner is dependent (changes); content and methods are intervening (carry the influence).

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Three functions of teaching

Many Indian textbooks describe three **functions of teaching**, which follow the order of a doctor's work.

1. **Diagnostic function** (निदानात्मक कार्य). Finding the learner's **entry behaviour** (प्रवेश व्यवहार) – what the learner already knows and can do – and his or her difficulties and weaknesses.

2. **Prescriptive function** (उपचारात्मक कार्य). Selecting the content, methods, strategies and aids that suit the diagnosis, and that remedy the weaknesses found.
3. **Evaluative function** (मूल्यांकनात्मक कार्य). Checking whether the objectives have been achieved, judging the outcomes, and using the result to improve teaching.

A pre-test before a new unit serves the diagnostic function. Choosing a demonstration for slow learners after the pre-test is prescriptive. A unit test at the end is evaluative.

More kinds of variable in research

Questions in the harder sets borrow terms from research design (Unit II).

- **Independent variable**: the factor the researcher manipulates (method of teaching).
- **Dependent variable**: the outcome measured (achievement).
- **Intervening variable**: a process on the causal path that carries the effect from the independent to the dependent variable, and is not directly observed (for example, interest aroused by the new method).
- **Moderator variable**: a factor that changes the strength or direction of the relation between independent and dependent variables (for example, the method works better for girls than boys: gender is a moderator).
- **Extraneous variables**: other factors (intelligence, motivation, home background) that could affect the outcome. **Random assignment** of learners to groups spreads these evenly over the groups, so that any difference in outcome can be put down to the method.

CHECK YOURSELF

1. Which of the following is NOT one of the phases of teaching?

(A) Pre-active phase
(B) Interactive phase
(C) Reactive phase
(D) Post-active phase

2. Match the operation in List I with the phase in List II.

List I (Operation)	List II (Phase)
A. Fixing the objectives of a lesson	I. Interactive
B. Perceiving the size and mood of the class	II. Post-active
C. Selecting suitable testing devices	III. Pre-active

Choose the correct answer from the options given below:

(A) A-III, B-I, C-II
(B) A-II, B-I, C-III
(C) A-III, B-II, C-I
(D) A-I, B-III, C-II

3. In the analysis of teaching as a process (the teacher, the learner and what lies between them), content and methods of teaching are treated as:

(A) independent variables
(B) dependent variables
(C) extraneous variables
(D) intervening variables

4. Given below are two statements, one labelled Assertion (A) and the other Reason (R).

Assertion (A): Decisions taken in the interactive phase are largely spontaneous.

Reason (R): In the interactive phase the teacher must respond at once to learners' verbal and non-verbal cues, with little time to deliberate.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

5. Given below are two statements.

Statement I: Post-active evaluation feeds into the planning of the next lesson.

Statement II: An experienced teacher no longer needs the pre-active phase.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

6. A teacher gives a short test before beginning a new unit to find out what her students already know. Which function of teaching is she performing?

(A) Diagnostic
(B) Prescriptive

(C) Evaluative
(D) Administrative

7. In an experiment comparing the lecture method with the seminar method on students' achievement, it is found that the seminar method helps only students with high interest. Here "interest" acts as:
(A) the dependent variable
(B) a moderator variable
(C) the independent variable
(D) a constant

1.3 Aims, goals, objectives and learning outcomes

Syllabus: Teaching: Objectives

EXAM FOOTPRINT

Asked occasionally; usually a multiple-select item on aims of teaching in higher education or a statement linking objectives to learning outcomes.

Core

From aims to objectives

Teachers speak of aims, goals and objectives as if they were the same. They are not. They differ in how broad they are and how long they take.

Term	What it is	Example
Aim (लक्ष्य)	A broad, long-term purpose of education; not directly testable	To produce responsible citizens
Goal	A narrower purpose for a stage, programme or course	To develop scientific temper in undergraduates
Instructional objective (शिक्षण उद्देश्य)	A precise statement of what the learner will be able to do at the end of a lesson or unit; achievable and testable	The student will be able to state three causes of the Revolt of 1857
Learning outcome (अधिगम परिणाम)	What the learner can actually demonstrate as a result of learning, stated in the same observable terms	The student explains, in a paragraph, how the Doctrine of Lapse caused unrest

The movement is from broad to narrow and from distant to near. An aim is never achieved in one lesson; an instructional objective should be.

What teaching aims at

Learning is a relatively permanent change in behaviour. Teaching therefore aims at **desirable change** in three areas: knowledge, skills, and attitudes and values. Covering the syllabus, getting good marks and keeping order are means or by-products, not the aim.

The usual list of the **functions of teaching** is: transmitting knowledge, developing skills, building values, and **socialisation** (helping learners adjust to the norms of society). Certification and holding examinations are outcomes of the system, not functions of teaching.

In **higher education** the aim is wider still. Teaching should not only pass on knowledge; it should develop critical thinking, inquiry, judgement, a research orientation and values, and the capacity to create new knowledge. This is why reflective-level teaching (1.5) matters most in universities.

Instructional objectives in behavioural terms

An **instructional objective** states what the **learner** will be able to do after instruction, in terms that can be observed and assessed. It does not state what the teacher will do, which topic will be covered, or how long the lesson will take.

- Not an objective: “The teacher will explain photosynthesis.” (teacher activity)
- Not an objective: “Photosynthesis.” (a topic)
- An objective: “The student will be able to label the parts of a chloroplast on a given diagram.”

Because such objectives describe behaviour, they are also called **behavioural objectives** (व्यवहारगत उद्देश्य). They make evaluation easier, since the teacher knows exactly what to look for.

Use action verbs that can be seen or checked: define, list, state, explain, compare, solve, label, design. Avoid vague verbs that describe an inner state: know, understand, appreciate, realise, be familiar with. You cannot see someone “understand”; you can see someone explain.

Mager’s three parts

Robert F. Mager’s book *Preparing Objectives for Programmed Instruction* (1962), later retitled *Preparing Instructional Objectives*, taught teachers how to write behavioural objectives. A Mager objective has three parts.

Part	Question it answers	Example
Performance	What will the learner do?	Locate the major rivers of India
Condition	Under what circumstances, with what materials?	Given an outline map
Criterion	How well? What is the standard of acceptable performance?	With at least 90 per cent accuracy; 8 out of 10 correct

Full objective: “Given an outline map of India (condition), the student will mark the major rivers (performance) with no more than one error (criterion).”

Do not confuse Mager with other names often placed in the options:

- Benjamin Bloom: the taxonomy of educational objectives (1.4).
- Ralph Tyler: principles of curriculum, *Basic Principles of Curriculum and Instruction*.
- Robert Gagné: *The Conditions of Learning*.

The ABCD format

A later format adds one letter to Mager’s three parts.

- **A**udience: who the learner is (“B.Sc. first-year students”).
- **B**ehaviour: what they will do (Mager’s performance).
- **C**ondition: the circumstances (Mager’s condition).
- **D**egree: how well (Mager’s criterion).

So ABCD = Mager’s three parts plus Audience.

Outcome-based education

In **outcome-based education** (OBE; परिणाम आधारित शिक्षा) planning begins at the end. The teacher first writes the **intended learning outcomes**. Then the content, the teaching activities and the assessment are derived from those outcomes and lined up with them. The older practice started with the content (the syllabus), or with the staff available, or with the hours in the timetable; OBE replaced that.

CLASSROOM EXAMPLE

A teacher of English at a college in Fatehabad writes three statements for a lesson on the sonnet. (1) “To teach the sonnet form.” This is a teacher activity. (2) “Students will understand the sonnet.” This uses a vague verb. (3) “Given an unseen sonnet, students will mark its rhyme scheme and identify the volta, with no more than one error.” This is a Mager objective: condition, performance and criterion. Only the third tells her what to test at the end of the hour.

OFTEN CONFUSED

- **Aim and objective.** An aim is broad, long-term and not directly testable; an objective is narrow, precise and testable at the end of a lesson or unit.
- **Objective and teacher activity.** An objective states what the learner will do; “the teacher will explain …” is not an objective.
- **Condition and criterion.** Condition is the circumstance (“given a map”); criterion is the standard (“90 per cent accuracy”).
- **Mager and Bloom.** Mager: how to write a single objective. Bloom: how to classify objectives into levels.
- **Mager’s three parts and ABCD.** ABCD adds the Audience to performance, condition and criterion.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Gronlund’s way of stating objectives

Norman E. Gronlund suggested a two-step way of writing objectives. First write a **general instructional objective** in broad words (“Understands the principles of demand”). Then clarify it with a sample of **specific learning outcomes** in behavioural terms (“States the law of demand in own words”; “Gives an example of an exception to the law”). The single-sentence format with condition and criterion is Mager’s, not Gronlund’s.

Terminal, enabling and expressive objectives

- A **terminal objective** states what the learner can do at the end of instruction.
- **Enabling objectives** are the smaller, intermediate skills that lead up to it. They are found by **task analysis**: breaking the final task into its component skills.
- Elliot Eisner proposed **expressive objectives**. These describe an educational encounter – a visit, a reading, a performance – without fixing in advance what the outcome must be. They suit the arts, where a unique response is the point.

The RCEM approach

Indian teacher-education texts describe the **RCEM approach** as developed at the Regional College of Education, Mysore (now the Regional Institute of Education, Mysuru, a constituent unit of NCERT). It sorts objectives into four categories: **knowledge, understanding, application and creativity**. Instead of Bloom-style action verbs, it states objectives through specified **mental processes**, such as recalling, recognising, seeing relationships, classifying, inferring and predicting.

Levels of outcome in OBE

Outcome-based programmes in Indian colleges, especially those accredited by the National Board of Accreditation, use four related terms.

Term	What it states
Course outcomes (COs)	What a student can do at the end of one course (paper)
Programme outcomes (POs)	What every graduate of the programme can do at the time of graduation
Programme-specific outcomes (PSOs)	What graduates of this particular programme can do at graduation
Programme educational objectives (PEOs)	What graduates are expected to achieve in their careers a few years after graduation

Constructive alignment

John Biggs called the lining-up of outcomes, teaching and assessment **constructive alignment**. The teaching activities and the assessment tasks must call for the same kind of performance as the intended learning outcome. If the outcome says “evaluate arguments” but the examination asks only for definitions, alignment has failed. That is a failure of alignment, not of reliability. Assessment in OBE exists to find out whether the intended outcomes have been reached, which is why it must be aligned with them.

CHECK YOURSELF

1. Which of the following is a properly stated instructional objective?

(A) The teacher will explain the causes of inflation.
(B) Students will appreciate the importance of savings.
(C) Given data for five years, the student will calculate the rate of inflation for each year without error.
(D) Inflation and its causes.

2. Match List I (part of a Mager objective) with List II (example).

List I	List II
A. Performance	I. Using a scientific calculator
B. Condition	II. Solve quadratic equations
C. Criterion	III. Eight out of ten correct

Choose the correct answer from the options given below:

(A) A-I, B-II, C-III
(B) A-II, B-I, C-III
(C) A-III, B-I, C-II
(D) A-II, B-III, C-I

3. Given below are two statements.

Statement I: An aim is broad and long-term, while an instructional objective is narrow and testable at the end of a lesson or unit.

Statement II: Verbs such as “know” and “appreciate” are preferred in behavioural objectives because they describe the learner’s mind.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

4. Given below are two statements, one labelled Assertion (A) and the other Reason (R).

Assertion (A): In outcome-based education, assessment tasks must be aligned with the stated learning outcomes.

Reason (R): The purpose of assessment in outcome-based education is to find out whether the intended outcomes have been attained.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

5. In the ABCD format of writing objectives, the letter that Mager’s three-part format does not have a separate part for is:

(A) Behaviour
(B) Condition
(C) Degree
(D) Audience

6. Which of the following are functions of teaching?

1. Transmitting knowledge
2. Socialisation of the learner
3. Issuing certificates
4. Developing skills and values

Choose the correct answer from the options given below:

(A) 1, 2 and 4 only
(B) 1 and 4 only
(C) 1, 2, 3 and 4
(D) 2, 3 and 4 only

7. The statements of what graduates are expected to achieve in their careers a few years after completing a programme are called:

(A) course outcomes
(B) programme-specific outcomes
(C) programme educational objectives
(D) programme outcomes

1.4 Taxonomies of educational objectives

Syllabus: Teaching: Objectives

EXAM FOOTPRINT

Asked in most cycles; usually an item ordering the cognitive levels, matching levels to verbs, or identifying the domain an outcome belongs to.

Core

Three domains

A **taxonomy** is a classification arranged in order. The **taxonomy of educational objectives** (शैक्षिक उद्देश्यों का वर्गीकरण) sorts what we want learners to achieve into three **domains**, or areas:

Domain	What it covers	Example objective
Cognitive (संज्ञानात्मक पक्ष)	Knowing and thinking, from simple recall to judgement and creation	Solves simultaneous equations
Affective (भावात्मक पक्ष)	Feelings, interests, attitudes and values	Shows respect for others' opinions in a debate
Psychomotor (मनोगत्यात्मक पक्ष)	Physical skills needing coordination of mind and muscle	Operates a microscope; handles laboratory apparatus

A fourth word, **conative**, is sometimes met. It refers to will, effort and striving. It is not one of Bloom's three domains.

To decide the domain, look at the verb that carries the objective. "Values honesty in research" is affective, even though research sounds intellectual, because the verb is "values". "Explains the causes of plagiarism" is cognitive.

The books and their authors

Work	Domain	Year
Bloom and others, *Taxonomy of Educational Objectives, Handbook I*	Cognitive	1956
Krathwohl, Bloom and Masia, *Handbook II*	Affective	1964
Anderson and Krathwohl (editors), revised taxonomy	Cognitive (revised)	2001
E. J. Simpson; R. H. Dave; A. J. Harrow	Psychomotor (three separate schemes)	Simpson 1972; Dave (usually dated 1970); Harrow 1972

Bloom's group did not finish a psychomotor handbook; the psychomotor schemes came from others. Robert Gagné, sometimes offered in the options, is known for *The Conditions of Learning*, not for a taxonomy of the affective domain.

The cognitive domain: Bloom's original (1956)

Six levels, from simple to complex. Each level builds on the ones below it.

No.	Level	Meaning	Typical verbs
1	Knowledge	Recall of facts, terms, methods, principles	define, list, name, state, recall
2	Comprehension	Grasping meaning; using material in a different form	explain, summarise, translate, interpret
3	Application	Using a learnt rule or method in a new concrete situation	apply, solve, use, demonstrate
4	Analysis	Breaking a whole into parts and finding the relations between them, e.g. separating facts from opinions	analyse, compare, differentiate, distinguish
5	Synthesis	Putting elements together to form a new whole, pattern or structure	design, compose, construct, formulate
6	Evaluation	Judging the worth of something against criteria	judge, justify, critique, appraise

In 1956 the highest level was **Evaluation**, with Synthesis just below it. The lowest was **Knowledge**.

Comprehension has three forms in the original handbook:

- **Translation**: putting a communication into another form – words into a graph, English into Hindi, a formula into words.
- **Interpretation**: explaining or summarising the meaning.
- **Extrapolation**: extending a trend beyond the given data, to predict.

So changing a paragraph of data into a bar graph is Comprehension (translation), not Application.

Knowledge, Comprehension and Application are called **lower-order** levels; Analysis, Synthesis and Evaluation are **higher-order** levels (उच्च-स्तरीय चिंतन).

The revised taxonomy (2001)

Lorin Anderson and David Krathwohl led a revision published in 2001. Three changes matter for the examination.

1. **Nouns became verbs.** Knowledge became **Remember**, Comprehension became **Understand**, and so on, because objectives describe what learners do.
2. **Synthesis became Create and moved to the top.** Evaluation became **Evaluate** and is now fifth.
3. **A second dimension was added**: the knowledge dimension (below).

The revised order, from lowest to highest: **Remember, Understand, Apply, Analyse, Evaluate, Create.**

1956 (nouns)	2001 (verbs)
Knowledge	Remember
Comprehension	Understand
Application	Apply
Analysis	Analyse
Synthesis (5th)	Create (6th, highest)
Evaluation (6th, highest)	Evaluate (5th)

"Recall" and "recognise" are not level names. They are processes inside Remember.

Verbs for the revised levels

Level	Verbs and cognitive processes
Remember	recognising, recalling; define, list, name, state
Understand	interpreting, exemplifying, classifying, summarising, inferring, comparing, explaining
Apply	executing, implementing; solve, use, compute by a known procedure
Analyse	differentiating, organising, attributing; distinguish, break down
Evaluate	checking, critiquing; judge, justify, defend a choice against criteria
Create	generating, planning, producing; design, compose, formulate, construct

Two sorting rules settle most questions:

- **Analyse or Evaluate?** Analyse stops at taking the thing apart and seeing how the parts relate. Evaluate goes on to judge it against criteria and justify a choice.
- **Apply or Create?** Apply uses a known procedure on a new case. Create produces something new: a plan, a design, a proposal, a hypothesis.

The knowledge dimension and the taxonomy table

The revision added a **knowledge dimension** (ज्ञान आयाम) with four types of knowledge:

- **Factual**: terminology and specific details (the date of an Act, the meaning of a term).
- **Conceptual**: classifications, categories, principles, theories, models.
- **Procedural**: how to do something; methods, techniques, algorithms, criteria for choosing a method.
- **Metacognitive** (अधिसंज्ञानात्मक): knowledge of thinking in general and awareness of one's own thinking, including strategies and self-knowledge. This type was new in 2001.

Put the six cognitive processes along one side and the four knowledge types along the other, and you get the **taxonomy table**, a grid of 24 cells. In an objective, the **verb** gives the cognitive process

and the **noun** gives the type of knowledge. “Apply the procedure for chi-square” sits at Apply x procedural. “Judge one’s own study strategies” sits at Evaluate x metacognitive.

The affective domain (Krathwohl, Bloom and Masia, 1964)

Five levels, from lowest to highest:

Level	Meaning	Sign
Receiving	Being aware of something and willing to attend to it	Listens; pays attention
Responding	Reacting actively and willingly	Answers, participates, volunteers a comment, complies
Valuing	Attaching worth to something and showing commitment; acting on it consistently	Shows concern; supports; holds to a value under pressure
Organisation	Bringing several values together, resolving conflicts between them, building a consistent value system	Weighs freedom against responsibility; forms a personal code
Characterisation by a value or value complex	The value system governs behaviour so consistently that it becomes the person’s character	Consistently acts with integrity over years

Receiving is the lowest; Characterisation is the highest. Valuing concerns a single value; Organisation brings several values into one system; Characterisation is when that system controls conduct, consistently, over time.

Affective outcomes are hard to evaluate: attitudes and values are inside the person, develop slowly, and can be faked on a questionnaire. Characterisation in particular is judged by conduct observed over a long time, not by one written test.

The psychomotor domain

Three schemes are asked about. Learn at least the first and last level of each.

Author	Levels	Order
E. J. Simpson	7	Perception, Set, Guided response, Mechanism, Complex overt response, Adaptation, Origination
R. H. Dave	5	Imitation, Manipulation, Precision, Articulation, Naturalisation
A. J. Harrow	6	Reflex movements, Basic fundamental movements, Perceptual abilities, Physical abilities, Skilled movements, Non-discursive communication

Simpson begins with Perception and ends with Origination; Dave begins with Imitation and ends with Naturalisation; Harrow begins with Reflex movements and ends with Non-discursive communication. “Imitation” belongs to Dave’s psychomotor scheme, never to the affective domain.

The organising principle of each domain

Each domain is ordered by a different principle.

- Cognitive: **complexity** (simple to complex).
- Affective: **internalisation** (आंतरीकरण) – how deeply a value has become one’s own.
- Psychomotor: **coordination** (from single movements to smooth, combined, automatic ones).

CLASSROOM EXAMPLE

In an M.A. economics class in Rohtak, the teacher sets tasks on the Green Revolution. “List the main inputs of the Green Revolution” (Remember). “Explain in your own words what a high-yielding variety is” (Understand). “Compare farming in Punjab and Bihar and show which inputs made the difference” (Analyse). “Judge whether the Green Revolution helped small farmers, and defend your view using three criteria” (Evaluate). A further task, “Draft a plan to reduce stubble burning in Sirsa district”, would be Create. If, by the end of term, a student who used to ignore rural issues now volunteers for a survey of small farmers, that change is in the affective domain (Responding, moving towards Valuing).

OFTEN CONFUSED

- **Original top and revised top.** 1956: Evaluation highest, Synthesis fifth. 2001: Create (old Synthesis) highest, Evaluate fifth.
- **Analyse and Evaluate.** Analyse breaks into parts and relates them; Evaluate judges against criteria.
- **Comprehension (translation) and Application.** Changing words into a graph is translation, a form of Comprehension; Application uses a rule in a new situation.
- **Evaluate (cognitive) and Valuing (affective).** Justifying a choice against criteria is Evaluate; attaching worth and commitment to a value is Valuing.
- **Valuing and Organisation.** Valuing: one value; Organisation: several values built into a system.
- **Organisation and Characterisation.** Organisation builds the value system; Characterisation is when it governs conduct consistently.
- **Receiving and Responding.** Mere attention is Receiving; active reaction (answering, complying, volunteering) is Responding.
- **Imitation and Receiving.** Imitation is Dave’s first psychomotor level; Receiving is the first affective level.
- **Simpson and Dave.** Simpson: seven levels, Perception to Origination. Dave: five levels, Imitation to Naturalisation.
- **Complexity, internalisation, coordination.** Cognitive, affective and psychomotor respectively.

FACT WATCH

- The revised cognitive taxonomy of Anderson and Krathwohl (2001), with Create at the top, is the version in current use (checked October 2026). Many older books still print only the 1956 order with Evaluation at the top; if a question names the “revised” taxonomy, use the 2001 order.
- The affective handbook (Krathwohl, Bloom and Masia) is dated 1964; the cognitive handbook (Bloom and others) 1956.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Finer divisions of the 1956 handbook

- **Knowledge** has three groups: knowledge of specifics (terms, specific facts); knowledge of ways and means of dealing with specifics (conventions, trends, classifications, **criteria**, methodology); knowledge of universals and abstractions (principles, theories). So merely recalling the criteria for judging a poem is still Knowledge; using those criteria to judge a poem is Evaluation.
- **Comprehension**: translation, interpretation, extrapolation (above). Predicting next year's figure from the trend in a graph is extrapolation.
- **Synthesis** has three forms: production of a unique communication (an essay, a speech); production of a plan or proposed set of operations (a research design); derivation of a set of abstract relations (a hypothesis to explain data). All three correspond to Create in 2001.
- **Evaluation** has two forms: judgements in terms of internal evidence (consistency, accuracy within the work) and judgements in terms of external criteria (comparison with standards).

Hierarchy: strict or overlapping?

The 1956 authors treated the six levels as a **cumulative hierarchy**: each level was assumed to need mastery of all levels below it. Anderson and Krathwohl kept the order of increasing complexity but allowed the categories to **overlap**. For example, the Understand process "explaining" can be more demanding than the Apply process "executing".

Sub-levels of the lower affective levels

- **Receiving**: awareness; willingness to receive; controlled or selected attention.
- **Responding**: acquiescence in responding (complying because told to); willingness to respond; satisfaction in response.
- **Valuing**: acceptance of a value; preference for a value; commitment.

Compliant action ("submits the assignment because it is required") is already Responding (acquiescence). Mere attention is Receiving.

More on the psychomotor levels

Level	Meaning
Dave: Imitation	Copying an action after watching a model
Dave: Manipulation	Performing by following instructions, not just copying
Dave: Precision	Performing accurately and independently, with few errors
Dave: Articulation	Coordinating a series of actions harmoniously and consistently
Dave: Naturalisation	Performing automatically, with little conscious effort
Simpson: Perception	Using the senses to pick up cues that guide action
Simpson: Set	Readiness – mental, physical and emotional – to act
Simpson: Guided response	Early stage: imitation and trial and error under guidance
Simpson: Mechanism	Learnt responses become habitual; performed with some confidence
Simpson: Complex overt response	Skilful performance of complex movement patterns
Simpson: Adaptation	Modifying well-developed skills to fit special requirements or new situations
Simpson: Origination	Creating new movement patterns (the highest level)

Harrow's six levels move from involuntary reflexes, through basic movements such as walking and grasping, perceptual and physical abilities (strength, endurance, agility), to skilled movements and finally non-discursive communication: expressive movement such as gesture, posture and dance.

Higher-order items: a short calculation

Teachers are often asked to check what share of a test demands higher-order thinking. The method:

1. Fix the convention: which levels count as higher-order. Usually Analyse, Evaluate and Create. Some lists of "higher-order thinking skills" also count Apply; if the question states the convention, follow the question.
2. Add the items at those levels.
3. Divide by the total number of items and multiply by 100.

WORKED EXAMPLE

A 20-item test has 5 Remember, 4 Understand, 4 Apply, 3 Analyse, 2 Evaluate and 2 Create items. Taking Analyse, Evaluate and Create as higher-order:

Higher-order items = 3 + 2 + 2 = 7

Percentage = 7 / 20 x 100 = 35 per cent

If Apply were also counted: (4 + 7) / 20 x 100 = 55 per cent.

USUAL SLIPS

- Counting Apply as higher-order when the question has not said so.
- Dividing by the number of higher-order items instead of the total.
- Using the 1956 names in a question set on the revised taxonomy (Synthesis is not a 2001 level).

CHECK YOURSELF

1. Arrange the levels of the revised cognitive taxonomy (2001) from the lowest to the highest:

1. Evaluate
2. Apply
3. Create
4. Understand
5. Analyse

Choose the correct answer from the options given below:

(A) 4, 2, 1, 5, 3
(B) 4, 2, 5, 1, 3
(C) 2, 4, 5, 3, 1
(D) 4, 5, 2, 1, 3

2. Match List I (task) with List II (level of the revised taxonomy).

List I	List II
A. Summarise the plot of a novel in your own words	I. Create
B. Distinguish relevant from irrelevant evidence in a report	II. Evaluate
C. Defend the choice of one research design over another using stated criteria	III. Understand
D. Write an original short story	IV. Analyse

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-II, D-I
(B) A-IV, B-III, C-II, D-I
(C) A-III, B-II, C-IV, D-I
(D) A-III, B-IV, C-I, D-II

3. "The student consistently refuses to copy in examinations, even when no invigilator is present, and has done so throughout the course." This outcome belongs to the:

(A) cognitive domain, Evaluate
(B) psychomotor domain, Naturalisation
(C) affective domain, Characterisation by a value
(D) affective domain, Receiving

4. Given below are two statements, one labelled Assertion (A) and the other Reason (R).

Assertion (A): Changing a table of rainfall data into a line graph is an example of Comprehension in Bloom's 1956 taxonomy.

Reason (R): Translation, interpretation and extrapolation are the three forms of Comprehension.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

5. Which of the following statements are correct?

1. The affective domain is ordered by the principle of internalisation.
2. Metacognitive knowledge was added to the knowledge dimension in the 2001 revision.
3. Dave's psychomotor scheme begins with Perception.
4. In the 1956 taxonomy, Evaluation was the highest level.

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 2, 3 and 4 only
(C) 1 and 4 only
(D) 1, 2 and 4 only

6. In the objective "Students will evaluate their own note-making strategies", the cell of the taxonomy table is:

(A) Evaluate x metacognitive knowledge
(B) Apply x procedural knowledge
(C) Evaluate x procedural knowledge
(D) Remember x metacognitive knowledge

7. Given below are two statements.

Statement I: Simpson's psychomotor taxonomy ends with Naturalisation.

Statement II: Harrow's psychomotor taxonomy begins with Reflex movements.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

8. A 25-item test has 6 Remember, 5 Understand, 4 Apply, 4 Analyse, 3 Evaluate and 3 Create items. Taking only Analyse, Evaluate and Create as higher-order, the percentage of higher-order items is:

(A) 56 per cent
(B) 40 per cent
(C) 24 per cent
(D) 60 per cent

1.5 Levels of teaching: memory, understanding and reflective

Syllabus: Levels of teaching (Memory, Understanding and Reflective)

EXAM FOOTPRINT

Asked in most cycles; usually a match-the-list of levels and thinkers or a two-statement item on the reflective level.

Core

Three levels on one ladder

Teaching can be pitched at three **levels of teaching** (शिक्षण के स्तर). From the lowest to the highest they are:

1. **Memory level** (स्मृति स्तर): learners commit facts to memory and reproduce them.
2. **Understanding level** (बोध स्तर): learners see the relationships between facts and the general principles behind them, and can explain and use them.
3. **Reflective level** (चिंतनशील स्तर): learners face a real problem, think it through critically, and reach and test their own conclusions.

The levels form a **ladder**. Each higher level presupposes the one below it and includes it. You cannot understand facts you do not have, and you cannot reflect on a problem you do not understand. So the memory level is the base for the other two; the understanding level requires the memory level and prepares the ground for the reflective level; the reflective level is the highest and includes both lower levels. Memory-level teaching is never thrown away; it is built upon.

What separates the levels is **the amount of thinking demanded of the learner**. Morris L. Bigge described teaching-learning situations as lying on a continuum from "thoughtless" to "thoughtful". Memory-level teaching demands the least thought, the understanding level more, and the reflective level the most. The levels are not defined by the length of the period, the size of the class, or the age of the teacher.

Only these three are the standard levels. Labels such as "insight level" or "autonomous development level" in an option are not among the three levels the syllabus names.

Who is linked with each level

Level	Name linked with it	Model in one phrase
Memory	Johann Friedrich Herbart	Herbartian formal steps
Understanding	Henry Clinton Morrison	Morrison's unit plan and mastery
Reflective	Maurice P. Hunt (also discussed by Morris L. Bigge)	Problem-centred inquiry

A word on the reflective level, because books differ. The examination and most Indian textbooks link the reflective-level model with **Hunt**. Maurice P. Hunt wrote, with Lawrence E. Metcalf, *Teaching High School Social Studies: Problems in Reflective Thinking and Social Understanding* (1955). Morris L. Bigge set out the three levels in *Learning Theories for Teachers* (1964), and Bigge and Hunt wrote *Psychological Foundations of Education* (1962) together. So you will see "Hunt", "Bigge", or "Bigge and Hunt". In the examination: if Hunt is among the options, choose Hunt; if the options are Bigge with names like Herbart, Morrison, Bloom or Pavlov, choose Bigge. Benjamin Bloom is known for the taxonomy of objectives (1.4), not for a level of teaching. Pavlov and Thorndike are learning theorists.

Note the spelling: **Herbart**, not "Herbert".

The memory level

What it is. Teaching at the memory level aims to fix facts, terms, dates and formulae in the learner's memory so that they can be recalled and reproduced. It is the lowest and most "thoughtless" level. Learning here is largely **rote learning** (रटंत अधिगम): learning by repetition without seeing why.

Teacher and learner. The teacher is dominant. The teacher chooses the content, fixes its order, presents it and tests it. The learner is passive: he or she receives, retains and reproduces. Learners do not choose the content or its sequence. Of the three levels, teacher control is greatest here.

Methods. Telling, drill, repetition, recitation (saying back what was learnt) and review. Debates, seminars and projects need understanding or reflection and belong to higher levels.

Motivation. Mostly **extrinsic** (बाह्य अभिप्रेरणा): marks, praise, rewards and reproof.

Evaluation. Oral questioning and written tests of **recall and recognition** – that is, objective-type items such as fill in the blanks, true/false, matching and multiple choice.

Its basis: Herbart's apperception. Johann Friedrich Herbart (1776–1841), a German philosopher and educationist, held that the mind is built up of ideas. The body of ideas already in the mind he called the **apperceptive mass** (पूर्व-ज्ञान पुंज). A new idea is learnt when it is linked with, and absorbed into, this existing mass. This process is **apperception** (अवबोधन). An idea with no link to anything already known has nothing to hold on to. Herbart's teaching steps are designed to make this link in an orderly way. (Learning by stimulus-response connections is Thorndike's connectionism; equilibration is Piaget's idea. Neither is Herbart's.)

The Herbartian steps

Herbart himself spoke of four stages of instruction: **clearness** (clarity), **association**, **system** and **method**. His followers, chiefly Tuiskon Ziller and Wilhelm Rein, turned them into the **five formal steps** that teacher-training colleges still teach:

1. **Preparation.** The teacher recalls the learners' related previous knowledge, arouses their curiosity, and ends with a **statement of the aim** of the lesson. This prepares the apperceptive mass for the new idea.
2. **Presentation.** The new material is introduced, clearly and in order, with examples, aids and questions.
3. **Comparison and association.** The new facts are compared with one another and with what learners already know, so that links form.
4. **Generalisation.** From the facts, the learners and teacher frame a rule, definition or principle.
5. **Application.** The rule is used on fresh examples and new situations. This fixes it in the mind and tests whether it was grasped.

Some lists split the first step into "preparation" and "statement of aim", or add **recapitulation** (a short revision) at the end. The five-step core remains: preparation, presentation, comparison/association, generalisation, application. Note that **recitation** is not a Herbartian step; it is the last step of Morrison's plan.

The Herbartian approach is orderly and teacher-centred. It rests on the doctrine of apperception and uses recitation and drill. Its strength is that a meaningful, well-organised presentation improves retention.

Merits of the memory level. It supplies the facts that higher levels need. It suits young learners and beginners in any subject.

Limits. Material learnt by rote has few links to other ideas, so it fades quickly and cannot be used in new situations. Learners stay passive. It does not develop independent, critical thinking – that is the outcome of reflective-level teaching.

The understanding level

What it is. To understand is to see relationships: between facts and the generalisations that explain them, between a principle and its uses. A learner at this level can restate an idea in his or her own words, give a fresh example, relate it to earlier ideas, and apply a principle in **familiar** situations. Exact repetition of the textbook's words shows only the memory level; producing an original theory goes beyond understanding to reflection.

This level is more thoughtful than the memory level, because learners see relationships and use principles. But the thinking is still largely the teacher's: the teacher explains the relationships and the learner grasps them. No originality is demanded.

Its basis: Morrison's unit plan. Henry Clinton Morrison (1871–1945), an American educationist at the University of Chicago, set out his plan in *The Practice of Teaching in the Secondary School* (1926). He organised content into **units** – significant wholes of related material, not scattered daily lessons – and aimed at **mastery** of each unit. His **mastery formula** runs: pre-test, teach, test the result, adapt

procedure, and teach and test again to the point of actual learning. If a learner has not mastered the unit, the teacher changes the teaching procedure; the teacher does not simply repeat the same lesson or move on.

Morrison's five steps

1. **Exploration.** The teacher finds out what the learners already know about the unit, through questions, discussion or a pre-test, so as to start at the right point. The teacher also plans the unit in logical order.
2. **Presentation.** The teacher presents the main points of the unit in outline, briefly, and checks with short tests that the class has grasped them; parts not grasped are presented again.
3. **Assimilation.** The longest step. Learners study the unit in depth on their own – reading, library work, assignments, worksheets – under the teacher's supervision. The teacher's role here is supervisory: guiding, helping with difficulties and testing. Assimilation ends with a **mastery test**. Learners who fall short of the mastery standard go back to supervised study with corrective help and are tested again.
4. **Organisation.** Each learner writes an organised outline of the whole unit, **without help** from books, notes or the teacher. This shows whether the learner can arrange the unit as a connected whole.
5. **Recitation.** The last step. Learners present the mastered unit **orally** to the class, in their own words, followed by discussion. Recitation here does not mean word-for-word reproduction.

Order to remember: **E-P-A-O-R** (exploration, presentation, assimilation, organisation, recitation).

Teacher and learner. The teacher is active and explains; the learner is more active than at the memory level and interacts with the teacher. Teacher control is less than at the memory level.

Evaluation. Both **essay-type** and **objective-type** tests. Understanding shows itself both in explanation (best seen in essay answers) and in recognition and application (which objective items can test).

The reflective level

What it is. Reflective-level teaching is **problem-centred**. The teacher creates a problematic situation. Learners examine it critically, form hypotheses, test them against evidence and reach conclusions. The aim is critical and creative thinking, independence of judgement and a scientific attitude. It is the highest and most "thoughtful" level.

It begins with a felt problem. The problem must be genuine: a real difficulty or perplexity that unsettles what the learners already believe – for example, data that contradict their expectations. This creates **cognitive conflict**. Recalling an earlier lesson, copying dictated definitions or hearing a conclusion announced in advance creates no such tension, so no reflection follows.

Steps of reflective-level teaching.

1. Recognising and defining the problem.
2. Formulating hypotheses – tentative solutions to be tested.
3. Testing the hypotheses against evidence.
4. Drawing conclusions.

These steps follow the scientific method, which John Dewey described as reflective thinking. A **hypothesis** (परिकल्पना) is a tentative answer. Even a hypothesis that is rejected is useful: it rules out one possibility, sharpens the problem and trains the habit of reasoning from evidence.

Teacher and learner. The teacher is a **democratic guide**: neither an authority who supplies the answers nor a passive spectator. The teacher raises the problem, helps learners clarify it, keeps the inquiry rigorous by asking for evidence and counter-examples, and lets learners revise their own

conclusions. The teacher does not hand over hypotheses or conclusions. The **learner is the most active** participant. Teacher control is least at this level.

Climate. The classroom must be open and democratic, so that learners are free to doubt, question, propose alternatives and test ideas. No source – not even the textbook – is treated as the final authority. An authoritarian classroom, a fixed sequence or imposed answers leave nothing to reflect on.

Motivation. Mostly **intrinsic** (आंतरिक अभिप्रेरणा): it comes from the learner's own wish to solve the felt problem. The learners' **level of aspiration** needs to be reasonably high.

Evaluation. Mainly **essay-type** tests and open-ended tasks, since reasoning, originality and judgement show only in extended answers. Evaluation also covers attitudes, beliefs, involvement and critical ability, not knowledge alone.

Where it suits. It needs a base of knowledge, mature reasoning and independence, so it suits higher classes, postgraduate students and research. It is slow and takes more time than memory-level teaching, and it does not suit young or unprepared learners.

The three levels side by side

Point	Memory level	Understanding level	Reflective level
Linked with	Herbart	Morrison	Hunt (and Bigge)
Focus	Committing facts to memory; retention	Seeing relationships; mastery of units	Solving problems; critical and creative thinking
Steps (syntax)	Preparation, presentation, comparison/association, generalisation, application	Exploration, presentation, assimilation, organisation, recitation	Felt problem, hypotheses, testing, conclusion
Teacher	Dominant; chooses and presents	Active; explains relationships	Democratic guide; raises problems
Learner	Passive; receives and reproduces	More active; grasps and applies	Most active; investigates
Motivation	Mainly extrinsic	Extrinsic and intrinsic	Mainly intrinsic; high level of aspiration
Typical methods	Drill, repetition, recitation, review	Explanation, discussion, generalisation	Problem-solving, inquiry, seminar, project
Evaluation	Oral and written recall and recognition tests (objective type)	Both essay-type and objective-type tests	Mainly essay-type; also attitudes, involvement, critical ability
Climate	Teacher-controlled	Teacher-guided	Open and democratic

Indian textbooks often describe each level as a model of teaching with four parts: **focus** (its aim), **syntax** (its steps), **social system** (roles of teacher and learner, and motivation) and **support system** (how outcomes are evaluated). The table follows the same order.

CLASSROOM EXAMPLE

In a B.Sc. chemistry class at a college in Bhiwani, three teachers take up “hard water”. Teacher A dictates: “Hardness is caused by salts of calcium and magnesium; temporary hardness is removed by boiling.” Students repeat it until they can write it word for word – memory level. Teacher B shows why boiling removes bicarbonate hardness but not sulphate hardness, asks students to write the reactions themselves and to predict what will happen to a sample from their own village – understanding level. Teacher C brings two samples: one from a canal and one from a tube-well, and shows that soap lathers in the canal water but not in the tube-well water, though both look clear. “Why? Is the tube-well water safe to drink?” Students propose causes, design tests, argue over the results and present a short report – reflective level. Teacher C did not tell them the answer; she kept asking, “What is your evidence?”

OFTEN CONFUSED

- **Herbart and Morrison steps.** Herbart: preparation, presentation, comparison/association, generalisation, application. Morrison: exploration, presentation, assimilation, organisation, recitation. Only “presentation” is common to both.
- **Preparation and exploration.** Preparation (Herbart) recalls related previous knowledge and states the aim. Exploration (Morrison) finds out what learners already know of the unit.
- **Application and recitation.** Application is Herbart’s last step (using the rule on new examples). Recitation is Morrison’s last step (oral presentation of the mastered unit).
- **Assimilation and organisation.** Assimilation is supervised individual study ending in a mastery test. Organisation is writing an unaided outline of the unit.
- **Recitation (memory-level method) and Recitation (Morrison’s step).** As a memory-level method, recitation is saying back what was learnt. As Morrison’s fifth step, it is presenting the unit in one’s own words.
- **Understanding and reflective.** Understanding: the teacher explains the generalisation and learners grasp and apply it in familiar situations. Reflective: learners face an unfamiliar problem and reach and test the generalisation by their own inquiry.
- **Hunt and Herbart.** Hunt: reflective level. Herbart: memory level.
- **Extrinsic and intrinsic motivation.** Extrinsic (marks, praise) marks the memory level; intrinsic (the felt problem) marks the reflective level.
- **Objective tests and essay tests.** Memory: recall and recognition items. Understanding: both. Reflective: mainly essay type.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Why rote learning fades: Ausubel’s point

David Ausubel distinguished **rote learning** from **meaningful learning**. In rote learning the new material is stored on its own, without being linked to the ideas the learner already has. With few anchors in the learner’s cognitive structure, it is soon forgotten and seldom transferred to new situations. In meaningful learning the new idea is linked to relevant ideas already present, so it lasts and can be used. This is why memory-level teaching alone is not enough, and why the understanding level, which links new ideas to old, is the next step. (Notice how close this is to Herbart’s apperception.)

More on Herbart's original stages

Herbart's own four stages were clearness (presenting the new idea clearly), association (linking it with old ideas), system (arranging the ideas in order) and method (applying the system). The five-step version, usually credited to Rein, gave the familiar names: preparation, presentation, association (comparison), generalisation and application. By the end of the nineteenth century the steps were often followed mechanically, which drew criticism, including from John Dewey.

More on Morrison

- The mastery test at the end of assimilation is the turning point. A learner who falls short does not move on to organisation; he or she returns to supervised study with corrective help and is tested again. Only when mastery is shown do organisation and recitation follow.
- Morrison's plan is an early form of the later movements for **mastery learning** and individualised instruction.

Bigge on understanding

Indian textbooks, following Bigge, describe understanding in two complementary ways. First, it is seeing the relationship between particulars (facts) and generalisations (principles). Second, it is seeing the **"tool use"** of a fact: what it can be used for in reaching a goal. A student understands a formula when he or she knows what it is for and when to use it. Remembering the exact words in which it was taught is a memory-level outcome.

Dewey's five steps of reflective thinking

John Dewey, in *How We Think* (1910), analysed a complete act of reflective thought into five logically distinct steps:

1. A **felt difficulty**.
2. Its **location and definition**.
3. **Suggestion of a possible solution** (a hypothesis). The suggestion arises by an inductive leap from the facts.
4. **Development by reasoning** of the bearings of the suggestion: working out, deductively, what would follow if the hypothesis were true.
5. **Further observation and experiment** leading to its acceptance or rejection.

Reflection therefore begins in doubt, not in fixed premises. Dewey himself noted that the first two steps often run together and that the steps need not always come in a fixed order. The reflective level of teaching follows this pattern, which is why it is said to parallel the scientific method.

When an option names a fourth level

Some questions add an "autonomous development level" to the three standard ones and describe it as a permissive approach in which learners follow their own feelings with little direction from the teacher. This guide could not confirm the label in Bigge's own text, and it is not one of the three levels the syllabus names. If you meet it in a matching question, fix the three standard levels first – memory: commit facts to memory; understanding: grasp relationships explained by the teacher; reflective: explore a problem critically – and the remaining description falls into place. Critical exploration of a problem always belongs to the reflective level, never to a permissive, feeling-led approach.

CHECK YOURSELF

1. Which one of the following is the correct sequence of the steps of Morrison's unit plan?

(A) Exploration, Assimilation, Presentation, Organisation, Recitation
(B) Exploration, Presentation, Assimilation, Organisation, Recitation
(C) Preparation, Presentation, Comparison, Generalisation, Application
(D) Presentation, Exploration, Organisation, Assimilation, Recitation

2. Match List I (level of teaching) with List II (feature).

List I	List II
A. Memory level	I. Felt problem, hypotheses and testing
B. Understanding level	II. Preparation, presentation, comparison, generalisation, application
C. Reflective level	III. Mastery of units through supervised study
D. Herbartian steps	IV. Drill, repetition and recall

Choose the correct answer from the options given below:

(A) A-II, B-III, C-I, D-IV
(B) A-IV, B-I, C-III, D-II
(C) A-IV, B-III, C-I, D-II
(D) A-III, B-IV, C-I, D-II

3. Given below are two statements, one labelled Assertion (A) and the other Reason (R).

Assertion (A): Reflective-level teaching needs an open and democratic classroom climate.

Reason (R): The reflective-level model is associated with Hunt.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. Which of the following statements about the levels of teaching are correct?

1. Teacher control is greatest at the memory level and least at the reflective level.
2. Motivation at the reflective level is mainly extrinsic.
3. Understanding-level outcomes can be evaluated with both essay-type and objective-type tests.
4. The reflective level presupposes the memory and understanding levels.

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 2, 3 and 4 only
(C) 1 and 4 only
(D) 1, 3 and 4 only

5. In Morrison's plan, the step in which each learner writes an outline of the whole unit without the help of books, notes or the teacher is:

(A) organisation
(B) assimilation
(C) recitation

(D) exploration

6. Given below are two statements.

Statement I: In the Herbartian steps, preparation ends with a statement of the aim of the lesson.

Statement II: Recitation is the last of the Herbartian steps.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

7. A teacher of sociology gives her M.A. class survey figures showing that, in their district, the sex ratio is lower in richer families than in poorer ones – the opposite of what most students expected. She asks them to suggest explanations and to say what evidence would test each. At which level is she teaching, and what is the first step she has secured?

(A) Understanding level; generalisation
(B) Memory level; preparation
(C) Reflective level; a felt problem
(D) Understanding level; assimilation

8. Arrange the following steps of reflective thinking, as given by Dewey, in order:

1. Development by reasoning of the bearings of the suggestion
2. A felt difficulty
3. Suggestion of a possible solution
4. Further observation and experiment leading to acceptance or rejection
5. Location and definition of the difficulty

Choose the correct answer from the options given below:

(A) 2, 5, 1, 3, 4
(B) 5, 2, 3, 1, 4
(C) 2, 3, 5, 1, 4
(D) 2, 5, 3, 1, 4

1.6 Maxims of teaching and general methods (analytic, synthetic, inductive, deductive)

Syllabus: Teaching: Concept, Objectives, Levels of teaching (Memory, Understanding and Reflective), Characteristics and basic requirements

EXAM FOOTPRINT

Rarely asked directly; when it appears it is a single-answer item asking which maxim an example illustrates. Mainly supports scenario questions.

Core

What a maxim of teaching is

A **maxim of teaching** (शिक्षण सूत्र) is a short working rule that tells a teacher in what order to present material. Maxims grew out of long classroom experience and out of the psychology of learning. They do not tell you what to teach. They tell you where to start and in which direction to move.

Every accepted maxim has the form “from A to B”, where A is the easier starting point for the learner and B is the harder end point. This gives you a simple test in the examination. If an option starts from the harder end (complex, abstract, unknown, general), it has been reversed, and the reversed form is not an accepted maxim. “From complex to simple”, “from abstract to concrete” and “from unknown to known” are all wrong as maxims.

The maxims one by one

- **From known to unknown** (ज्ञात से अज्ञात की ओर). Begin with what learners already know and use it as a bridge to new material. A lesson on democracy that starts from the students’ own class elections follows this maxim. It rests on Herbart’s idea of apperception: new ideas are understood through old ones already in the mind (see 1.5). It also agrees with Ausubel’s stress on prior knowledge (see Chapter 2).
- **From simple to complex** (सरल से जटिल की ओर). Present easy elements before difficult ones, so that difficulty rises in steps. Simple sentences come before compound and complex sentences. The basis is the graded difficulty of tasks.
- **From concrete to abstract** (मूर्त से अमूर्त की ओर). Begin with things learners can see, touch, handle or picture, and only then give definitions, rules and theories, which are abstract. Counting beads before learning number symbols, or visiting a local fort before discussing feudalism, follows this maxim. Starting with a definition reverses it.
- **From particular to general** (विशिष्ट से सामान्य की ओर). Present several particular cases first and let the general rule be drawn from them. Measuring several triangles before stating that the angles add up to 180 degrees follows this maxim. It is the basis of the inductive method.
- **From whole to part** (पूर्ण से अंश की ओर). Give the complete picture first, then the details. Read a poem right through before studying its stanzas; show the map of India before taking up the states. The basis is Gestalt psychology, which holds that we perceive organised wholes before their parts. Starting with the parts and building up to the whole is the reverse.
- **From analysis to synthesis.** First break a whole into its elements so that each can be understood (analysis), then put the elements together again into a meaningful whole (synthesis). Breaking a complex sentence into clauses and then building new complex sentences follows this maxim. Learning is incomplete if it stops at analysis.
- **From empirical to rational** (अनुभवजन्य से तर्कसंगत की ओर). Empirical knowledge rests on observation and experience; rational knowledge rests on reasons and proof. Students first observe that the sun always rises in the east; later the teacher explains it by the earth’s rotation. This is one observed fact followed by its reasoned explanation, not a rule built from many cases.
- **From indefinite to definite** (अनिश्चित से निश्चित की ओर). A learner’s first ideas are vague. Teaching makes them clear and exact.
- **From psychological to logical** (मनोवैज्ञानिक से तार्किक की ओर). The psychological order follows the learner’s interests, needs, readiness and level of development. The logical order follows the structure of the subject as an expert sees it, for example the chapter order of a textbook. Teaching begins with the psychological order and moves gradually to the logical order.
- **From near to far.** Begin with the learner’s own surroundings, such as the village, district or state, and move outwards to distant places and times. “From remote to near” is the reversed form.
- **From induction to deduction.** First let learners reach a rule from examples, then let them apply it. This links the maxims to the methods described below.

Maxim	Classroom example	Idea it rests on
Known to unknown	Class elections before democracy	Herbart’s apperception; prior knowledge

Maxim	Classroom example	Idea it rests on
Simple to complex	Simple sentences before compound sentences	Graded difficulty
Concrete to abstract	Beads before number symbols	Things before ideas
Particular to general	Several triangles measured before the angle-sum rule	Inductive reasoning
Whole to part	Whole poem before its stanzas	Gestalt view of perception
Analysis to synthesis	Clauses separated, then new sentences built	Understanding parts, then rebuilding the whole
Empirical to rational	Sunrise observed, then explained	Experience before reasons
Psychological to logical	Learner's interest before textbook order	Learner before subject structure

Two maxims that look alike

"Concrete to abstract" and "particular to general" often work together in one lesson, so students take them to be one maxim. They are different. Concrete to abstract is about the **kind of experience**: from things to ideas. Particular to general is about **scope**: from single cases to a rule that covers all cases. A particular case can itself be abstract. For example, three worked algebra examples are particular cases, but none of them is a concrete object.

Inductive and deductive methods

The **inductive method** (आगमन विधि) moves from particular examples to a general rule. The teacher gives several examples, the learners observe what is common, and the learners themselves state the rule. A science teacher who lets students watch iron, copper and brass rods expand on heating, and only then states that metals expand on heating, is teaching inductively. Induction follows the maxims "particular to general", "concrete to abstract" and "known to unknown". It suits discovery and gives real understanding, but it is slow.

The **deductive method** (निगमन विधि) moves the other way, from the general rule to particular cases. The teacher states the rule, law or formula first, and the learners apply it to examples. Stating "metals expand on heating" and then verifying it with rods is deductive. Deduction moves from general to particular and from abstract to concrete. It covers content quickly and gives practice, but by itself it encourages memorising.

Point	Inductive method	Deductive method
Direction	Particular to general; examples to rule	General to particular; rule to examples
Also moves	Concrete to abstract	Abstract to concrete
Learner's role	Active; discovers the rule	Receives the rule; applies it
Time	Slow	Quick; covers more content
Best for	Understanding a new rule; discovery	Practice, verification, problem-solving, revision

The two methods are not rivals. Good teaching usually joins them in one order: **induction first, then deduction**. A teacher lets students infer the rule for forming plurals from a set of examples (induction), then gives exercises in which they apply the rule to new words (deduction). Induction secures understanding; deduction gives practice and tests the rule. It is therefore wrong to say that deduction has no place in the classroom.

CLASSROOM EXAMPLE

In a B.A. first-year English class at a college in Sirsa, the teacher writes on the board: “She has finished her work”, “They have gone home”, “I have read the notice”. She asks what the sentences share. Students notice “has” or “have” followed by a past participle, and that each action is complete but still matters now. Together they state the rule for the present perfect. This is induction: particular to general, and known to unknown, because the sentences come from the students’ own college life. The teacher then gives ten new situations and asks students to write present perfect sentences for each. This is deduction: the rule is applied to fresh cases.

OFTEN CONFUSED

- **Concrete to abstract vs particular to general.** The first is about the kind of experience (things to ideas). The second is about scope (cases to a rule). They are two separate maxims.
- **Particular to general vs empirical to rational.** Particular to general needs several cases leading to one rule. Empirical to rational takes an observed fact and gives the reasoned explanation for it.
- **Whole to part vs analysis to synthesis.** Whole to part says present the whole picture first. Analysis to synthesis says break the whole into elements and then rebuild it; the stress is on recombining.
- **Psychological order vs logical order.** Psychological follows the learner (interest, readiness). Logical follows the subject (its expert structure, the textbook sequence). Teaching moves from psychological to logical.
- **Inductive vs deductive.** Inductive: examples, then rule; slow; suits discovery. Deductive: rule, then examples; quick; suits practice and application.
- **Maxim “known to unknown” vs analytic method “unknown to known”.** The maxim is about linking new knowledge to old. The analytic method (see Going Further) is a direction of proof, from what is to be proved back to the given data. Do not mix them.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Analytic and synthetic methods

These two methods are used especially in mathematics, for proving theorems and solving problems. They put the maxim “analysis to synthesis” into practice.

The **analytic method** (विश्लेषण विधि) starts from the **unknown**, that is, the conclusion to be proved or the quantity to be found, and works backwards to the **known**, that is, the given data. At each step the teacher asks, “What must be true for this to hold?” The chain is followed back until it reaches something given. The method shows learners why each step is taken and how a proof is discovered. It is slow and the record of it can be untidy, but it develops reasoning.

The **synthetic method** (संश्लेषण विधि) starts from the **known** data and builds forward, step by step, to the **unknown** conclusion. It is the form in which a finished proof is written in a textbook. It is short, neat and logical, but it hides how the proof was found, so learners may memorise it without understanding.

Good practice again combines the two. The teacher first finds the path by analysis, discussing it with the class, and then writes the proof out by synthesis. Hence the maxim “from analysis to synthesis”.

Point	Analytic method	Synthetic method
Direction	Unknown to known (conclusion back to data)	Known to unknown (data forward to conclusion)
Shows	Why each step is taken; how the proof is found	The finished proof
Length	Long	Short
Learner	Discovers; thinks	Follows; may memorise

Now the four methods can be matched with their direction of reasoning, a favourite matching item:

Method	Direction
Inductive	Particular to general
Deductive	General to particular
Analytic	Unknown to known
Synthetic	Known to unknown

Two traps. First, deduction applies a general rule to cases; it is not the same as analysis, which works backwards from a conclusion. Second, the synthetic method, not the analytic, is the one that runs from known to unknown.

The bases of the maxims

Harder items ask on what idea a maxim rests. The explanations usually given are these. Known to unknown rests on Herbart's doctrine of apperception (new ideas are learnt through the mass of ideas already in the mind). Whole to part rests on the Gestalt view that wholes are perceived before parts. Simple to complex rests on the grading of tasks by difficulty. Particular to general rests on inductive reasoning. Known to unknown is also consistent with Ausubel's stress on the learner's prior knowledge.

CHECK YOURSELF

1. Which of the following is an accepted maxim of teaching?

(A) From general to particular
(B) From abstract to concrete
(C) From psychological to logical
(D) From complex to simple

2. Match the maxims in List I with the classroom practices in List II.

List I	List II
A. Known to unknown	I. Showing the whole map of Haryana before its districts
B. Whole to part	II. Starting a lesson on inflation from the price of vegetables last year and this year
C. Concrete to abstract	III. Handling coins of different weights before defining mass
D. Empirical to rational	IV. Noticing that ice floats, then explaining it by density

(A) A-II, B-I, C-III, D-IV
(B) A-I, B-II, C-III, D-IV
(C) A-II, B-I, C-IV, D-III
(D) A-III, B-I, C-II, D-IV

3. Statement I: The deductive method suits a lesson in which learners are expected to discover a rule for themselves. Statement II: The synthetic method moves from the known data to the unknown conclusion.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

4. Assertion (A): Good teaching usually follows induction with deduction. Reason (R): Induction helps learners understand a rule, and deduction gives practice in applying it.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

5. Which of the following statements are correct? 1. The analytic method moves from the unknown to the known. 2. “From particular to general” is the basis of the inductive method. 3. “From whole to part” rests on Gestalt psychology. 4. The logical order follows the interests of the learner. Choose the correct answer from the options given below:

(A) 2, 3 and 4 only
(B) 1, 2 and 3 only
(C) 1 and 4 only
(D) 1, 2, 3 and 4

6. A mathematics teacher writes the finished proof of a theorem neatly from the given data to the conclusion, without discussing how the proof was found. The method is:

(A) inductive
(B) analytic
(C) heuristic
(D) synthetic

7. Arrange the steps of a combined lesson on a grammar rule in the order good practice recommends: 1. Learners apply the rule to new sentences. 2. Learners study several example sentences. 3. Learners state the rule in their own words. 4. Learners notice what the examples have in common.
(A) 3, 2, 4, 1
(B) 2, 3, 4, 1
(C) 2, 4, 3, 1
(D) 1, 2, 4, 3

1.7 Characteristics of effective teaching and of a good teacher

Syllabus: Teaching: Concept, Objectives, Levels of teaching (Memory, Understanding and Reflective), Characteristics and basic requirements

EXAM FOOTPRINT

Asked often; usually a classroom scenario asking what the teacher should do, or a multiple-select item on qualities of an effective teacher.

Core

Effective teaching is judged by learning

Teaching exists to produce learning. So the final test of **effective teaching** (प्रभावी शिक्षण) is the change it brings about in learners: in their knowledge, skills and attitudes. The number of lectures delivered, the length of the teacher's notes, the speed of syllabus coverage and the silence of the class are not measures of effectiveness. Lectures and notes are inputs. A syllabus can be "covered" with very little learning, and a silent class may or may not be a learning class.

The characteristics examiners ask about

Most questions on this topic list some true characteristics of good teaching and mix in features of poor teaching. Learn the true list, and learn the opposite of each item, because the opposite is the usual distractor.

Good teaching is...	Poor teaching (the distractor) is...
Planned and purposeful: clear objectives and an organised plan	Unplanned; aimless
Interactive and two-way: the teacher explains and asks, learners respond and question	One-way; dominated by teacher talk
Based on feedback (प्रतिपुष्टि): the teacher adjusts to what learners have understood	Blind to how the class is responding
Flexible and adaptive: pace, example or method changes with the class	Rigid; one fixed procedure whatever happens

Good teaching is…	Poor teaching (the distractor) is…
Democratic: respects learners, invites questions and differing views	Authoritarian; discourages questions
Diagnostic and remedial: finds difficulties and helps overcome them	Assumes all learners learn in the same way and at the same pace
Linked with what learners already know	Unconnected with previous knowledge
Aimed at an independent learner	Keeps the learner dependent on the teacher
Judged by the learning achieved	Judged by content covered or lectures given

Now the reasons behind some of these, because Assertion-Reason items test the reasons.

- **Two-way communication and feedback.** Teaching is an interactive process. The teacher can match pace, examples and method to the class only by knowing how the class is responding. Questions, facial expressions, answers and test results supply this feedback. One-way talk gives the teacher no way of knowing whether the class has understood.
- **Flexibility.** When a teacher sees puzzled faces and replaces her planned example with a simpler one, she shows flexibility. Sticking to the plan whatever the response would put the plan above the learning.
- **Democratic teaching** (लोकतांत्रिक शिक्षण). It means treating learners with respect, giving them freedom to ask, to differ and to take part, and reasoning with them instead of commanding. It does not mean handing over the teacher's responsibilities (letting students fix examination dates, voting before every lesson) or leaving errors uncorrected.
- **Diagnostic and remedial** (निदानात्मक एवं उपचारात्मक). Good teaching locates learners' difficulties and gives help to overcome them. It does so precisely because learners differ in pace, style and background. If all learners learnt alike, diagnosis would be unnecessary. So a statement that "good teaching assumes all learn alike" is false, and it cannot be the reason for diagnostic teaching.
- **Independence as the aim.** Good teaching gives guidance suited to the learner's needs, but it gives that guidance so that it can be withdrawn. The aim is a learner who can learn alone. A statement that good teaching keeps learners permanently dependent is false.

The good teacher

Some characteristics belong to the teacher rather than to the lesson.

- **Reflective practice** (चिंतनशील अभ्यास). The teacher looks back on her own teaching, asks what worked and what did not, and changes future lessons. This is self-examination. Subject mastery and verbal fluency are good qualities too, but they describe what a teacher knows or how she speaks, not self-examination.
- **Non-verbal communication** (अशाब्दिक संप्रेषण). A teacher's message is carried not only by words but by voice, face, gesture, eye contact and movement. Learners read these cues for emphasis, interest and attitude. That is why the teacher's non-verbal behaviour affects what students learn.
- **Classroom management** (कक्षा प्रबंधन). Its aim is to create conditions in which students are engaged in learning for as much of the period as possible. It covers the organisation of space, time, routines and relationships. Good management prevents most problems before they arise. Its aim is not punishment, not silence for its own sake, and not finishing the syllabus early.

Leadership styles: Lewin, Lippitt and White

In classic experiments reported in 1939, Kurt Lewin, Lippitt and White studied groups of boys in after-school clubs led by adults who used three different **leadership styles** (नेतृत्व शैली).

Style	What happened
Authoritarian (autocratic)	Boys worked hard, but mainly while the leader was present and watching; there was hostility or submissiveness
Laissez-faire	The leader left the boys to themselves; little was done and the work was poor
Democratic	Work went on even when the leader was away; members were friendly and satisfied; the best combination of work and morale

The lesson for the classroom: a democratic climate gives both good work and good morale. A "bureaucratic" style was not one of the three styles studied.

CLASSROOM EXAMPLE

In a B.Com. class at a government college in Hisar, Dr Meena is explaining compound interest. Half-way through she notices that several students have stopped writing and look puzzled. She stops, asks one student to say what he has understood, and hears that the class has not grasped the idea of interest on interest. She drops her prepared formula and takes a simpler case: a savings account of 1,000 rupees left in a bank for three years. After the lesson she writes in her diary that next year she will start with the bank example. In this one lesson she showed feedback-based, flexible and diagnostic teaching, and, in the diary, reflective practice.

OFTEN CONFUSED

- **Reflective practice vs subject mastery.** Reflective practice is examining one's own teaching to improve it. Subject mastery is knowing the subject well.
- **Democratic vs laissez-faire.** Democratic: the teacher guides, reasons and invites participation, and still corrects errors. Laissez-faire: the teacher gives up direction, and little work is done.
- **Authoritarian vs democratic (Lewin, Lippitt and White).** Authoritarian groups worked mainly under watch. Democratic groups kept working in the leader's absence.
- **Classroom management vs discipline by punishment.** Management creates conditions for engaged learning time and prevents problems. Punishment reacts to problems after they occur.
- **Input vs outcome.** Lectures, notes and syllabus coverage are inputs. Learning is the outcome, and it is the outcome that judges teaching.
- **Withitness vs overlapping (Kounin).** Withitness is showing awareness of everything happening in the room. Overlapping is attending to two matters at the same time.
- **Presage vs process vs product.** Presage is what the teacher brings (training, traits). Process is classroom behaviour. Product is the outcome in pupils.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Kounin's findings on classroom management

Jacob Kounin's research on classroom management (published as Discipline and Group Management in Classrooms, 1970) found that good managers prevent problems rather than react to them. His terms are often asked:

- **Withitness**: the teacher communicates, by her behaviour, that she knows what is going on in every part of the room. A teacher helping one group who notices two students at the back drifting off task and checks them with a glance is showing withitness.
- **Overlapping**: attending to two or more matters at the same time, for example continuing to help one group while answering another student's question.
- **Momentum**: keeping the lesson moving at a suitable pace, without slowdowns such as dwelling on a point longer than needed.
- **Smoothness**: moving through the lesson and from one activity to the next without abrupt breaks or irrelevant digressions.

Presage, process and product variables

Research on teacher effectiveness often uses the model set out by Dunkin and Biddle in The Study of Teaching (1974). The terms presage, process and product had been used earlier by Harold E. Mitzel (1960, in the Encyclopedia of Educational Research) for criteria of teacher effectiveness; Dunkin and Biddle's model also includes context variables. It classifies variables as follows.

Variable	Meaning and examples
Presage	What the teacher brings to teaching: formative experiences, training, qualifications, personality traits
Context	The setting the teacher works in: the pupils (their abilities, background), the classroom, the school and the community
Process	What actually happens in the classroom: the observable behaviour of teacher and pupils
Product	The outcomes in pupils: achievement, attitudes, skills

So a teacher's qualifications, training and personality are presage variables, not context variables. Context is the setting, not what the teacher brings to it.

A department that judges its teachers only by pass percentage makes a common mistake. Pass percentage is a product measure, and product measures have many causes besides the teacher: the pupils' entry level and motivation, class size and facilities. Comparing teachers on raw results is therefore unfair. It is fairer to take such factors into account, or to measure the gain made by each class (a "value-added" measure) rather than the final level alone.

Seven principles of good practice in undergraduate education

Chickering and Gamson published "Seven Principles for Good Practice in Undergraduate Education" in the AAHE Bulletin in March 1987. Good practice:

1. encourages contact between students and faculty;
2. develops reciprocity and cooperation among students;
3. encourages active learning;
4. gives prompt feedback;
5. emphasises time on task;
6. communicates high expectations;
7. respects diverse talents and ways of learning.

Public ranking of students to sharpen competition is not among them; it works against cooperation among students.

Immediacy, clarity, credibility and efficacy

Immediacy is a concept from the psychologist Mehrabian: behaviours that reduce the physical or psychological distance between people. Andersen applied it to teaching in 1979. **Teacher immediacy** covers non-verbal behaviours such as smiling, eye contact, a relaxed posture and moving among students, and verbal behaviours such as calling students by name. It is associated with students' liking for the course and their motivation. Do not confuse it with three neighbouring terms. **Teacher clarity** is how understandable the teacher's explanations are. **Teacher credibility** is how competent and trustworthy the teacher appears to students. **Teacher efficacy** is the teacher's own belief in her capability to bring about learning.

CHECK YOURSELF

1. Which of the following are characteristics of good teaching? 1. It provides feedback to learners. 2. It is judged by the amount of syllabus covered. 3. It adapts to the responses of learners. 4. It treats all learners as if they learn alike. Choose the correct answer from the options given below:

(A) 1 and 2 only
(B) 1 and 3 only
(C) 2 and 4 only
(D) 1, 3 and 4 only

2. Assertion (A): Good teaching is diagnostic and remedial. Reason (R): Learners in any class differ in pace, style and background.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

3. Match the terms in List I with their meanings in List II.

List I	List II
A. Withitness	I. Doing two things at the same time
B. Overlapping	II. Keeping the lesson moving without slowdowns
C. Momentum	III. Showing awareness of all that happens in the room
D. Smoothness	IV. Moving between activities without abrupt breaks

(A) A-I, B-III, C-II, D-IV
(B) A-III, B-I, C-IV, D-II
(C) A-III, B-I, C-II, D-IV
(D) A-II, B-I, C-III, D-IV

4. Statement I: In the studies of Lewin, Lippitt and White, work in the authoritarian groups continued steadily when the leader left. Statement II: Democratic teaching means that the teacher leaves students' errors uncorrected.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

5. A researcher records the number of years of training of each teacher in a district. In the Dunkin and Biddle model, this is a:

(A) product variable
(B) process variable
(C) context variable
(D) presage variable

6. Which of the following is NOT one of Chickering and Gamson's seven principles?

(A) Emphasising time on task

(B) Reducing contact between students and faculty to save time
(C) Developing cooperation among students
(D) Respecting diverse talents and ways of learning

7. A teacher learns every student's name in the first week, smiles often and walks between the rows while teaching. She is showing:
(A) teacher credibility
(B) teacher clarity
(C) teacher immediacy
(D) teacher efficacy

1.8 Basic requirements of teaching: teaching skills, micro-teaching and classroom interaction

Syllabus: Teaching: Concept, Objectives, Levels of teaching (Memory, Understanding and Reflective), Characteristics and basic requirements

EXAM FOOTPRINT

Asked fairly often; usually a single-answer item on Flanders' interaction categories or a sequence of the micro-teaching cycle.

Core

The basic requirements of teaching

A teacher needs a small number of things before any teaching can work. Questions usually give a list and ask which items are basic requirements. Learn these:

- **Mastery of the content.** The teacher must know the subject well.
- **Pedagogical skill.** The teacher must know how to teach it: planning, explaining, questioning, managing the class and assessing. Knowledge of content without pedagogical skill is not enough; a scholar who cannot communicate does not teach effectively.
- **Knowing the learner.** The teacher must find out what learners already know, how they differ and what they need, and fit the lesson to them. The knowledge and skill a learner brings to a new lesson is called **entry behaviour** (प्रवेश व्यवहार). It must be known so that new learning can start where the learners are. Knowledge of learners' psychology is as much a requirement as knowledge of the content.
- **Communication skill.** Teaching is carried on through communication, so the teacher's message must reach learners clearly.
- **Planning.** Objectives, content, method and assessment are thought out in advance.
- **Fair evaluation.** Assessment is part of teaching, and it must be transparent and accountable.

These are not requirements: costly equipment (helpful, but not essential), seniority, salary grade, appearance, an administrative post, authority to award grades without scrutiny, and a belief that learners' ability is fixed. That last belief lowers the teacher's expectations and works against the learner.

A teacher's **professional competence**, then, consists chiefly of subject knowledge joined to pedagogical skill. A teacher shows that she "knows the learner" when she gauges prior knowledge and adjusts examples and pace; finishing the syllabus early, dictating notes or setting a paper few can pass show nothing of the kind.

Two separate requirements do not explain each other. “A teacher must know the entry behaviour of learners” and “A teacher must master the subject” are both true, but the second is not the reason for the first.

Teaching skills

A **teaching skill** (शिक्षण कौशल) is a small, definable group of teacher behaviours that helps learning, for example asking a good question or closing a lesson well. Teacher-training programmes break teaching into such skills so that each can be practised and improved. The skills most often asked are these.

Skill	Purpose	Typical components or signs
Set induction (प्रस्तावना कौशल), the skill of introducing a lesson	Prepares learners at the start: arouses interest and links the new topic with previous knowledge	Linking with previous knowledge; a question, story or problem that leads into the topic
Explaining	Makes an idea, process or relationship understood	Clear beginning and concluding statements; explaining links; fluency; checking understanding
Illustrating with examples	Clarifies an abstract idea by connecting it with the familiar	Simple, relevant, interesting examples; examples from the learners’ own life
Questioning	Makes learners think and take part	Clear and relevant questions; pausing; distribution of questions over the class
Probing questioning	Deepens a student’s partial or shallow answer	Prompting, seeking further information, refocusing, redirection, asking for justification
Reinforcement (पुनर्बलन)	Encourages desired responses so that students take part more	Praise words (“Good”, “Well explained”), nods, smiles, accepting and using answers
Stimulus variation (उद्दीपन परिवर्तन)	Secures and holds attention throughout the lesson	Movement, gestures, change in speech pattern, pausing, focusing, change in interaction style, switching between oral and visual channels
Closure (समापन), achieving closure	Consolidates learning at the end of the lesson	Summarising the main points; relating them to earlier learning; pointing ahead

Some points that decide many questions:

- Set induction and stimulus variation both concern attention. Set induction works **only at the opening** of a lesson and links the topic with previous knowledge. Stimulus variation works **throughout** the lesson, by changes in the teacher’s behaviour.
- A teacher who says “Good, well explained” after a correct answer is using reinforcement. A teacher who, after a partly correct answer, asks “Can you say more? Why do you think so?” is using **probing** (खोजपूर्ण प्रश्न). Explaining would mean the teacher supplies the missing part herself.
- Probing works on a student’s answer. Illustrating with examples works on the idea itself, for the whole class.
- In questioning, put the question to the whole class, pause, and only then name a student. If you name the student first, the rest relax because they know they will not be asked. Asking first and pausing keeps everyone thinking.

Convergent and divergent questions

A **convergent question** has one correct answer, which is recalled or computed: “Who issued the guidelines for …?”, “What is meant by …?”, or a calculation with one result. A **divergent question** has many acceptable answers and calls for prediction, imagination or reasoning: “What might happen to university teaching if examinations were abolished?” Divergent questions are usually higher-order. A question that needs a calculation is still convergent if it has a single right answer.

Micro-teaching

Micro-teaching (सूक्ष्म शिक्षण) is a scaled-down teaching encounter used in teacher training. It was developed at Stanford University, USA, by Dwight W. Allen and his colleagues; the year usually given is 1963. Its purpose is to give trainee teachers controlled practice in one teaching skill at a time, with immediate feedback. It was later adapted for Indian teacher education. It has nothing to do with teaching small children, with very short quizzes or with miniature models.

Micro-teaching scales teaching down in three ways:

- **Class size**: about 5 to 10 pupils, who may be real pupils or fellow trainees acting as pupils.
- **Lesson length**: about 5 to 10 minutes. (A normal college period is about 40 to 45 minutes.)
- **Task**: one teaching skill at a time.

One thing is **not** scaled down: **feedback** (प्रतिपुष्टि). Feedback is the heart of the technique. It is immediate and detailed, and it comes from the supervisor, from peers, or from a video recording.

The **micro-teaching cycle** has six steps, always in this order:

1. Plan: the trainee plans a short lesson on one skill.
2. Teach: she teaches it to a small group.
3. Feedback: supervisor and peers tell her how she used the skill.
4. Re-plan: she revises the lesson in the light of the feedback.
5. Re-teach: she teaches the revised lesson, usually to a different group of pupils.
6. Re-feedback: she receives feedback again.

Feedback must come before re-planning; otherwise there is nothing to revise against. When the trainee is satisfied, the cycle is repeated with the next skill.

The Indian model of micro-teaching

In India the technique was adapted for conditions of limited resources. The adaptation is associated with B. K. Passi and NCERT. Its features: fellow trainees act as pupils (simulated conditions), feedback comes from supervisors and peer observers using observation schedules, and no costly equipment is needed. The full cycle and the one-skill principle are kept. The timings of one cycle, after the lesson has been planned, are:

Step	Minutes
Teach	6
Feedback	6
Re-plan	12
Re-teach	6
Re-feedback	6
Total for one cycle	36

Learn it as “6, 6, 12, 6, 6”. Re-planning is the only step that takes 12 minutes. The initial planning is done before the timed cycle and is not counted in the 36 minutes.

WORKED EXAMPLE

Method for any micro-teaching timing question.

1. Write the five timed steps: teach 6, feedback 6, re-plan 12, re-teach 6, re-feedback 6.
2. Add them: one cycle = 36 minutes.
3. Convert the session to minutes (hours x 60).
4. Divide by 36. The whole number is the number of complete cycles; the remainder is time left over.

Example 1. How long is one cycle of the Indian model?

6 + 6 + 12 + 6 + 6 = 36 minutes

Example 2. How many complete cycles fit into a three-hour laboratory session?

3 hours = 3 x 60 = 180 minutes

180 / 36 = 5 exactly

Answer: 5 complete cycles

Example 3. How many complete cycles fit into a two-hour session, and how much time is left?

2 hours = 120 minutes

120 / 36 = 3, remainder 12 (because 3 x 36 = 108 and 120 - 108 = 12)

Answer: 3 complete cycles, with 12 minutes left

Example 4. What share of one cycle goes to re-planning?

12 / 36 = 1/3, that is, about 33.3 per cent

USUAL SLIPS

- Leaving out re-feedback and taking the cycle as 30 minutes. With 30, a three-hour session seems to give 6 cycles; the correct answer is 5.
- Giving re-planning 6 minutes like the other steps. It takes 12.
- Counting the initial planning inside the 36 minutes. It is done before the cycle.
- Rounding up. If 120 minutes give 3.33 cycles, only 3 cycles are complete.
- Putting re-plan before feedback in a sequencing item. Feedback always comes first.

Simulated teaching and link practice

In **simulated teaching** (अनुरूपित शिक्षण) a small group of trainees practise among themselves. One teaches, some act as pupils and others observe and record; then the roles are rotated. Each trainee thus plays, by turns, the roles of teacher, pupil and observer.

Micro-teaching trains skills one at a time, but real teaching needs them together. **Link practice**, also called integration of skills, bridges this gap. After practising several skills separately, the trainee combines them in lessons of intermediate length (about 20 minutes) with groups of intermediate size, before going to a real classroom.

Flanders' Interaction Analysis

Interaction analysis (अंतःक्रिया विश्लेषण) is the systematic observation and recording of classroom interaction. The best-known system is Flanders' **Interaction Analysis Categories** (FIAC, often written FIACS for the category system), developed at the University of Minnesota in the late 1950s. It studies **verbal interaction in the classroom** only. It does not record non-verbal behaviour, written work, textbooks or test items; recording only talk is one of its accepted limitations.

The system has **ten categories**:

Category	Group	Behaviour
1	Teacher talk: indirect	Accepts feelings
2	Teacher talk: indirect	Praises or encourages
3	Teacher talk: indirect	Accepts or uses ideas of students
4	Teacher talk: indirect	Asks questions
5	Teacher talk: direct	Lectures (gives facts or own opinions)
6	Teacher talk: direct	Gives directions or commands
7	Teacher talk: direct	Criticises or justifies authority
8	Student talk	Response (answers the teacher)
9	Student talk	Initiation (student starts the talk)
10	Silence or confusion	Pauses, short silences, talk that cannot be understood

So there are seven teacher categories (four indirect, three direct), two student categories and one for silence or confusion. **Indirect influence** (categories 1 to 4) invites and widens students' participation. **Direct influence** (categories 5 to 7) restricts students' freedom to respond. Two traps: asking questions (4) comes from the teacher but is classed as indirect, because it invites participation; lecturing (5) feels neutral but is classed as direct.

How the observer records. Every **three seconds** the observer writes down the number of the category that best describes the talk at that moment. That gives about 20 tallies a minute. The sequence of numbers is then entered, in pairs, into a ten-by-ten matrix, from which percentages and ratios are worked out. The record shows how far the teacher's influence was direct or indirect.

WORKED EXAMPLE

Method for Flanders calculations.

1. Add the tallies in each group: indirect (1 to 4), direct (5 to 7), student talk (8 and 9), silence (10).
2. Teacher talk = indirect + direct, that is, categories 1 to 7.
3. Total = all ten categories.
4. Teacher talk per cent = teacher talk / total x 100. Student talk per cent = (8 + 9) / total x 100. Silence per cent = 10 / total x 100.
5. I/D ratio = indirect / direct = (tallies in 1 to 4) / (tallies in 5 to 7). Indirect goes on top.
6. Time observed in minutes = total tallies x 3 / 60, that is, total / 20.

Example 1. Indirect tallies 120, direct 240. Find the I/D ratio.

I/D = 120 / 240 = 0.5

A ratio below 1 means the teacher's influence was more direct than indirect.

Example 2. A record shows 150 tallies in categories 1 to 4, 300 in 5 to 7, 120 in 8 and 9, and 30 in 10.

Teacher talk = 150 + 300 = 450

Total = 150 + 300 + 120 + 30 = 600

Teacher talk = 450 / 600 x 100 = 75 per cent

Student talk = 120 / 600 x 100 = 20 per cent

Silence or confusion = 30 / 600 x 100 = 5 per cent

I/D = 150 / 300 = 0.5

Time observed = 600 / 20 = 30 minutes

Example 3. In a lesson at a university department the tallies are: category 1: 5, 2: 25, 3: 30, 4: 60, 5: 300, 6: 40, 7: 20, 8: 90, 9: 30, 10: 40.

Indirect = 5 + 25 + 30 + 60 = 120

Direct = 300 + 40 + 20 = 360

Teacher talk = 120 + 360 = 480

Student talk = 90 + 30 = 120

Total = 480 + 120 + 40 = 640

Teacher talk = 480 / 640 x 100 = 75 per cent

Student talk = 120 / 640 x 100 = 18.75 per cent

Silence = 40 / 640 x 100 = 6.25 per cent

I/D = 120 / 360 = 0.33 (one-third)

Time observed = 640 / 20 = 32 minutes

Example 4. How many tallies does a 40-minute observation produce?

40 minutes x 60 = 2,400 seconds; 2,400 / 3 = 800 tallies

USUAL SLIPS

- Counting only the direct categories (5 to 7) as teacher talk. Indirect categories 1 to 4 are teacher talk too. In Example 2 this slip gives 50 per cent instead of 75.
- Inverting the ratio. I/D puts indirect on top. 240 / 120 = 2.0 is D/I, not I/D.
- Putting category 4 (asking questions) under direct influence. It is indirect.
- Leaving category 10 out of the total when finding percentages. The total includes all ten categories.
- Taking the interval as 10, 30 or 60 seconds. It is 3 seconds.

CLASSROOM EXAMPLE

At a college of education in Rohtak, B.Ed. trainees practise the skill of reinforcement. Priya plans a six-minute lesson on the parts of a flower and teaches it to five classmates acting as pupils, while two others fill in an observation schedule. In the six-minute feedback the supervisor reports that she praised only two answers, always with the same word, “Good”. In twelve minutes she re-plans, adding varied praise words and a nod or smile for each attempt. She re-teaches to a different group of five classmates, and the re-feedback shows that she reinforced nine responses in varied ways. The whole cycle took 36 minutes.

OFTEN CONFUSED

- **Set induction vs closure.** Set induction opens the lesson and links it with previous knowledge. Closure ends the lesson and consolidates.
- **Set induction vs stimulus variation.** Both serve attention; set induction only at the start, stimulus variation throughout the lesson.
- **Reinforcement vs probing.** Reinforcement praises or accepts a response. Probing asks follow-up questions to deepen it.
- **Probing vs illustrating.** Probing works on a student's answer. Illustrating uses examples to clarify the idea for the whole class.
- **Convergent vs divergent questions.** Convergent: one correct answer, even if it needs calculation. Divergent: many acceptable answers; usually higher-order.
- **Micro-teaching vs simulated teaching.** Micro-teaching is the scaled-down cycle for one skill with feedback. Simulated teaching is role-play among trainees (teacher, pupil, observer, rotated); the Indian model of micro-teaching uses simulated conditions.
- **Micro-teaching vs link practice.** Micro-teaching: one skill at a time. Link practice: several skills combined in a longer lesson with a larger group.
- **Indirect vs direct influence (Flanders).** Indirect: categories 1 to 4 (accepting feelings, praising, using students' ideas, asking questions). Direct: 5 to 7 (lecturing, directing, criticising).
- **Student response (8) vs student initiation (9).** Response answers the teacher. Initiation is talk the student starts.
- **Stanford model vs Indian model.** Stanford: real pupils and videotape feedback. Indian: peers as pupils, feedback from supervisors and peers, no costly equipment; same cycle and one-skill principle.
- **PCK vs TPACK.** PCK (Shulman) joins content and pedagogy. TPACK (Mishra and Koehler) adds technology.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Pedagogical content knowledge and TPACK

Lee Shulman, in "Those Who Understand: Knowledge Growth in Teaching" (Educational Researcher, 1986), distinguished three kinds of content knowledge a teacher needs: subject matter knowledge, **pedagogical content knowledge** (शैक्षणिक विषयवस्तु ज्ञान) and curricular knowledge. Pedagogical content knowledge, PCK for short, is the special blend of content and pedagogy that a good teacher has for a particular topic: the most useful examples, analogies and explanations, and the difficulties learners usually have with that topic. It is distinct from knowing the subject and from general teaching skill. It is knowing how to make this subject understandable to these learners.

TPACK (Technological Pedagogical Content Knowledge) was proposed by Mishra and Koehler in Teachers College Record in 2006. It extends Shulman's idea by adding technology: the knowledge a teacher needs lies at the intersection of technology, pedagogy and content. Do not confuse it with three other acronyms. **SAMR**, proposed by Ruben Puentedura, classifies levels of technology use (substitution, augmentation, modification, redefinition); it does not describe teacher knowledge. **ADDIE** and **ASSURE** are models of instructional design.

More on the skills

Explaining. Explaining means showing how statements are related. Words and phrases such as "because", "therefore", "hence" and "as a result" carry these relations and are called **explaining links**. An explanation that is a string of correct statements with no such links leaves listeners to guess the connections. Other components are clear beginning and concluding statements, fluency, and testing the learners' understanding. Planned repetition, which stresses key points for retention, does not supply causal links.

Probing. Its components are:

- **Prompting**: giving hints when the student gives no answer or a wrong answer.
- **Seeking further information**: asking the student to add to or clarify a partly correct answer.
- **Refocusing**: asking the student to relate a correct answer to another situation or topic.
- **Redirection**: putting the same question to other students, to obtain more answers or viewpoints.
- **Increasing critical awareness**: asking the student to justify the answer ("Why?", "How do you know?").

Wait time. Mary Budd Rowe's studies of science classrooms found that teachers typically waited less than one second after asking a question before speaking again. When teachers waited about three seconds or more, students gave longer and more thoughtful answers, and more students answered. Calling only on volunteers does not ensure fair participation: volunteers are usually the same few confident students. Questions should be distributed across the class.

Phases and feedback in micro-teaching

Micro-teaching as a training programme has three phases:

1. **Knowledge acquisition phase**: the trainee learns about the skill by reading, discussion and observing a model lesson that demonstrates it.
2. **Skill acquisition phase**: the trainee plans, teaches, receives feedback, re-plans and re-teaches, that is, goes through the cycle.
3. **Transfer phase**: the trainee uses the skill in real classroom teaching.

So watching and discussing a model lesson before attempting the skill belongs to the knowledge acquisition phase, not to skill acquisition.

Of the sources of feedback, a **video recording** is regarded as the most objective. It shows exactly what happened, can be replayed and stopped at any point, and is free from the observer's memory and bias. Supervisors and peers give useful judgements, but these are filtered through their perception.

Useful feedback is **specific, descriptive and tied to the skill practised**, and it points to a concrete next step. "Ten of your twelve questions asked for recall; add two that ask why" is useful feedback. "Your lesson was quite good" or "You need more confidence" gives the trainee nothing to act on.

Flanders: the rule of two-thirds and ratios

From observations with his system Flanders stated a **rule of two-thirds**:

1. about two-thirds of classroom time someone is talking;
2. about two-thirds of that talk is teacher talk;
3. about two-thirds of the teacher's talk is direct influence (lecturing, directing, criticising).

So about two-thirds of the talk in a typical classroom is teacher talk, and the third part of the rule is about direct influence, not indirect. The rule shows how teacher-dominated ordinary classrooms are.

WORKED EXAMPLE

Using the rule of two-thirds.

Share of class time that is teacher talk = 2/3 x 2/3 = 4/9, about 44 per cent

Share of class time that is direct teacher talk = 2/3 x 2/3 x 2/3 = 8/27, about 30 per cent (29.6)

Reading an I/D ratio.

Indirect 200, direct 160: I/D = 200 / 160 = 1.25. Above 1, so the teacher's influence was more indirect than direct.

Indirect 90, direct 270: I/D = 90 / 270 = 0.33. Below 1, so more direct.

CHECK YOURSELF

1. Which of the following is NOT reduced in micro-teaching?

(A) The number of pupils
(B) The detail of feedback
(C) The duration of the lesson
(D) The number of skills practised

2. Arrange the steps of the micro-teaching cycle in order: 1. Re-feedback 2. Teach 3. Re-plan 4. Plan 5. Feedback 6. Re-teach

(A) 4, 2, 3, 5, 6, 1
(B) 4, 2, 5, 3, 6, 1
(C) 2, 4, 5, 3, 6, 1
(D) 4, 5, 2, 3, 6, 1

3. A college of education runs Indian-model micro-teaching cycles (teach 6, feedback 6, re-plan 12, re-teach 6, re-feedback 6 minutes) in a session of 2 hours 24 minutes. How many complete cycles can it run?

(A) 3
(B) 5
(C) 4
(D) 6

4. Match the skills in List I with their components in List II.

List I	List II
A. Probing	I. Moving about the room and changing the pitch of the voice
B. Stimulus variation	II. Redirecting the same question to another student
C. Explaining	III. Summarising the main points at the end
D. Closure	IV. Using links such as “therefore” and “as a result”

(A) A-II, B-I, C-IV, D-III
(B) A-I, B-II, C-IV, D-III
(C) A-II, B-I, C-III, D-IV
(D) A-IV, B-I, C-II, D-III

5. A Flanders record has 90 tallies in categories 1 to 4, 270 in 5 to 7, 180 in 8 and 9, and 60 in 10. Which of the following are correct? 1. Teacher talk is 60 per cent. 2. Student talk is 30 per cent. 3. The I/D ratio is 3.0. 4. About 30 minutes were observed. Choose the correct answer from the options given below:

(A) 1 and 3 only
(B) 2, 3 and 4 only
(C) 1, 2 and 3 only
(D) 1, 2 and 4 only

6. Assertion (A): In Flanders’ system, asking questions is classified as indirect influence. Reason (R): Questions invite students to participate in the lesson.

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

7. Statement I: Shulman's pedagogical content knowledge is the same thing as a teacher's knowledge of the latest research in the discipline. Statement II: In TPACK, technology is added to Shulman's pairing of content and pedagogy.
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

8. Which question is divergent?
(A) "What is the full form of SWAYAM?"
(B) "In which year was the University Grants Commission Act passed?"
(C) "How might a village library change the reading habits of its young people?"
(D) "What is 6 + 6 + 12 + 6 + 6?"

1.9 Models of teaching

Syllabus: Teaching: Concept, Objectives, Levels of teaching (Memory, Understanding and Reflective), Characteristics and basic requirements

EXAM FOOTPRINT

Asked occasionally; usually a match-the-list of models with proponents or families, or a sequence of steps in a basic teaching model.

Core

What a model of teaching is

A **model of teaching** (शिक्षण प्रतिमान) is a planned pattern for teaching that sets out what the lesson aims at, the steps it goes through, the roles of teacher and learners, how the teacher responds to learners, and what materials are needed. It is more complete than a method. The standard work is Models of Teaching by Bruce Joyce and Marsha Weil, first published in 1972 and revised many times since (later editions add Emily Calhoun as an author). Do not confuse the authors with other famous pairs: Bloom and Krathwohl are linked with the taxonomy of objectives, Flanders with interaction analysis, and Allen and Ryan with micro-teaching.

The elements of a model

Joyce and Weil describe each model by a set of elements. Questions ask you to match each element with its meaning.

Element	Meaning
Focus (केंद्र-बिंदु)	The central aim or frame of reference of the model: what it is for
Syntax (संरचना)	The sequence of phases or steps through which the model proceeds
Social system (सामाजिक प्रणाली)	The roles and relationships of teacher and students, and the norms of the class

Element	Meaning
Principles of reaction (प्रतिक्रिया के सिद्धांत)	How the teacher responds to what learners say and do
Support system (सहायक प्रणाली)	The special materials, facilities and conditions the model needs, beyond the usual skills of a teacher

Joyce and Weil also describe each model's instructional and nurturing effects, that is, what it teaches directly and what it develops along the way. The trap: "system" occurs in two names. The social system is about **people** (roles and relationships); the support system is about **materials** and facilities.

The four families

Joyce and Weil grouped the models into four families, according to what each family mainly develops. The names in brackets in the last column are the developers that Indian teacher-education books and examination papers attach to each model; learn them in that form. (Group investigation in its later cooperative-learning form is credited to Shlomo and Yael Sharan; see Section 4.3.)

Family	What it develops	Examples
Information-processing	How learners take in, organise and use information; concepts and thinking	Concept attainment (Bruner); inquiry training (Suchman); advance organiser (Ausubel); inductive thinking
Social (social interaction)	Learning through working with others; democratic process and social values	Group investigation (Thelen); role playing; jurisprudential inquiry (Oliver and Shaver)
Personal	The individual's self-understanding, self-worth and creativity	Non-directive teaching (Rogers); synectics (Gordon)
Behavioural (behavioural systems)	Observable changes in behaviour through reinforcement	Contingency management, based on Skinner's operant conditioning

Pair each model with the person that Indian textbooks and examination papers link with it:

- **Concept attainment model** (संप्रत्यय उपलब्धि प्रतिमान): based on Bruner's work on how people form concepts. Learners compare positive examples and negative examples (non-examples) to find the defining attributes of a concept.
- **Inquiry training model** (पृच्छा प्रशिक्षण प्रतिमान): developed by Suchman. Learners are shown a puzzling event and investigate it by asking questions, so that they learn the process of inquiry.
- **Advance organiser model** (अग्रिम संगठक प्रतिमान): based on Ausubel's theory of meaningful verbal learning. The teacher presents a broad introductory framework before the detailed material.
- **Jurisprudential inquiry model**: by Oliver and Shaver, for analysing public policy issues and the values in conflict in them.

Bruner and Ausubel are easily exchanged. Bruner goes with concept attainment; Ausubel goes with advance organisers.

CLASSROOM EXAMPLE

In a political science class at a university in Kurukshetra, the teacher wants students to form the concept "federation". She writes two lists on the board without naming the concept. The "yes" list has India, the USA, Australia and Canada. The "no" list has France, Sri Lanka and Japan. Students guess what the "yes" countries share, test their guesses against new countries she adds, and finally state the idea: power divided by a constitution between a central government and units such as states. She names the concept, asks them to give their own examples, and then asks how they reached their answer. This is the concept attainment model, an information-processing model.

OFTEN CONFUSED

- **Social system vs support system.** Social system: roles and relationships of people. Support system: materials and facilities.
- **Syntax vs focus.** Syntax: the order of phases. Focus: the central aim.
- **Syntax vs principles of reaction.** Syntax is the sequence of steps. Principles of reaction are how the teacher responds to learners within those steps.
- **Concept attainment (Bruner) vs advance organiser (Ausubel).** Concept attainment uses examples and non-examples. Advance organiser gives a general framework before the detail.
- **Inquiry training vs group investigation.** Inquiry training (Suchman, information-processing) trains an individual's questioning about a puzzling event. Group investigation (Thelen, social) rests on democratic group process.
- **Inquiry training vs synectics.** Both begin with something unusual. Inquiry training trains questioning and explanation; synectics (Gordon, personal) develops creativity through analogy.
- **Joyce and Weil vs Flanders vs Allen.** Joyce and Weil: models of teaching. Flanders: interaction analysis. Allen: micro-teaching.
- **Glaser's model vs Gagné's events.** Glaser: four components of a teaching system. Gagné: nine events of a single lesson.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Concept attainment in more detail

The syntax of the concept attainment model has three phases:

1. **Presentation of data and identification of the concept**: the teacher presents labelled examples and non-examples; learners compare them, form hypotheses about the defining attributes, and state the concept.
2. **Testing attainment of the concept**: learners identify further unlabelled examples as "yes" or "no", the teacher confirms and names the concept, and learners give examples of their own.
3. **Analysis of thinking strategies**: learners describe how they reached the concept.

Synectics

Synectics was developed by William J. J. Gordon (his book Synectics appeared in 1961). It is a personal-family model that develops creativity. Its aim is to "make the strange familiar and the

familiar strange", breaking habitual ways of seeing. It uses three kinds of analogy: **direct analogy** (a simple comparison of two things), **personal analogy** (imagining oneself to be the object or idea) and **compressed conflict** (a two-word description whose words seem to fight each other).

Glaser's Basic Teaching Model

Robert Glaser described (1962) a basic teaching model with four components in sequence:

1. **Instructional objectives**: what the learner should be able to do at the end.
2. **Entering behaviour**: what the learner already knows and can do at the start.
3. **Instructional procedures**: the methods chosen to move the learner from entering behaviour to the objectives.
4. **Performance assessment**: testing whether the objectives have been reached.

The components are linked by **feedback loops**: the results of assessment are used to correct the earlier components. The component that comes immediately after objectives is entering behaviour. Procedures come third because they can be chosen only after the gap between entering behaviour and objectives is known.

Gagné's nine events of instruction

Robert Gagné, in The Conditions of Learning (1965), set out nine events that a well-designed lesson passes through:

1. Gaining attention
2. Informing learners of the objective
3. Stimulating recall of prior learning
4. Presenting the stimulus (the new content)
5. Providing learning guidance
6. Eliciting performance
7. Providing feedback
8. Assessing performance
9. Enhancing retention and transfer

The list begins with gaining attention. The objective is stated before the content is presented, and feedback can only follow a performance.

CHECK YOURSELF

1. Match the models in List I with the persons in List II.

List I	List II
A. Inquiry training	I. Thelen
B. Group investigation	II. Rogers
C. Non-directive teaching	III. Suchman
D. Synectics	IV. Gordon

(A) A-III, B-II, C-I, D-IV
(B) A-I, B-III, C-II, D-IV
(C) A-III, B-I, C-IV, D-II
(D) A-III, B-I, C-II, D-IV

2. In Joyce and Weil's description of a model, the special materials and facilities it requires form its:
(A) syntax
(B) support system
(C) social system
(D) focus

3. Which of the following models belong to the information-processing family? 1. Concept attainment 2. Advance organiser 3. Jurisprudential inquiry 4. Inquiry training Choose the correct answer from the options given below:
(A) 1, 2 and 4 only
(B) 1 and 3 only
(C) 2, 3 and 4 only
(D) 1, 2, 3 and 4

4. Arrange the components of Glaser's Basic Teaching Model in order: 1. Performance assessment 2. Instructional objectives 3. Instructional procedures 4. Entering behaviour
(A) 2, 3, 4, 1
(B) 4, 2, 3, 1
(C) 2, 4, 3, 1
(D) 2, 4, 1, 3

5. Statement I: In Joyce and Weil's classic grouping, synectics is a model of the personal family that uses analogies to develop creativity. Statement II: In the concept attainment model, learners compare examples and non-examples.
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

6. Assertion (A): Gagné places "informing learners of the objective" before "presenting the stimulus". Reason (R): Gagné's first event of instruction is providing feedback.
(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

1.10 Professional ethics and the teacher's professional growth

Syllabus: Teaching: Concept, Objectives, Levels of teaching (Memory, Understanding and Reflective), Characteristics and basic requirements

> **EXAM FOOTPRINT**
>
> Rarely asked directly; it supports scenario questions on the good teacher and Unit X items on higher-education governance.

Core

Teaching as a profession

A **profession** (व्यवसाय) is an occupation marked by four things: a body of specialised knowledge, a long period of training, a motive of service to society, and a **code of conduct** (आचार संहिता) that its members accept. Teaching has all four, which is why it is called a profession. A monthly salary, a post, a pay scale, a pension, long vacations, a classroom or a timetable are conditions of employment or of work. They are not marks of a profession.

Professional ethics (व्यावसायिक नैतिकता) are the standards of right conduct that a profession expects of its members. For university and college teachers in India, the University Grants Commission has set out a Code of Professional Ethics, most recently as part of the UGC Regulations of 2018 on the appointment and qualifications of teachers. The code describes the teacher's duties under heads such as teachers and their responsibilities, teachers and students, teachers and colleagues, and teachers and authorities. Questions do not ask you to recite it. They describe a teacher's action and ask which value it breaks.

The values most often tested

- **Impartiality** (निष्पक्षता) towards students. Every student must be treated justly, whatever their background, and above all in assessment. A teacher who gives extra internal marks to students who attend his private tuition breaks this value. The UGC code asks teachers to refrain from private tuition and coaching that interfere with their duties, because such work creates a conflict of interest.
- **Confidentiality** (गोपनीयता). Teachers learn personal things about students, for example in guidance and counselling. They must keep these private, unless disclosure is needed to protect the student or others. Betraying a confidence destroys the trust on which counselling depends.
- **Intellectual honesty.** Using another person's work as one's own is **plagiarism** (साहित्यिक चोरी). A teacher who puts a colleague's unpublished lecture notes into his own book without acknowledgement breaks intellectual honesty and the respect owed to colleagues.
- **Integrity in examination work.** Teachers set papers, invigilate and evaluate. They must do this honestly and keep it confidential. A paper-setter who reveals questions to a relative commits a serious breach; personal relationships give no exemption, and "teacher autonomy" does not cover disclosure of confidential papers.
- **Accountability.** Teachers answer for their work to students, colleagues and the institution. Evaluation in particular must be transparent and open to scrutiny.

Academic freedom

Academic freedom (शैक्षणिक स्वतंत्रता) protects a teacher's freedom to inquire, to teach, to research and to publish: to present views supported by evidence even when they are controversial, to criticise

established theories in the field, and to choose the methods by which a topic is taught. It is exercised within institutional obligations. It does **not** include freedom to ignore the prescribed curriculum, to skip classes, or to disregard the rules of evaluation. Nor does it give any right over other people's work: using a colleague's work without acknowledgement is plagiarism, not academic freedom. Freedom and accountability go together.

Professional growth

A professional keeps learning. A teacher grows through reading in the subject, attending refresher and orientation programmes and workshops, doing and publishing research, reflecting on her own teaching (see 1.7), and learning from colleagues and students. This lifelong learning is called **continuous professional development** (सतत व्यावसायिक विकास), CPD for short.

CLASSROOM EXAMPLE

At a college in Fatehabad, a student tells her English teacher, in a private meeting, that she is struggling because of trouble at home. That afternoon another teacher asks why the girl looks upset. The teacher says only that the student needs some support and that she is helping her, and keeps the details to herself. The same week a colleague offers her a set of unpublished notes on Indian English poetry; when she later uses some of the ideas in an article, she acknowledges him by name. In one week she has shown confidentiality and intellectual honesty.

OFTEN CONFUSED

- **Academic freedom vs freedom from rules.** Academic freedom covers inquiry, expression and choice of method. It does not cover ignoring the curriculum or evaluation norms.
- **Academic freedom vs plagiarism.** Freedom to publish one's own ideas gives no right to publish another's ideas as one's own.
- **Confidentiality vs accountability.** Confidentiality keeps a student's personal information private. Accountability means answering for one's work to others.
- **Marks of a profession vs conditions of employment.** Specialised knowledge, long training, service motive and a code of conduct are marks of a profession. Salary, post and facilities are conditions of employment.
- **Teacher autonomy vs integrity in examinations.** Autonomy covers professional decisions in teaching. It never covers disclosing confidential examination material.
- **Gift authorship vs ghost authorship.** Gift: a non-contributor is put in. Ghost: a real contributor is left out.
- **Pre-service vs in-service (NEP 2020).** The four-year integrated B.Ed. is pre-service preparation. Fifty hours of CPD a year is in-service growth.

FACT WATCH

- The National Education Policy 2020 expects every teacher to take part in at least 50 hours of continuous professional development each year, driven by the teacher's own interests. School principals and school complex leaders are also expected to take part in 50 hours or more a year. (Checked October 2026.)
- The National Education Policy 2020 states that by 2030 the minimum degree qualification for school teaching will be a four-year integrated B.Ed. degree. (Checked October 2026; this is a target date, so check its progress before your examination.)
- The UGC Code of Professional Ethics for university and college teachers that students now meet is the one in the UGC Regulations of 2018. Older books may quote an earlier version of the code.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

NEP 2020 and the teacher's growth

Two provisions of the National Education Policy 2020 are often paired in Assertion-Reason items. The policy expects continuous professional development for every teacher, at least 50 hours a year. Separately, it sets the four-year integrated B.Ed. as the minimum degree qualification for school teachers by 2030. Both are true, but they are separate provisions: one concerns in-service growth, the other pre-service preparation. Neither is the reason for the other.

Ethics of authorship

Teachers who do research also owe honesty in publication. Authorship should reflect a real intellectual contribution to the work and carries responsibility for it.

- **Gift (honorary) authorship**: a person who made no substantial contribution is listed as an author, for example a supervisor who places his name first on a paper written entirely by his research scholar.
- **Ghost authorship**: the opposite fault; a person who did contribute is left off the list of authors.
- **Salami slicing**: one study is cut into several small papers to increase the number of publications.
- **Self-plagiarism**: reusing one's own published work as if it were new, without citing it.

All four breach publication ethics. Research and publication ethics are taught more fully under Unit II (Research Aptitude).

CHECK YOURSELF

1. Which of the following are marks of a profession? 1. Specialised body of knowledge 2. Long period of training 3. A pension scheme 4. A code of conduct accepted by members Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 2, 3 and 4 only
(C) 1, 2 and 4 only
(D) 1, 2, 3 and 4

2. A teacher tells the class what a student had shared with her privately about his health. The value she has broken is:

(A) academic freedom
(B) confidentiality
(C) punctuality
(D) accountability

3. Statement I: Academic freedom allows a teacher to present evidence-based views that are controversial. Statement II: Academic freedom allows a teacher to set aside the evaluation rules of the university.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

4. Match the practices in List I with their names in List II.

List I	List II
A. A real contributor is left off the author list	I. Gift authorship
B. A head of department who did no work is added as author	II. Salami slicing
C. One study is split into five short papers	III. Self-plagiarism
D. A teacher republishes her own old paper as new without citing it	IV. Ghost authorship

(A) A-I, B-IV, C-II, D-III
(B) A-IV, B-I, C-III, D-II
(C) A-II, B-I, C-IV, D-III
(D) A-IV, B-I, C-II, D-III

5. Assertion (A): A teacher should not award marks in favour of students who attend his private coaching. Reason (R): The code of professional ethics requires just and impartial treatment of all students.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

6. Assertion (A): NEP 2020 expects every teacher to take part in at least 50 hours of continuous professional development each year. Reason (R): NEP 2020 makes the four-year integrated B.Ed. the minimum degree qualification for school teachers by 2030.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

Answers to “Check yourself”: Chapter 1

Section 1.1

1. (C), because teaching is defined by its purpose of producing learning in another person. Syllabus coverage and discipline are means, and (D) includes accidental learning, which is not teaching.

2. (A), because conditioning by-passes understanding, training builds skill by practice, instruction passes on organised knowledge, and teaching aims at understanding and judgement.

3. (D), because Morrison speaks of mature and less mature personalities, Gage of interpersonal influence, Smith of a system of actions, and Amidon of classroom talk.

4. (C), because indoctrination is decided by manner and aim, not subject; instruction can come through a programmed text, computer or broadcast with no teacher present.

5. (A), because the learner must do his or her own learning, which is exactly why the teacher must arouse and keep up motivation.

6. (D), because statement 3 is false: much learning happens through experience and self-study without any teaching.

7. (C), because training and conditioning shape behaviour, and training is the nearer to teaching as it still engages intelligence. Instructing is closest on the belief side, not the behaviour side.

Section 1.2

1. (C), because the three phases are pre-active, interactive and post-active; there is no reactive phase.

2. (A), because fixing objectives is planning (pre-active), sizing up the class happens while facing it (interactive), and choosing tests comes after teaching (post-active).

3. (D), because content and methods stand between teacher and learner and carry the teacher’s influence. The teacher is independent and the learner dependent.

4. (A), because the need to respond on the spot to learners’ cues is precisely what makes interactive decisions spontaneous.

5. (C), because the phases form a cycle in which evaluation feeds planning; experience shortens planning but does not remove it.

6. (A), because finding entry behaviour is the diagnostic function. Choosing methods to remedy what she finds would be prescriptive.

7. (B), because interest changes the strength of the relation between method (independent) and achievement (dependent), which is what a moderator does.

Section 1.3

1. (C), because it states the learner’s observable performance with a condition and a criterion. (A) is a teacher activity, (B) uses a vague verb, and (D) is only a topic.

2. (B), because solving is what the learner does, the calculator is the circumstance, and eight out of ten is the standard.

3. (C), because Statement I is true and Statement II is false: vague verbs cannot be observed; behavioural objectives need verbs like state, explain or solve.

4. (A), because assessment exists to check whether outcomes are attained, and that is exactly why it must be aligned with them.

5. (D), because Behaviour, Condition and Degree match Mager's performance, condition and criterion; Audience is the added letter.

6. (A), because certification is an outcome of the examination system, not a function of teaching.

7. (C), because programme educational objectives look a few years into the career; programme outcomes and programme-specific outcomes refer to the time of graduation.

Section 1.4

1. (B), because the order is Understand, Apply, Analyse, Evaluate, Create (Remember, not listed, comes first).

2. (A), because summarising is Understand, distinguishing is Analyse, defending a choice against criteria is Evaluate, and writing something original is Create.

3. (C), because a value that governs conduct consistently over a long time is Characterisation, the highest affective level. Evaluate (A) is a cognitive judgement, not a settled value.

4. (A), because changing data into another form is translation, which is one of the three forms of Comprehension; R explains A.

5. (D), because Dave begins with Imitation; Perception is Simpson's first level.

6. (A), because "evaluate" gives the process and "one's own strategies" is metacognitive knowledge.

7. (D), because Simpson ends with Origination (Naturalisation is the last of Dave's levels), while Harrow does begin with Reflex movements.

8. (B), because higher-order items are 4 + 3 + 3 = 10, and 10 / 25 x 100 = 40 per cent. (A) wrongly counts the Apply items: 14 / 25 = 56 per cent.

Section 1.5

1. (B), because Morrison's order is exploration, presentation, assimilation, organisation, recitation. (C) gives the Herbartian steps.

2. (C), because memory-level methods are drill and recall, the understanding level aims at mastery of units (Morrison), the reflective level works through a felt problem and hypotheses, and the five named steps are Herbart's.

3. (B), because both are true, but the link with Hunt does not explain the need for an open climate; that need comes from learners' freedom to doubt and test ideas.

4. (D), because statement 2 is false: reflective-level motivation is mainly intrinsic, arising from the felt problem.

5. (A), because organisation is the unaided written outline. Assimilation is supervised study; recitation is oral presentation.

6. (C), because Statement I is true, but recitation is Morrison's last step; Herbart's last step is application.

7. (C), because data that contradict the learners' beliefs create a genuine felt problem, the starting point of reflective-level teaching; asking for explanations and evidence moves towards hypotheses and testing.

8. (D), because Dewey's order is felt difficulty, location and definition, suggestion, reasoning out its bearings, and observation and experiment.

Section 1.6

1. (C), because teaching starts from the learner's interests and readiness (psychological order) and moves towards the structure of the subject (logical order). The other three options reverse accepted maxims: particular to general, concrete to abstract, simple to complex.
2. (A), because the price of vegetables is already known (known to unknown), the whole map comes before districts (whole to part), coins are handled before the definition (concrete to abstract), and an observed fact is given its reasoned explanation (empirical to rational).
3. (D), because discovery of a rule suits the inductive method, not the deductive; Statement II correctly describes synthesis, which runs from known data to the unknown conclusion.
4. (A), because both are true and the Reason gives exactly why the order induction-then-deduction is preferred.
5. (B), because 1, 2 and 3 are correct. Statement 4 is wrong: the logical order follows the structure of the subject; the psychological order follows the learner.
6. (D), because the synthetic method presents the finished proof from data to conclusion. The analytic method would work backwards from the conclusion and show how the proof is found.
7. (C), because induction comes first (examples, noticing what is common, stating the rule) and deduction follows (applying the rule).

Section 1.7

1. (B), because good teaching gives feedback and adapts to learners. Coverage is an input, not the test of teaching, and good teaching assumes that learners differ.
2. (A), because learners differ, a teacher must find each one's difficulties and help with them; the Reason explains the Assertion.
3. (C), because withitness is awareness of the whole room, overlapping is handling two matters at once, momentum is pace without slowdowns, and smoothness is flow without abrupt breaks.
4. (B), because authoritarian groups worked mainly while watched, and democratic teaching still corrects errors; both statements are false.
5. (D), because training is something the teacher brings to teaching, which makes it a presage variable. Context would be the pupils, school or community.
6. (B), because the first principle is to encourage contact between students and faculty, not to reduce it.
7. (C), because using names, smiling and moving among students reduce psychological distance, which is immediacy. Clarity concerns explanations, credibility concerns perceived competence and trust.

Section 1.8

1. (B), because class size, lesson length and number of skills are all scaled down, but feedback stays immediate and detailed; it is the heart of the technique.
2. (B), because the order is plan, teach, feedback, re-plan, re-teach, re-feedback. Option (A) puts re-planning before feedback.
3. (C), because 2 hours 24 minutes = 144 minutes and 144 / 36 = 4 exactly. Taking the cycle as 30 minutes would give 4.8, and taking it as 24 would give 6.
4. (A), because redirection is a component of probing, movement and voice changes are stimulus variation, explaining links belong to explaining, and summarising ends the lesson in closure.

5. (D), because total = 600, teacher talk = 360 / 600 = 60 per cent, student talk = 180 / 600 = 30 per cent, and 600 tallies at 20 a minute = 30 minutes. The I/D ratio is 90 / 270 = 0.33, not 3.0; 3.0 is the inverted ratio.

6. (A), because Flanders put asking questions (category 4) among the indirect categories precisely because questions open the way for student participation.

7. (D), because PCK is knowing how to make a particular topic understandable to learners (examples, analogies, typical difficulties), not knowledge of new research; Statement II is true.

8. (C), because it has many acceptable answers and asks students to predict and reason. The other three each have a single correct answer, even the one that needs a calculation.

Section 1.9

1. (D), because Suchman developed inquiry training, Thelen group investigation, Rogers non-directive teaching and Gordon synectics.

2. (B), because the support system lists materials and facilities. The social system is about roles and relationships; syntax is the order of phases.

3. (A), because concept attainment, advance organiser and inquiry training deal with how learners process information. Jurisprudential inquiry belongs to the social family.

4. (C), because objectives come first, then entering behaviour, then procedures, then performance assessment.

5. (A), because both statements correctly describe the two models.

6. (C), because the objective is indeed stated before the content, but the first event is gaining attention; feedback is the seventh.

Section 1.10

1. (C), because specialised knowledge, long training and a code of conduct mark a profession. A pension is a condition of employment.

2. (B), because personal information shared in confidence must be kept private unless disclosure protects the student or others.

3. (C), because academic freedom protects evidence-based expression, but it is exercised within the institution’s curriculum and evaluation rules.

4. (D), because leaving out a contributor is ghost authorship, adding a non-contributor is gift authorship, splitting one study is salami slicing, and reusing one’s own work uncited is self-plagiarism.

5. (A), because favouring tuition students breaks impartiality, and the Reason states exactly the duty that is broken.

6. (B), because both provisions are in the policy, but one concerns in-service growth and the other pre-service preparation; one does not explain the other.

Chapter 2
Learner's Characteristics

The syllabus asks you to know the "characteristics of adolescent and adult learners (academic, social, emotional and cognitive), individual differences". This chapter takes that one sentence apart, piece by piece, in nine sections. A learner (शिक्षार्थी) in a college or university is usually a late adolescent (किशोर) or a young adult, so we begin with who these learners are, and then look at each kind of characteristic in turn.

Syllabus phrase	Where this chapter teaches it
adolescent and adult learners	2.1 (who the learner in higher education is: adolescence and early adulthood); 2.7 (adult learners and andragogy)
cognitive characteristics	2.2 (how the learner thinks: Piaget, Vygotsky, Bruner, Ausubel, memory); 2.3 (how learners learn: the theories of learning)
academic characteristics	2.4 (learning styles, study habits and self-regulation)
social characteristics	2.5 (peers, groups and moral reasoning)
emotional characteristics	2.6 (emotions and motivation)
individual differences	2.8 (individual differences and intelligence); 2.9 (learners with diverse needs and inclusive teaching)

Read 2.1 first. Its picture of the late adolescent and the young adult is used in every later section.

2.1 The learner in higher education: adolescence and early adulthood

Syllabus: Characteristics of adolescent and adult learners

> **EXAM FOOTPRINT**
>
> Asked occasionally; usually a multiple-select item on characteristics of adolescent or adult learners, or a match of stages of education with age groups.

Core

Who sits in a college classroom

Most students join an undergraduate course at about seventeen or eighteen. This age falls in **late adolescence** (उत्तर किशोरावस्था). **Adolescence** (किशोरावस्था) is the period between childhood and adulthood. It begins with the bodily changes of puberty and ends when the young person takes on adult roles. The World Health Organization uses the range 10 to 19 years for adolescents. Postgraduate and research students, in their twenties, are mostly **young adults** (युवा वयस्क). So a college teacher teaches late adolescents and young adults, often in the same building.

What do late adolescents bring to the classroom? Five things matter for the examination.

- They can think in the abstract. They can reason about ideas, possibilities and hypotheses, not only about things they can see. (Piaget's formal operational stage; see 2.2.)
- They are searching for an identity. They ask, "Who am I? What will I become?"
- They want independence from parents, and this often brings conflict at home.
- They are very sensitive to the opinion of their peers (friends of their own age), and they care about how they look. Body image matters to them.

- Their emotions are intense and they change quickly. Emotional stability is not a mark of this age.

From these follows a classroom rule. Because adolescents are self-conscious and hurt easily by criticism in front of others, correct them privately and with respect. Do not shame a student before the class.

Adolescents also look for heroes. They idealise a sports star, a film actor, a leader or a teacher, and they day-dream about their own future roles. **Hero-worship** and **day-dreaming** are typical of adolescence. Both come from the search for identity and from strong feelings.

G. Stanley Hall: storm and stress

G. Stanley Hall (Granville Stanley Hall), an American psychologist, wrote a large book called *Adolescence* (1904). He described adolescence as a period of **storm and stress** (तूफ़ान और तनाव). He meant rapid bodily growth, ups and downs of mood, the search for identity and conflict with parents and other authority. The phrase comes from the German *Sturm und Drang*. If a question asks which phrase "commonly describes" adolescence, the answer is storm and stress, not rest, stability, play or decline.

Later research softened Hall's view. Not every adolescent goes through storm and stress, but such stress is more likely in adolescence than at other ages.

Erik Erikson: eight psychosocial stages

Erik H. Erikson described the whole life span as eight stages in his book *Childhood and Society* (1950). His theory is called **psychosocial** (मनोसामाजिक) because each stage is a conflict between the person's needs and the demands of society. Each stage has a "crisis", named as a pair of opposites. Examiners use the other stages as distractors, so learn all eight in order.

No.	Crisis	Life period
1	Trust versus mistrust	Infancy
2	Autonomy versus shame and doubt	Early childhood (toddler)
3	Initiative versus guilt	Pre-school years
4	Industry versus inferiority	School age
5	Identity versus role confusion	Adolescence
6	Intimacy versus isolation	Young adulthood
7	Generativity versus stagnation	Middle adulthood
8	Integrity versus despair	Old age

For adolescence the crisis is **identity versus role confusion** (पहचान बनाम भूमिका भ्रम). The young person tries to build a clear sense of who he or she is: values, beliefs, career, relationships. If this fails, the result is confusion about one's role in life. Erikson called this struggle the **identity crisis**.

The next stage, **intimacy versus isolation**, belongs to young adulthood. It is about forming close, lasting relationships. Many postgraduate students are in this stage. But it is not the stage of adolescence. This is a favourite trap.

Robert Havighurst: developmental tasks

Robert J. Havighurst gave the idea of **developmental tasks** (विकासात्मक कार्य). A developmental task is a task that society expects a person to master at a certain period of life. If the person masters it, he or she is happy and succeeds with later tasks. If not, there is unhappiness, disapproval by society and difficulty with later tasks. Havighurst set out these tasks in *Developmental Tasks and Education* (first edition 1948).

His tasks for adolescence include:

- forming new and more mature relations with age-mates of both sexes;
- accepting one's body (physique) and using it effectively;
- achieving emotional independence from parents and other adults;
- preparing for a career (an economic career);
- preparing for marriage and family life;
- acquiring a set of values and an ethical system to guide behaviour;
- desiring and achieving socially responsible behaviour.

Keep the three names apart. **Havighurst**: developmental tasks. **Erikson**: psychosocial stages and the identity crisis. **Hall**: storm and stress.

James Marcia: four identity statuses

James Marcia (1966) took Erikson's idea of identity and asked two questions about a young person.

1. Has there been **exploration**? Has the person actively looked at different choices (courses, careers, beliefs)?
2. Has there been **commitment**? Has the person firmly chosen?

The answers give four **identity statuses** (पहचान स्थितियाँ).

Status	Exploration	Commitment	Example
Identity diffusion	No	No	A student who has not thought about a career and has not chosen one
Identity foreclosure	No	Yes	A student firmly set on the career his parents chose, without looking at any other
Identity moratorium	Yes	No	A first-year student trying out courses, clubs and career ideas, not yet decided
Identity achievement	Yes	Yes, after exploring	A student who explored several fields and then chose one with conviction

The word **moratorium** means a period of delay. Moratorium is common, and healthy, among students in their first years of college. **Foreclosure** means a door closed early: commitment without exploration.

University students are still forming an identity

Erikson and Marcia both show that identity formation does not end at eighteen. It continues into the early twenties, especially for students who are still studying. So many university students are still consolidating their sense of identity.

Do not mix this up with Kohlberg. Lawrence Kohlberg described stages of **moral reasoning** (see 2.5). Moral reasoning is a different part of development from identity. A true statement about Kohlberg does not explain a fact about identity.

Jeffrey Arnett: emerging adulthood

Jeffrey Jensen Arnett proposed in 2000 that the years from about **eighteen to twenty-five** form a separate stage of life. He called it **emerging adulthood** (उभरती वयस्कता). In societies where marriage,

parenthood and settled work come late, young people in these years are neither adolescents nor full adults. Arnett named five features:

- identity explorations (in love, work and values);
- instability (changes of residence, job and relationships);
- self-focus;
- feeling in-between (neither adolescent nor adult);
- possibilities (a sense that many futures are still open).

Most university students fall in this period. If a question asks who proposed "emerging adulthood", the answer is Arnett. Erikson described the identity crisis, Marcia the identity statuses, and Daniel Levinson the "seasons" of adult life (*The Seasons of a Man's Life*, 1978).

CLASSROOM EXAMPLE

In a B.A. first-year class in Sirsa, Meena reads out her answer and a classmate laughs at a mistake. Meena goes silent for the rest of the week. Her teacher speaks to her alone after class, praises the good part of her answer and shows her the error kindly. Next week the teacher forms small groups so that every student speaks first to three peers before speaking to the whole class. The teacher has used two facts about late adolescents: they are sensitive to peer opinion, and they learn readily with peers when they feel safe.

OFTEN CONFUSED

- **Identity versus role confusion** and **intimacy versus isolation.** Identity is the crisis of adolescence; intimacy is the crisis of young adulthood, the next stage.
- **Industry versus inferiority** and **identity versus role confusion.** Industry is the school-age stage; identity is adolescence.
- **Hall**, **Havighurst** and **Erikson.** Hall: storm and stress. Havighurst: developmental tasks. Erikson: psychosocial stages.
- **Moratorium** and **diffusion.** Both lack commitment. In moratorium the person is actively exploring; in diffusion there is no exploration either.
- **Foreclosure** and **achievement.** Both show commitment. In foreclosure there was no exploration before it; in achievement there was.
- **Imaginary audience** and **personal fable** (Going Further). Imaginary audience: "everyone is watching me". Personal fable: "I am unique and nothing bad can happen to me".
- **Arnett** and **Levinson.** Arnett: emerging adulthood (18 to 25). Levinson: seasons of adult life.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

David Elkind: adolescent egocentrism

David Elkind (1967) described a special kind of self-centredness in adolescents, which he called **adolescent egocentrism** (किशोर अहंकेंद्रिता). It appears when the young person first gains abstract thought and can think about what others think, but then wrongly assumes that others are thinking about him or her. It has two forms.

- **Imaginary audience** (काल्पनिक दर्शक). The adolescent feels "on stage", as if others are always watching and judging. This explains self-consciousness about appearance, and the dread of being laughed at.
- **Personal fable** (व्यक्तिगत कल्पित कथा). The adolescent believes that he or she is special, unique and invulnerable (cannot be harmed). "Accidents happen to other people, not to me." This lies behind risk-taking, such as riding without a helmet.

Imaginary audience and personal fable are Elkind's terms. Foreclosure and moratorium are Marcia's identity statuses. A question may mix the two sets of terms in its options.

Applying Marcia to cases

To place a student in Marcia's scheme, ask the two questions in order. First: is the student committed? If no, the answer is moratorium (if exploring) or diffusion (if not exploring). If yes, ask: did the commitment come after exploration? If yes, achievement; if no, foreclosure. A student who obeys the family's choice of career without considering anything else is in foreclosure, even if the choice is a good one.

Statuses are not fixed

Marcia's statuses are not a ladder that everyone climbs once. A person can be achieved in one area (career) and foreclosed in another (religion), and can move from achievement back to moratorium when life changes. This is why identity work continues through the university years.

CHECK YOURSELF

1. Which pairing of Erikson's crisis with its life period is correct?

(A) Industry versus inferiority – adolescence
(B) Intimacy versus isolation – adolescence
(C) Identity versus role confusion – adolescence
(D) Integrity versus despair – young adulthood

2. Match List I with List II.

List I	List II
A. G. Stanley Hall	I. Developmental tasks
B. Robert Havighurst	II. Emerging adulthood
C. James Marcia	III. Storm and stress
D. Jeffrey Arnett	IV. Identity statuses

(A) A-III, B-I, C-IV, D-II
(B) A-I, B-III, C-IV, D-II
(C) A-III, B-I, C-II, D-IV
(D) A-IV, B-I, C-III, D-II

3. A second-year student has firmly decided to become a lawyer because her father is one. She has never considered any other career. In Marcia's terms her status is:

(A) identity achievement
(B) identity moratorium
(C) identity diffusion
(D) identity foreclosure

4. Which of the following are typical of late adolescent learners? Choose the correct answer from the options given below:

1. Ability to reason about hypotheses
2. Steady, unchanging emotions
3. Strong concern for peer opinion
4. Wish for independence from parents

(A) 1, 2 and 3 only
(B) 1, 3 and 4 only
(C) 2, 3 and 4 only
(D) 1, 2, 3 and 4

5. Assertion (A): A teacher should point out an undergraduate's mistakes privately rather than before the whole class.

Reason (R): Adolescents are self-conscious and highly sensitive to how their peers see them.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

6. Statement I: Elkind's "personal fable" is the adolescent's feeling of being constantly watched by others.

Statement II: Arnett placed emerging adulthood roughly between eighteen and twenty-five

years.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

2.2 Cognitive characteristics: how the learner thinks

Syllabus: Characteristics of adolescent and adult learners (cognitive)

EXAM FOOTPRINT

Asked fairly often; usually a sequence of Piaget's stages or a match of thinkers with ideas such as ZPD, scaffolding, advance organisers.

Core

Cognitive (संज्ञानात्मक) means to do with knowing: thinking, understanding, reasoning and remembering. This section gives the four thinkers the examination asks about most (Piaget, Vygotsky, Bruner and Ausubel) and then the basic facts of memory.

Jean Piaget: the child builds knowledge in stages

Jean Piaget, a Swiss psychologist, studied how children's thinking grows. His central idea is that the child is an **active constructor** of knowledge, not a passive receiver. The child builds mental structures called **schemas** (स्कीमा, मानसिक संरचना). A schema is an organised pattern of thought or action, such as the child's idea of "dog".

Three processes change schemas.

- **Assimilation** (आत्मसात्करण): fitting a new experience into an existing schema. A child who knows "dog" sees a cow and calls it a "big dog".
- **Accommodation** (समायोजन): changing a schema, or making a new one, because the experience does not fit. When corrected, the child forms a new category, "cow".
- **Equilibration** (संतुलनीकरण): the self-regulating process that balances the two. When a new experience will not fit, the child feels a mental imbalance, **disequilibrium**. Accommodation restores balance at a higher level. Piaget saw equilibration as the chief motor of cognitive development.

Piaget described four stages. Learn the order and the rough ages.

Stage	About	What marks it
Sensorimotor	Birth to 2	The infant knows the world through the senses and through actions. Object permanence develops: knowing that an object still exists when out of sight.
Pre-operational	2 to 7	Language and symbols grow, but thinking is not yet logical. Egocentrism (cannot see another person's point of view), animism (treating objects as alive) and centration (attending to one feature only). Fails conservation tasks.
Concrete operational	7 to 11	Logical thinking about real, concrete objects and events. Conservation, reversibility, classification and seriation (arranging in order).

Stage	About	What marks it
Formal operational	11 or 12 onwards	Abstract and hypothetical thinking. Reasoning about possibilities and propositions. Hypothetico-deductive reasoning: forming a hypothesis and testing it systematically.

Key terms in the table:

- **Conservation** (संरक्षण): understanding that quantity, number or mass stays the same when only its appearance changes. Water poured from a short wide glass into a tall narrow glass is still the same amount. Conservation is the chief achievement of the concrete operational stage.
- **Reversibility** (उत्क्रमणीयता): understanding that an action can be undone in the mind (the water can be poured back). Conservation depends on it.
- **Egocentrism** (अहंकेंद्रिता): difficulty in seeing things from another's point of view. It marks the pre-operational stage.

Three rules about the stages are often tested.

1. The **order** of the stages is fixed (invariant). No child skips a stage.
2. The **ages** are only approximate. They vary with the child, the culture and schooling.
3. Formal operations begin at about eleven or twelve, not earlier.

For a college teacher the point is this: most undergraduates can handle abstract ideas, propositional logic and hypotheses, because formal thought frees reasoning from concrete objects. Dependence on concrete objects is the limit of the earlier, concrete operational stage.

Piaget's view is called **cognitive constructivism** (or individual constructivism): the individual learner builds knowledge by acting on the world. For Piaget, development comes first and largely sets the limits of what a child can learn at a given stage.

Lev Vygotsky: learning is first social

Lev Vygotsky, a Russian psychologist, also held that learners build knowledge. But he stressed that they build it **with other people**, inside a culture. His view is called **social constructivism** (सामाजिक रचनावाद) or socio-cultural theory.

His central claim: every higher mental function appears twice, first between people (on the social plane) and then inside the individual. A child first solves a problem by talking with a parent or teacher, and later solves it alone in thought. So learning is first social, then individual. This is why classroom dialogue is central in a Vygotskian classroom.

Language is the chief tool of thought. Speech develops in three steps:

1. **Social speech**: talking with others to communicate.
2. **Private speech**: talking aloud to oneself to guide one's own action (a child talking while doing a puzzle).
3. **Inner speech**: private speech goes silent and becomes verbal thought.

The zone of proximal development

Vygotsky's best-known idea is the **zone of proximal development**, or ZPD (समीपस्थ विकास का क्षेत्र). Look at three kinds of task.

- Tasks the learner can do **alone**. These show the **level of actual development**.
- Tasks the learner **cannot do alone but can do with help** from an adult or a more capable peer. The upper edge of these shows the **level of potential development**.
- Tasks the learner cannot do **even with help**. These lie beyond the zone.

The ZPD is the **distance (gap) between the actual level and the potential level**. It is neither level itself. It is the gap between what a learner can do alone and what he or she can do with guidance. Vygotsky wrote about this in work published in English as *Mind in Society* (1978).

Teaching is most fruitful when it is aimed at the zone. If a learner cannot solve a problem alone but solves it after a few hints, that problem lies within the ZPD.

Vygotsky argued that good learning runs **ahead of** development and leads it. Instruction aimed at the zone awakens abilities that are still maturing. Compare Piaget, for whom development sets the limits of learning.

A classic illustration, usually credited to Vygotsky himself, makes the zone clear. Two children both solve problems at the eight-year level when alone. With help, one reaches the nine-year level and the other the twelve-year level. Their actual development is the same, but their zones differ: one year and four years. So they need different teaching, and a test of unaided work alone would hide the difference.

The more knowledgeable other and scaffolding

The person who gives the help is often called the **more knowledgeable other** (MKO). It may be a teacher, a parent, a capable classmate, or even a tool such as a computer program. The term is widely used by later writers to describe Vygotsky's idea. Do not confuse it with the **generalised other**, which is G. H. Mead's term for the attitudes of society that a person takes in.

The help itself is called **scaffolding** (आलंबन, पाड़). Scaffolding is temporary support: hints, prompts, models, questions, breaking a task into steps. Like a builder's scaffold, it is removed gradually as the learner becomes able to work alone. So scaffolding is not extra homework, a fixed syllabus or a grade.

Peer tutoring works on the same principle. A peer who is slightly ahead can give hints within the tutee's zone, so peer tutoring can promote the tutee's cognitive development.

Jerome Bruner: discovery and three modes

Jerome Bruner, an American psychologist, is linked with **discovery learning** (खोज अधिगम). He held that learners understand and remember best what they find out for themselves. So the teacher should arrange situations in which learners discover principles.

Bruner described three **modes of representation**, three ways in which knowledge is held in the mind.

Mode	Represents knowledge through	Example
Enactive	Action, handling objects	Placing weights on a beam with one's own hands to learn the balance
Iconic	Images, pictures, diagrams	Looking at a diagram of the balance
Symbolic	Language, numbers, formulae	Writing the principle of moments as a formula

Bruner recommended that new ideas be met in this order: enactive, then iconic, then symbolic. The "concrete – pictorial – abstract" sequence used in mathematics teaching follows exactly this order. Do not confuse the three modes with the VAK learning styles (visual, auditory, kinaesthetic), with the affective levels of Bloom's group (receiving, responding, valuing), or with the Herbartian steps (preparation, presentation, application).

Bruner made a famous claim in *The Process of Education* (1960): any subject can be taught effectively in some intellectually honest form to any child at any stage of development. So he rejected the idea of waiting until the child is "ready". From this claim came the **spiral curriculum**: the

curriculum returns to the basic ideas of a subject again and again, each time at a higher level of complexity.

David Ausubel: meaningful reception learning

David Ausubel, an American psychologist, argued that most classroom learning is the **reception** of material that the teacher presents in words, as in a lecture or a textbook. His theory is called **meaningful verbal learning**, or meaningful reception learning.

Learning is **meaningful** (सार्थक अधिगम) when the learner relates the new material to what he or she already knows. It is **rote** (रटंत अधिगम) when it is memorised without such links. Ausubel's famous principle: the most important single factor in learning is what the learner already knows.

His best-known teaching device is the **advance organiser** (अग्रिम संगठक). It is a brief, general introduction given **before** the new material. It gives learners a framework to which the details can be attached, so that learning becomes meaningful.

Term	What it is
Advance organiser (Ausubel)	A general introduction given before the lesson, as a framework for new material
Scaffolding (Wood, Bruner and Ross)	Support given during a task, then withdrawn
Schema (Piaget)	The learner's own mental structure
Mnemonic	A memory aid, such as an acronym or rhyme

Keep the names apart: discovery learning is Bruner; meaningful reception learning and advance organisers are Ausubel; conditions of learning are Gagné (2.3); observational learning is Bandura (2.3).

Memory: holding and keeping what is learnt

Working memory (कार्यकारी स्मृति), also called short-term memory, holds the information we are using at this moment. George A. Miller (1956) showed that it holds only about **seven items, plus or minus two**. **Chunking** (खंडीकरण) gets round this limit by grouping items into larger meaningful units. A twelve-digit number is hard to hold; four groups of three digits are easy. Chunking deals with the limited capacity of working memory.

Some well-established facts about retention (holding learning over time):

- Meaningful material is retained far better than nonsense material.
- **Distributed practice** (spaced over days) gives better long-term retention than **massed practice** (all at once).
- **Overlearning**, that is, continuing practice after the first correct recall, improves retention.
- Hermann Ebbinghaus's **forgetting curve** (1885) shows that forgetting is fastest soon after learning and then slows down. So the first revision should come soon.

Mnemonic devices (स्मृति सहायक युक्तियाँ) such as acronyms, rhymes and vivid images help us remember unrelated facts. Unrelated facts have no links of their own. A mnemonic imposes an artificial organisation on them and so supplies cues for retrieval (recall). For example, "VIBGYOR" for the colours of the spectrum.

CLASSROOM EXAMPLE

A teacher of B.Sc. physics in Hisar begins a unit on energy with a five-minute overview of what energy is and how its forms are related (an advance organiser). Students then roll a ball down a ramp and feel it push a block (enactive), draw an energy bar chart (iconic) and finally write the conservation equation (symbolic). While they work in pairs, she gives hints to the pairs who are stuck and stops giving them as they succeed (scaffolding within the zone of proximal development). She ends by asking them to recall the main points the next day and again after a week (distributed practice).

OFTEN CONFUSED

- **Assimilation** and **accommodation.** Assimilation fits new experience into an old schema; accommodation changes the schema.
- **Equilibration** and **conservation.** Equilibration is the balancing process behind all development; conservation is one achievement of the concrete operational stage.
- **Centration** and **conservation.** Centration (attending to one feature, such as the height of the water) is why the pre-operational child fails conservation.
- **Concrete operational** and **formal operational.** Concrete: logic on real objects (classification, seriation, conservation). Formal: abstract and hypothetical reasoning.
- **Zone of proximal development** and **level of potential development.** The zone is the gap; the potential level is its upper edge (what the learner does with help).
- **Cognitive constructivism** (Piaget) and **social constructivism** (Vygotsky).
- **More knowledgeable other** and **generalised other** (G. H. Mead).
- **Scaffolding** and **advance organiser.** Scaffolding is help during the task, withdrawn later; the advance organiser is a framework given before the lesson.
- **Enactive** and **iconic.** Enactive is action; iconic is images.
- **Bruner** (discovery) and **Ausubel** (meaningful reception).

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

How later research judged Piaget

- Piaget **underestimated young children**. With simpler, more familiar tasks, children show abilities earlier than he reported (Margaret Donaldson, *Children's Minds*, 1978).
- He gave **less weight to social and cultural influences** than Vygotsky did.
- The **ages are not fixed and universal**. They vary with culture, schooling and experience.
- **Not all adults use formal operations in all fields.** Many adolescents and adults reason formally in fields they know well but less so in unfamiliar ones. So it is false to say that every adolescent reaches formal operations at eleven and uses them uniformly in every subject. Chronological age is only a rough guide, and many adolescents still reason concretely in unfamiliar subjects.

Piaget and Vygotsky on children's self-talk

A five-year-old talks aloud while doing a puzzle. **Piaget** called this **egocentric speech**: a sign that the child cannot yet take another's point of view, which fades with development. **Vygotsky** called it

private speech: a tool by which the child plans and regulates action, which later turns inward to become inner speech, or thought. Know the two readings and who holds each.

More on Bruner

In *Toward a Theory of Instruction* (1966) Bruner said that a theory of instruction should deal with four things: the **predisposition** to learn, the **structure** of knowledge, the most effective **sequence** of material, and the nature and pacing of **reinforcement** (rewards). Because he held that any subject can be taught in some honest form at any stage, he did not think discovery unsuitable for young children.

The word **scaffolding** was introduced by David Wood, Jerome Bruner and Gail Ross in a 1976 paper on how a tutor helps a child solve problems. It was later linked with Vygotsky's zone of proximal development. So scaffolding is not Vygotsky's own word, though it describes help within his zone.

More on Ausubel

Two kinds of advance organiser. A **comparative organiser** is used when the new material can be related to something familiar. It points out similarities and differences (a lesson on Buddhism introduced by comparing it with Hinduism, which the class knows). An **expository organiser** is used when the material is wholly new. It supplies a general framework.

Subsumption. Ausubel saw the learner's **cognitive structure** (the organised body of knowledge in the mind) as a hierarchy, with broad ideas at the top. Meaningful learning happens when new, more specific material is **subsumed** (placed under) a relevant inclusive idea already present, called a **subsumer**. Advance organisers supply or activate subsumers. Subsumption is Ausubel's term; accommodation is Piaget's.

Three conditions for meaningful learning.

1. The material is logically meaningful (organised, not arbitrary).
2. The learner has relevant anchoring ideas in his or her cognitive structure.
3. The learner has a **meaningful learning set**: the intention to relate the new material to what is known.

If the learner intends only to memorise word for word, learning will be rote, even when the first two conditions are met. A lecture does not make learning rote.

Two separate dimensions. Ausubel treated "reception versus discovery" (how the content reaches the learner) and "rote versus meaningful" (how the learner takes it in) as two independent dimensions. Crossing them gives four kinds of learning.

	Reception	**Discovery**
Meaningful	A well-organised lecture that the learner links to what she knows	A student finds a principle herself and understands why it works
Rote	Memorising a lecture word for word	A student stumbles on a solution by trial and error and cannot say why it works

So reception learning is not necessarily rote, and discovery learning is not necessarily meaningful.

Sequencing principles. Progressive differentiation: present the most general, inclusive idea first, then differentiate it into details (begin with "energy", then its forms, then their details). **Integrative reconciliation**: explicitly relate new ideas to earlier ones and resolve apparent contradictions between them.

More on memory

The multi-store model of Richard Atkinson and Richard Shiffrin (1968) has three stores: the **sensory register** (holds input for a very brief moment), **short-term (working) memory** (limited capacity) and **long-term memory** (large and lasting).

Maintenance rehearsal is plain repetition to keep an item in short-term memory. **Elaborative rehearsal** links the item to what one already knows, for example to a personal example and to related concepts. Elaborative rehearsal is far better for long-term retention. It matches the "deep processing" of Fergus Craik and Robert Lockhart's levels of processing idea (1972): the more deeply, meaningfully, we process information, the better we remember it.

Interference explains much forgetting. In **proactive interference** old learning disturbs the recall of new learning; in **retroactive interference** new learning disturbs the recall of old learning. Interference concerns forgetting, not how much can be held at one time.

Four effects with names:

Effect	What it says
Serial position effect	Items at the beginning (primacy) and at the end (recency) of a list or lecture are recalled best; those in the middle worst. So put key points at the start and the end.
Von Restorff effect	An item that stands out (is distinctive) is recalled better, whatever its position.
Zeigarnik effect	Unfinished tasks are recalled better than finished ones.
Hawthorne effect	People change their behaviour because they know they are being observed (named after studies at the Hawthorne Works of the Western Electric Company).

CHECK YOURSELF

1. Arrange Piaget's stages in the order in which they appear.

1. Concrete operational
2. Sensorimotor
3. Formal operational
4. Pre-operational

(A) 2, 1, 4, 3
(B) 2, 4, 1, 3
(C) 4, 2, 1, 3
(D) 2, 4, 3, 1

2. Statement I: According to Vygotsky, the zone of proximal development is what a learner can do alone.

Statement II: Scaffolding is gradually withdrawn as the learner becomes able to work alone.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

3. Match List I with List II.

List I	List II
A. Bruner	I. Advance organiser
B. Ausubel	II. Equilibration
C. Piaget	III. Zone of proximal development
D. Vygotsky	IV. Spiral curriculum

(A) A-IV, B-I, C-III, D-II
(B) A-I, B-IV, C-II, D-III
(C) A-IV, B-I, C-II, D-III
(D) A-IV, B-II, C-I, D-III

4. A student says that a long row of five coins has more coins than a short row of five coins. In Piaget's terms, she has not yet attained:

(A) object permanence
(B) conservation
(C) animism
(D) egocentrism

5. Assertion (A): Peer group work can help a student solve problems she cannot yet solve alone.

Reason (R): Vygotsky held that development proceeds without any need for social interaction.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

6. Which of the following statements about Ausubel are correct? Choose the correct answer

from the options given below:

1. A lecture can produce meaningful learning.
2. Discovery learning is always meaningful.
3. An advance organiser is presented before the new material.
4. A comparative organiser relates new material to something familiar.

(A) 1, 2 and 3 only
(B) 2, 3 and 4 only
(C) 1 and 3 only
(D) 1, 3 and 4 only

7. A teacher of history groups thirty dates into five periods so that students can hold them in mind. The device and the limit it overcomes are:

(A) mnemonic; decay of long-term memory
(B) elaboration; proactive interference
(C) chunking; limited capacity of working memory
(D) overlearning; serial position effect

8. A student remembers the opening and the closing remarks of a lecture but little of the middle. This is explained by the:

(A) Hawthorne effect
(B) Zeigarnik effect
(C) von Restorff effect
(D) serial position effect

2.3 How learners learn: theories of learning

Syllabus: Characteristics of adolescent and adult learners (academic and cognitive)

EXAM FOOTPRINT

Asked often; usually a match of theories and theorists, a sequence from Gagné's hierarchy or events, or an item on memory types.

Core

A **theory of learning** explains how learning (अधिगम) happens. The examination asks you to match each theory with its psychologist, its experiment and its key terms. The theories fall into four families.

Family	Main names	Central idea
Behaviourist (connectionist)	Pavlov, Thorndike, Skinner	Learning is a link between a stimulus and a response, built by pairing or by consequences
Gestalt and cognitive	Wertheimer, Köhler, Koffka; Tolman; Lewin	Learning is a change in how the learner perceives and organises a situation
Social learning	Bandura	Learning by observing others
Constructivist and experiential	Piaget, Vygotsky, Bruner (2.2); Dewey	Learners build knowledge through activity and reflection

Ivan Pavlov: classical conditioning

Ivan Pavlov, a Russian physiologist, worked with dogs. A dog salivates naturally when food is put in its mouth. Pavlov rang a bell just before giving food, many times. In the end the bell alone made the dog salivate. This is **classical conditioning** (शास्त्रीय अनुबंधन): a neutral stimulus, paired again and again with a stimulus that naturally produces a response, comes to produce that response by itself.

Term	In Pavlov's experiment
Unconditioned stimulus (UCS)	Food: it produces salivation without any learning
Unconditioned response (UCR)	Salivation to food
Conditioned stimulus (CS)	The bell, after pairing with food (before pairing it is a neutral stimulus)
Conditioned response (CR)	Salivation to the bell alone

Four further terms:

- **Extinction** (विलोपन): if the bell is rung again and again without food, salivation to the bell fades.
- **Spontaneous recovery**: after a rest, the extinguished response returns for a while.
- **Stimulus generalisation** (उद्दीपक सामान्यीकरण): a response conditioned to one stimulus is also produced by similar stimuli. A child frightened by one stern teacher becomes afraid of all teachers.
- **Stimulus discrimination**: learning to respond only to the original stimulus, not to similar ones (fearing only that one teacher).
- Do not confuse stimulus generalisation with **response generalisation**, in which one stimulus comes to produce several similar responses. In stimulus generalisation, several similar stimuli produce the same response.

Classical conditioning explains **involuntary emotional reactions**. A student humiliated in a mathematics class may feel anxious on entering any mathematics classroom. The classroom, at first neutral, was paired with humiliation, which naturally produces distress; the classroom is now a conditioned stimulus for anxiety. This is classical, not operant, conditioning.

Edward Thorndike: trial and error and the three laws

E. L. Thorndike (Edward Lee Thorndike), an American psychologist, put hungry cats in a **puzzle box** with food outside. A cat tried many movements, and by chance one movement opened the door. Over many trials it dropped the useless movements and escaped faster. Thorndike concluded that learning is the "stamping in" of correct stimulus-response (S-R) connections by **trial and error** (प्रयास और त्रुटि). His theory is called **connectionism**.

His three main laws:

Law	What it says
Law of readiness (तत्परता का नियम)	When a learner is ready to act, acting is satisfying and not acting is annoying. When not ready, being forced to act is annoying.
Law of exercise (अभ्यास का नियम)	A connection is strengthened by use and weakened by disuse. "Practice makes perfect."
Law of effect (प्रभाव का नियम)	A response followed by a satisfying result is strengthened; one followed by an annoying result is weakened.

Satisfaction and annoyance appear in both readiness and effect. The deciding difference: readiness is about being prepared to act **before** the response; effect is about the consequences **after** the response.

B. F. Skinner: operant conditioning

B. F. Skinner (Burrhus Frederic Skinner), an American psychologist, studied rats and pigeons in a box (often called the Skinner box). A rat pressed a lever and got food, so it pressed more often. Skinner's principle: behaviour is shaped by its consequences. Responses that are reinforced tend to be repeated. This is **operant conditioning** (क्रियाप्रसूत अनुबंधन). "Operant" means a voluntary action that operates on the environment.

Learn the four terms with one rule. **Reinforcement** (पुनर्बलन) always increases a behaviour; **punishment** (दंड) always decreases it. **Positive** means something is added; **negative** means something is taken away.

	Something added (positive)	**Something removed (negative)**
Behaviour increases (reinforcement)	Positive reinforcement: praise after a correct answer	Negative reinforcement: an unpleasant thing is removed, e.g. students who submit every assignment on time are excused from a tedious test, and timely submission rises
Behaviour decreases (punishment)	Positive punishment: a reprimand is given	Negative punishment (response cost): something pleasant is taken away, e.g. cancelling recess for talking

Extinction in operant conditioning: a behaviour weakens when the reinforcer that kept it going is withheld. A teacher who stops paying attention to a student's attention-seeking interruptions sees them die out. Nothing unpleasant is added (so it is not punishment) and nothing unpleasant is removed (so it is not negative reinforcement).

Schedules of reinforcement. Reinforcement can be given after every response (continuous) or only sometimes (partial). Partial schedules are of four kinds. **Ratio** schedules depend on the **number** of responses; **interval** schedules depend on **time**.

Schedule	**Reinforcement comes**
Fixed-ratio	After a set number of responses (praise after every fifth correct answer)
Variable-ratio	After a varying, unpredictable number of responses around an average
Fixed-interval	After a set time (a weekly test)
Variable-interval	After varying, unpredictable times (surprise tests)

Skinner applied operant conditioning to teaching through **programmed instruction** (see 4.2): small steps, active response and immediate reinforcement.

Gestalt psychology and Köhler's insight

Gestalt (गेस्टाल्ट) is a German word meaning "form" or "whole". The Gestalt psychologists were Max Wertheimer, Wolfgang Köhler and Kurt Koffka. Their central principle: the whole is different from, and more than, the sum of its parts. We perceive and learn organised wholes, not separate pieces. Behaviourism and connectionism, by contrast, break learning into S-R units, and structuralism (Wundt and Titchener) broke consciousness into elements.

Wolfgang Köhler studied chimpanzees. His chimpanzee Sultan could not reach a banana outside the cage with either of two short sticks. After a pause he suddenly joined the two sticks and pulled the

banana in. Köhler called this **insight** (अंतर्दृष्टि): the learner perceives the whole situation and suddenly grasps how its parts are related. An insightful solution appears all at once, not through slow removal of errors, and it is easily repeated and transferred to new situations.

Edward Tolman: latent learning

Edward C. Tolman let rats explore a maze without any food reward. They seemed to learn nothing. But when food was later placed in the goal box, they quickly ran the maze as well as rats that had been rewarded all along. Learning had occurred without reinforcement and stayed hidden until it was needed. Tolman called this **latent learning** (अव्यक्त अधिगम).

Albert Bandura: observational learning

Albert Bandura, a Canadian-American psychologist, showed that people learn by watching others. In his **Bobo doll** experiment, children who watched an adult hit and kick an inflated doll later did the same when left alone with it. Learning by watching a **model** is **observational learning** (अवलोकनात्मक अधिगम). His theory is called **social learning theory**.

Bandura named four processes, which run in this order:

1. **Attention**: the observer must notice the model.
2. **Retention**: the observer must remember what was seen.
3. **Reproduction** (motor reproduction): the observer must be able to perform the behaviour.
4. **Motivation**: the observer must have a reason to perform it.

Vicarious reinforcement (प्रतिनिधिक पुनर्बलन) is reinforcement experienced at second hand. A student starts keeping her notebook neat after she sees a classmate praised for neat work. She herself received nothing. If she had been praised, it would be **direct reinforcement**; if she rewarded herself for meeting her own standard, **self-reinforcement**.

Bandura separated **acquisition** (learning a behaviour) from **performance** (showing it). A behaviour can be acquired simply by watching, with no reinforcement at all. Reinforcement, direct or vicarious, mainly decides whether the learnt behaviour is performed. So it is false that, for Bandura, no learning can occur without direct reinforcement.

Robert Gagné: eight types of learning

Robert M. Gagné set out the **conditions of learning** in his book *The Conditions of Learning* (1965). He arranged eight types of learning in a hierarchy, from simplest to most complex. Each type is a prerequisite for the next.

1. Signal learning (like Pavlov's conditioning)
2. Stimulus-response learning
3. Chaining (linking a series of S-R units)
4. Verbal association
5. Discrimination learning
6. Concept learning
7. Rule (principle) learning
8. Problem solving

For a sequencing question, remember the top four: discrimination, then concept, then rule, then problem solving. Concepts must be learnt before rules, and rules before problems can be solved.

John Dewey: learning by doing

John Dewey, the American philosopher and leader of **progressive education**, held that learners learn best through active experience and reflection on it, not by passively receiving information. The principle of **learning by doing** is linked with him. His ideas underlie the project method and the

problem-solving method (see 4.4). Herbart (formal steps of teaching), Pavlov (conditioning) and Spearman (general intelligence, see 2.8) are not linked with learning by doing.

Constructivism and the learner's errors

Constructivism (रचनावाद) holds that learners build their own knowledge by testing their existing ideas against experience. If that is so, an error is useful: it shows what the learner's present idea is and where it breaks down. In a constructivist classroom errors are treated as sources of insight, for the teacher and as an opportunity for the learner to rebuild understanding.

CLASSROOM EXAMPLE

In a B.Com. class in Rohtak, students who come late are greeted coldly and fall silent. The teacher changes her approach. She praises the first correct answer each day (positive reinforcement), stops reacting to a student's attention-seeking jokes (extinction), and lets a student who has earned five correct-answer points skip one routine worksheet (negative reinforcement). Before a hard topic she asks the best student to work one problem aloud on the board while others watch (observational learning, with attention, retention, reproduction and motivation).

OFTEN CONFUSED

- **Classical** (Pavlov) and **operant** (Skinner) conditioning. Classical: involuntary responses, learnt by pairing stimuli. Operant: voluntary behaviour, shaped by consequences.
- **Negative reinforcement** and **punishment.** Negative reinforcement removes something unpleasant and increases behaviour. Punishment decreases behaviour.
- **Negative punishment** and **negative reinforcement.** Both remove something. Negative punishment removes something pleasant to reduce behaviour; negative reinforcement removes something unpleasant to increase behaviour.
- **Extinction** and **punishment.** Extinction withholds the reinforcer; nothing unpleasant is applied.
- **Extinction** and **spontaneous recovery.** Recovery is the return of a response after extinction.
- **Stimulus generalisation** and **discrimination.** Generalisation: similar stimuli produce the response. Discrimination: only the original stimulus does.
- **Law of readiness** and **law of effect.** Readiness: before acting. Effect: after acting.
- **Trial and error** (Thorndike) and **insight** (Köhler). Gradual versus sudden.
- **Insight** (Köhler) and **latent learning** (Tolman). Both are cognitive ideas; insight comes from chimpanzees, latent learning from rats in mazes.
- **Ratio** and **interval** schedules. Number of responses versus time.
- **Direct** and **vicarious** reinforcement. Received oneself versus seen given to a model.
- **Bruner** (discovery), **Gagné** (conditions and hierarchy of learning) and **Ausubel** (meaningful reception). Bruner and Gagné are often exchanged.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

More on Thorndike

Thorndike later added subsidiary laws, among them the **law of belongingness**: a connection is learnt more easily when the response seems to belong with the stimulus. He also revised the law of exercise: he found that mere repetition does little to strengthen a connection unless the response has some effect, such as the learner getting knowledge of results. Drill and revision in classrooms still rest on the law of exercise, but practice works best with feedback.

Schedules and resistance to extinction

Continuous reinforcement gives the fastest learning at first, but the behaviour dies out fastest once reinforcement stops, because its absence is noticed at once. **Variable-ratio** schedules produce responding that is most **resistant to extinction**, because the learner cannot tell when reinforcement has stopped. Fixed-ratio and fixed-interval schedules are more predictable.

A language-learning app that awards points immediately after every correct answer is using **immediate positive reinforcement**, as Skinner recommended for programmed learning. If it also gives an occasional surprise bonus after an unpredictable number of answers, that bonus follows a **variable-ratio schedule**, which keeps the learner responding.

Techniques built on operant conditioning

Technique	What it does
Shaping	Reinforcing successive approximations (closer and closer steps) to a target behaviour, raising the requirement gradually. Praise a shy student first for raising her hand, then only for saying a word, then only for a full answer.
Chaining	Linking separate responses into a sequence
Modelling	Learning by imitating a model (Bandura)
Fading	Gradually withdrawing prompts
Token economy	Giving tokens (points, stars) that can be exchanged for rewards
Premack principle	A high-frequency (preferred) activity is used to reinforce a low-frequency (less preferred) one. “Finish the assignment first; then you may go to the playground.” Named after David Premack and popularly called “Grandma’s rule”.

Kurt Lewin: field theory

Kurt Lewin’s **field theory** states that behaviour is a function of the person and the psychological environment: **B = f(P, E)**. The person and the environment as that person perceives it together form the **life space**. Learning is a change in the cognitive structure of the life space. Teachers should therefore attend to the learner’s perceived situation, goals and barriers. Lewin’s field theory opposed Thorndike’s connectionism (practice and effect). Lewin is also known for studies of leadership styles.

Tolman: cognitive maps

Rats that had explored a maze took a new, shorter route to the food when their usual path was blocked. Tolman explained that they had learnt the layout of the maze, a mental picture he called a **cognitive map**, not a fixed chain of turns. A chain of habits would have sent them down the old, blocked path.

Bandura: social cognitive theory

Bandura later called his theory **social cognitive theory**. Its central principle is **reciprocal determinism**: personal factors (beliefs, expectations), behaviour and environment all influence one another. A student’s belief that she can do well makes her work harder; her work changes how

teachers treat her; their treatment in turn shapes her belief. Bandura rejected both one-way explanations, environment alone (the strict behaviourist view) and inner drives alone. His idea of **self-efficacy** is taught in 2.6.

More on Gagné

Five learned capabilities. Gagné named five kinds of learning outcome.

Capability	Example
Verbal information	Stating the dates of historical events
Intellectual skills	Using rules and concepts, e.g. applying a grammatical rule to new sentences
Cognitive strategies	Managing one's own learning and thinking, e.g. devising one's own way of remembering
Motor skills	Writing neatly, using a microscope
Attitudes	Choices of personal action, e.g. choosing to read beyond the syllabus

Nine events of instruction. Gagné listed nine steps for a lesson, each supporting an internal process in the learner.

Event (external)	Internal process it supports
1. Gaining attention	Reception
2. Informing learners of the objective	Expectancy
3. Stimulating recall of prior learning	Retrieval of earlier learning into working memory
4. Presenting the stimulus (content)	Selective perception
5. Providing learning guidance	Semantic encoding (into long-term memory)
6. Eliciting performance	Responding
7. Providing feedback	Reinforcement
8. Assessing performance	Retrieval and reinforcement
9. Enhancing retention and transfer	Retrieval and generalisation

In a sequencing question, the objective is stated before the content is presented, and feedback can only follow a performance.

Learning hierarchy and task analysis. A rule is a relation among concepts, so a rule can be learnt only if its concepts are mastered first. A student who cannot use "area equals length multiplied by breadth" should be checked first on the concepts of length, breadth and multiplication. Problem solving lies **above** rule learning; failure at a rule sends the teacher **down** to its prerequisites. Working downward from the target skill to its prerequisites is called **task analysis**.

Connectivism

Connectivism was proposed by George Siemens and Stephen Downes as a learning theory for the digital age. It sees learning as creating and moving through **networks** of people, information sources and devices. Among its principles: learning may reside in non-human appliances, and the capacity to know more is more important than what is known now. Knowing where to find knowledge matters more than holding it. The first MOOCs, later called **cMOOCs** (connectivist MOOCs), were built on this idea (see 4.6). Constructivism, by contrast, places knowledge in the learner's own constructions.

CHECK YOURSELF

1. Match List I with List II.

List I	List II
A. Pavlov	I. Insight learning
B. Skinner	II. Observational learning
C. Köhler	III. Classical conditioning
D. Bandura	IV. Operant conditioning

(A) A-IV, B-III, C-I, D-II
(B) A-III, B-IV, C-II, D-I
(C) A-III, B-I, C-IV, D-II
(D) A-III, B-IV, C-I, D-II

2. A teacher stops giving extra marks for neat answer sheets, and neatness gradually declines. This is:

(A) negative punishment
(B) extinction
(C) negative reinforcement
(D) spontaneous recovery

3. Arrange Bandura's processes of observational learning in order.

1. Reproduction
2. Attention
3. Motivation
4. Retention

(A) 2, 4, 1, 3
(B) 4, 2, 1, 3
(C) 2, 1, 4, 3
(D) 2, 4, 3, 1

4. Statement I: In Pavlov's experiment, the bell before any pairing with food is the unconditioned stimulus.

Statement II: Thorndike's law of exercise states that use strengthens a connection and disuse weakens it.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

5. A student who is late three times loses the privilege of choosing her project topic, and her late-coming falls. In operant terms this is:

(A) positive reinforcement
(B) negative reinforcement
(C) negative punishment
(D) positive punishment

6. Assertion (A): Bandura held that a behaviour can be learnt by a child who has never been rewarded for it.

Reason (R): Bandura distinguished the acquisition of a behaviour from its performance.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

7. Which of the following are correctly paired? Choose the correct answer from the options given below:

1. Tolman – latent learning
2. Gagné – hierarchy of eight types of learning
3. Thorndike – insight
4. Dewey – learning by doing

(A) 1, 2 and 4 only
(B) 1, 2 and 3 only
(C) 2, 3 and 4 only
(D) 1 and 4 only

8. Which schedule of reinforcement is most resistant to extinction?

(A) Continuous
(B) Fixed-interval
(C) Fixed-ratio
(D) Variable-ratio

2.4 Academic characteristics: learning styles, study habits and self-regulation

Syllabus: Characteristics of adolescent and adult learners (academic)

EXAM FOOTPRINT

Rarely asked directly; it supports items on learner characteristics and learner-centred methods.

Core

Academic characteristics are the ways a learner goes about study: how he or she prefers to take in information, how he or she studies, and how far he or she can plan and check his or her own learning. Three ideas cover this: learning styles, study habits and self-regulation.

Learning styles

A **learning style** (अधिगम शैली) is a learner's preferred way of taking in and working with information. Two models are asked most often.

Fleming's VARK. Neil Fleming, a teacher in New Zealand, developed the VARK questionnaire in 1987. VARK stands for:

Preference	The learner likes
Visual	Diagrams, flow charts, graphs, maps
Aural (auditory)	Listening, lectures, talking a topic through in a group
Read/Write	Information as words: lists, written notes, handouts, textbooks

Preference	The learner likes
Kinaesthetic	Real experience, practice, examples, field work and experiments

So V is Visual, not verbal, vocal or virtual. (VAK is an older, three-part version without Read/Write.)

Kolb's experiential learning cycle. David A. Kolb, in *Experiential Learning* (1984), described learning as a cycle of four stages. Learning is the process of creating knowledge from experience, so his theory is called **experiential learning** (अनुभवात्मक अधिगम).

1. **Concrete experience (CE)**: the learner has an experience.
2. **Reflective observation (RO)**: the learner reflects on it.
3. **Abstract conceptualisation (AC)**: the learner draws general ideas from it.
4. **Active experimentation (AE)**: the learner tests the ideas in new situations, which leads to new experience.

The stage that follows concrete experience is reflective observation.

Kolb's four **learning styles** each combine two neighbouring stages of the cycle.

Style	Combines	Strength
Diverging (diverger)	Concrete experience + reflective observation ("feeling and watching")	Imagination, seeing a situation from many angles
Assimilating (assimilator)	Abstract conceptualisation + reflective observation ("thinking and watching")	Building theories and logical models
Converging (converger)	Abstract conceptualisation + active experimentation ("thinking and doing")	Practical application of ideas; solving problems with one answer
Accommodating (accommodator)	Concrete experience + active experimentation ("feeling and doing")	Hands-on work, learning by trial, relying on intuition

A memory aid: diverger and assimilator both **watch** (reflective observation); converger and accommodator both **do** (active experimentation). Diverger and accommodator both start from **feeling** (concrete experience); assimilator and converger from **thinking** (abstract conceptualisation).

What the evidence says about learning styles

Learners do have **preferences**. But they do not have one exclusive channel, and most people can learn well through several modes. Reviews of research have found little sound evidence that teaching each learner in his or her diagnosed style raises achievement (see Going Further). Yet teachers are rightly advised to use a variety of methods and media. The reasons are that a class contains diverse learners and that the content itself often demands a particular mode (a map for geography, a demonstration for a laboratory skill). The defensible lesson is: vary methods for everyone; do not label students by style.

Study habits

Study habits (अध्ययन आदतें) are a learner's regular ways of studying: when, where and how. Good study habits include a fixed timetable, a quiet place, making notes in one's own words, revising soon after learning and at spaced intervals, and testing oneself. A well-known study method is **SQ3R**, given by Francis P. Robinson: Survey, Question, Read, Recite, Review. It turns passive reading into active reading. (For why spaced revision and self-testing work, see the memory facts in 2.2.)

Self-regulation and metacognition

Metacognition (अधिसंज्ञान) means thinking about one's own thinking: knowing how one learns, and checking whether one has understood. The term is linked with John Flavell.

A **self-regulated learner** (स्व-नियमित शिक्षार्थी) plans, monitors and evaluates his or her own learning. Barry Zimmerman described self-regulation as a cycle of three phases.

1. **Forethought**: setting goals and planning, with belief in one's ability.
2. **Performance** (performance control): using strategies, managing time and attention, monitoring progress.
3. **Self-reflection**: judging the result, finding causes, and adjusting for next time.

Self-regulation matters more in college than in school, because the college student has more free time and less daily supervision. Teachers can build it by sharing objectives, teaching study strategies openly, and asking students to reflect on how they learnt, not only on what they learnt.

CLASSROOM EXAMPLE

A teacher of B.A. Geography in Fatehabad teaches the monsoon. She shows a map of wind directions (visual), explains the process and lets pairs discuss it (aural), gives a one-page written summary (read/write), and asks students to record the rainfall in the college gauge for a week (kinaesthetic). At the end she asks each student to write two lines: "What helped me understand? What will I revise tomorrow?" This builds metacognition and self-regulation. She does not divide the class by style; every student gets all four modes.

OFTEN CONFUSED

- **VARK** (Fleming) and **Bruner's modes** (enactive, iconic, symbolic). VARK is about preferences; Bruner's modes are ways knowledge is represented.
- **Diverger** and **assimilator.** Both use reflective observation. Diverger starts from concrete experience; assimilator from abstract ideas.
- **Converger** and **accommodator.** Both use active experimentation. Converger: abstract ideas, practical application. Accommodator: concrete experience, hands-on intuition.
- **Field-independent** and **field-dependent** (Going Further). Analytical and self-directed versus global and socially oriented.
- **Impulsive** and **reflective** tempo (Going Further). Fast and inaccurate versus slow and accurate.
- **Reflector** and **theorist** (Going Further). Reflector watches and gathers data; theorist fits facts into a logical system.
- **Pragmatist** and **activist** (Going Further). Pragmatist tests ideas in practice; activist plunges into new experience.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Witkin: field dependence and field independence

Herman A. Witkin studied a **cognitive style** (संज्ञानात्मक शैली), a person's typical way of perceiving and thinking. **Field-independent** (क्षेत्र-स्वतंत्र) people can separate an item from its background (the "field"). They tend to analyse a situation into parts and work well on their own. **Field-dependent** (क्षेत्र-आश्रित) people see the situation as a whole, rely more on outside cues, attend to social cues and prefer group learning with clear guidance. Witkin measured this with tests such as the Embedded Figures Test, in which a simple shape must be found hidden in a complex figure.

Kagan: reflective and impulsive tempo

Jerome Kagan distinguished two **cognitive tempos**. **Impulsive** learners answer fast and make many errors. **Reflective** learners take longer and are usually right. He measured this with the **Matching Familiar Figures Test**, in which the learner must find which of several similar pictures exactly matches a standard. Impulsive learners can be taught to pause and check alternatives. Do not confuse tempo with field independence (separating figure from background) or with convergent thinking (J. P. Guilford's term for thinking that seeks one right answer; see 2.8).

Honey and Mumford

Peter Honey and Alan Mumford built four learning styles on Kolb's cycle. Each style matches one stage.

Style	Kolb stage	The learner
Activist	Concrete experience	Plunges into new experiences
Reflector	Reflective observation	Stands back, gathers data, observes from several angles before concluding
Theorist	Abstract conceptualisation	Integrates observations into logical theories
Pragmatist	Active experimentation	Tries ideas out to see whether they work in practice

Kolb's roots and the stability of styles

Kolb built his cycle on the work of John Dewey, Kurt Lewin and Jean Piaget. Like Dewey, he places experience followed by reflection at the centre of learning. Kolb described the styles as relatively stable preferences, shaped by personality, education and career, which can change over time. They are not fixed traits. He held that mature learners can use all four modes.

The evidence on matching teaching to style

In 2008 Harold Pashler, Mark McDaniel, Doug Rohrer and Robert Bjork reviewed the research on learning styles. They tested the **meshing hypothesis**: the claim that teaching matched to a learner's diagnosed style improves learning. They found little sound evidence for it. So Fleming did develop VARK, but research has not firmly established that matching teaching to style greatly raises achievement. Varying methods for all learners is the defensible practice.

CHECK YOURSELF

1. Match List I (Kolb's styles) with List II (the modes that define them).

List I	List II
A. Diverger	I. Abstract conceptualisation and active experimentation
B. Assimilator	II. Concrete experience and active experimentation
C. Converger	III. Concrete experience and reflective observation
D. Accommodator	IV. Abstract conceptualisation and reflective observation

(A) A-IV, B-III, C-I, D-II
(B) A-III, B-IV, C-I, D-II
(C) A-III, B-IV, C-II, D-I
(D) A-II, B-IV, C-I, D-III

2. In Fleming's model, a student who learns best by talking a topic through with classmates shows which preference?

(A) Visual
(B) Read/Write
(C) Aural
(D) Kinaesthetic

3. Arrange the stages of Kolb's cycle, starting from concrete experience.

1. Active experimentation
2. Abstract conceptualisation
3. Reflective observation
4. Concrete experience

(A) 4, 2, 3, 1
(B) 4, 3, 1, 2
(C) 4, 1, 3, 2
(D) 4, 3, 2, 1

4. Assertion (A): A teacher should present a topic through several modes, such as diagrams, discussion and practice.

Reason (R): Research has shown that each learner can learn through one fixed style only.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

5. Which of the following are correct? Choose the correct answer from the options given below:

1. Field-independent learners tend to analyse a situation into its parts.
2. Kagan's impulsive learners respond slowly but accurately.
3. Honey and Mumford's pragmatist corresponds to Kolb's active experimentation.

4. Metacognition means thinking about one's own thinking.

(A) 1, 2 and 3 only
(B) 2, 3 and 4 only
(C) 1, 3 and 4 only
(D) 1 and 4 only

6. Statement I: Kolb held that learning styles can change with education and career.

Statement II: SQ3R stands for Survey, Question, Read, Recite, Review.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

2.5 Social characteristics

Syllabus: Characteristics of adolescent and adult learners (social)

EXAM FOOTPRINT

Rarely asked directly; it supports multiple-select items on learner-related factors affecting teaching.

Core

Social characteristics (सामाजिक विशेषताएँ) are the ways a learner relates to other people: friends, family, teachers and society. For the college learner, two things matter most in the examination: the pull of the peer group, and the way moral reasoning develops.

The peer group

A **peer group** (समवयस्क समूह) is a group of people of about the same age and status, such as classmates and friends. During adolescence the strongest social influence on behaviour usually comes from the peer group, not from grandparents, siblings or neighbours. Adolescents seek the acceptance and approval of their peers. Peer norms (the group's unwritten rules) shape their dress, speech, interests and even their attitude to study.

The social adolescent shows these features:

- a strong need to belong and to be accepted;
- loyalty to friends and a wish to conform to the group;
- growing independence from parents, with peers becoming a major reference group;
- interest in relationships, including with the other sex;
- concern for fairness and for social causes, with a wish to take part in them.

For the teacher, peer influence is a resource, not only a risk. Late adolescents seek the company and approval of peers and learn readily from one another. So group projects, discussions, seminars and peer learning (see 4.3) suit undergraduate learners. **Cooperative learning** (सहकारी अधिगम), in which small groups work towards a shared goal and each member is responsible for a part, uses this characteristic directly. The teacher must also watch for exclusion, so that no student is left out of every group.

Lawrence Kohlberg: stages of moral reasoning

Lawrence Kohlberg, an American psychologist, studied how people reason about right and wrong. He presented moral dilemmas (the best known is the Heinz dilemma, in which a man must decide whether to steal a drug to save his dying wife) and studied the **reasons** people gave, not only their answers. He described three levels of **moral development** (नैतिक विकास), each with two stages.

Level	Stage	Reasoning is guided by
Pre-conventional (पूर्व-परंपरागत)	1	Avoiding punishment; obedience to authority
	2	Self-interest; reward; "you scratch my back, I scratch yours"
Conventional (परंपरागत)	3	Approval of others: the "good boy, good girl" orientation
	4	Law and order: upholding rules and duty to keep society working
Post-conventional (उत्तर-परंपरागत)	5	Social contract: laws are agreements that can be changed for the common good
	6	Universal ethical principles (justice, human dignity) that may override particular rules

How to place a statement:

- If the reason is punishment, reward or personal gain, it is **pre-conventional**.
- If the reason is approval, or keeping rules and social order, it is **conventional**. "Rules are rules; if everyone broke them, the system would collapse" is stage 4, law and order, within the conventional level.
- If the reason judges rules by higher principles, it is **post-conventional**.

Most adolescents and adults reason at the conventional level. Kohlberg's stages describe moral **reasoning**, a separate part of development from identity (2.1).

Piaget's two moralities

Before Kohlberg, Piaget studied children's moral judgement. He described **heteronomous morality** (rules are fixed and come from authority; judge an act by its consequences) in younger children, and **autonomous morality** (rules are agreements that can be changed; judge an act by its intention) in older children. "Heteronomous" is Piaget's term, not a level in Kohlberg's scheme.

> **CLASSROOM EXAMPLE**
>
> In an M.A. Political Science class in Kurukshetra, the teacher gives a case: a student finds the answer key of an internal test on a printer. Should she tell the teacher, use it quietly, or share it with friends? Students discuss in groups of four and must give reasons. One says, "I would not use it because I might be caught" (pre-conventional). Another says, "Our class has agreed to be honest, and the rules protect everyone" (conventional). A third says, "Fair assessment is a matter of justice to every student, whatever the rules say" (post-conventional). The teacher uses peer discussion, which suits the age group, to move reasoning towards higher stages.

OFTEN CONFUSED

- **Pre-conventional** and **conventional.** Punishment and self-interest versus approval and law and order.
- **Conventional stage 4** and **post-conventional stage 5.** Stage 4 upholds rules as such; stage 5 sees rules as social agreements that can be changed for the common good.
- **Kohlberg's levels** and **Piaget's moralities.** Pre-conventional, conventional and post-conventional are Kohlberg's; heteronomous and autonomous are Piaget's.
- **Moral reasoning** (Kohlberg) and **identity** (Erikson, Marcia).
- **Peer group** and **family.** In adolescence the peer group becomes the strongest everyday influence on behaviour, though family values remain important.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Criticisms of Kohlberg

Carol Gilligan, in *In a Different Voice* (1982), argued that Kohlberg's scheme rests on an **ethic of justice** and overlooks an **ethic of care**, which stresses relationships and responsibility for others. Others have noted that stage 6 is rarely found, and that the scheme reflects Western, individualist values. The examination may ask you to name Gilligan as the critic who proposed the ethic of care.

Using dilemmas in college teaching

Discussing moral dilemmas in groups, with students hearing reasoning one stage above their own, is the classroom method that grew out of Kohlberg's work. It suits subjects such as law, management, medicine, journalism and teacher education, where professional ethics is taught (see 1.10).

CHECK YOURSELF

1. "I will not cheat because the teacher may catch me and fail me." In Kohlberg's scheme this reasoning is:

(A) post-conventional
(B) conventional, law and order
(C) pre-conventional
(D) conventional, good boy-good girl

2. Assertion (A): Group discussions and projects suit undergraduate learners.

Reason (R): Late adolescents seek the approval of peers and learn readily from one another.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

3. Match List I with List II.

List I	List II
A. Stage 3	I. Universal ethical principles
B. Stage 4	II. Approval of others
C. Stage 5	III. Law and order
D. Stage 6	IV. Social contract

(A) A-II, B-III, C-I, D-IV
(B) A-III, B-II, C-IV, D-I
(C) A-II, B-IV, C-III, D-I
(D) A-II, B-III, C-IV, D-I

4. Statement I: "Heteronomous morality" is one of Kohlberg's three levels.

Statement II: Most adolescents and adults reason at Kohlberg's conventional level.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

5. Who criticised Kohlberg for neglecting the ethic of care?

(A) Carol Gilligan
(B) Jean Piaget
(C) Erik Erikson
(D) James Marcia

2.6 Emotional characteristics and motivation

Syllabus: Learner's characteristics: characteristics of adolescent and adult learners (academic, social, emotional and cognitive)

> **EXAM FOOTPRINT**
>
> Asked occasionally; usually a single-answer item separating intrinsic from extrinsic motivation, or ordering a needs hierarchy.

Core

A learner is not only a thinking mind. A learner also feels: hope, fear, boredom, pride, shame. These feelings, or emotions (संवेग), decide whether the learner starts a task, how hard she tries and how long she keeps going. The force that starts, directs and sustains behaviour towards a goal is called **motivation** (अभिप्रेरणा). This section explains the main theories of motivation that the examination asks about, and the emotional states, such as anxiety, that help or harm learning.

Intrinsic and extrinsic motivation

Intrinsic motivation (आंतरिक अभिप्रेरणा) comes from within the learner. She studies because the subject is interesting, because she is curious, or because doing the task well is satisfying in itself. The reward is inside the activity.

Extrinsic motivation (बाह्य अभिप्रेरणा) comes from outside the activity. The learner studies for marks, a prize, a scholarship, praise, a job, or to avoid punishment. The reward lies outside the activity of studying.

A simple test decides the type: ask “would the learner still do it if the outside reward were removed?” If yes, the motivation is intrinsic. A student who studies hard only to win a scholarship is extrinsically motivated. A student who reads history at night because the past fascinates her is intrinsically motivated.

Intrinsic motivation generally leads to deeper and longer-lasting learning. Extrinsic rewards work in the short run, but if they are overused they can weaken a learner’s own interest. Going Further explains why.

Point	Intrinsic	Extrinsic
Source	Inside the activity: interest, curiosity, satisfaction	Outside the activity: marks, prizes, praise, fear
Example	Reading poetry because it moves you	Reading poetry because it is in the examination
Effect	Deeper, more lasting learning	Quick results; interest may fade when rewards stop

Maslow’s hierarchy of needs

Abraham Maslow, a humanistic psychologist, arranged human needs in a **hierarchy of needs** (आवश्यकता पदानुक्रम), usually drawn as a pyramid. From the base upwards the five levels are:

1. Physiological needs: food, water, sleep, shelter.
2. Safety needs: security, freedom from danger and fear.
3. Love and belonging needs: friendship, acceptance, being part of a group.
4. Esteem needs: self-respect, recognition, respect from others.
5. Self-actualisation (आत्मसिद्धि): realising one’s full potential, becoming all that one can be.

Physiological needs form the base, and self-actualisation stands at the top. The main idea is that lower needs must be reasonably satisfied before higher needs begin to motivate behaviour. A

hungry or frightened student finds it hard to attend to lessons. This is the thinking behind giving a mid-day meal to school children, and behind making a college campus safe and welcoming.

Maslow divided the needs into two kinds. The four lower needs (physiological, safety, belonging and esteem) are **deficiency needs**. They arise from a lack, and once they are met they stop motivating; a person who has eaten is no longer driven by hunger. Self-actualisation is a **growth need** (also called a being need). It does not arise from a lack, and it goes on motivating, even growing stronger, as it is satisfied. Note the trap: esteem is the highest of the deficiency needs, but it is still a deficiency need.

Other names that appear beside Maslow

Questions on Maslow often list other motivation theorists as distractors. Learn one line for each.

- **Frederick Herzberg**: two-factor theory of work motivation (hygiene factors and motivators).
- **David McClelland**: theory of achievement motivation (उपलब्धि अभिप्रेरणा); the need for achievement, written n-Ach.
- **Carl Rogers**: humanistic psychology; the teacher as a facilitator of learning.
- **Kurt Lewin**: field theory; studies of leadership styles (autocratic, democratic, laissez-faire).

McClelland's need for achievement is the desire to do things well and to succeed against a standard of excellence. People high in n-Ach prefer tasks of moderate difficulty, where success is possible but not certain, and they want feedback on how they are doing. Very easy tasks give no sense of achievement, and very hard ones offer little chance of success.

Emotional intelligence

Emotional intelligence (संवेगात्मक बुद्धि) is the ability to recognise, understand and manage one's own emotions and those of others. The term was introduced by Peter Salovey and John Mayer (1990). It was made popular by Daniel Goleman's book *Emotional Intelligence* (1995). Goleman's components are usually listed as five:

- self-awareness: knowing one's own emotions;
- self-regulation: managing them;
- motivation: using emotions to pursue goals;
- empathy: recognising and understanding the feelings of others;
- social skills: handling relationships well.

Do not confuse this with neighbouring names. Howard Gardner's interpersonal and intrapersonal intelligences (section 2.8) are related ideas but belong to a different theory, the theory of multiple intelligences. L. L. Thurstone proposed primary mental abilities, and Alfred Binet built the first intelligence test. If the question asks who popularised emotional intelligence, the answer is Goleman; if it asks who introduced the term, it is Salovey and Mayer.

Self-efficacy

Self-efficacy (आत्म-प्रभावकारिता) is Albert Bandura's term for a person's belief in his or her own capability to succeed at a particular task. It is a belief, not the actual ability. Bandura showed that learners with high self-efficacy choose harder tasks, try harder and persist longer when they meet difficulty, whereas learners who doubt themselves give up early. The belief in one's capability is the mechanism: it influences how much effort is put in and how long it lasts. (Bandura's observational learning is taught in section 2.3.)

Anxiety and performance: the Yerkes-Dodson law

Anxiety (दुश्चिंता) is a feeling of worry and tension about what may happen. A little anxiety before an examination is normal and can even help. The relation between arousal (alertness, tension) and performance is described by the **Yerkes-Dodson law**, named after Robert Yerkes and his co-author Dodson (1908).

The law describes an inverted U. As arousal rises from a low level, performance improves; it reaches a peak at a moderate, optimum level; beyond that, more arousal makes performance worse. The optimum is lower for difficult tasks than for easy ones. A bored, sleepy student does poorly; a moderately keyed-up student does best; a panicking student does poorly again.

So two statements are true: moderate anxiety can help performance, and very high anxiety harms it. The statement “performance rises steadily as arousal rises” is false, because it would make extreme panic the best state of all.

CLASSROOM EXAMPLE

In a B.A. first-year class at a district college in Haryana, Ms Kaur notices that several students from distant villages come without breakfast after a long bus journey and doze in the first period. Reading Maslow, she sees that their physiological needs are unmet. She persuades the college to open the canteen early with low-cost tea and snacks. Next, she makes the class safe for mistakes: no ridicule when an answer is wrong (safety and belonging). She praises specific improvement (esteem). Only then does she introduce an optional reading club for those who want to go further (growth). Before the semester test she tells the class, “Some nervousness is normal; it keeps you alert”, and gives a short practice test so that anxiety stays moderate, not overwhelming.

OFTEN CONFUSED

- Intrinsic and extrinsic motivation: is the reward inside the activity (interest, satisfaction) or outside it (marks, scholarship, fear)?
- Deficiency and growth needs: physiological, safety, belonging and esteem stop motivating once met; self-actualisation keeps motivating. Esteem is the top deficiency need, not a growth need.
- Introduced and popularised emotional intelligence: Salovey and Mayer introduced the term; Goleman popularised it.
- Emotional intelligence and Gardner’s personal intelligences: Goleman’s EI is a separate theory; interpersonal and intrapersonal intelligences belong to Gardner’s multiple intelligences.
- Self-efficacy and self-esteem: self-efficacy is belief in capability for a task; self-esteem is a general feeling of one’s worth.
- Yerkes-Dodson: moderate arousal is best (inverted U), not “more is always better”.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Weiner’s attribution theory

Bernard Weiner asked how learners explain their own successes and failures. The cause a learner names is an **attribution** (गुणारोपण). Weiner classified causes along two main dimensions: locus (is the cause inside the person or outside?) and stability (does it stay the same from one occasion to the next, or change?). He later added a third, controllability. The four classic causes fall as follows.

Cause	Locus	Stability
Ability	Internal	Stable
Effort	Internal	Unstable

Cause	Locus	Stability
Task difficulty	External	Stable
Luck	External	Unstable

A student who says “I failed because I did not work hard enough” names effort: internal and unstable. This is the healthiest attribution after failure, because effort can be changed next time. A student who says “I failed because I am not clever” names ability: internal and stable, which leads to hopelessness.

Locus of control and learned helplessness

Julian Rotter’s **locus of control** (नियंत्रण-स्थल) is where a person places the control of what happens to him. A learner with an internal locus believes results depend on his own effort and decisions; such learners take responsibility, work harder and persist. A learner with an external locus believes results depend on luck, fate, the teacher’s favour or other people.

Learned helplessness (अधिगत असहायता), described by Martin Seligman, is the state in which, after repeated failure that one could not control, a person comes to believe that nothing he does makes any difference. He then stops trying, even on tasks at which he could succeed. In students it shows as passivity and giving up. Graded tasks that guarantee early success, and teaching students to attribute results to effort and strategy, help to reverse it.

Do not confuse learned helplessness with: test anxiety (fear that interferes with performance during a test; the student still tries); field dependence (a cognitive style, the tendency to perceive things as a whole and be influenced by surroundings); identity diffusion (one of James Marcia’s identity statuses, an adolescent who has neither explored nor committed to an identity; see section 2.1).

Carol Dweck’s mindsets

Carol Dweck distinguished two beliefs about ability. In a **fixed mindset** (also called the entity theory), ability is a fixed quantity that cannot be changed. Learners with this belief avoid difficult tasks, to protect their image, and treat failure as proof of low ability. In a **growth mindset** (also called the incremental theory), ability grows with effort and good strategy. These learners welcome challenge and treat failure as information. Dweck found that praising effort and strategy (“you tried a new method and it worked”) encourages a growth mindset, while praising intelligence (“you are so clever”) encourages a fixed one. Remember: “incremental” is the technical name for growth, the opposite of fixed.

Achievement goals

Achievement goal theory asks what a learner is trying to achieve when she works hard. A **mastery goal** (also called a learning goal) aims to understand and to improve one’s own competence. A **performance goal** aims to demonstrate ability compared with others: to be seen as the brightest, or not to look stupid. A **work-avoidance goal** aims to get by with the least effort. Two students may work equally hard; the deciding question is why. Mastery goals go with deeper study strategies and with persistence after failure.

Atkinson and expectancy-value theory

John Atkinson treated achievement behaviour as the result of two opposing motives: the motive to succeed (hope of success) and the motive to avoid failure (fear of failure). He also held that motivation depends on the expectancy of success and the value (incentive) of success.

Expectancy-value theory (developed further by Jacquelynne Eccles and colleagues for school learning) states the idea simply: motivation = expectancy of success × value of the goal. Because the two are multiplied, if either is near zero the motivation is near zero. A student who values a course

highly but is certain she cannot pass it will make little effort. This is why building confidence matters as much as showing the value of a subject.

Vroom's expectancy theory: a small calculation

Victor Vroom stated a similar idea for work motivation, with three terms:

- **Expectancy**: the belief that effort will lead to good performance (a probability from 0 to 1).
- **Instrumentality**: the belief that good performance will lead to the outcome (0 to 1).
- **Valence**: the value the person places on the outcome.

Motivational force = expectancy × instrumentality × valence.

WORKED EXAMPLE

A student is almost sure that effort will bring good marks (expectancy 0.9), fairly sure that good marks will bring a scholarship (instrumentality 0.5), but places no value at all on the scholarship (valence 0).

Force = 0.9 × 0.5 × 0 = 0

Another student has expectancy 0.8, instrumentality 0.5 and valence 0.9.

Force = 0.8 × 0.5 × 0.9 = 0.36

The terms are multiplied, not added or averaged. A zero anywhere makes the whole product zero: nobody strives for an outcome that has no value for him, however sure the path.

USUAL SLIPS

- Adding the terms (0.9 + 0.5 + 0 = 1.4) or averaging them (about 0.47). The model multiplies.
- Leaving the zero out of the product (0.9 × 0.5 = 0.45). A zero term cannot be dropped.

More needs theories: Herzberg, Alderfer, McClelland

Herzberg's two-factor theory (motivation-hygiene theory) came from studies of people at work. **Hygiene factors**, such as salary, job security, supervision, company policy and working conditions, prevent dissatisfaction when they are adequate, but do not by themselves motivate. **Motivators**, such as achievement, recognition, responsibility, the work itself and growth, produce satisfaction and real motivation. For a college: a decent classroom and fair rules prevent complaints; interesting work and recognition make students want to learn.

Clayton Alderfer's ERG theory condensed Maslow's five levels into three: existence, relatedness and growth.

McClelland's acquired-needs theory names three needs: achievement, affiliation and power. They are learned through experience, not inborn, which is why McClelland held that achievement motivation can be developed through training.

Maslow, too, did not say that a lower need must be completely satisfied before the next appears. He spoke of relative satisfaction: a need must be reasonably satisfied, and a person may be partly satisfied at several levels at once.

Theorist	Key terms to recognise
Maslow	Physiological, safety, love and belonging, esteem, self-actualisation
Alderfer	Existence, relatedness, growth (ERG)

Theorist	Key terms to recognise
Herzberg	Hygiene factors and motivators
McClelland	Needs for achievement, affiliation, power
Vroom	Valence, instrumentality, expectancy
Deci and Ryan	Autonomy, competence, relatedness
Weiner	Attribution: locus, stability (and controllability)
Atkinson	Motive to succeed and motive to avoid failure
Bandura	Self-efficacy

Self-determination theory and the overjustification effect

Edward Deci and Richard Ryan's **self-determination theory** holds that intrinsic motivation flourishes when three basic psychological needs are met: **autonomy** (feeling that one acts by choice), **competence** (feeling effective) and **relatedness** (feeling connected to others). Note that "relatedness" also appears in Alderfer's ERG; the deciding word is "autonomy", which only Deci and Ryan use.

Research in this tradition gives a balanced picture of rewards:

- Expected tangible rewards (money, prizes) for a task that is already interesting can weaken intrinsic motivation.
- Unexpected praise that carries information about performance ("your argument in the second paragraph is clear") tends to sustain intrinsic motivation.
- Teaching that supports learners' autonomy (choices, reasons, respect for their views) promotes intrinsic motivation.
- Extrinsic motivation is not always harmful. It is useful for tasks that are not interesting in themselves, and extrinsic reasons that the learner has accepted as his own work well.

The **overjustification effect** is the name for the first point. When people are rewarded for something they already enjoy, they may come to see the reward as their reason for doing it; when the reward stops, their interest falls below its original level. Keep it apart from three other named effects:

- **Hawthorne effect**: people change their behaviour because they know they are being observed or studied.
- **Pygmalion effect**: learners tend to perform in line with what teachers expect of them (expectations become self-fulfilling).
- **Zeigarnik effect**: unfinished tasks are remembered better than finished ones.

Bandura's four sources of self-efficacy

Bandura named four sources from which learners build self-efficacy:

1. **Mastery experiences**: one's own past successes. This is the strongest source, because it gives direct evidence of capability.
2. **Vicarious experiences**: seeing similar others succeed ("if she can, so can I").
3. **Verbal persuasion**: encouragement from a respected person.
4. **Physiological and emotional states**: reading one's own calm or tension as a sign of capability or weakness.

Keller's ARCS model

John Keller's **ARCS model** gives teachers four conditions for motivating learners:

- **Attention**: gain and hold it, for example by opening with a puzzling case or a surprising fact.
- **Relevance**: link the topic to learners' goals, such as their careers.
- **Confidence**: arrange tasks so that learners expect to succeed, for example graded tasks with early success.
- **Satisfaction**: let learners feel good about what they achieved, through rewards, recognition and chances to use what they have learnt.

Confidence is about the expectation of success before and during the task; satisfaction is about the feeling after achievement.

CHECK YOURSELF

1. A student reads Premchand's stories in her free time only because she enjoys them, though they are not in her syllabus. Her motivation is best described as:

(A) extrinsic
(B) intrinsic
(C) instrumental
(D) competitive

2. Match List I with List II.

List I (Theorist)	List II (Concept)
A. Herzberg	I. Autonomy, competence and relatedness
B. Deci and Ryan	II. Hygiene factors and motivators
C. Alderfer	III. Locus and stability of causes
D. Weiner	IV. Existence, relatedness and growth

Choose the correct answer from the options given below:

(A) A-II, B-I, C-IV, D-III
(B) A-II, B-IV, C-I, D-III
(C) A-III, B-I, C-IV, D-II
(D) A-IV, B-I, C-II, D-III

3. Assertion (A): A student who is in a state of panic before a difficult examination is likely to perform below her best.

Reason (R): According to the Yerkes-Dodson law, performance falls when arousal rises beyond an optimum level, and the optimum is lower for difficult tasks.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. Which of the following are deficiency needs in Maslow's theory?

1. Safety
2. Esteem
3. Self-actualisation
4. Belongingness

Choose the correct answer from the options given below:

(A) 1 and 3 only
(B) 2, 3 and 4 only
(C) 1, 2 and 4 only
(D) 1, 2, 3 and 4

5. A learner has expectancy 0.6, instrumentality 0.5 and valence 0.8. By Vroom's model her motivational force is:

(A) 0.24
(B) 0.63
(C) 1.90

(D) 0.30

6. Statement I: Daniel Goleman introduced the term emotional intelligence in 1990.

Statement II: Self-awareness and empathy are among the components of emotional intelligence as Goleman described it.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

7. After a poor result a student says, “The paper was set far above our level.” In Weiner’s scheme the cause named is:

(A) internal and stable
(B) internal and unstable
(C) external and unstable
(D) external and stable

8. A teacher wants to strengthen the self-efficacy of a student who has failed twice. Which step uses the strongest source of self-efficacy?

(A) Telling him that he is capable
(B) Showing him a video of a senior who once failed and later topped
(C) Giving him a sequence of tasks on which he actually succeeds
(D) Teaching him to relax his breathing before tests

2.7 Adult learners: andragogy and beyond

Syllabus: Learner’s characteristics: characteristics of adolescent and adult learners (academic, social, emotional and cognitive)

EXAM FOOTPRINT

Asked occasionally; usually a two-statement or assertion-reason item on how adults learn differently from children.

Core

Most students in a university are adults. Many in distance and evening programmes are working men and women, parents and teachers themselves. The examination asks how adults learn differently from children, and the answer is built almost entirely on one thinker, Malcolm Knowles.

Pedagogy and andragogy

Pedagogy (शिक्षाशास्त्र) comes from Greek roots meaning “leading the child”. It refers to the teaching of children. In the pedagogical model the teacher decides what is to be learnt, how and when; the learner is dependent and follows.

Andragogy (प्रौढ़ शिक्षाशास्त्र) is the art and science of helping adults learn. The word was first used in 1833 by the German teacher Alexander Kapp, but it was the American adult educator **Malcolm Knowles** who popularised it and built a theory on it. Distractors in such questions are usually Piaget (cognitive development), Skinner (operant conditioning) and Bloom (taxonomy of objectives), none of whom is linked with andragogy.

Knowles's assumptions about adult learners

Knowles described the adult learner through a set of assumptions. He began with four and later added two more, so most books now list six. Learn them with their meaning.

1. **Self-concept.** As a person matures, the self-concept moves from dependence towards self-direction. Adults see themselves as responsible for their own decisions, so they resist being treated as dependent recipients of instruction.
2. **Role of experience.** Adults bring a large and varied store of experience. It is their richest resource for learning, so methods that draw on it (discussion, case studies, problem-solving, simulations) work better than one-way transmission.
3. **Readiness to learn.** Adults become ready to learn when the tasks of their social roles, as worker, parent or citizen, demand it. A new clerk becomes ready to learn office software when his job requires it.
4. **Orientation to learning.** Adults want to apply what they learn at once. Their orientation is problem-centred (or life-centred), not subject-centred. Learning organised around subjects for later use is typical of pedagogy.
5. **Motivation.** Adults respond to some external motivators, such as promotion or a better salary, but the strongest motivators are internal: self-esteem, job satisfaction, quality of life and the wish to grow. Outside pressure, fear of punishment and small prizes are typical of school children.
6. **Need to know.** Adults need to know why they should learn something before they set out to learn it.

Assumption	Pedagogy (children)	Andragogy (adults)
Self-concept	Dependent learner	Self-directing learner
Experience	Of little use; the teacher's experience counts	A rich resource for learning
Readiness	Ready when told, by age and grade	Ready when life roles demand
Orientation	Subject-centred, for later use	Problem-centred, for immediate use
Motivation	Mainly external (marks, praise, fear)	Mainly internal (self-esteem, satisfaction, growth)

A useful key: readiness is about when adults become ready (life roles); orientation is about how content should be organised (around problems). Students often swap these two.

Methods that suit andragogy

Because adults are self-directing and experienced, Knowles preferred participatory, problem-centred methods: case discussion, problem-solving, projects, role play, peer discussion and learner involvement in planning. Dictation of notes, repetitive drill of definitions and lecture followed by a recall quiz treat learners as dependent receivers of content, so they fit the pedagogical model.

Self-directed learning

Self-directed learning (स्व-निर्देशित अधिगम) is, in Knowles's description, a process in which individuals take the initiative, with or without help from others, in diagnosing their learning needs, setting goals, finding resources, choosing and carrying out learning strategies, and evaluating the outcomes. A working professional who identifies what she needs to learn, chooses online resources and judges her own progress is a self-directed learner. Contrast two other terms: incidental learning is unplanned learning that happens by the way; programmed learning follows a path fixed in advance by the programme writer.

Experience: resource and obstacle

Knowles called experience the adult's richest resource. He also saw its other side. Long experience produces fixed habits of mind, biases and assumptions that can close the mind to new ideas. So adults sometimes have to unlearn before they can learn. Any statement that experience is "always" an aid and "never" a hindrance is false.

Each assumption is separate

Knowles's assumptions are separate statements about adults. One does not cause another. For example, "readiness to learn arises from the tasks of social roles" and "adults have a large reservoir of experience" are both true, but the reservoir of experience does not explain readiness. In an Assertion-Reason item built from two different assumptions, the usual answer is that both are true but the Reason does not explain the Assertion. The exception is when the Reason states the very cause given by Knowles, as in "adults resist being treated as dependants" (Assertion) because "the self-concept moves from dependence to self-direction" (Reason).

CLASSROOM EXAMPLE

At a university's Centre for Distance and Online Education, Dr Yadav teaches a contact class of in-service school teachers for an M.A. in Education. Instead of dictating notes on classroom management, she asks each teacher to describe one discipline problem from his or her own school (experience). The group analyses three of the cases and proposes solutions (problem-centred orientation). She explains at the start why the unit matters for their promotion examination and their daily work (need to know). The teachers choose which two problems to study further (self-concept). A teacher who has just been made headmistress is especially keen, because her new role demands it (readiness).

OFTEN CONFUSED

- Pedagogy and andragogy: teaching children (dependent, subject-centred, external motivation) against helping adults learn (self-directing, problem-centred, internal motivation).
- Readiness and orientation: readiness is when adults are ready (role demands); orientation is how content is organised (around problems, for immediate use).
- Self-directed and incidental learning: self-directed learning is planned by the learner; incidental learning is unplanned.
- Self-directed and programmed learning: in programmed learning the path is fixed by the programme, not the learner.
- Experience as resource and as obstacle: both are Knowles's view; "never a hindrance" is the trap.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Knowles's later view: a continuum

In his early writing Knowles presented andragogy as opposed to pedagogy. Later he treated the two as the ends of a continuum, to be used as the situation requires. Where adults are complete novices in an unfamiliar skill, a teacher-directed approach is suitable at first, moving towards self-direction as competence grows. So andragogy is not a rule that fits every adult in every situation, and pedagogy is not forbidden after the age of eighteen.

Applying the assumptions

Each assumption has a matching implication for teaching:

Assumption	Implication for the teacher
Self-concept (self-directing)	Involve learners in planning their own learning, including diagnosing their needs
Experience	Use discussion and case methods that draw on what learners have lived
Readiness to learn	Time learning to coincide with the demands of learners' roles
Orientation to learning	Organise content around real problems, not subjects

The learning contract

Knowles recommended the **learning contract** (अधिगम अनुबंध) as a tool of self-directed learning. It is a written agreement, drawn up in advance by the adult learner in consultation with a facilitator, that states the learning objectives, the resources and strategies to be used, the evidence of accomplishment and the criteria by which that evidence will be judged. A portfolio is different: it collects evidence of work already done. The contract looks forward; the portfolio looks back.

Mezirow's transformative learning

Jack Mezirow described **transformative learning** (रूपांतरणकारी अधिगम), in which an adult changes a deep-seated way of seeing. It begins with a **disorienting dilemma**, an experience, often a crisis, that does not fit the learner's existing beliefs. It proceeds through **critical reflection** (आलोचनात्मक चिंतन) on assumptions long taken for granted, and ends in a changed frame of reference, which Mezirow called perspective transformation.

Keep the neighbouring theories apart. David Kolb's experiential learning cycle explains how people learn from experience in general (see section 2.4), without the idea of overturning a frame of reference. Knowles's andragogy concerns assumptions about adult learners. Jean Lave and Etienne Wenger's situated learning concerns learning through participation in communities of practice.

Freire: against banking education

The Brazilian educator Paulo Freire, in *Pedagogy of the Oppressed*, criticised what he called **banking education**, in which the teacher deposits knowledge in passive learners as money is deposited in a bank. He proposed instead education through dialogue, in which teacher and learners investigate problems of their own lives together and become critically conscious of their situation. Freire's work is a classic of adult literacy.

Cross's barriers to adult learning

K. Patricia Cross grouped the barriers that keep adults from learning into three kinds:

- **Situational barriers** arise from a person's circumstances at a given time: job and family duties, shift work, care of children, cost, transport.
- **Institutional barriers** are created by providers: inconvenient schedules, rigid rules, fees, lack of suitable courses.
- **Dispositional barriers** are the learner's own attitudes and self-perceptions: "I am too old to learn", fear of failure.

Ask where the obstacle lies: in the learner's life (situational), in the institution's practice (institutional) or in the learner's mind (dispositional).

Heutagogy and peeragogy

Heutagogy (स्व-निर्धारित अधिगम), a term introduced by Stewart Hase and Chris Kenyon, is self-determined learning. The learner decides not only how to learn but what to learn, sets and revises her own goals as the work proceeds, and develops the capability to learn in unfamiliar situations. It emphasises capability and **double-loop learning**, in which the learner questions her own assumptions and goals, not only her methods. In heutagogy the learner, not the teacher, determines the path.

Heutagogy goes a step beyond andragogy. In andragogy the learner is self-directed, but the teacher still largely designs the course and its objectives. Pedagogy, andragogy and heutagogy are therefore often arranged as a continuum of increasing learner autonomy.

Peeragogy is learning produced by peers together, without a single teacher at the centre.

Thinker	Idea
Malcolm Knowles	Andragogy: assumptions about the self-directing adult learner; learning contract
Jack Mezirow	Transformative learning; perspective transformation
Paulo Freire	Critique of banking education; dialogue
Hase and Kenyon	Heutagogy: self-determined learning
K. Patricia Cross	Situational, institutional and dispositional barriers

CHECK YOURSELF

1. Which of the following are among Knowles's assumptions about adult learners?

1. Adults need to know why they should learn something.
2. Adults are motivated mainly by fear of punishment.
3. Adults use their experience as a resource for learning.
4. Adults become ready to learn when their social roles demand it.

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 2 and 4 only
(C) 1, 3 and 4 only
(D) 1, 2, 3 and 4

2. A training officer arranges a course on new pension rules for office staff in the month the rules come into force, when the staff must start applying them. The assumption of Knowles she is using is:

(A) self-concept
(B) readiness to learn
(C) role of experience
(D) need to know

3. Match List I with List II.

List I (Thinker)	List II (Idea)
A. Mezirow	I. Banking education criticised
B. Freire	II. Situational, institutional and dispositional barriers
C. Cross	III. Disorienting dilemma and critical reflection
D. Hase and Kenyon	IV. Self-determined learning

Choose the correct answer from the options given below:

(A) A-III, B-II, C-I, D-IV
(B) A-I, B-III, C-II, D-IV
(C) A-IV, B-I, C-II, D-III
(D) A-III, B-I, C-II, D-IV

4. Assertion (A): Andragogy suits adults better than dictation of notes does.

Reason (R): Adults bring a large store of experience that participatory methods can draw upon.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

5. Statement I: In Knowles's later view, pedagogy and andragogy are ends of a continuum.

Statement II: In andragogy the learner decides what to learn and revises the goals of the course, while the teacher plays no part in its design.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

6. A woman wishes to join an evening course but believes she is “too old to study now”. In Cross’s classification this is a:

(A) situational barrier
(B) institutional barrier
(C) informational barrier
(D) dispositional barrier

7. Arrange the phases of transformative learning, as Mezirow described it, in order:

1. Changed frame of reference
2. Disorienting dilemma
3. Critical reflection on assumptions

Choose the correct answer from the options given below:

(A) 2, 3, 1
(B) 3, 2, 1
(C) 2, 1, 3
(D) 1, 3, 2

2.8 Individual differences and intelligence

Syllabus: Learner’s characteristics: characteristics of adolescent and adult learners (academic, social, emotional and cognitive); individual differences

EXAM FOOTPRINT

Asked occasionally; usually a match of intelligence types with examples or a single-answer item on individual differences.

Core

What individual differences means

No two learners are alike. They differ in intelligence, aptitude, interests, motivation, background and the speed at which they learn. These variations are called **individual differences** (वैयक्तिक भिन्नताएँ). Recognising them is the reason a teacher varies methods, examples and pace instead of treating the class as one uniform group. The term does not mean that learners prefer to study alone, are taught one at a time, or compete for ranks.

Individual differences are found in every aspect of a person: physical (height, health, energy), mental (intelligence, memory, reasoning), emotional (temperament, anxiety) and social (sociability, leadership). They are differences of degree, that is quantitative (more or less of the same ability), and differences of kind, that is qualitative (different interests, styles, talents). Teaching every student in the same way does not remove them; it usually widens the gaps, because those who are ahead move further ahead and those who are behind fall further behind.

There are two kinds of difference:

- **Inter-individual differences**: differences between one person and another. Ravi is better at mathematics than Sunita.

- **Intra-individual differences**: differences among the abilities of the same person. Sunita is strong in language and weak in mathematics.

Both matter, the second especially in guiding a student towards suitable courses.

Causes: heredity and environment

Individual differences arise from the interaction of **heredity** (वंशानुक्रम), what a person inherits, and **environment** (वातावरण), what a person experiences: family, culture, schooling, nutrition, health and opportunity. Neither alone explains them. Heredity sets a range of possibilities; environment decides where in that range a person ends up.

Four terms: aptitude, achievement, interest, attitude

Term	Meaning	Time it looks at
Aptitude (अभिक्षमता)	Potential to acquire a skill or knowledge if given training	Future
Achievement (उपलब्धि)	What has already been learnt	Past and present
Interest (रुचि)	Liking for an activity	Present
Attitude (अभिवृत्ति)	A settled tendency to respond favourably or unfavourably to a person, object or idea	Present

A note on Hindi: the title of this unit, Teaching Aptitude, is rendered शिक्षण अभिवृत्ति in the Hindi version of the paper. In psychology books, however, अभिवृत्ति usually translates attitude, and aptitude is अभिक्षमता. Read the English meaning in each question to avoid confusion.

The first intelligence test: Binet

Intelligence (बुद्धि) is the general capacity to learn, reason, solve problems and adapt to new situations. The first workable intelligence scale was produced in France in 1905 by **Alfred Binet** with **Théodore Simon**, to identify children who needed special help in school. It introduced the idea of **mental age** (मानसिक आयु): the level of performance typical of children of a given age. A ten-year-old who does what typical twelve-year-olds do has a mental age of twelve.

The German psychologist William Stern suggested dividing mental age by chronological age. Lewis Terman at Stanford University multiplied the result by 100 and used it in his American revision of Binet's test, the Stanford-Binet. This gave the **intelligence quotient** (बुद्धि-लब्धि), or IQ.

Theories of intelligence

Psychologists disagree on whether intelligence is one ability or many. The examination asks you to match each theory with its author.

- **Charles Spearman: two-factor theory.** Every intellectual task draws on a general factor, **g**, common to all tasks, and a specific factor, **s**, peculiar to that task.
- **L. L. Thurstone: group-factor theory, or primary mental abilities.** Thurstone rejected a single general factor in favour of seven primary mental abilities: verbal comprehension, word fluency, number, space, memory, perceptual speed and reasoning.
- **Godfrey Thomson: sampling theory.** Each task samples a number of small, independent abilities.
- **J. P. Guilford: structure-of-intellect model.** Guilford classified abilities along three dimensions: the **operation** performed (such as cognition, memory, convergent production, divergent production, evaluation), the **content** operated on (such as figural, symbolic, semantic, behavioural) and the **product** that results (such as units, classes, relations). The combinations give 120 or more distinct abilities.

- **Raymond Cattell: fluid and crystallised intelligence. Fluid intelligence** (तरल बुद्धि) is the capacity to reason and solve problems in new situations. **Crystallised intelligence** (क्रिस्टलीकृत बुद्धि) is the store of knowledge and skills built up through experience, such as vocabulary and general information.
- **Howard Gardner: multiple intelligences.** See below.
- **Robert Sternberg: triarchic theory.** Three aspects: analytical, creative and practical.

Gardner, Sternberg and Guilford, in their different ways, all rejected the idea of intelligence as one single ability. Spearman is the name linked with g.

Gardner's multiple intelligences

In *Frames of Mind* (1983) Howard Gardner argued that intelligence is not one ability but several relatively independent ones. He proposed seven, and later added an eighth, the naturalistic. This is the theory of **multiple intelligences** (बहु-बुद्धि).

Intelligence	What it is	Who shows it
Linguistic	Skill with words, spoken and written	Poet, journalist
Logical-mathematical	Reasoning, numbers, logical patterns	Mathematician, scientist
Spatial	Perceiving the visual world accurately and transforming it in the mind	Architect, sculptor, navigator, chess player
Musical	Sensitivity to pitch, rhythm, melody	Musician, composer
Bodily-kinaesthetic	Skilled use of the body and of objects	Dancer, athlete, surgeon
Interpersonal	Understanding other people's moods, motives and intentions, and working well with them	Teacher, counsellor, leader
Intrapersonal	Understanding oneself: one's own feelings, strengths and goals	Reflective thinker, writer
Naturalistic (added later)	Recognising and classifying plants, animals and other features of nature	Botanist, farmer

Interpersonal and intrapersonal are easily swapped. "Inter" means between people: understanding others. "Intra" means within: understanding oneself. A student who quickly senses classmates' moods and works well in teams shows interpersonal intelligence.

Gardner criticised schools for valuing mainly linguistic and logical-mathematical intelligence. His theory is a reaction against describing a learner by one IQ score.

Sternberg's three intelligences

- **Analytical** intelligence: solving academic problems, analysing, judging. It is what ordinary tests measure.
- **Creative** intelligence: dealing with new situations and producing new ideas.
- **Practical** intelligence: adapting to, shaping and selecting one's everyday environment. This is "street-smartness". A student with average grades who handles real-life situations with unusual skill is strong in practical intelligence.

Guilford: convergent and divergent production

Guilford's two most-asked operations are these. **Convergent production** (अभिसारी चिंतन) seeks the one correct answer, as in most ordinary test items. **Divergent production** (अपसारी चिंतन) generates many different answers and is the basis of creativity. A test that asks "list as many uses of a brick as you can" measures divergent production. Such unusual-uses tests are scored for **fluency** (number of ideas), **flexibility** (variety of kinds of idea) and **originality** (unusualness of ideas).

Cattell and age

Fluid intelligence tends to decline from early adulthood. Crystallised intelligence is maintained, or even grows, with age. So a teacher of fifty may do better on vocabulary and general information than she did at twenty-five, while being slower on novel abstract puzzles. Remember: "fluid" means flexible reasoning on new problems, not knowledge that "flows" from experience.

Calculating IQ: the method

The **ratio IQ** (classical IQ) is:

IQ = (mental age ÷ chronological age) × 100

Chronological age (कालानुक्रमिक आयु) is the actual age since birth. Mental age is the age level of the person's test performance. When the two are equal, IQ = 100, which is average. When mental age is ahead of chronological age, IQ is above 100; when it is behind, IQ is below 100.

Steps:

1. Write mental age (MA) and chronological age (CA) in the same unit, both in years or both in months.
2. Divide MA by CA. MA goes on top.
3. Multiply by 100.
4. To find MA from a known IQ: MA = IQ × CA ÷ 100. To find CA: CA = MA × 100 ÷ IQ.

WORKED EXAMPLE

Example 1. MA = 12 years, CA = 10 years.

IQ = 12 ÷ 10 × 100 = 1.2 × 100 = 120

Example 2. MA = 9 years, CA = 12 years.

IQ = 9 ÷ 12 × 100 = 0.75 × 100 = 75

Example 3 (years and months). MA = 10 years 6 months, CA = 9 years. Convert to months.

MA = 10 × 12 + 6 = 126 months; CA = 9 × 12 = 108 months

IQ = 126 ÷ 108 × 100 = 116.67, about 117

Example 4 (finding MA). A 10-year-old has an IQ of 110.

MA = 110 × 10 ÷ 100 = 11 years

Example 5 (finding CA). A child with MA 15 years has an IQ of 125.

CA = 15 × 100 ÷ 125 = 12 years

USUAL SLIPS

- Dividing the wrong way round. With MA 12 and CA 10, 10 ÷ 12 × 100 gives about 83, not 120. Mental age always goes on top.
- Mixing units: dividing 10 years 6 months by 9 years as "10.6 ÷ 9". Six months is 0.5 of a year, not 0.6. Convert both ages to months.
- Forgetting to multiply by 100, and writing an IQ of 1.2.
- Reading IQ 100 as "perfect". It means mental age equals chronological age: average for one's age.

Why the ratio IQ fails for adults: the deviation IQ

Mental age stops rising in the late teens, but chronological age keeps increasing. Suppose, for illustration, that a person's mental age levels off at 16. At age 16 the ratio IQ is 16 ÷ 16 × 100 = 100; at 20 it is 16 ÷ 20 × 100 = 80; at 40 it would be 16 ÷ 40 × 100 = 40. The person has not become less intelligent; the formula has broken down.

David Wechsler therefore introduced the **deviation IQ**, used in his adult and children's scales and now in almost all modern tests. A deviation IQ expresses a person's standing relative to others of the same age. Scores in each age group are scaled so that the mean is 100 and the standard deviation is 15.

Deviation IQ = 100 + 15 × z, where z = (person's score − mean of age group) ÷ SD of age group

WORKED EXAMPLE

A 30-year-old scores 65 on a test where adults of her age have a mean of 50 and a standard deviation of 10.

z = (65 − 50) ÷ 10 = 1.5

Deviation IQ = 100 + 15 × 1.5 = 122.5

Because IQs are distributed normally with mean 100 and SD 15, about 68 per cent of people have IQs between 85 and 115 (within one SD of the mean), and about 95 per cent between 70 and 130 (within two SD). About 34 per cent lie between 100 and 115 alone.

CLASSROOM EXAMPLE

In a B.Ed. psychology practical at a university in Rajasthan, the class looks at an anonymised case. A boy of 8 years (96 months) has a mental age of 10 years 4 months (124 months) on an individual test. The students compute 124 ÷ 96 × 100 = 129.17, about 129. Then the teacher shows a second case: a woman of 35 whose ratio IQ would look absurdly low, and explains why the test manual gives her a deviation IQ instead. She ends by noting that the same boy may be weak at drawing and strong at music: one IQ number does not describe the whole learner.

OFTEN CONFUSED

- Inter- and intra-individual differences: between persons, or within one person's own abilities.
- Aptitude and achievement: what one could learn with training (future), or what one has already learnt (past).
- Attitude and aptitude: a settled favourable or unfavourable tendency, or a potential to learn. In Hindi books, अभिवृत्ति is usually attitude and अभिक्षमता aptitude.
- Spearman and Thurstone: one general factor plus specifics (g and s), or seven primary mental abilities.
- Thurstone and Guilford: both proposed many abilities; the three-dimensional operations-contents-products model is Guilford's.
- Fluid and crystallised: reasoning on new problems (declines with age), or stored knowledge (maintained or grows).
- Interpersonal and intrapersonal: understanding others, or understanding oneself.
- Sternberg's creative and practical: dealing with novelty, or adapting to everyday settings ("street-smart").
- Convergent and divergent production: one right answer, or many answers (creativity).
- Ratio and deviation IQ: MA ÷ CA × 100, or standing among same-age peers (mean 100, SD 15).

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

The Flynn effect and other named effects

The **Flynn effect**, documented by James Flynn, is the steady rise in average scores on intelligence tests from one generation to the next, observed in many countries during the twentieth century. It is usually attributed to better nutrition, more schooling and a more stimulating environment. The rise is too fast to be genetic, so it shows that measured intelligence is not fixed by heredity alone.

Keep it apart from three other effects:

- **Matthew effect**: early advantages accumulate over time, so the gap between individuals widens ("the rich get richer"). It concerns widening gaps, not a rise in the average.
- **Pygmalion effect**: learners tend to perform in line with teachers' expectations.
- **Halo effect**: one good (or bad) quality colours the rater's judgement of all other qualities.

The PASS theory

J. P. Das, Jack Naglieri and John Kirby, drawing on the Russian neuropsychologist A. R. Luria's work on the brain, described intelligence as four interdependent cognitive processes. The initials give the name **PASS**:

- **P**lanning: deciding how to solve a problem and checking the solution;
- **A**ttention-arousal: staying alert and focusing on what matters;
- **S**imultaneous processing: combining pieces of information into a whole at once;
- **S**uccessive processing: handling information in a sequence.

The theory underlies a test called the Cognitive Assessment System (CAS). The theory is often asked in Indian examinations.

Links to other units

Mean, standard deviation and z-scores are taught as methods in section 6.5. The normal curve that gives the 68 and 95 per cent figures is the same one used in reading test scores. Teaching for individual differences (differentiation, enrichment, remediation) follows in section 2.9.

CHECK YOURSELF

1. Using the classical formula, a child of 8 years with a mental age of 6 years has an IQ of:

(A) 133
(B) 75
(C) 80
(D) 125

2. A child aged 7 years 6 months has a mental age of 9 years 4 months. Taking both in months, her ratio IQ is closest to:

(A) 124
(B) 80
(C) 112
(D) 131

3. A 15-year-old has an IQ of 80. His mental age is:

(A) 18 years 9 months
(B) 13 years
(C) 12 years
(D) 10 years

4. Match List I with List II.

List I (Psychologist)	List II (Idea)
A. Cattell	I. Operations, contents and products
B. Guilford	II. Fluid and crystallised intelligence
C. Gardner	III. Analytical, creative and practical
D. Sternberg	IV. Relatively independent intelligences, including the naturalistic

Choose the correct answer from the options given below:

(A) A-II, B-III, C-IV, D-I
(B) A-I, B-II, C-IV, D-III
(C) A-II, B-I, C-IV, D-III
(D) A-IV, B-I, C-II, D-III

5. Statement I: The ratio IQ would show a steady fall in adulthood even if a person's abilities did not change.

Statement II: Deviation IQs are scaled so that each age group has a mean of 100 and a standard deviation of 15.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

6. Assertion (A): Teaching all students of a class in exactly the same way tends to widen the gaps among them.

Reason (R): Individual differences are differences of degree only, not of kind.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false
(D) A is false but R is true

7. A girl can recognise and name dozens of local birds and trees and sorts them into families with ease. Gardner would call this strength:
(A) spatial intelligence
(B) naturalistic intelligence
(C) intrapersonal intelligence
(D) logical-mathematical intelligence

8. If IQs are normal with mean 100 and SD 15, roughly what percentage of people score above 115?
(A) 34 per cent
(B) 68 per cent
(C) 2 per cent
(D) 16 per cent

2.9 Learners with diverse needs and inclusive teaching

Syllabus: Learner's characteristics; individual differences; inclusive education (Rights of Persons with Disabilities Act, 2016)

EXAM FOOTPRINT

Asked now and then; usually a scenario on accommodating a learner with special needs or a multiple-select item on inclusive practice.

Core

Section 2.8 showed that learners differ. This section asks what a teacher does about it: how to teach the gifted, the slow learner, the underachiever, the learner with a specific learning disability and the learner with a sensory or physical impairment, all in one institution. The answer the examination looks for is almost always the same: vary the teaching to suit the learner, and change the institution so that every learner can take part.

Gifted learners

Gifted learners (प्रतिभाशाली शिक्षार्थी) perform, or can perform, far above their age group in intellectual, creative or other fields. If they are given only the ordinary work they finish early, become bored and may even disturb the class. They need work that stretches them:

- **Enrichment** (संवर्धन): deeper and wider tasks while staying in the same class with age-mates, such as an extra research question or a reading beyond the syllabus.
- **Acceleration** (त्वरण): moving through the curriculum faster than usual, by early entry, skipping a class or grade, or taking advanced courses early.
- Independent projects and mentoring by an expert.

Giving a gifted student more of the same exercises increases boredom; exempting her from work leaves her ability unused; placing her with slow learners for drill wastes it.

Ability grouping places learners of similar ability together. **Remediation** (remedial teaching, उपचारात्मक शिक्षण) corrects deficiencies; it is for learners who are behind, not for the gifted.

Slow learners

Slow learners (मंद अधिगमकर्ता) have somewhat below-average general ability, and their attainment is in line with it. They can learn, but they need:

- a slower pace and more time;
- simple language and concrete examples;
- repetition and practice;
- frequent encouragement and feedback;
- peer tutoring, in which a classmate helps.

Speeding up, giving fewer examples or withholding feedback widens the gap between them and the rest of the class.

Underachievers

An **underachiever** has adequate or high measured ability but attains far below what that ability predicts. The cause is usually motivational, emotional or environmental: lack of interest, anxiety, family trouble, poor study habits. The key is the gap between potential and performance. A slow learner also gets low marks, but there is no such gap, because his marks match his limited ability.

Specific learning disabilities

A **specific learning disability** (विशिष्ट अधिगम अक्षमता) is a difficulty in one area of learning that occurs despite average or above-average intelligence and adequate teaching. Four are asked again and again:

Term	Main difficulty	Remember by
Dyslexia	Accurate and fluent word reading, and spelling	“lexis” = word
Dysgraphia	Handwriting and written expression: letter formation, spacing, speed	“graph” = write
Dyscalculia	Number and calculation	“calculate”
Dyspraxia	Planning and coordination of movement	“praxis” = action

A student who reads well but whose handwriting is illegible, unevenly spaced and very slow shows dysgraphia, not dyslexia. Such learners need extra time, alternative formats (oral answers, typed work) and specific help in the weak area.

Teaching a mixed class

Because learners differ in ability, interest and speed, one method at one pace suits only a few. A teacher meets individual differences by:

- varying methods and pace;
- **graded** (tiered) tasks, so that each learner has an appropriate challenge;
- **flexible grouping**, in which students sometimes work with others of similar level and sometimes in mixed groups;
- extra help for those behind and enrichment for those ahead.

This is the basis of **differentiated instruction** (विभेदित अनुदेशन), developed further in Going Further. Uniform treatment (one common lecture, one pace, identical homework for all) looks fair but ignores real differences in readiness, leaving some bored and others lost. Public ranking after every test harms weaker learners.

Learners with sensory impairments

The rule is simple: send information through the channel the learner can use.

- **Visual impairment** (दृष्टि बाधिता). **Braille** is a system of raised dots read by touch, which lets blind learners read and write. Others use large print, audio material and screen-reader software. On a course web page, **alternative text** (alt text) describes each image in words, so that a screen reader can read it aloud.
- **Hearing impairment** (श्रवण बाधिता). These learners need information through the eyes: **captions** on videos, written outlines of lectures, visual aids, a front seat that allows lip-reading and, where needed, sign language interpretation. Audio recordings and extra oral questions depend on hearing and do not help.

So alt text serves learners with visual impairment, and captions serve learners with hearing impairment. Questions often reverse the pair.

Exclusion, segregation, integration and inclusion

These four words describe how an education system treats learners with disabilities.

Term	Meaning
Exclusion	The learner is denied education altogether.
Segregation	The learner is educated in a separate setting, such as a separate special school or class.
Integration	The learner is placed in the regular setting but must fit into a system that stays as it is.
Inclusion	The institution itself changes, removing barriers and providing accommodation, so that every learner takes part fully.

A college that admits a wheelchair user to regular classes but keeps her laboratory on a floor she cannot reach is practising integration, not inclusion.

Inclusive education (समावेशी शिक्षा) means that learners of all abilities and backgrounds, including those with disabilities, learn together in the same institution, with the supports each needs. It is not admitting only high achievers, separating learners strictly by ability, or teaching everyone by one method.

The Rights of Persons with Disabilities Act, 2016

India's main law on disability is the **Rights of Persons with Disabilities Act, 2016** (RPwD Act). Learn these points.

- It replaced the Persons with Disabilities (Equal Opportunities, Protection of Rights and Full Participation) Act, 1995.
- It raised the number of specified disabilities from 7 under the 1995 Act to **21**, and the Central Government can add more. The 21 include, for example, blindness, low vision, hearing impairment, locomotor disability, intellectual disability, autism spectrum disorder, specific learning disabilities, mental illness, cerebral palsy, dwarfism, acid attack victims, thalassaemia and multiple disabilities.
- A **person with benchmark disability** is one with not less than 40 per cent of a specified disability, as certified.
- The Act defines inclusive education as a system of education in which students with and without disability learn together and the system of teaching and learning is suitably adapted to their different needs.
- It requires **reasonable accommodation** (उचित सुविधा): necessary and appropriate changes and adjustments, without a disproportionate or undue burden, so that persons with disabilities can enjoy their rights equally with others.

- Section 16 asks governments and local authorities to make sure that the educational institutions they fund or recognise provide inclusive education.
- Section 31: every child with benchmark disability between 6 and 18 years has the right to free education.
- Section 32: all government institutions of higher education, and other higher education institutions receiving aid from the government, must reserve **not less than 5 per cent** of seats for persons with benchmark disabilities, and such persons get an **upper age relaxation of 5 years** for admission.
- Section 34: in government employment the reservation is **not less than 4 per cent** of vacancies. Do not mix this 4 per cent with the 5 per cent for higher education seats.

The Right of Children to Free and Compulsory Education Act (RTE Act) is a different law, of 2009.

CLASSROOM EXAMPLE

At a state university in Punjab, an M.A. English class has 48 students. One is blind, one is hard of hearing, one is a wheelchair user, and two are exceptionally able. The teacher uploads her lecture notes in a format a screen reader can read, adds alt text to every picture and captions to every video clip, and seats the student with hearing impairment in front where he can see her face. The department moves the class from the second floor to the ground floor, because the building has no lift; this is a change in the institution, not in the student, and it is what makes the arrangement inclusion rather than integration. For the two able students she sets an enrichment task, a comparison of two translations of a Tagore poem. A student who is struggling gets a weekly meeting with a peer tutor. At admission the student with benchmark disability got a seat under the 5 per cent reservation for higher education.

OFTEN CONFUSED

- Enrichment and acceleration: deeper and wider work at the same level, or faster movement through the curriculum (skipping a class, early entry).
- Slow learner and underachiever: low ability with matching attainment, or adequate ability with attainment far below it.
- Slow learner and learning disability: generally below-average ability, or a specific difficulty despite average or higher intelligence.
- Dyslexia, dysgraphia, dyscalculia, dyspraxia: reading, writing, number, movement.
- Alt text and captions: for learners who cannot see, and for learners who cannot hear.
- Integration and inclusion: the learner fits into an unchanged system, or the system changes for the learner.
- 5 per cent and 4 per cent under the RPwD Act: seats in government and aided higher education, and vacancies in government employment.
- RPwD Act 2016 and RTE Act 2009: rights of persons with disabilities, and free and compulsory education for children aged 6 to 14.

> **FACT WATCH**
>
> Rights of Persons with Disabilities Act, 2016: 21 specified disabilities; not less than 5 per cent of seats reserved in government and government-aided higher education institutions, with 5 years' upper age relaxation; not less than 4 per cent of vacancies in government employment (checked October 2026). Books written before the Act still give the figures of the 1995 Act: 7 disabilities and 3 per cent reservation. The 2016 Act received assent in December 2016 and came into force in April 2017.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Differentiated instruction (Tomlinson)

Differentiated instruction is associated with Carol Ann Tomlinson. It means planning varied routes to the same goals according to each learner's **readiness** (current level of knowledge and skill), **interest** and **learning profile** (preferred ways of learning). Tomlinson names four things a teacher can differentiate:

Element	What varies
Content	What is learnt, and the materials through which learners reach it
Process	How learners make sense of the content: the activities
Product	How learners show what they have learnt
Learning environment	The arrangement and climate of the classroom

If all students read the same passage but may show their understanding through an essay, a concept map or an oral presentation, the teacher is differentiating the product. Direct instruction, programmed instruction and mass instruction do not differentiate in this sense.

Universal Design for Learning

Universal Design for Learning (UDL), developed by the organisation CAST in the United States, asks teachers to plan variety into a course from the start, instead of adapting it later for individual learners. It rests on three principles:

- multiple means of **representation**: present content in several ways (text, audio, video, diagrams);
- multiple means of **engagement**: interest and motivate learners in several ways;
- multiple means of **action and expression**: let learners show their learning in several ways.

UDL benefits all learners and reduces the need for later individual accommodation. It is the opposite of one uniform method and one test format for all, of separate special classes, and of waiting until a learner has failed before adapting anything.

Differentiation and UDL compared

Differentiation is mostly a teacher's response to the learners in front of her. UDL is mostly design done before the learners arrive. Both reject the single fixed method. In a question, "planned from the start for all learners" points to UDL; "content, process and product varied by readiness, interest and learning profile" points to differentiation.

CHECK YOURSELF

1. A college allows an exceptionally able first-year student to take a second-year paper in the same semester. This provision is:

(A) acceleration
(B) enrichment
(C) remediation
(D) integration

2. Under the Rights of Persons with Disabilities Act, 2016, which of the following are correct?

1. The number of specified disabilities is 21.
2. Government and government-aided higher education institutions must reserve not less than 5 per cent of seats for persons with benchmark disabilities.
3. The reservation in government employment is not less than 7 per cent of vacancies.
4. Persons with benchmark disabilities get an upper age relaxation of five years for admission to higher education.

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 1, 2 and 4 only
(C) 2 and 4 only
(D) 1, 2, 3 and 4

3. Match List I with List II.

List I (Difficulty)	List II (Main area affected)
A. Dyslexia	I. Coordination of movement
B. Dysgraphia	II. Number work
C. Dyscalculia	III. Reading
D. Dyspraxia	IV. Handwriting

Choose the correct answer from the options given below:

(A) A-III, B-II, C-IV, D-I
(B) A-IV, B-III, C-II, D-I
(C) A-III, B-IV, C-I, D-II
(D) A-III, B-IV, C-II, D-I

4. Assertion (A): Placing a learner with a disability in a regular classroom is by itself enough to make the classroom inclusive.

Reason (R): Inclusion requires the institution to remove barriers and provide reasonable accommodation.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

5. Statement I: An underachiever is a learner whose attainment is far below his measured ability.

Statement II: A learner with dyslexia usually has below-average general intelligence.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

6. A teacher records her lectures as videos for an online course. Which pair of measures serves learners with visual impairment and learners with hearing impairment respectively?

(A) Captions; a front seat
(B) Alt text and screen-reader-friendly notes; captions
(C) Captions; alt text
(D) Large print; Braille

7. In Tomlinson's terms, a teacher who gives weaker readers a simplified version of a text while stronger readers read the original is differentiating the:

(A) product
(B) process
(C) content
(D) environment

Answers to “Check yourself”: Chapter 2

Section 2.1

1. (C), because identity versus role confusion is Erikson’s adolescent stage. Industry belongs to the school years, intimacy to young adulthood and integrity to old age.

2. (A), because Hall described storm and stress, Havighurst developmental tasks, Marcia identity statuses and Arnett emerging adulthood.

3. (D), because she is committed but has not explored any alternative. Commitment without exploration is foreclosure; achievement needs exploration first.

4. (B), because abstract reasoning, peer sensitivity and the wish for independence mark late adolescence. Emotions in adolescence are intense and changeable, so statement 2 is wrong.

5. (A), because the reason for private correction is exactly the adolescent’s self-consciousness before peers.

6. (D), because the feeling of being watched is the imaginary audience; the personal fable is the belief in being unique and invulnerable. Statement II is correct.

Section 2.2

1. (B), because the order is sensorimotor, pre-operational, concrete operational, formal operational. Pre-operational must come before both operational stages.

2. (D), because what a learner can do alone is the level of actual development, the lower edge of the zone. Statement II correctly describes scaffolding.

3. (C), because the spiral curriculum is Bruner’s, the advance organiser Ausubel’s, equilibration Piaget’s and the zone of proximal development Vygotsky’s.

4. (B), because understanding that number does not change when the row is spread out is conservation. Object permanence is a sensorimotor achievement; animism and egocentrism are pre-operational features, not achievements.

5. (C), because peer help within the zone does assist learning, but Vygotsky held the opposite of the Reason: mental functions arise first in social interaction.

6. (D), because Ausubel held that reception learning can be meaningful and that discovery can be rote. Statement 2 is the popular belief he argued against.

7. (C), because grouping items into larger units is chunking, which keeps them within the small capacity of working memory.

8. (D), because better recall of first and last items is the serial position effect (primacy and recency). The von Restorff effect concerns a distinctive item.

Section 2.3

1. (D), because Pavlov is classical conditioning, Skinner operant conditioning, Köhler insight and Bandura observational learning. Swapping Pavlov and Skinner is the commonest slip.

2. (B), because the reinforcer (extra marks) is withheld and the behaviour fades. Nothing pleasant the student already had is taken away as a penalty, so it is not negative punishment.

3. (A), because the observer attends, retains, reproduces and then is motivated to perform.

4. (D), because before pairing the bell is a neutral stimulus; food is the unconditioned stimulus. Statement II is correct.

5. (C), because something pleasant (a privilege) is removed and the behaviour decreases. Negative reinforcement would increase behaviour.

6. (A), because the acquisition-performance distinction is exactly what allows learning without reward; reward affects whether the behaviour is shown.

7. (A), because insight belongs to Köhler, not Thorndike, whose theory is trial and error.

8. (D), because with a variable-ratio schedule the learner cannot tell when reinforcement has stopped. Continuous reinforcement extinguishes fastest.

Section 2.4

1. (B), because the diverger combines concrete experience with reflective observation, the assimilator abstract conceptualisation with reflective observation, the converger abstract conceptualisation with active experimentation, and the accommodator concrete experience with active experimentation.

2. (C), because aural learners prefer listening and discussion.

3. (D), because experience is followed by reflection, then general ideas, then testing in practice.

4. (C), because variety is good practice, but the Reason overstates learning-style theory; learners have preferences, not single exclusive channels.

5. (C), because impulsive learners respond fast and make errors; it is reflective learners who are slow and accurate.

6. (A), because Kolb saw styles as relatively stable but changeable, and SQ3R is correctly expanded.

Section 2.5

1. (C), because fear of punishment is the reason, and that marks the pre-conventional level.

2. (A), because the peer orientation of late adolescents is the reason group methods suit them.

3. (D), because stage 3 seeks approval, stage 4 upholds law and order, stage 5 sees laws as a social contract and stage 6 follows universal principles.

4. (D), because heteronomous morality is Piaget's term; Statement II is correct.

5. (A), because Gilligan proposed the ethic of care in *In a Different Voice*.

Section 2.6

1. (B), because the reward lies in the reading itself; nothing outside the activity drives it.

2. (A), because Herzberg is hygiene and motivators, Deci and Ryan is autonomy-competence-relatedness, Alderfer is ERG and Weiner is attribution. "Relatedness" appears in both I and IV, but only Deci and Ryan have "autonomy".

3. (A), because panic is arousal well past the optimum, and the optimum is lower for a difficult task; R states exactly why A happens.

4. (C), because safety, esteem and belongingness are deficiency needs; self-actualisation is the growth need.

5. (A), because $0.6 \times 0.5 \times 0.8 = 0.24$. Adding gives 1.90 and averaging gives about 0.63; both are slips.

6. (D), because the term was introduced by Salovey and Mayer; Goleman popularised it in 1995. Statement II is true.

7. (D), because task difficulty lies outside the learner and stays the same from one occasion to the next.

8. (C), because mastery experience, actual success, is the strongest source. (A) is verbal persuasion, (B) vicarious experience and (D) works on physiological states.

Section 2.7

1. (C), because need to know, experience and readiness are Knowles's assumptions; fear of punishment is typical of children, while adult motivation is mainly internal.

2. (B), because readiness is about timing learning to the demands of a social role, here the job.

3. (D), because Mezirow is the disorienting dilemma, Freire the critique of banking education, Cross the three barriers and Hase and Kenyon heutagogy.

4. (A), because the experience assumption is exactly why participatory methods work better with adults than one-way dictation.

5. (C), because Statement I is Knowles's later position; Statement II describes heutagogy, not andragogy, in which the teacher still largely designs the course.

6. (D), because the obstacle is the learner's own attitude. A shift duty or a child to care for would be situational; a college fee or timetable would be institutional.

7. (A), because the dilemma comes first, reflection follows, and the changed perspective is the result.

Section 2.8

1. (B), because 6 ÷ 8 × 100 = 75. Dividing the other way, 8 ÷ 6 × 100, gives about 133, the usual slip.

2. (A), because CA = 90 months and MA = 112 months; 112 ÷ 90 × 100 = 124.4. Reading 9 years 4 months as 9.4 years leads to wrong answers.

3. (C), because MA = 80 × 15 ÷ 100 = 12 years.

4. (C), because Cattell is fluid and crystallised, Guilford the three dimensions, Gardner multiple intelligences and Sternberg the triarchic theory.

5. (A), because mental age levels off while chronological age keeps rising, and deviation IQs use mean 100 and SD 15 within each age group.

6. (C), because A is true but R is false: individual differences are differences of both degree (more or less of the same ability) and kind (different interests, styles and talents). Older books that say "of degree, not of kind" give only half the picture.

7. (B), because classifying plants and animals is naturalistic intelligence, the eighth, added later.

8. (D), because about 68 per cent lie between 85 and 115, so about 32 per cent lie outside, half of them (about 16 per cent) above 115.

Section 2.9

1. (A), because the student moves ahead faster than the usual sequence. Enrichment would deepen work within the first-year course.

2. (B), because statements 1, 2 and 4 follow the Act; the employment figure is not less than 4 per cent, not 7.

3. (D), because dyslexia is reading, dysgraphia handwriting, dyscalculia number and dyspraxia movement.

4. (D), because placement alone is integration; R is true and describes what inclusion needs.

5. (C), because underachievement is a gap between ability and attainment, while dyslexia occurs despite average or above-average intelligence.

6. (B), because alt text and screen-reader-friendly notes reach those who cannot see, and captions reach those who cannot hear. Option (C) reverses the pair.

7. (C), because the material through which learners reach the ideas is varied; the activities (process) and the way of showing learning (product) stay the same.

Chapter 3
Factors Affecting Teaching

The same lesson, taught by the same teacher, can succeed in one classroom and fail in another. This chapter explains why. The syllabus groups the influences on teaching under six heads: the teacher, the learner, support material, instructional facilities, the learning environment and the institution. Most questions on this topic ask you to place a given influence in the right group, so learn the boundaries between the groups as carefully as the groups themselves.

3.1 The six groups of factors

Syllabus: Factors affecting teaching related to: Teacher, Learner, Support material, Instructional facilities, Learning environment and Institution

> **EXAM FOOTPRINT**
>
> Asked occasionally; usually a multiple-select item sorting factors into teacher, learner, support, environment or institutional groups.

Core

A **factor affecting teaching** (शिक्षण को प्रभावित करने वाले कारक) is anything that makes teaching more or less successful in bringing about learning (अधिगम). The syllabus names six groups of such factors. Learn them in this order, because the order moves outward from the people in the room to the organisation around them.

Group	What it covers	Typical examples
Teacher (शिक्षक)	What the teacher brings: knowledge, skill, personality, attitudes, expectations	Subject mastery, communication skill, warmth, enthusiasm, questioning skill, expectations of students
Learner (शिक्षार्थी)	What the students bring	Prior knowledge, intelligence, aptitude, interest, motivation, maturation, anxiety, health, home background
Support material (सहायक सामग्री)	The aids and resources used to teach a particular content	Charts, models, handouts, worksheets, slides, films, e-journals
Instructional facilities (अनुदेशात्मक सुविधाएँ)	The infrastructure the institution provides for all teaching	Classrooms, laboratories, libraries, projectors, computer terminals, Wi-Fi network
Learning environment (अधिगम वातावरण)	The physical and psychological surroundings of learning	Light, air, noise, seating; warmth, respect, freedom from fear
Institution (संस्था)	The policies, leadership and organisation of the college or university	Timetable, academic calendar, class size, workload, evaluation policy, leadership, faculty development

How to place a factor in its group

Ask two questions. First, where does the factor live: in the teacher, in the students, in the things used for one lesson, in the building and equipment, in the surroundings of the class, or in the rules and leadership of the institution? Second, who controls it? The teacher controls his or her own preparation and style. The institution controls admissions, timetables and equipment.

- A projector is equipment provided for all classes, so it is an **instructional facility**. A set of slides prepared for one lesson is **support material**.
- Noise from a nearby road is a condition around the class, so it belongs to the **learning environment**. A classroom where students fear ridicule also has a poor learning environment, on its psychological side.
- The head's support for trying a new method is a matter of leadership, so it is an **institution** factor.
- Home background, prior knowledge and motivation live in the students, so they are **learner** factors.

No single factor is enough

The factors work together. A teacher may know the subject thoroughly and still fail, if he cannot explain it, if the method does not suit the class, or if the students lack the background to follow. So subject mastery is **necessary but not sufficient** for effective teaching. In the same way, the effect of a method depends partly on the **readiness** (तत्परता) of the learners, that is, whether they have the maturity and prior knowledge the method assumes. When a statement says that any one factor "alone ensures" or "is sufficient for" good teaching, doubt it.

CLASSROOM EXAMPLE

In a government college in Hisar, a newly appointed lecturer in Economics has an excellent research record. Her first-year B.A. students come mostly from Hindi-medium schools and have never studied the subject. Her lectures, full of graphs and English terms, leave the class silent. Her subject mastery (a teacher factor) is not in question. The gap lies in the learners' prior knowledge and language (learner factors). Her lectures improve when she starts each topic with an example from the village market and adds a bilingual handout (support material). The college then gives her a smaller tutorial group on Saturdays (an institution decision).

OFTEN CONFUSED

- **Support material vs instructional facilities.** Support material is made or chosen for a particular content (a chart, a worksheet, a film). Instructional facilities are lasting infrastructure the institution provides for all teaching (a laboratory, a library, a projector, the network).
- **Learning environment vs instructional facilities.** The environment is the condition of the surroundings (light, noise, seating, emotional climate). Facilities are the equipment and spaces themselves. A dim room is an environment problem; the absence of a laboratory is a facilities problem.
- **Learner factor vs teacher factor.** Ask in whom the quality resides. Prior knowledge belongs to the student; subject mastery belongs to the teacher.
- **Necessary vs sufficient.** Necessary means "cannot succeed without it"; sufficient means "enough on its own". Subject mastery is necessary, not sufficient.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Interactions between the groups

Examiners at the harder levels like cases where two groups meet. Three patterns recur.

1. **Facilities need teachers.** A projector in every room does nothing if teachers are not trained or willing to use it (see 3.5).
2. **Material needs facilities.** E-journals are useless without terminals and a network to reach them (see 3.5).
3. **Methods need learners.** The same method gives different results with different learners; this is the aptitude-treatment interaction (see 3.3).

Links to other units

The learner factors here connect with Chapter 2 (Learner's Characteristics). The communication side of the teacher and of noise in the classroom connects with Unit IV (Communication), where noise is treated as a barrier to communication. The institution factors connect with Unit X (Higher Education System), where bodies such as NAAC and the statutory councils of universities are studied.

CHECK YOURSELF

1. Which of the following is an instructional facility and not support material?

(A) A worksheet prepared for a lesson on demand curves
(B) A computer laboratory with forty terminals
(C) A chart of the parts of a flower
(D) A set of slides on the Mughal administration

2. Given below are two statements:

Statement I: A teacher's subject mastery is sufficient for effective teaching.

Statement II: The same method can work differently with learners who differ in prior knowledge.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

3. Match List I (influence on teaching) with List II (group of factors):

List I	List II
A. Noise from a construction site	I. Teacher
B. Principal's encouragement of innovation	II. Learning environment
C. Students' interest in the subject	III. Institution
D. Teacher's skill in questioning	IV. Learner

Choose the correct answer from the options given below:

(A) A-II, B-III, C-IV, D-I
(B) A-III, B-II, C-IV, D-I
(C) A-II, B-III, C-I, D-IV
(D) A-IV, B-III, C-II, D-I

4. Which of the following are learner-related factors?

1. Motivation of the students
2. Seating arrangement of the room
3. Aptitude of the students
4. Teacher's enthusiasm
5. Prior knowledge of the students

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 2, 4 and 5 only
(C) 1, 3 and 5 only
(D) 1, 3, 4 and 5 only

5. Assertion (A): A well-equipped classroom does not by itself ensure good teaching.

Reason (R): The effect of teaching depends on several groups of factors acting together.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false
(D) A is false but R is true

6. The timetable and the size of each section in a college are best classed as factors related to the:
(A) learner
(B) learning environment
(C) support material
(D) institution

3.2 Factors related to the teacher

Syllabus: Factors affecting teaching related to: Teacher

EXAM FOOTPRINT

Asked occasionally; usually a multiple-select item sorting factors into teacher, learner, support, environment or institutional groups.

Core

The teacher is the factor closest to the lesson. Teacher-related factors are the qualities that the teacher carries into every class. They fall into four groups.

1. Knowledge: subject mastery

Subject mastery (विषय पर अधिकार) means a deep and accurate command of the content. A teacher can explain clearly only what he or she understands well. Mastery lets the teacher choose the right example, answer an unexpected question and correct a misconception. But mastery is necessary, not sufficient. A brilliant scholar can still be a poor teacher.

2. Skill: communication and questioning

Communication skill (संप्रेषण कौशल) is the ability to put an idea into language, voice and examples that reach these particular learners. The clarity of an explanation depends most directly on two teacher factors together: command of the subject and command of the language of instruction. The teacher's age, length of service, social status, handwriting or dress have no direct bearing on clarity.

Questioning skill (प्रश्न कौशल) also belongs to the teacher. Well-framed questions, spread across the whole class, with enough time to think, draw students into the lesson. Poor questioning, or questions put only to the front row, leaves most students silent.

3. Personality and attitude

A teacher's **personality** (व्यक्तित्व) shows in warmth, enthusiasm, fairness, patience and humour. Enthusiasm is infectious. A teacher who is warm, approachable and visibly interested in the subject arouses interest and effort, which is to say that he or she **motivates** the learners (अभिप्रेरणा), and builds trust. Warmth does not shorten the syllabus or reduce class size; its effect is on the learners' willingness to learn.

4. Expectations: the Pygmalion effect

What a teacher expects of a student can change what the student achieves. In *Pygmalion in the Classroom* (1968), Rosenthal and Jacobson reported a study in an American elementary school which they called "Oak School". They told teachers that certain pupils, in fact chosen at random, were

likely to “bloom” intellectually during the year. When the pupils were retested, those named as bloomers had gained more on an intelligence test than the others. The explanation offered was that teachers, without knowing it, treated these pupils differently: more warmth, more attention, more chances to answer, more feedback.

This is the **Pygmalion effect** (पिग्मेलियन प्रभाव): high teacher expectations tend to raise performance. It is a case of the **self-fulfilling prophecy** (स्वतः पूर्ण होने वाली भविष्यवाणी), an expectation that causes itself to come true. The lesson for you as a future teacher is plain: expect every student to learn, and show it in how you treat them.

Teacher factor	What it affects
Subject mastery	Accuracy, choice of examples, ability to answer questions
Communication and language skill	Clarity of explanation
Questioning skill	Participation of students
Warmth and enthusiasm	Motivation, interest, trust
Expectations	Students’ performance, through the self-fulfilling prophecy

CLASSROOM EXAMPLE

A teacher of B.Com. in a Sirsa college notices that she calls on the same five students every time because they “always know”. After reading about teacher expectations, she starts a rule: every question goes to a name drawn from a box, and every student gets at least five seconds to think and a hint if needed. Within a month, several students from the back benches begin to answer. Nothing about their ability changed; her expectations and her questioning did.

OFTEN CONFUSED

- **Pygmalion effect vs Golem effect.** Same mechanism, opposite direction. High expectations raise performance (Pygmalion); low expectations lower it (Golem).
- **Pygmalion effect vs Hawthorne effect.** Pygmalion works through the teacher’s expectations. Hawthorne works through the students’ own awareness of being observed or given attention (see 3.6).
- **Pygmalion effect vs halo effect.** Pygmalion changes the student’s actual performance. The halo effect changes only the rater’s judgement: one good trait colours the rating of others (see 3.6).
- **Teacher efficacy vs teacher autonomy.** Efficacy is the teacher’s belief in his or her own power to bring about learning. Autonomy is the freedom to take professional decisions.
- **Subject mastery vs communication skill.** Mastery is knowing; communication is making it known. Clear explanation needs both.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

The Golem effect

The negative side of the self-fulfilling prophecy is the **Golem effect** (गोलेम प्रभाव). Babad, Inbar and Rosenthal named it in a 1982 study of biased and unbiased student teachers. When teachers expect

little of some students, they tend to give them less time to answer, fewer hints, less praise and less challenging work. Their performance then falls, which seems to confirm the low expectation. A typical examination case: “A teacher expects little of a group, gives them less time and fewer hints, and their performance declines.” That is the Golem effect, not the Pygmalion effect.

How the original study is judged today

The 1968 study was criticised from the start; the testing expert Robert Thorndike, for example, questioned the quality of its test data. Later research accepts that teacher expectations do affect students, though usually less dramatically than the 1968 report suggested. For the examination, the attribution (Rosenthal and Jacobson, 1968) and the meaning (expectations act as a self-fulfilling prophecy) are what matter.

Teacher efficacy and related terms

Self-efficacy is Bandura’s term (1977) for a person’s belief in his or her capacity to carry out the actions needed to reach a goal. Applied to teachers, **teacher efficacy** (शिक्षक प्रभावकारिता) is the teacher’s confidence that he or she can bring about learning, even among difficult or unmotivated students. Tschannen-Moran and Woolfolk Hoy reviewed this idea in an influential 2001 paper. Teachers with high efficacy persist longer with struggling students and are more willing to try new methods. Keep the neighbouring terms apart:

Term	Meaning
Teacher efficacy	Belief in one’s own capacity to bring about learning
Teacher autonomy	Freedom to make professional decisions about teaching
Teacher accountability	Answerability to others for results
Teacher burnout	Exhaustion and loss of enthusiasm after long stress

Other teacher factors

Books also list the teacher’s attitude towards the profession and towards students, mental health, professional preparation and continuing professional development (see 1.10). These matter, but questions usually test the five factors in the Core table and the expectation effects.

CHECK YOURSELF

1. A teacher's belief that she can help even unmotivated students to learn is called:

(A) teacher accountability
(B) teacher efficacy
(C) teacher autonomy
(D) teacher burnout

2. Consider the following statements about teacher-related factors:

1. A teacher's warmth and enthusiasm can raise students' motivation.
2. A teacher's length of service directly determines the clarity of his explanations.
3. A teacher's expectations can influence how well students perform.

Which of the statements given above are correct?

(A) 1 and 2 only
(B) 2 and 3 only
(C) 1 and 3 only
(D) 1, 2 and 3

3. Assertion (A): Students whom a teacher expects to do well often do better.

Reason (R): Teachers tend to give more attention, time and encouragement to students of whom they expect more.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. Match List I with List II:

List I	List II
A. Rosenthal and Jacobson	I. Self-efficacy
B. Babad, Inbar and Rosenthal	II. Pygmalion in the Classroom
C. Bandura	III. Golem effect
D. Tschannen-Moran and Woolfolk Hoy	IV. Review of teacher efficacy (2001)

Choose the correct answer from the options given below:

(A) A-III, B-II, C-I, D-IV
(B) A-II, B-III, C-IV, D-I
(C) A-II, B-III, C-I, D-IV
(D) A-I, B-III, C-II, D-IV

5. A lecturer rarely calls on students from rural schools, waits only a second for their answers, and finds that their marks keep falling. This illustrates the:

(A) Hawthorne effect
(B) halo effect
(C) Pygmalion effect in its positive form
(D) Golem effect

6. Given below are two statements:

Statement I: Subject mastery is a teacher-related factor.

> Statement II: Subject mastery alone is enough to make teaching effective.
>
> In the light of the above statements, choose the correct answer:
>
> (A) Both Statement I and Statement II are true
> (B) Both Statement I and Statement II are false
> (C) Statement I is true but Statement II is false
> (D) Statement I is false but Statement II is true

3.3 Factors related to the learner

Syllabus: Factors affecting teaching related to: Learner

> **EXAM FOOTPRINT**
>
> Asked occasionally; usually a multiple-select item sorting factors into teacher, learner, support, environment or institutional groups.

Core

Learner-related factors are those that lie in the students themselves. The test is simple: if the quality would go with the student to another college, another teacher and another room, it is a learner factor. Prior knowledge, motivation and aptitude are learner factors. The teacher's subject mastery is a teacher factor, and the seating arrangement belongs to the learning environment.

Prior knowledge and entry behaviour

Prior knowledge (पूर्व ज्ञान) is what the learner already knows about the topic. New material is always interpreted in the light of it. The psychologist David Ausubel put this at the centre of his theory of meaningful learning. In *Educational Psychology: A Cognitive View* (1968) he wrote that the most important single factor influencing learning is what the learner already knows, and that the teacher should find this out and teach accordingly.

In lesson planning the same idea is called **entry behaviour** (प्रवेश व्यवहार): the knowledge and skills the learner has before the lesson begins. Before planning a lesson, a teacher should first find out the learners' previous knowledge. It tells the teacher where to begin and which examples will make sense. A lesson pitched far above or far below this level fails.

Three kinds of learner characteristics

Kind	What it includes	Example in a question
Cognitive (संज्ञानात्मक)	Intelligence, aptitude, prior knowledge, reasoning, memory, reading speed	"Level of intelligence", "previous knowledge"
Affective (भावात्मक)	Interest, attitude, motivation, values, emotions such as anxiety	"Interest in the subject", "test anxiety"
Psychomotor (मनोगत्यात्मक) and physical	Health, fatigue, sight and hearing, motor skill	"A tired learner", "poor eyesight"

Interest is a feeling of liking that draws a learner towards a subject, so it is **affective**. Intelligence, previous knowledge and reading speed are cognitive. Many books add **social** factors: home background, family education, language at home, peer group.

Emotional factors: anxiety

Emotions are learner factors too. Severe **anxiety** (दुश्चिंता) uses up attention and working memory (the small mental workspace in which we hold and handle information at the moment). A capable student with severe test anxiety may therefore perform well below her ability. The factor at work is **emotional**: her cognitive ability is not in question, no physical cause is involved, and institutional factors lie outside the learner.

Physical condition

A tired, hungry or unwell learner cannot sustain attention, and without attention little is learned. Fatigue reduces the efficiency of learning. Poor eyesight or hearing, if unnoticed, can look like inattention or slowness.

Maturation and readiness

Maturation (परिपक्वता) is the natural unfolding of physical and mental capacities with age, independent of training. A learner must be mature enough for a task before teaching can succeed. This is the idea of **readiness** (तत्परता). Coaching, discipline and testing are outside influences on the learner, not maturation. In higher education, maturation shows in the capacity for abstract and reflective thinking, which develops through adolescence into early adulthood (see 2.1).

CLASSROOM EXAMPLE

A teacher of Political Science at a university campus in Haryana begins the unit on federalism with a five-question quiz in Hindi and English. It shows that most students know the three lists of the Seventh Schedule by name, but few can say why a subject was placed in one list and not another. She drops the planned revision of the lists and begins with a case from the students' own district: who maintains the village road, the State or the Union? She has used their entry behaviour to decide where to begin.

OFTEN CONFUSED

- **Maturation vs learning.** Maturation comes from growth with age; learning comes from experience and practice. Coaching produces learning, not maturation.
- **Aptitude vs achievement (prior knowledge).** Aptitude is the potential to learn a field; prior knowledge is what has already been learned.
- **Cognitive vs affective learner factors.** Intelligence, aptitude and prior knowledge are cognitive. Interest, attitude, motivation and anxiety are affective.
- **Learner factor vs learning environment.** Anxiety inside the student is a learner factor; a fearful classroom climate that causes anxiety is a learning environment factor.
- **Cultural capital vs aptitude.** Cultural capital is familiarity with educated culture gained at home; aptitude is the potential to learn. A first-generation student may have high aptitude and little cultural capital.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Cultural capital

The French sociologist Pierre Bourdieu used the term **cultural capital** (सांस्कृतिक पूँजी) for the knowledge, language habits, tastes and familiarity with educated culture that families pass on and that schools and universities reward. In "The Forms of Capital" (1986) he described three states of it:

embodied (lasting habits of mind and speech), objectified (cultural goods such as books and instruments) and institutionalised (educational qualifications).

First-generation college students often struggle with the unwritten conventions of academic life: how to address a professor, how to write an assignment, how to use a library. In Bourdieu's terms, what they lack is cultural capital. They are not less able; they have had less exposure. Bridge courses, mentoring and explicit teaching of academic conventions address this home-background factor.

Aptitude-treatment interaction

An **aptitude-treatment interaction** (अभिक्षमता-उपचार अंतःक्रिया, ATI) exists when the effect of a teaching method (the "treatment") depends on a characteristic of the learner (the "aptitude", used here in a wide sense for any learner characteristic). Lee J. Cronbach and Richard E. Snow surveyed this research in *Aptitudes and Instructional Methods* (1977). A pattern often reported is that highly structured teaching suits anxious learners and those with lower prior ability, while able, confident learners often do better with more freedom.

The consequence is that the same method gives different results with different learners, and no method is best for all. Read a typical research case carefully. Suppose highly anxious students do better under structured teaching, less anxious students do better under unstructured teaching, and the overall averages of the two methods are equal. The sound conclusion is that method and learner trait interact. It is wrong to conclude that one method is better for all, and equally wrong to conclude that method has no effect: each method helped one group and hindered the other. The practical lesson is to match or vary methods.

CHECK YOURSELF

1. Which of the following learner factors is affective?

(A) Reasoning ability
(B) Attitude towards the subject
(C) Knowledge of basic terms
(D) Speed of reading

2. Which of the following are learner-related factors?

1. A student's prior knowledge of the topic
2. The college's library hours
3. A student's maturation
4. A student's anxiety before tests
5. The teacher's command of English

Choose the correct answer from the options given below:

(A) 1, 3 and 4 only
(B) 1, 2 and 3 only
(C) 2, 4 and 5 only
(D) 1, 3, 4 and 5 only

3. Given below are two statements:

Statement I: Maturation depends mainly on the amount of coaching a learner receives.

Statement II: A learner's entry behaviour helps the teacher decide where to begin a lesson.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

4. Assertion (A): No single method of teaching is the best for all learners.

Reason (R): The effect of a method depends on its interaction with learner characteristics.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

5. Match List I with List II:

List I	List II
A. Ausubel	I. Cultural capital
B. Bourdieu	II. Aptitude-treatment interaction
C. Cronbach and Snow	III. What the learner already knows
D. Readiness	IV. Maturity sufficient for a task

Choose the correct answer from the options given below:

(A) A-III, B-II, C-I, D-IV
(B) A-I, B-III, C-II, D-IV

(C) A-III, B-I, C-IV, D-II
(D) A-III, B-I, C-II, D-IV

6. A first-generation student with high aptitude finds the conventions of academic writing unfamiliar. The most useful response is:
(A) to move him to a lower course, since he lacks ability
(B) to wait for maturation to remove the difficulty
(C) to ignore the matter, since home background does not affect learning
(D) to teach the conventions explicitly through mentoring or a bridge course

7. In a study, two methods give equal overall average scores, but one helps anxious students and the other helps confident students. This shows that:
(A) method has no effect on learning
(B) method and learner characteristics interact
(C) the first method is better for everyone
(D) anxiety does not affect learning

3.4 Factors related to support material

Syllabus: Factors affecting teaching related to: Support material

EXAM FOOTPRINT

Asked occasionally; usually a multiple-select item sorting factors into teacher, learner, support, environment or institutional groups.

Core

Support material (सहायक सामग्री) is the set of aids and resources a teacher uses to teach a particular content: charts, maps, models, handouts, worksheets, slides, films, recordings, and digital resources such as e-journals and videos. Chapter 5 describes the kinds of aid. This section asks a different question: when does support material help teaching, and when does it get in the way?

Support material is distinct from instructional facilities

Support material is made or chosen for a particular lesson or content. A worksheet prepared for one lesson is support material. **Instructional facilities** (अनुदेशात्मक सुविधाएँ) are the lasting infrastructure that the institution provides for all teaching: seminar halls, laboratories, libraries, computer terminals, the campus network. A computer laboratory with thirty terminals is a facility; the worksheet the students fill in on those terminals is support material.

Principles for selecting support material

Good support material meets these criteria. Each is a separate principle.

1. **Relevance to the objective.** The aid must serve what the lesson is meant to achieve. It is a means, not an end.
2. **Suitability to the learners' level.** Material that is too difficult, or too childish, fails to communicate.
3. **Accuracy and currency.** The content must be correct and up to date. An old map with outdated boundaries teaches error.
4. **Visibility and legibility.** Every student, including those at the back, must be able to see and read it. Letters must be large and the layout uncrowded.

5. **Simplicity.** One main idea per chart or slide. A crowded, text-heavy chart violates both legibility and simplicity.
6. **Availability, cost and durability.** These are practical considerations, but they are secondary. A cheap, durable or novel aid that the class cannot read or understand has failed.

Note that suitability and accuracy are two separate criteria. Neither explains the other: a chart may be perfectly accurate and still too advanced for the class.

Principles for using support material

- **Prepare.** Check the aid beforehand; preview a film; know what you will point out.
- **Show it at the right moment.** Display the aid when the point arises.
- **Remove it afterwards.** An aid left on display all period competes with what follows.
- **Do not overload.** Too many aids, however attractive, divide attention. More is not better.
- **Integrate.** Talk about the aid, ask questions about it and link it to the objective. An aid shown without discussion leaves learning to chance.

CLASSROOM EXAMPLE

A teacher of Geography at a college in Rohtak used to fill the board with six wall maps before each lecture on Indian agriculture. Students kept looking at the wrong map. Now she uses one large, clearly lettered map of rainfall zones, unrolls it when she reaches that point, asks three questions about it, and rolls it up before moving to cropping patterns. The aids are fewer, but they now do their work.

OFTEN CONFUSED

- **Support material vs instructional facility.** A worksheet, chart or set of slides for a lesson is support material. A laboratory, seminar hall, projector or network is a facility.
- **Attractiveness vs relevance.** An aid should be relevant first. Colour and novelty help only when the aid is already relevant, legible and suited to the learners.
- **Suitability vs accuracy.** Suitability is about the learners' level; accuracy is about the truth of the content. Both are needed; neither is the reason for the other.
- **Aid as means vs aid as end.** An aid used to serve the objective is a means. An aid displayed for its own sake has become an end and distracts.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Why too many aids hinder learning

Each aid asks for the learner's attention. Working memory can handle only a few new elements at once, so when aids multiply, the load exceeds what the learner can process and the point of the lesson is lost. This idea, from cognitive load theory, is explained more fully in 5.1. It is the reason for the rule "one idea per aid, and one aid at a time".

Support material and the learner

Support material connects with learner factors. The same chart may suit postgraduates and baffle first-year students; the same English handout may help some students and shut out others. Suitability to the learners' level is therefore not a separate concern from the learners; it is where support material and learner factors meet.

CHECK YOURSELF

1. A teacher displays a chart with dense small print that most students cannot read. Which principle has been violated?

(A) Economy of cost
(B) Durability for reuse
(C) Novelty of design
(D) Visibility and legibility

2. Which of the following is support material and not an instructional facility?

(A) A seminar hall with a sound system
(B) A handout of key terms for a lecture on inflation
(C) A library with open shelves
(D) A wireless network covering the campus

3. Assertion (A): Support material should be accurate and up to date.

Reason (R): Support material should be suited to the learners' level.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. Which of the following are sound practices in using teaching aids?

1. Showing the aid when the point it illustrates arises
2. Keeping every aid on display throughout the period
3. Choosing the aid according to the objective of the lesson
4. Using as many aids as the time allows

Choose the correct answer from the options given below:

(A) 1 and 3 only
(B) 2 and 4 only
(C) 1, 3 and 4 only
(D) 1, 2, 3 and 4

5. Given below are two statements:

Statement I: An aid is a means to the objective of the lesson and not an end in itself.

Statement II: The cost of an aid is the first criterion in choosing it.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

6. A teacher of first-year students uses a chart copied from a research monograph. It is accurate but full of technical symbols the students have never met. The chart fails mainly on:

(A) accuracy
(B) suitability to the learners' level
(C) durability
(D) availability

3.5 Factors related to instructional facilities

Syllabus: Factors affecting teaching related to: Instructional facilities

> **EXAM FOOTPRINT**
>
> Asked occasionally; usually a multiple-select item sorting factors into teacher, learner, support, environment or institutional groups.

Core

Instructional facilities (अनुदेशात्मक सुविधाएँ) are the physical resources an institution provides so that teaching can take place: classrooms and seminar halls, laboratories, workshops, libraries and reading rooms, language laboratories, computer terminals, projectors and smart boards, internet access and the campus network. They belong to the institution and serve all teachers and all courses.

Facility	What it makes possible
Classroom and furniture	A place to meet, see, hear and write
Laboratory and workshop	Handling apparatus, measuring, observing: practical skills
Library and reading room	Independent reading, reference, research
Language laboratory	Individual listening and speaking practice
Projectors, smart boards	Showing images, video and computer output to a class
Network, terminals, Wi-Fi	Access to e-journals, online courses, learning management systems

Keep facilities apart from the other groups. A sense of humour is a teacher characteristic. A high aptitude is a learner characteristic. A policy for revising the syllabus is an institutional or administrative matter. A well-equipped laboratory is an instructional facility.

Facilities limit what can be achieved

Facilities decide which objectives can be reached at all. Suppose a science course has to be taught entirely through lectures because no laboratory exists. Knowledge of facts and principles (the **cognitive domain** (संज्ञानात्मक पक्ष)) can still be gained, though less vividly. But manipulative skills, such as handling apparatus, measuring and observing, belong to the **psychomotor domain** (मनोगत्यात्मक पक्ष) and cannot be developed at all without a laboratory. So the objectives most affected by a missing laboratory are psychomotor.

Facilities enable; they do not cause

Facilities are **enabling conditions**: they make good teaching possible, but they do not produce it. Three common situations show this.

1. **Equipment unused.** A college fits a projector in every room, yet teachers seldom use them. Facilities help only when teachers are trained and willing to use them. So institutions must provide training and support along with equipment.
2. **ICT does not change methods by itself.** Many rooms with projectors continue with unchanged lecturing. Providing ICT facilities does not by itself change the method of teaching. Access is a **necessary but not sufficient** condition: use depends further on the teacher's skill, willingness and time, and on the institution's support.
3. **Libraries unused.** A library can stand nearly empty when teaching does not require independent reading, when its hours are inconvenient, or when students lack library skills. The fact that libraries are classified as instructional facilities is true, but it explains none of this.

Material needs facilities

Facilities and support material are distinct, and one cannot substitute for the other. Suppose an institution subscribes to a rich collection of e-journals, but students have neither campus Wi-Fi nor computer terminals. The support material (the e-journals) exists; what is missing is the means of access. The factor limiting learning is instructional facilities.

CLASSROOM EXAMPLE

A university in Haryana receives a grant and installs smart boards in twenty classrooms. A year later a survey finds that most are used only as ordinary projector screens. The university then runs a three-day workshop in which teachers prepare one interactive lesson each, and sets up a help desk. Use rises in the following term. The facility was the same; the training and support made the difference.

OFTEN CONFUSED

- **Facility vs support material.** The computer laboratory is a facility; the e-journal or worksheet used in it is support material.
- **Facility vs learning environment.** The laboratory is a facility. Whether that laboratory is well lit, quiet and free from fear is a matter of the learning environment.
- **Necessary vs sufficient.** Access to a facility is necessary for its use; it is not sufficient, because use also needs skill, willingness and support.
- **Classification vs explanation.** In Assertion-Reason items, the statement “X is an instructional facility” is usually true but rarely explains why X is or is not used.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Facilities in Indian higher education

Facilities vary widely between a well-funded central university and a rural affiliated college. Students in under-equipped colleges may still reach cognitive objectives through good lectures and reading, but practical skills and independent research suffer most. This is one reason why digital resources, such as online courses and open educational resources, matter so much for such colleges; they are discussed in 4.6 and 5.4.

Facilities and the teacher together

In questions that combine two groups, the deciding idea is usually that facilities need teachers. A facility multiplies what a skilled, willing teacher can do; it cannot replace the teacher’s planning and judgement.

CHECK YOURSELF

1. Which of the following is an instructional facility?

(A) A teacher's patience
(B) A learner's interest
(C) A rule fixing the number of working days
(D) A reading room in the college library

2. Assertion (A): Installing projectors in every classroom does not by itself change the way teachers teach.

Reason (R): The use of a facility depends on teachers' skill, willingness and the support they receive.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

3. A Physics course is taught without any laboratory work. The objectives most seriously affected are:

(A) knowledge of laws and definitions
(B) skills of handling apparatus and measuring
(C) appreciation of the history of science
(D) recall of formulae

4. Given below are two statements:

Statement I: Access to a facility is a sufficient condition for its educational use.

Statement II: Facilities and support material are distinct factors.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

5. A college buys access to an online database of journals, but its only computer room is locked most of the day. The factor limiting learning is:

(A) support material
(B) learner characteristics
(C) teacher's subject mastery
(D) instructional facilities

6. Which of the following are likely reasons why a well-stocked library is little used?

1. Teaching does not require independent reading
2. Library hours clash with class hours
3. Students have not been taught how to search the catalogue
4. Libraries are instructional facilities

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 1 and 4 only
(C) 2, 3 and 4 only

(D) 1, 2, 3 and 4

3.6 Factors related to the learning environment

Syllabus: Factors affecting teaching related to: Learning environment

EXAM FOOTPRINT

Asked occasionally; usually a multiple-select item sorting factors into teacher, learner, support, environment or institutional groups.

Core

The **learning environment** (अधिगम वातावरण) is everything that surrounds the act of learning. It has two sides, and almost every question on this topic asks you to tell them apart.

Side	What it includes	What goes wrong
Physical environment (भौतिक वातावरण)	Space, light, ventilation, temperature, furniture, seating arrangement, noise, cleanliness	Glare, heat, overcrowding, noise from the road, seats from which the board cannot be seen
Psychological climate (मनोवैज्ञानिक वातावरण)	Relationships between teacher and students and among students; warmth, respect, trust, freedom from fear	Fear of ridicule or punishment, humiliating comparison, hostility

The psychological side is also called the **socio-emotional climate** or classroom climate (कक्षा का वातावरण). Lighting, ventilation and seating belong to the physical environment, not to the psychological climate, the curriculum or the evaluation system.

The physical environment: noise

Attention is the gateway to learning, and noise competes for it. **Noise** (शोर) is therefore treated as a physical barrier in the learning environment. It lowers learning by distracting attention. It does not change intelligence or aptitude, which are relatively stable characteristics of the learner.

In the language of communication (Unit IV), noise is a **barrier** that distorts or blocks the reception of the teacher's message. Parts of the message are lost on the way from sender to receiver, and attention is diverted. That is how excessive noise reduces learning: the communication explanation and the attention explanation describe the same mechanism.

The physical environment: seating

Seating decides who can see whom, and so who talks to whom.

- **Rows facing the board** and **tiered lecture halls** direct all attention to the front. They suit lectures and demonstrations.
- **Circular or U-shaped seating** lets students see one another's faces, so they talk to one another and not only to the teacher. It suits discussion and seminars.
- **Clusters of tables** suit group work.
- Separate **individual cubicles** suit private study or testing, not discussion.

The psychological climate

A **positive climate** shows mutual respect, warmth, open communication and shared purpose. Students feel safe to ask, to answer and to make mistakes. A negative climate is marked by fear, strict silence enforced by threat, constant competition and public ranking, or minimal interaction. These raise anxiety and reduce participation.

A classroom in which students are afraid to ask questions has a poor psychological climate, even if the room is comfortable and the curriculum is sound. Fear of ridicule or punishment silences students.

Psychological safety (मनोवैज्ञानिक सुरक्षा) is the shared feeling that one will not be humiliated for a question or a mistake. A teacher builds it by treating errors as useful steps in learning, not as failures to be ridiculed. It is not the same as a **laissez-faire** climate (हस्तक्षेप-रहित, in which the teacher withdraws guidance), nor an **authoritarian** climate (in which the teacher rules by command and fear), nor a climate of keen competition. A safe climate still has high expectations and active guidance.

Cooperation and competition

Research generally favours **cooperative** classroom structures, in which students work towards shared goals, over **competitive** ones, in which one student's success is another's failure. Cooperation tends to give higher achievement and better relationships. But competition is not always harmful. Moderate, friendly competition, especially between groups whose members cooperate within the group, or a student competing against his or her own earlier record, can motivate without harm. A statement that competition "always" harms or "has no place" in a classroom is too absolute and therefore false.

Effects that colour what happens and how it is judged

Four named effects appear often in questions. They are about the social and psychological side of the classroom and of classroom research.

Effect	What happens	Cause
Hawthorne effect (हॉथोर्न प्रभाव)	Performance improves because people know they are being studied or given special attention	Awareness of being observed
Pygmalion effect	Performance rises because the teacher expects it to (see 3.2)	Teacher's expectations
John Henry effect	Members of a control group work harder to compete with the experimental group	Rivalry with the experimental group
Halo effect (प्रभामंडल प्रभाव)	A favourable impression of one trait spreads to the rating of other, unrelated traits	Rater's general impression

The **Hawthorne effect** takes its name from studies made between 1924 and 1932 at the Hawthorne Works of the Western Electric Company in the United States. It threatens classroom experiments: students placed in a new programme may do better simply because they know they are being observed. The **halo effect** was named by the psychologist Edward Thorndike in 1920. A teacher impressed by neat handwriting who also rates the content of the answer too highly is showing the halo effect.

CLASSROOM EXAMPLE

In a postgraduate seminar room at a university in Rohtak, the chairs are in rows and only three students speak. The teacher moves the chairs into a U-shape, sits among the students, and begins by saying, “There are no foolish questions here; a wrong answer is how we find the right one.” When a student mixes up two concepts, she thanks him and asks the class how the two differ. By the third week most students speak. She changed one physical factor (seating) and one psychological factor (safety to err).

OFTEN CONFUSED

- **Physical vs psychological environment.** Light, air, noise and seating are physical. Relationships, warmth and fear are psychological.
- **Psychological safety vs laissez-faire.** Safety keeps guidance and high expectations; laissez-faire removes guidance.
- **Hawthorne vs John Henry effect.** Hawthorne: the group that is studied improves because it is observed. John Henry: the control group improves because it competes.
- **Hawthorne vs Pygmalion.** Hawthorne works through the students’ awareness; Pygmalion through the teacher’s expectations.
- **Halo effect vs leniency error vs central tendency error.** Halo: one trait colours the rating of others in the same person. Leniency: rating everyone high. Central tendency: rating everyone near the middle.
- **Cooperative vs competitive climate.** Cooperation is generally better for achievement and relationships; moderate friendly competition can still motivate.

FACT WATCH

- The **Hawthorne effect** is named after studies conducted from 1924 to 1932 at the Western Electric Company’s Hawthorne Works. Later re-analyses of the original data found weaker evidence than the popular story suggests, but the term remains standard in research methods (checked October 2026).
- The **John Henry effect** was named by Gary Saretsky in 1972, after the legendary American steel-driver who tried to outwork a steam drill.
- The **halo effect** was named by Edward Thorndike in “A Constant Error in Psychological Ratings” (1920).

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Climate and participation: two separate components

Keep the two sides of the environment apart in Assertion-Reason items. “A threatening classroom climate lowers participation” is true, and the reason is psychological: students fear ridicule or punishment. “Lighting and ventilation affect attention” is also true, but it concerns the physical side. Both statements are true, but the second does not explain the first.

The action zone

Observational studies of traditional classrooms, notably Adams and Biddle's *Realities of Teaching* (1970), found that most teacher-student interaction takes place with students seated at the front and down the centre of the room. This T-shaped region is called the **action zone** (क्रिया क्षेत्र). Students outside it tend to receive fewer questions and less feedback. Later studies have not always confirmed the effect equally strongly, but the general finding stands: front and centre students interact most with the teacher, while circular seating increases student-to-student talk. The claim that back-row students "always participate more" reverses the usual finding.

Cooperation research

A widely cited review by Johnson, Maruyama, Johnson, Nelson and Skon (1981) pooled 122 studies and found that cooperation produced higher achievement than interpersonal competition or individual work. Cooperation combined with competition between groups did about as well as pure cooperation. This is the evidence behind the rule "cooperation within groups, friendly competition between them".

Other rating errors

Rating errors matter wherever teachers judge students: in viva voce, practical examinations, internal assessment and observation of teaching. Besides the halo effect, the **leniency error** (rating everyone generously), the **severity error** (rating everyone harshly), and the **error of central tendency** (avoiding the extremes and rating everyone near the middle) are the common ones. Clear rubrics and rating each trait for all students before moving to the next trait reduce them (see Chapter 6).

CHECK YOURSELF

1. Which of the following belong to the physical environment of a classroom?

1. Ventilation
2. Fear of ridicule
3. Seating arrangement
4. Noise from the street
5. Mutual respect

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 1, 3 and 4 only
(C) 2, 4 and 5 only
(D) 1, 3, 4 and 5 only

2. Students in a new remedial programme score higher in the first month mainly because they know researchers are watching their class. This is the:

(A) halo effect
(B) Pygmalion effect
(C) Hawthorne effect
(D) John Henry effect

3. Assertion (A): Students participate more when the teacher treats errors as steps in learning.
Reason (R): Psychological safety removes the fear of humiliation that silences students.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. Match List I with List II:

List I	List II
A. Halo effect	I. Rating all students near the middle
B. Leniency error	II. One impressive trait colours other ratings
C. Central tendency error	III. Control group works harder to compete
D. John Henry effect	IV. Rating all students generously

Choose the correct answer from the options given below:

(A) A-II, B-IV, C-I, D-III
(B) A-IV, B-II, C-I, D-III
(C) A-II, B-I, C-IV, D-III
(D) A-II, B-IV, C-III, D-I

5. Given below are two statements:
Statement I: Circular seating encourages students to talk to one another.

Statement II: In traditional rows, students at the back usually interact most with the teacher.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

6. Assertion (A): Cooperative classroom structures tend to favour learning.
Reason (R): Competition of every kind is harmful and should be removed from classrooms.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

7. Noise around a classroom mainly reduces learning by:
(A) lowering the intelligence of the learners
(B) changing the aptitude of the learners
(C) distracting attention and blocking reception of the message
(D) altering the curriculum

8. A teacher who wants a debate among forty students should arrange the seating in:
(A) fixed rows facing the board
(B) a tiered lecture gallery
(C) separate study cubicles
(D) a U-shape or circle

3.7 Factors related to the institution

Syllabus: Factors affecting teaching related to: Institution

EXAM FOOTPRINT

Asked occasionally; usually a multiple-select item sorting factors into teacher, learner, support, environment or institutional groups.

Core

Institution-related factors (संस्थागत कारक) are the policies, structures, leadership and climate of the college or university. They set the limits within which every teacher plans. A teacher chooses examples, questions, tone of voice and style. The institution fixes admissions, sections, class size, the timetable, the academic calendar, the teaching load, the rules of evaluation and the support given to teachers. These factors operate in every institution, public or private.

Institutional factor	How it affects teaching
Academic calendar and timetable	How much time there is, and when; the length and spacing of periods
Class size and teacher-student ratio	How much individual attention is possible
Workload and non-teaching duties	Time and energy left for preparation and students
Curriculum and evaluation policy	What is taught and how it is taught

Institutional factor	How it affects teaching
Leadership and administrative support	Freedom and resources to innovate
Institutional climate	Teacher morale
Faculty development	Growth of teachers' knowledge and skill
Quality assurance (for example the IQAC)	Planning, monitoring and improvement of teaching

Class size

As **class size** (कक्षा का आकार) grows, the teacher has less time for each student, so individual guidance, interaction and feedback suffer. A very large class mainly makes it difficult to give individual attention. A lecture, an announcement of the syllabus or a common written test can still be given to a large group, which is why the lecture dominates large classes. Class size is decided by the institution through its admission and sectioning policies.

Time: calendar, timetable and workload

The academic calendar and timetable are decided mainly by the institution, not by learners, parents or publishers. **Workload** (कार्यभार) is also institutional. Heavy clerical and administrative duties take time from lesson preparation, feedback on assignments and contact with students, and they add to fatigue. They do not change admissions or the syllabus; their effect is on the time and energy left for teaching.

Curriculum: who frames the syllabus

In an Indian university the syllabus of a subject is usually framed and revised by the **Board of Studies** (अध्ययन मंडल) of that subject. It consists of teachers of the subject, usually with outside experts. Its recommendations go to the **Academic Council** (विद्या परिषद), the university's main academic body, for approval. The Finance Committee deals with money, the Building Committee with construction, and a Selection Committee with appointments; none of them frames syllabi. The exact composition and powers are laid down in each university's statutes, so details differ from university to university.

Evaluation policy and teaching to the test

What the institution examines, and how, strongly shapes what happens in class. When everything depends on a final examination, teachers and students concentrate on likely questions and neglect skills and topics that are not examined. This narrowing is called **teaching to the test** (परीक्षा-केंद्रित शिक्षण). It is an institution-related constraint on method. In the other direction, an assessment policy that rewards projects, presentations or internal assessment can encourage innovation in teaching.

Leadership, climate and morale

A supportive head and administration help teaching mainly by giving teachers freedom and resources to innovate, and by encouraging professional development. They cannot remove the need for teachers to plan lessons, use their judgement or evaluate students; those remain the teacher's work.

Institutional climate (संस्थागत वातावरण) is the general atmosphere of the institution: the style of leadership, recognition of good work, participation of teachers in decisions and relations among staff. It shapes teacher morale. Note that the institution also fixes the academic calendar, but that function says nothing about morale.

Subject mastery is not institutional

Subject mastery is a characteristic of the teacher. An institution can support it, by recruiting well and providing faculty development, but the mastery itself resides in the teacher. So it remains a teacher-related factor.

CLASSROOM EXAMPLE

A college in Fatehabad has two sections of B.A. Part I English, of 140 students each. Teachers lecture and set the same test; no one reads individual essays. After the college splits each section into two groups of 70, reduces its teachers' admission-duty hours, and allows 25 per cent of marks for a portfolio of short essays, teachers begin returning written comments on every essay. The teachers did not change; the institution changed what was possible.

OFTEN CONFUSED

- **Board of Studies vs Academic Council.** The Board of Studies frames the syllabus of one subject; the Academic Council approves courses, syllabi and academic regulations for the whole university.
- **Academic Council vs IQAC.** The Academic Council approves academic matters; the IQAC plans, guides and monitors quality on a continuing basis.
- **Institutional factor vs teacher factor.** Class size, timetable, workload and evaluation policy are institutional; examples, questions, tone and subject mastery are the teacher's.
- **Institutional climate vs classroom climate.** Institutional climate is the atmosphere among staff and leadership, which shapes teacher morale. Classroom climate is the atmosphere in one class, which shapes student participation (see 3.6).

FACT WATCH

- **IQAC.** The National Assessment and Accreditation Council (NAAC) asks every accredited institution to set up an **Internal Quality Assurance Cell** (आंतरिक गुणवत्ता आश्वासन प्रकोष्ठ, IQAC) as a measure for sustaining quality after accreditation. Through the IQAC the institution submits an **Annual Quality Assurance Report** (AQAR) to NAAC each year (checked October 2026).
- NAAC's accreditation framework has been under revision. Do not rely on the grading scale printed in older books; check the current position on NAAC's website before the examination.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

The Internal Quality Assurance Cell

The IQAC is the institution's own body for quality. Its tasks include developing benchmarks for academic and administrative work, collecting and analysing feedback from students and other stakeholders, organising workshops and faculty development, documenting good practices, and preparing the AQAR. It is an institution-related factor in the quality of teaching: where the IQAC works well, teaching is planned, observed and improved; where it exists only on paper, it changes nothing.

Faculty development

Opportunities for faculty development, such as orientation and refresher programmes, workshops, conference leave and online courses, are an institutional factor. An institution that supports its teachers' growth improves teaching over time. This links with 1.10 (the teacher's professional growth).

Teacher-student ratio

The teacher-student ratio of a whole institution is not the same as class size: a college may have a good overall ratio and still run some very large sections. Both are set by institutional policy, and both affect how much individual attention a student receives.

CHECK YOURSELF

1. Which of the following is decided at the level of the institution?

(A) The analogy a teacher uses to explain a concept
(B) The follow-up question a teacher asks
(C) The pace of a teacher's speech
(D) The number of students admitted to each section

2. In a university, the syllabus of a subject is usually framed by the:

(A) Finance Committee
(B) Selection Committee
(C) Board of Studies of that subject
(D) Building Committee

3. Consider the following statements about institution-related factors:

1. Institution-related factors operate only in government colleges.
2. The teacher-student ratio is an institutional factor.
3. An examination policy that rewards only memorised answers can discourage innovation in teaching.
4. A teacher's mastery of the subject is an institution-related factor.

Which of the statements given above are correct?

(A) 1 and 2 only
(B) 2 and 3 only
(C) 2, 3 and 4 only
(D) 1, 2, 3 and 4

4. Assertion (A): Teacher morale is higher in institutions where teachers take part in decisions and good work is recognised.

Reason (R): The institution fixes the dates of the academic calendar.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

5. Match List I with List II:

List I	List II
A. Board of Studies	I. Approves courses and academic regulations of the university
B. Academic Council	II. Prepares the Annual Quality Assurance Report
C. IQAC	III. Frames the syllabus of a subject
D. Selection Committee	IV. Recommends candidates for appointment

Choose the correct answer from the options given below:

(A) A-I, B-III, C-II, D-IV
(B) A-III, B-I, C-II, D-IV
(C) A-III, B-II, C-I, D-IV
(D) A-III, B-I, C-IV, D-II

6. When a college judges its teachers only by their students' final examination results, the most likely effect on teaching is:

(A) teaching narrowed to likely examination questions
(B) wider use of projects and fieldwork
(C) smaller classes
(D) higher salaries for teachers

7. Given below are two statements:

Statement I: A very large class makes it hard to give individual feedback.

Statement II: A very large class makes it impossible to deliver a lecture.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

Answers to “Check yourself”: Chapter 3

Section 3.1

1. (B), because a laboratory is lasting infrastructure provided for all teaching. The worksheet, chart and slides are made for particular content, so they are support material.

2. (D), because mastery is necessary but not sufficient, which makes Statement I false; Statement II states the accepted position that the effect of a method depends on the learners’ readiness.

3. (A), because noise is part of the learning environment, the principal’s encouragement is institutional leadership, interest lives in the learner and questioning skill belongs to the teacher.

4. (C), because motivation, aptitude and prior knowledge reside in the students. Seating belongs to the learning environment and enthusiasm to the teacher.

5. (A), because equipment is only one of six groups of factors; without a capable teacher, ready learners and a supportive climate it cannot produce learning, so R explains A.

6. (D), because the timetable and sectioning are fixed by institutional policy, not by the learner or the surroundings of a particular class.

Section 3.2

1. (B), because efficacy is the teacher’s belief in his or her own capacity to bring about learning. Accountability is answerability, autonomy is freedom of decision, and burnout is exhaustion.

2. (C), because warmth motivates and expectations act as self-fulfilling prophecies. Length of service has no direct bearing on clarity, which depends on mastery and language.

3. (A), because differential treatment is the mechanism through which expectations become self-fulfilling, so R explains A.

4. (C), because Rosenthal and Jacobson wrote Pygmalion in the Classroom (1968), Babad, Inbar and Rosenthal named the Golem effect (1982), Bandura introduced self-efficacy (1977), and Tschannen-Moran and Woolfolk Hoy reviewed teacher efficacy (2001).

5. (D), because low expectations shown through less attention and less wait time lower performance, which is the Golem effect. Pygmalion is the positive direction.

6. (C), because mastery belongs to the teacher but is necessary, not sufficient; communication, method and the learners also decide the result.

Section 3.3

1. (B), because attitude is a feeling towards something, which makes it affective. Reasoning, knowledge and reading speed are cognitive.

2. (A), because prior knowledge, maturation and anxiety lie in the student. Library hours are institutional, and the teacher’s English is a teacher factor.

3. (D), because maturation is growth with age independent of training, so Statement I is false; entry behaviour does tell the teacher where to begin.

4. (A), because the interaction between method and learner is exactly why no method suits all, so R explains A.

5. (D), because Ausubel stressed what the learner already knows, Bourdieu named cultural capital, Cronbach and Snow surveyed aptitude-treatment interactions, and readiness means maturity sufficient for a task.

6. (D), because the student lacks exposure, not ability; explicit teaching of the conventions supplies the missing cultural capital.

7. (B), because equal averages hide opposite effects in two groups, which is an aptitude-treatment interaction, not an absence of effect.

Section 3.4

1. (D), because an aid that cannot be read fails in its purpose; cost, durability and novelty are secondary considerations.

2. (B), because a handout prepared for a particular lecture is support material; halls, libraries and networks are infrastructure, that is, facilities.

3. (B), because both are sound selection criteria, but suitability to level is not the reason for accuracy; they stand side by side.

4. (A), because aids should serve the objective and be shown when needed. Constant display and overloading distract.

5. (C), because an aid serves the objective, so Statement I is true; cost is a secondary consideration, so Statement II is false.

6. (B), because the chart is accurate but pitched above the learners' level, so it does not communicate.

Section 3.5

1. (D), because a reading room is infrastructure provided by the institution. Patience is a teacher factor, interest a learner factor, and the rule on working days an institutional policy.

2. (A), because the dependence of use on skill, willingness and support is exactly why equipment alone does not change teaching, so R explains A.

3. (B), because manipulative skills belong to the psychomotor domain and need a laboratory; knowledge and recall can still be gained from lectures.

4. (D), because access is necessary but not sufficient, so Statement I is false; facilities and material are indeed distinct.

5. (D), because the material (the database) exists; the bottleneck is access to terminals, which is a facility.

6. (A), because the first three explain non-use. Statement 4 is true but explains nothing about use.

Section 3.6

1. (B), because ventilation, seating and street noise are physical. Fear of ridicule and mutual respect are psychological.

2. (C), because the improvement comes from awareness of being observed. Pygmalion would need teacher expectations, and John Henry concerns a control group competing.

3. (A), because safety from humiliation is the reason students dare to take part, so R explains A.

4. (A), because halo spreads one impression, leniency rates everyone high, central tendency rates everyone in the middle, and John Henry is the control group's rivalry.

5. (C), because seeing faces encourages student-to-student talk; in rows it is the front and centre (the action zone), not the back, that interacts most.

6. (C), because the Assertion is supported by research, but the Reason is too absolute: moderate, friendly competition can motivate.

7. (C), because noise is a physical barrier to attention and communication; it does not change stable learner traits or the curriculum.

8. (D), because seeing one another's faces encourages discussion; rows and galleries focus attention on the front.

Section 3.7

1. (D), because admission and sectioning are institutional policy. The analogy, pace and follow-up question are the teacher's choices.

2. (C), because the Board of Studies frames the syllabus and sends it to the Academic Council for approval; the other committees deal with money, buildings and appointments.

3. (B), because ratio and examination policy are institutional. Such factors operate in all institutions, and subject mastery resides in the teacher.

4. (B), because morale depends on institutional climate; fixing the calendar is a true institutional function but does not explain morale.

5. (B), because the Board of Studies frames syllabi, the Academic Council approves academic matters, the IQAC prepares the AQAR, and the Selection Committee recommends appointments.

6. (A), because exam-driven judgement pushes teachers towards teaching to the test.

7. (C), because a lecture can still be delivered to a large group; what suffers is individual attention and feedback.

Chapter 4
Methods of Teaching in Institutions of Higher Learning

This chapter is about the ways a college or university teacher can organise learning. It begins with the words used to talk about methods and with the great divide of the syllabus, teacher-centred against learner-centred methods. It then takes the methods group by group: the lecture and its relatives, the group and discussion methods, the activity methods, and finally off-line and on-line teaching, ending with SWAYAM, SWAYAM PRABHA and the other national digital initiatives. The last section is full of dates and figures that change, so read its Fact watch boxes with care.

4.1 Approach, method, strategy and technique; teacher-centred and learner-centred methods

Syllabus: Methods of teaching in Institutions of higher learning: Teacher centred vs. Learner centred methods

> **EXAM FOOTPRINT**
>
> Asked in most cycles; usually a multiple-select item picking learner-centred methods from a list, or a scenario contrasting the two approaches.

Core

Four words that are often mixed up

Before we compare methods, we need four words. Examiners sometimes test them, and books use them loosely, so fix their meanings now.

- An **approach** (उपागम) is the broad view of teaching and learning that lies behind everything a teacher does. “Learning is something the learner builds through activity” is an approach.
- A **method of teaching** (शिक्षण विधि) is an organised, step-by-step way of teaching a whole lesson or unit. The lecture, the project method and the seminar are methods.
- A **strategy** (शिक्षण व्यूह रचना) is the overall plan a teacher makes for reaching an objective with a particular group, often combining several methods. “Introduce the topic by lecture, then divide the class into groups, then hold a quiz” is a strategy.
- A **technique** (प्रविधि) is a smaller device used inside a method: asking a question, giving an example, a buzz group of five minutes, writing on the board.

The order from broad to narrow is: approach, then strategy and method, then technique. The safe rule for the examination is that a technique is the smallest unit and an approach is the widest.

Teacher-centred methods

In a **teacher-centred method** (शिक्षक-केंद्रित विधि) the teacher (शिक्षक) is the main actor. The teacher decides the content, controls the pace and does most of the talking or showing. The learners (शिक्षार्थी) mainly listen, watch and take notes. The teacher is a transmitter of information.

Typical teacher-centred methods are the lecture, dictation of notes, the teacher’s demonstration and the lecture-cum-demonstration. They are economical: one teacher can cover much content for a large class in little time. Their weakness is that learners stay comparatively passive.

Learner-centred methods

In a **learner-centred method** (शिक्षार्थी-केंद्रित विधि) the learner is the main actor. Learners inquire, discuss, plan, solve problems and make things. The teacher arranges the conditions, designs the tasks, guides and supports. The teacher becomes a **facilitator** (सुगमकर्ता) of learning.

Typical learner-centred methods are the project method, the problem-solving method, group discussion, role play, the case method, the heuristic (discovery) method, peer tutoring and cooperative learning.

Behind learner-centred methods lies a view of learning (अधिगम) called constructivism: knowledge is built (constructed) by the learner through activity, not simply received. The shift from teacher-centred to learner-centred teaching is therefore a shift from transmitting content to helping learners construct knowledge.

Point of comparison	Teacher-centred	Learner-centred
Main actor	Teacher	Learner
Role of teacher	Transmitter of information, controller of content and pace	Facilitator, designer of tasks, guide
Role of learner	Listens, watches, takes notes (passive)	Discovers, discusses, plans, does (active)
Communication	Mostly one-way	Many-way: learner to teacher and learner to learner
Pace	Same for the whole class, set by the teacher	Varies with learners
Assessment	Mostly at the end	Continuous and formative, used to guide learning
Relations among learners	Often individual or competitive	Cooperative
Strength	Economy of time; covers much content; good for introducing new topics	Deeper understanding, longer retention, skills of inquiry and cooperation
Weakness	Little activity, little feedback, no adjustment to individual pace	Slow; needs planning, resources and smaller groups
Examples	Lecture, dictation, teacher demonstration	Project, problem-solving, discussion, role play, case method

Why learner-centred methods are favoured

What learners work out, discuss and apply for themselves is processed more deeply, so it is understood better and retained longer. This is the main reason learner-centred methods are favoured. Note the cause carefully: retention is better because the learner is active. A statement that “in learner-centred methods the learner is passive and the teacher active” is false; it describes teacher-centred methods.

Three cautions that examiners test

1. Learner-centred teaching does not make the teacher unimportant. The teacher’s role changes from presenter to designer and guide, and it becomes, if anything, more demanding.
2. Learner-centred teaching uses **formative evaluation** (रचनात्मक मूल्यांकन) a great deal: frequent checks whose results guide the next step of learning.
3. Teacher-centred methods are not “wrong”. A clear lecture is often the best way to open an unfamiliar topic or give a large class an overview. The choice of method depends on the objectives, the learners, the content, the size of the class and the time available. There is no single best method.

Inductive and deductive methods: a first look

Two general methods cut across this divide. In the **inductive method** (आगमन विधि) learners study several particular examples and are led to frame the general rule themselves: from examples to rule, from particular to general. In the **deductive method** (निगमन विधि) the teacher states the rule or definition first and learners apply it to examples: from rule to examples, from general to particular. Induction suits discovery and is the more learner-centred; deduction saves time. Section 4.4 compares them more fully.

What the National Education Policy 2020 asks for

The National Education Policy 2020 (NEP 2020) asks all levels of education, higher education included, to move away from rote learning. The pedagogy it wants is described in the policy as experiential, holistic, integrated, inquiry-driven, discovery-oriented, learner-centred, discussion-based, flexible and enjoyable; for higher education it adds freedom for teachers to design their own approaches and greater use of formative assessment. If an option reads "teacher-centred, lecture-driven and examination-based", that is the pattern the policy wants to leave behind.

CLASSROOM EXAMPLE

In a B.A. Political Science class of 90 students at a government college in Sirsa, the teacher opens the unit on the Panchayati Raj system with a 20-minute lecture giving the structure of the three tiers. That part is teacher-centred, and it is the right choice for a first overview. Then she divides the class into groups of six. Each group studies the records of one nearby gram panchayat, discusses what works and what does not, and reports back. Now the learners are the main actors and she moves from group to group as facilitator. One strategy, two methods, several techniques.

OFTEN CONFUSED

- **Approach and method.** An approach is the underlying view of learning; a method is the organised way of teaching a lesson. Constructivism is an approach; the project method is a method.
- **Method and technique.** A method organises a whole lesson; a technique is a small device inside it, such as questioning or a buzz group.
- **Teacher demonstration and group discussion.** Both can happen in one lesson, but the demonstration is teacher-centred (learners watch) and the discussion is learner-centred (learners talk and think).
- **Facilitator and transmitter.** Facilitator is the teacher's role in learner-centred methods; transmitter is the role in teacher-centred methods.
- **Inductive and deductive.** Inductive goes from examples to rule (slower, discovery); deductive goes from rule to examples (quicker, practice).

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Where the methods of this chapter sit

The following grouping is the one used in this chapter. Read it as a map.

Group	Methods (section)
Teacher-centred, presentation	Lecture, demonstration, lecture-cum-demonstration (4.2)

Group	Methods (section)
Individualised and self-paced	Programmed instruction (4.2); Personalised System of Instruction or Keller Plan, mastery learning (4.4)
Group and interactive	Tutorial, seminar, symposium, panel, workshop, conference, debate, brainstorming, buzz groups, cooperative learning, team teaching (4.3)
Learner-centred, activity	Project, problem-solving, problem-based learning, heuristic, case method, role play, simulation (4.4)
Off-line and on-line	Face-to-face teaching, synchronous and asynchronous on-line learning, blended learning, flipped classroom, MOOCs (4.5, 4.6)

A few methods sit between the two poles. Programmed instruction is controlled by the programmer, who fixes the content and its order, yet the learner works alone, responds at every step and sets the pace. It is usually called individualised or self-instructional rather than simply teacher-centred or learner-centred. The tutorial is led by a tutor but is built around the learner's own difficulties.

Questions that test the reason, not the fact

Many harder items pair two true statements in an Assertion and Reason where the Reason does not explain the Assertion. Ask yourself: does this Reason say why the Assertion is true? "Learner-centred methods improve retention" is explained by "learners process material actively", not by "learner-centred methods are recommended by NEP 2020", though both are true.

CHECK YOURSELF

1. In which of the following is the teacher acting mainly as a facilitator?

(A) Reading out a chapter summary to the class
(B) Dictating definitions for students to copy
(C) Setting a problem and guiding groups as they work on it
(D) Performing an experiment while students watch silently

2. Arrange the following from the broadest to the narrowest:

1. Technique
2. Approach
3. Method

(A) 3, 2, 1
(B) 1, 3, 2
(C) 2, 1, 3
(D) 2, 3, 1

3. Consider the following methods:

1. Lecture
2. Role play
3. Case method
4. Dictation of notes
5. Peer tutoring

Choose the correct answer from the options given below. The learner-centred methods are:

(A) 1, 2 and 3 only
(B) 2, 3 and 5 only
(C) 2, 4 and 5 only
(D) 3, 4 and 5 only

4. Given below are two statements:

Statement I: In learner-centred teaching the teacher's role becomes less important.

Statement II: A teacher-centred method may be the better choice for giving a large class a first overview of a new topic.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

5. Given below are an Assertion (A) and a Reason (R):

Assertion (A): Learning gained through learner-centred methods is usually retained longer.

Reason (R): In learner-centred methods learners work on, discuss and apply the material themselves.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

6. Match List I with List II:

List I (Feature)	List II (Belongs to)
A. Pace set for the whole class	I. Inductive method
B. Assessment used continuously to guide learning	II. Teacher-centred method
C. Rule reached from examples	III. Deductive method
D. Rule stated first, then applied	IV. Learner-centred method

(A) A-II, B-IV, C-I, D-III
(B) A-IV, B-II, C-I, D-III
(C) A-II, B-IV, C-III, D-I
(D) A-III, B-IV, C-I, D-II

7. The pedagogy that the National Education Policy 2020 recommends for higher education is best described as:
(A) uniform, timetable-driven and recall-based
(B) lecture-driven and examination-centred
(C) inquiry-driven, discussion-based and learner-centred
(D) textbook-driven and teacher-controlled

4.2 Teacher-centred methods: lecture, demonstration and programmed instruction

Syllabus: Methods of teaching in Institutions of higher learning: Teacher centred vs. Learner centred methods

EXAM FOOTPRINT

Asked occasionally; usually a single-answer item on lecture or demonstration features, or matching programmed-instruction types with proponents.

Core

The lecture method

In the **lecture method** (व्याख्यान विधि) the teacher presents a body of organised content by talking, while the learners listen and take notes. The teacher decides what is said, in what order and at what speed. That is why the lecture is the standard example of a teacher-centred method.

The great strength of the lecture is **economy**. One teacher can address a large class, even several hundred students, at the same time, and can cover much organised content in one hour. The ratio of learners to teacher is high, so the cost and time spent per student are low. This is the correct explanation of the statement “the lecture is economical”.

The lecture is useful for:

- giving a large group an overview of a topic in limited time;
- introducing a new topic and showing how its parts fit together;
- presenting recent or scattered material that is not yet in any single textbook;
- covering the syllabus within the time allowed;
- arousing interest, when the speaker is skilled.

Its limitations are the other side of the same coin:

- Communication is mostly **one-way**. Learners listen without much chance to question, discuss or apply, so they remain comparatively passive.
- The lecture gives little or no **feedback** (प्रतिपुष्टि) from every learner, so the teacher cannot tell who has understood.
- It moves at one pace for all and makes no adjustment to individual differences.
- It is weak for developing **psychomotor skills** (manual and practical skills), because skills need practice by the learner, and in a lecture the learner only listens. Note the cause: the lecture is weak for skills because there is no learner practice, not because the class is large.
- Attention falls as the lecture goes on.

Attention and the "fifteen to twenty minutes" advice

Attention is high at the start of a lecture. It is commonly said to decline after about fifteen to twenty minutes. Lecturers are therefore advised to divide the hour into segments and to break it with a question, a short task, an example, a visual aid or a change of medium. This advice is found in most textbooks and is what examination questions expect. You should know, however, that the research behind the exact figure is disputed: some psychologists (for example Wilson and Korn, 2007) have argued that the evidence for a fixed attention span of this length is weak. The safe answer in the examination remains "about fifteen to twenty minutes".

Making a lecture more effective

A lecture improves when it becomes partly two-way. The teacher should:

- state the plan of the lecture at the start and sum up at the end;
- pause to ask questions and invite questions;
- give examples from the learners' own experience;
- use the board, slides, charts or objects at intervals;
- insert short activities, such as a two-minute discussion with a neighbour;
- keep eye contact and speak at a pace the class can follow.

Speaking very fast to cover more, reading straight from the book and avoiding eye contact all make a lecture worse.

The lecture as communication

In terms of the communication process, a lecture has a **sender** (the teacher), a **message** (the content, encoded in words), a **channel** (voice, board, slides) and **receivers** (the students). What it largely lacks is **feedback**, the return flow from the receivers to the sender. Without it, the lecturer cannot tell whether the message has been understood as intended. Questions, pauses and short tasks restore the feedback loop.

The demonstration method

In the **demonstration method** (प्रदर्शन विधि) the teacher shows, step by step, how something is done, while explaining what is being done and why. Typical demonstrations are a laboratory experiment, the use of an instrument, a dissection, a surveying technique, a computer procedure, or a dance step. It suits practical skills and procedures, because learners see the process before trying it themselves.

For a demonstration to work:

- every student must be able to see and hear each step clearly;
- the teacher must rehearse it beforehand;
- the apparatus should be simple and the steps taken at a moderate pace;
- the teacher should question the class during the demonstration.

The best way to keep students thinking is to ask them to **predict** what will happen, **observe** it, and then **explain** it. Announcing the conclusion before starting, performing silently, or letting only the front rows see, all leave students passive.

A demonstration is more concrete than a lecture, because learners see as well as hear. But it is still teacher-centred: the teacher performs and the learners watch. It does not give hands-on practice. A manipulative skill, such as titration, is learnt only when the demonstration is followed by **laboratory practice** by each learner, with feedback.

Lecture-cum-demonstration

In the **lecture-cum-demonstration method** the teacher explains a principle and at the same time demonstrates it with apparatus, specimens or models. It combines telling and showing, so learners both hear and see. It is widely used in science teaching in Indian colleges. It is more concrete than a pure lecture, though the learners still mostly watch.

Matching method to objective

The method should follow the objective (उद्देश्य), the size of the class and the time. For an overview of a policy document for 180 students in one period, the lecture is suitable. For the skill of titration for 20 students, a demonstration must be followed by laboratory practice. Watching alone does not give a skill.

Programmed instruction

Programmed instruction (अभिक्रमित अनुदेशन) is a form of self-instruction. The content is broken into a sequence of small units called **frames**. Each frame gives a small piece of information and asks the learner to respond, for example by filling in a blank. The learner then checks the answer at once and moves to the next frame. The sequence of frames is called a **programme**. It can be printed as a programmed text or presented by a teaching machine or a computer.

Programmed instruction is based on the operant conditioning (क्रियाप्रसूत अनुबंधन) of **B. F. Skinner**. In operant conditioning, behaviour that is followed by reinforcement (पुनर्बलन) becomes more likely. In a programme, the immediate knowledge that one's response is correct acts as the reinforcement.

Its principles are usually listed as five:

1. **Small steps.** Content is broken into small frames.
2. **Active responding.** The learner must respond at every frame, not just read.
3. **Immediate confirmation** (or immediate feedback, knowledge of results). The learner knows at once whether the response is right. This reinforces right answers and prevents wrong ones from being practised.
4. **Self-pacing.** Each learner works alone, quickly or slowly as needed. A pace fixed by the teacher for the whole class is the mark of conventional teaching, not of programmed instruction.
5. **Student testing** (also called learner verification or evaluation). The programme is tried out on learners and revised wherever they make errors; the learner's record shows how well the programme works.

If a learner checks each answer immediately after writing it, the principle at work is immediate confirmation. If a question asks which principle allows each learner to go at his or her own speed, the answer is self-pacing.

Linear and branching programmes

There are two main styles of programme.

In **linear programming** (रेखीय अभिक्रमण), developed by Skinner, every learner passes through the same fixed sequence of small frames, one after another. The steps are kept very small so that the

learner's responses are almost always correct and therefore reinforced; errors are kept to a minimum. Responses are **constructed**: the learner writes the answer. Only the pace varies from learner to learner; the path is the same for all.

In **branching programming** (शाखीय अभिक्रमण), developed by **Crowder**, frames are larger. Each frame ends with a **multiple-choice** question. The choice the learner makes decides the next frame. A correct choice takes the learner forward; a wrong choice leads to a **remedial frame** that explains the error and then returns the learner to the main sequence. Errors are not avoided; they are used for diagnosis. Crowder called this intrinsic programming. Because the route depends on each learner's responses, both the path and the pace are individualised, so branching adjusts to individual differences better than linear programming.

Feature	Linear (Skinner)	Branching (Crowder)
Size of frame	Small	Larger
Type of response	Constructed (learner writes the answer)	Multiple-choice (learner selects)
Errors	Kept to a minimum	Used for diagnosis; lead to remedial frames
Path	Same for all learners	Depends on each learner's response
What varies	Pace only	Pace and path
Basis	Operant conditioning, reinforcement of correct responses	Diagnosis and correction of errors

CLASSROOM EXAMPLE

In a B.Sc. Chemistry class of 60 at a college in Hisar, the teacher first gives a 15-minute lecture on acid-base titration. Then, at the front bench raised on a platform so that all can see, she performs a titration, stopping before each step to ask, "What colour do you expect now?" Students write a prediction, watch, and explain. Next week, each pair of students performs the titration in the laboratory while she moves from bench to bench. For students who missed the class, the department has a programmed booklet on titration calculations: each page gives one small step, a blank to fill and the answer overleaf.

OFTEN CONFUSED

- **Lecture and demonstration.** Both are teacher-centred. In a lecture learners hear; in a demonstration they also see a process performed. Neither gives hands-on practice.
- **Demonstration and laboratory practice.** In a demonstration the teacher performs; in laboratory practice each learner performs. Skills come from practice.
- **Economy and participation.** The lecture is economical (many learners per teacher, much content per hour); it is not participatory. An answer that the lecture "ensures active participation" is false.
- **Immediate confirmation and self-pacing.** Immediate confirmation is knowing at once whether the response is right; self-pacing is going at one's own speed.
- **Linear and branching.** Linear: small frames, written responses, minimal errors, same path (Skinner). Branching: larger frames, multiple choice, errors lead to remedial frames, path depends on response (Crowder).

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Why the five principles hang together

Each principle of programmed instruction follows from operant conditioning. Small steps make correct responses likely. Active responding produces the behaviour to be reinforced. Immediate confirmation is the reinforcement. Self-pacing lets each learner respond only when ready. Student testing makes sure that the steps really are small enough for real learners. In an Assertion-Reason item, “programmed instruction allows learners to proceed at their own pace” is true, while “in programmed instruction the pace is fixed by the teacher for the whole class” is false.

Mathetics

A third style of programming, **mathetics**, was developed by Thomas F. Gilbert, who defined it as the systematic application of reinforcement theory to the analysis and construction of complex behaviour. It is best known for **retrogressive (backward) chaining**: a chain of steps is taught backwards, so that the final step is mastered first, then the step before it, and so on. In this way the learner always ends a practice by completing the whole task. (Gilbert used backward chaining where a task is a chain of steps; not every mathetics lesson is taught backwards.) If a matching item asks for “the final step is mastered first, working backwards”, the answer is mathetics. The other self-paced plan often listed with programmed instruction, the Personalised System of Instruction (the Keller Plan), is taught in section 4.4.

When the lecture still wins

Two true statements are often paired in a way that tempts you to link them. “The lecture is suitable for introducing a new topic to a large class” is true because of economy, not because of participation. “The lecture is weak for skills” is true because the learner does not practise, not because the class is large. Read every Reason and ask whether it gives the cause.

CHECK YOURSELF

1. Consider the following statements about the lecture method:

1. It allows one teacher to present organised content to a large class in a short time.
2. It gives the teacher feedback from every learner.
3. It can present recent material not yet available in textbooks.
4. It is the best way to develop laboratory skills.

Choose the correct answer from the options given below:

(A) 1 and 3 only
(B) 1, 2 and 3 only
(C) 2 and 4 only
(D) 1, 3 and 4 only

2. Given below are an Assertion (A) and a Reason (R):

Assertion (A): The lecture method is weak in developing manipulative skills.

Reason (R): Skills develop through practice by the learner, which the lecture does not provide.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

3. A teacher of Physics stops before each step of an experiment and asks the class to write down what they expect to see, then asks them to explain what they saw. The teacher is mainly:

(A) saving time by covering more steps
(B) turning a demonstration into an active experience for the learners
(C) using a branching programme
(D) replacing laboratory practice altogether

4. Match List I with List II:

List I (Principle of programmed instruction)	List II (Description)
A. Small steps	I. The learner knows at once whether the response is right
B. Active responding	II. Each learner moves at his or her own speed
C. Immediate confirmation	III. Content is divided into short frames
D. Self-pacing	IV. The learner must answer at every frame

(A) A-III, B-I, C-IV, D-II
(B) A-II, B-IV, C-I, D-III
(C) A-III, B-IV, C-I, D-II
(D) A-IV, B-III, C-II, D-I

5. Given below are two statements:

Statement I: In Crowder's branching programme, a wrong answer leads the learner to a remedial frame.

Statement II: In Skinner's linear programme, the learner chooses an answer from several options.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

6. In the communication model, the element that a straight lecture most lacks is:
(A) the sender
(B) the channel
(C) the message
(D) feedback from the receivers

7. Which of the following is the best plan for teaching 25 students to use a microscope?
(A) A lecture with slides of the parts of the microscope
(B) A demonstration followed by each student using a microscope with guidance
(C) A demonstration alone, performed carefully by the teacher
(D) A reading assignment from the textbook

8. In a programme, the learner masters the final step of a procedure first and then works backwards through the earlier steps. This style of programming is:
(A) linear programming
(B) branching programming
(C) mathetics
(D) mastery learning

4.3 Group and interactive methods

Syllabus: Methods of teaching in Institutions of higher learning: Teacher centred vs. Learner centred methods

EXAM FOOTPRINT

Asked fairly often; usually a match-the-list of group methods (seminar, symposium, panel, workshop) with their defining features.

Core

Group methods bring learners together so that they talk, question and work with one another and with experts. Most of them are learner-centred or at least interactive. Examiners test them mainly by asking you to tell one format from another, so for each method learn the one feature that decides it.

The tutorial

A **tutorial** (अनुशिक्षण) is a meeting of a tutor with a small group of students, or with one or two students. The tutor clears doubts, checks understanding, discusses written work and gives personal feedback. Its function is mainly **remedial**: it deals with individual and small-group difficulties that a lecture to a large class cannot reach. When a tutor finds that six students share one misconception and gives them exercises aimed at it, the tutorial is serving a remedial function, supported by diagnosis. Nothing is being graded (summative), placed or predicted (prognostic).

Tutorials **supplement** lectures; they do not replace them. A lecture addresses a large group; a tutorial addresses a small group and its particular problems.

Peer tutoring

In **peer tutoring** (सहपाठी शिक्षण) students teach fellow students, usually a more advanced student helping a less advanced one. It benefits the tutor as well as the tutee: to explain something, the tutor has to organise it and understand it more deeply. Do not confuse peer tutoring (students as teachers) with team teaching (several teachers working together).

The seminar

A **seminar** (संगोष्ठी) centres on a prepared paper. A student or scholar presents a paper on a topic, and the group then discusses and questions it in depth, under a **chairperson**. A seminar develops skills of study, presentation, argument and criticism. The usual order of a seminar session is:

1. The chairperson introduces the theme and the speaker.
2. The paper is presented.
3. Participants discuss it and raise questions; the speaker responds. Often a designated **discussant** first comments critically on the paper.
4. The chairperson sums up and closes the session.

Summing up comes last, because the chairperson must take account of the points raised in the discussion. The defining feature of a seminar is one paper (or a few) examined in depth.

The symposium

A **symposium** (परिसंवाद) is a series of short, prepared talks by several speakers, each dealing with a **different aspect** of one theme, one after another. The speakers do not converse with one another. At the end, the chairperson usually invites questions from the audience.

A symposium gives the audience a many-sided view of a topic in a short time. Note the cause: the breadth comes from having several speakers on different aspects. The question session at the end is customary, but it is not what produces the breadth. And questions from the audience are not prohibited; they normally close a symposium.

The panel discussion

In a **panel discussion** (पैनल चर्चा) a small group of experts, usually three to six, sit before an audience and **converse informally among themselves** on a topic, without prepared speeches, guided by a **moderator**. Questions from the audience usually follow.

The moderator introduces the topic and the panel, puts questions to the members, gives each a fair chance to speak, prevents digression and sums up.

Symposium or panel: the deciding difference

Both have several experts and an audience. Ask: do the speakers present or converse?

- Prepared statements on different aspects, delivered one after another, without conversation: **symposium**.
- Spontaneous exchange of views among the experts, under a moderator: **panel discussion**.

The same four specialists can do either. If they set out four positions in turn, it is a symposium; if they challenge one another freely before the students, it is a panel.

The workshop

A **workshop** (कार्यशाला) is a working session in which participants learn by doing. They practise a skill or produce something, such as a question bank, test items, a research proposal or teaching materials, under the guidance of **resource persons**. Talks by experts are short and serve the work. The defining mark of a workshop is practical work by the participants leading to a product.

The conference

A **conference** (सम्मेलन) is a larger gathering on a broad theme. It usually has plenary lectures (to all participants) and many parallel sessions in which papers are presented, and it may run over several days. A conference involves more participants and a broader theme than a seminar, but both include the presentation of papers. A statement that a conference has no place for papers is false.

The debate

A **debate** (वाद-विवाद) has a **motion** (a stated proposition), two opposing sides (for and against), a fixed order of speakers and time limits. It trains students to gather evidence and argue logically. In a panel, members exchange views without taking fixed opposing sides. A **colloquium** is simply an academic meeting at which scholars discuss a subject, often after a talk; it has no motion and no opposing teams.

Who does what

Role	Function
Convener	Organises the event: plans it, invites participants, arranges the venue
Chairperson	Conducts a seminar or symposium session: introduces, keeps time, invites questions, sums up
Moderator	Conducts a panel discussion: puts questions, keeps balance, prevents digression, sums up
Paper presenter / speaker	Presents the prepared paper or talk
Discussant	Comments critically on a paper before the open discussion
Rapporteur	Records the main points of the proceedings and prepares the report
Resource person	Guides the practical work in a workshop
Invigilator	Supervises an examination; has no role in a group discussion

Format	What happens	Deciding feature
Seminar	A paper is presented and discussed in depth under a chairperson	Depth on one paper
Symposium	Several prepared talks on different aspects of one theme, in turn; questions at end	Several speakers, several aspects, no conversation among them
Panel discussion	Experts converse informally before an audience under a moderator	Spontaneous exchange among experts
Workshop	Participants practise and produce under resource persons	Hands-on work and a product
Conference	Large meeting on a broad theme; plenary and parallel paper sessions	Size and breadth
Debate	Two sides argue for and against a motion	Opposing sides
Tutorial	Tutor with a small group or one or two students	Remedial, personal help

Brainstorming

Brainstorming (मस्तिष्क उद्वेलन) is a technique for producing as many ideas as possible on a problem, in a group, without criticism at first. It was introduced by **Alex F. Osborn**, an American advertising

executive, who described it in his book *Applied Imagination*. It encourages creative thinking and participation. It is not a technique for memorising, testing or revising content.

Osborn's rules for the idea-generation stage are:

1. **Defer judgement.** No criticism or evaluation while ideas are being produced.
2. **Aim at quantity.** The more ideas, the more likely a good one is among them.
3. **Welcome unusual ideas.** Wild or odd ideas are encouraged.
4. **Combine and build on others' ideas.**

Evaluation and the sorting of ideas for quality come in a separate, later stage. So the statements "ideas are evaluated as soon as they are offered" and "quantity is discouraged in favour of quality" are both false.

Do not confuse Osborn with others named in options: Edward de Bono is known for lateral thinking and the six thinking hats; J. P. Guilford for the structure of intellect and divergent thinking; Frank Lyman for Think-Pair-Share.

Buzz sessions and group discussion

In a **buzz session** a large class is divided on the spot into small groups of about four to six, which discuss a question for a few minutes and then report back. The name comes from the hum of many conversations at once. It brings participation into large classes.

In a **group discussion** learners do most of the talking and thinking on a topic while the teacher guides. It is a learner-centred method.

Cooperative learning

In **cooperative learning** (सहकारी अधिगम) students work in small, mixed groups towards a **shared goal**. Each member is responsible both for his or her own learning and for the success of the group. It develops academic and social skills together. Its opposites are individual competition for rank and solitary, silent work.

In Elliot Aronson's **jigsaw** technique, a lesson is divided into parts. Each member of a "home group" receives one part. Members holding the same part meet as an "expert group" to master it, then return to their home groups and teach it. Every member depends on the others for the whole lesson; this is called positive interdependence.

Team teaching

In **team teaching** (दल शिक्षण) two or more teachers pool their strengths and share responsibility for **planning, teaching and evaluating** the same group of students. Each topic can be taught by the teacher best qualified for it. A common pattern combines large-group lectures, small-group discussion with different teachers, and independent study, with all the teachers setting and marking the test together.

Team teaching makes use of each teacher's special competence and requires joint planning. It demands far more coordination than individual teaching, which is its main practical difficulty. It is different from a tutorial system, in which tutors only supplement one teacher's lectures, and from peer tutoring, in which students teach students.

CLASSROOM EXAMPLE

In the M.A. English department of a university in Rohtak, a unit on Partition literature is taught by three teachers as a team: one lectures to all 120 students on the history, another leads small groups on the novels, and the third on the poetry; together they set the unit test. In week two, two students present seminar papers, each followed by a discussant's comments and open discussion. Week three ends with a symposium in which four invited scholars speak in turn on memory, gender, language and migration, and the audience questions them. To start a project, the class brainstorms possible topics for twenty minutes, writing every idea on the board before any is judged.

OFTEN CONFUSED

- **Seminar and symposium.** Seminar: one paper discussed in depth. Symposium: several speakers, each on a different aspect, in turn.
- **Symposium and panel.** Symposium speakers present prepared talks without conversing; panel members converse informally under a moderator.
- **Seminar and workshop.** Seminar: paper and discussion. Workshop: practical work and a product.
- **Seminar and conference.** Conference is larger and broader, with plenary and parallel sessions; both include papers.
- **Chairperson, moderator, rapporteur, convener.** Chairperson conducts a seminar or symposium; moderator conducts a panel; rapporteur records and reports; convener organises.
- **Tutorial and lecture.** Tutorial: small group, remedial, supplements; lecture: large group, presentation.
- **Team teaching and peer tutoring.** Team teaching: several teachers share one course; peer tutoring: students teach students.
- **Brainstorming and discussion.** Brainstorming defers all judgement while ideas are produced; ordinary discussion evaluates as it goes.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Kinds of tutorial

Tutorials are often classified into three kinds:

- **Group tutorial:** a tutor meets a small group to clear common difficulties after lectures.
- **Supervision tutorial:** a tutor works with one or two students on their own written work, usually an essay, as in the tradition of Oxford and Cambridge.
- **Practical tutorial:** skills are practised in a laboratory or workshop under a tutor.

Forum and fishbowl

A **forum** is an open discussion in which the whole audience may ask questions and make comments from the floor, under a chairperson, usually after a lecture, film, panel or symposium. In a panel or symposium the main talking is done by a few people on the platform; in a forum the audience talks.

In a **fishbowl**, a small inner circle of students discusses an issue while the rest sit in an outer circle and observe both the content and the process. Observers comment afterwards, and the two circles may change places. A buzz group, in contrast, is a brief discussion in many small groups at once; in a round robin each member of a group speaks in turn.

Brainwriting

Brainwriting is a variant of brainstorming in which ideas are first written silently, on cards or sheets, before any talk. Quiet members then contribute as much as vocal ones, and nobody waits for a turn. Deferred judgement is kept. The written ideas are then pooled, grouped and discussed.

More cooperative learning techniques

- **Think-Pair-Share**, devised by **Frank Lyman**: each student thinks alone about a question, then discusses it with a partner, then pairs share with the class. It gives every student a chance to take part, even in a large class.
- **STAD (Student Teams-Achievement Divisions)**, developed by **Robert Slavin**: the teacher presents the lesson; students study in mixed teams; each takes an individual quiz; team scores are based on **improvement points**, that is, how far each member improves on his or her own past average. Weak students can contribute as much as strong ones.
- **Group Investigation**, developed by Shlomo and Yael Sharan: groups choose sub-topics, plan and carry out an inquiry, and present their findings.
- **Jigsaw** (Aronson), described in the Core.

What makes cooperation work

David W. Johnson and Roger T. Johnson list five essentials of cooperative learning:

1. **Positive interdependence:** each member succeeds only if the others succeed.
2. **Individual accountability:** each member's contribution and learning is assessed, so nobody can leave the work to others (the problem called free riding).
3. **Face-to-face (promotive) interaction:** members help and encourage one another directly.
4. **Social skills:** leadership, communication, trust and handling conflict are taught and used.
5. **Group processing:** the group reflects on how well it is working and how to improve.

The Johnsons also distinguish three goal structures:

- **Positive interdependence (cooperation):** I reach my goal only if the others reach theirs.
- **Negative interdependence (competition):** I reach my goal only if others fail to reach theirs. A rule that only the top five may get grade A creates negative interdependence.
- **No interdependence (individualistic work):** my result is unaffected by others.

From his review of research, Robert Slavin concluded that group work raises achievement most reliably when two conditions are present: **group goals** (or group rewards) and **individual accountability**, so that the team's success depends on the learning of every member. If only the group product is assessed, some members do the work and others ride free.

The Trump Plan

Team teaching is associated with **J. Lloyd Trump**, who proposed reorganising the secondary-school timetable around **large-group instruction** by the most competent teacher, **small-group discussion**, and **independent study**. Reports of the plan from the time give only about 20 per cent of students' time to small-group discussion, with the larger shares going to large-group instruction and independent study. The split usually taught for the examination is roughly 40 per cent large-group, 20 per cent small-group and 40 per cent independent study; of these figures only the 20 per cent for small groups could be confirmed in a source of the period. Teachers work as a team and are deployed according to their strengths.

CHECK YOURSELF

1. Match List I with List II:

List I (Format)	List II (Deciding feature)
A. Seminar	I. Experts converse freely under a moderator
B. Symposium	II. Participants produce something under resource persons
C. Panel discussion	III. Several prepared talks on different aspects, in turn
D. Workshop	IV. One paper examined in depth

(A) A-IV, B-III, C-I, D-II
(B) A-III, B-IV, C-I, D-II
(C) A-IV, B-I, C-III, D-II
(D) A-II, B-III, C-I, D-IV

2. Arrange the stages of a seminar session in the order in which they normally occur:

1. Open discussion by participants
2. Chairperson's concluding remarks
3. Introduction of the speaker by the chairperson
4. Presentation of the paper

(A) 3, 1, 4, 2
(B) 4, 3, 1, 2
(C) 3, 4, 2, 1
(D) 3, 4, 1, 2

3. Given below are two statements about brainstorming:

Statement I: Members are asked to produce as many ideas as they can.

Statement II: Each idea is judged for quality before the next idea is taken.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

4. Given below are an Assertion (A) and a Reason (R):

Assertion (A): Team teaching makes better use of the special abilities of individual teachers.

Reason (R): Team teaching requires less coordination among teachers than individual teaching.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

5. In a teacher-education programme, the person who listens through all sessions and then presents a summary report of the proceedings is the:

(A) convener
(B) rapporteur
(C) moderator

(D) discussant

6. Consider the following statements about cooperative learning:

1. Each member is responsible for his or her own learning and for the group's success.
2. Individual accountability prevents some members from leaving the work to others.
3. A rule allowing only the top three students to receive grade A promotes cooperation.
4. In the jigsaw technique, each member teaches the home group the part he or she has mastered.

Choose the correct answer from the options given below:

(A) 1, 2 and 4 only
(B) 1 and 3 only
(C) 2, 3 and 4 only
(D) 1, 2, 3 and 4

7. A teacher with 140 students asks a question, gives each student one minute to think alone, then two minutes to talk with a neighbour, and then calls on several pairs to report. The technique is:

(A) fishbowl
(B) debate
(C) Think-Pair-Share
(D) supervision tutorial

8. In the plan associated with J. Lloyd Trump, the smallest share of students' time goes to:

(A) small-group discussion
(B) large-group instruction
(C) independent study
(D) written examinations

4.4 Learner-centred and activity methods

Syllabus: Methods of teaching in Institutions of higher learning: Teacher centred vs. Learner centred methods

EXAM FOOTPRINT

Asked occasionally; usually a match of methods (project, Dalton, Keller plan) with originators or a single-answer item on the project method.

Core

The methods in this section put the learner to work: finding out, solving, making, deciding, acting. Most questions ask who devised a method, what its steps are in order, or how two similar methods differ.

The project method

The **project method** (परियोजना विधि) was set out by **William Heard Kilpatrick** in an essay of 1918. Kilpatrick was a follower of **John Dewey**, and the method rests on Dewey's pragmatism and his idea of **learning by doing**. Kilpatrick defined a project as "a wholehearted purposeful activity proceeding in a social environment".

In the project method students carry out a purposeful, real-life task from start to finish: they choose it, plan it, do it and judge the result. It replaces rote learning and passive listening with the learners' own activity. A project normally ends in a tangible product: a report, a survey, a model, an exhibition.

Kilpatrick's four steps, in order, are:

1. **Purposing:** the students choose a project that has a real purpose for them.
2. **Planning:** they decide how to carry it out.
3. **Executing:** they do the work.
4. **Evaluating** (judging): they review the result against the purpose with which they started.

A project draws on several subjects at once. A project on village sanitation, for example, uses statistics, sociology, health science and report writing together. So the project method **promotes correlation** among subjects. This is possible because it does not follow a timetable of separate subject periods.

Its limitations: projects take much time, and the knowledge gained is **incidental** and not in systematic order. So the method is hard to use as the only method for covering a prescribed syllabus. (That Kilpatrick propounded it is true, but it does not explain this difficulty.)

Do not confuse Kilpatrick with others: **Maria Montessori** and **Friedrich Froebel** devised methods for early childhood education, and **Helen Parkhurst** devised the **Dalton Plan** (डाल्टन योजना), in which each pupil works through monthly assignments, or "contracts", at his or her own pace in subject rooms called laboratories.

The problem-solving method

In the **problem-solving method** (समस्या समाधान विधि) learners meet a real problem and work towards a solution by the steps of scientific inquiry:

1. Recognising and defining the problem.
2. Collecting relevant data.
3. Forming hypotheses (possible solutions).
4. Testing the hypotheses.
5. Drawing a conclusion.

The first step is always to identify the problem: nothing can be collected or tested before that. The method develops reasoning and a scientific attitude; it makes little use of rote memory or mechanical drill.

The method is based on John Dewey's analysis of reflective thinking in his book *How We Think*. Dewey's five steps are:

1. A felt difficulty.
2. Its location and definition.
3. Suggestion of possible solutions.
4. Reasoning out the bearings (implications) of each suggestion.
5. Further observation or experiment, leading to acceptance or rejection of the suggestion.

Problem-solving and project compared. Both involve purposeful activity by learners. But problem-solving is chiefly mental and ends in a solution or conclusion; a project is carried out in a real-life setting and ends in a tangible product. They are not the same in every respect.

Problem-based learning

Problem-based learning (PBL) (समस्या आधारित अधिगम) is a modern form in which small groups of students learn through working on complex, real problems. Its marks are:

- the problem comes **first**, before the relevant content is taught, and drives the learning;
- the problems are **ill-structured**: open, with incomplete information and more than one possible answer;
- the tutor acts as a **facilitator** of the group's process, not as a lecturer.

A physical product is not required; that is a mark of the project method.

The heuristic method

The **heuristic method** (अन्वेषण विधि) was advocated by **H. E. Armstrong** (Henry Edward Armstrong), a professor of chemistry in England, for the teaching of science. The word comes from the Greek for "I find out". The pupil is placed in the position of a **discoverer** and finds facts and principles by his or her own observation and experiment. The teacher sets problems and guides but tells as little as possible. A statement that "in the heuristic method the teacher supplies all the facts and conclusions" is false: that is the very practice the method opposes.

Inductive, deductive and heuristic methods

Method	Direction	When it suits
Inductive	From particular examples to a general rule	Discovery and understanding; slower, because many examples are needed
Deductive	From a general rule to particular cases	Application, practice and revision of an established rule; quicker
Heuristic	The learner finds out by own inquiry	Developing scientific attitude; slow
Analytic	From the unknown (what is to be proved) back to the known	Discovering how to prove or solve
Synthetic	From the known to the unknown	Setting out a proof neatly

If a teacher states the formula for the area of a triangle and then has students solve problems with it, the method is deductive. If students measure many triangles and arrive at the formula themselves, it is inductive. Induction takes more time; deduction is suitable for practice and revision.

There is also a point of logic. Induction goes from a limited number of instances to a general rule, so its conclusion claims more than the evidence contains and is only **probable**: a new case may always contradict it. Deduction applies a general premise to a particular case; if the premises are true and the form is valid, the conclusion is **certain**. "All metals expand on heating; copper is a metal; therefore copper expands on heating" is deduction. Two other forms appear as distractors: **abduction** infers the best explanation of what is observed, and **analogy** argues from resemblance.

The case method

In the **case method** (व्यष्टि अध्ययन विधि) students study a **case**: a written account of a real or realistic problem situation. They study the facts, identify the issues, discuss alternatives and defend a decision. The method links theory with practice and develops analysis, judgement and decision-making.

The case method was developed at **Harvard**: first in the Law School, and then, from the 1920s, at **Harvard Business School**, which made it famous for management education. It is now used in management, law, medicine and teacher education.

Case or project? If students analyse a case already written up for them, it is the case method. If they go out, collect data and produce a report on a real task, it is a project.

Role play

In **role play** (भूमिका निर्वाह) learners take the parts of people in a real or imagined situation and act it out **without a script**. Players are given a situation and a role and improvise. A memorised script would turn it into a drama (dramatisation).

By acting as another person, a learner experiences that person's situation and feelings. Role play is therefore used for empathy, attitudes, communication and interpersonal skills, which belong to the affective domain, not for recall or calculation.

The enactment only provides experience. Learning comes in the **debriefing**: players and observers discuss what happened, how the characters felt and what else could have been done, and link this to the objectives of the lesson. Debriefing is the most important step.

Simulation

In **simulation** (अनुरूपण) learners practise a task in an artificial situation made to resemble the real one: a mock court (moot court) for law students, a flight simulator for pilots, a mock parliament, or simulated teaching among fellow trainees. Mistakes cost nothing, so learners can try, fail and get feedback before facing the real thing. Simulation is used above all for skills, decisions and attitudes that are costly or risky to practise in reality, such as flying, surgery or classroom teaching. Like role play, it must be followed by **debriefing**, where the experience is analysed and turned into learning. Working in a real workplace under a supervisor is an internship or apprenticeship, not a simulation.

Mastery learning

Mastery learning (निपुणता अधिगम) was developed by **Benjamin S. Bloom**. Its idea is that nearly all learners can master what is taught if they are given enough time and the right help. The course is divided into units, and each unit has a fixed standard of mastery, for example 80 per cent, that every learner is expected to reach before moving on.

After each unit, a short **formative test** shows each learner which objectives have and have not been mastered. These tests are not graded. Learners who fall short receive **correctives** (re-teaching, different materials, peer help) and then take a second, parallel test. Grades are given only by a later **summative test** (योगात्मक मूल्यांकन). Because performance is judged against a fixed standard, not against other learners, mastery learning relies on **criterion-referenced tests**, not norm-referenced ranking.

The Personalised System of Instruction (Keller Plan)

The **Personalised System of Instruction** (PSI), also called the **Keller Plan** (वैयक्तिकृत अनुदेशन प्रणाली), was developed by the American psychologist **Fred S. Keller** and colleagues in the 1960s and described in his paper "Good-bye, Teacher…" (1968). It applies behavioural principles to a whole university course. Its features are:

1. **Self-pacing:** each student moves through the course at his or her own speed.
2. **Unit mastery:** the course is divided into units; a student must pass a test on one unit before going to the next. A student who fails simply studies again and retakes the test, with no penalty.
3. **Proctors:** the unit tests are marked at once, and discussed with the student, by **proctors**, usually advanced students who have already done the course. This gives immediate feedback and a personal, social element.
4. **Written materials:** the main content is in written study guides and readings, not in lectures.
5. **Lectures for motivation only:** the few lectures and demonstrations are meant to inspire, not to convey essential content, and attendance is optional.

Keep the names apart: Skinner is linked with linear programming, Crowder with branching programming, Bloom with mastery learning, and Keller with PSI. PSI and mastery learning share the

idea of unit mastery, but in PSI each student goes at his or her own pace, while in Bloom's mastery learning the class usually moves together with group teaching, and correctives are given to those who need them.

CLASSROOM EXAMPLE

In a B.Ed. college in Bhiwani, trainees run a project: they survey the number of out-of-school girls in two villages (purposing and planning with the teacher), collect the data (executing), present a report to the gram sabha and judge it against their aim (evaluating). In a management class at a university in Gurugram, students analyse a written case on a cooperative milk plant and defend a decision. In a law college, students argue a moot court, a simulation. In a sociology class, students role-play a meeting between a family and a panchayat member about a dowry dispute, and then spend twenty minutes in debriefing.

OFTEN CONFUSED

- **Project and case method.** Project: students carry out a real task in the field and produce something. Case: students analyse a situation already written up.
- **Project and problem-solving.** Both purposeful. Problem-solving is mainly mental and ends in a solution; a project is a real-life activity ending in a product.
- **Project method and PBL.** PBL presents an ill-structured problem first, with a facilitator; a product is not required. A product is the mark of the project method.
- **Role play and simulation.** Role play: acting a person's part, chiefly for empathy and attitudes. Simulation: practising a task in a situation built to resemble reality, chiefly for skills and decisions. Both need debriefing.
- **Role play and dramatisation.** Role play is unscripted; dramatisation uses a learnt script.
- **Heuristic and inductive.** Both move towards discovery, but heuristic means the learner finds things out by own experiment with minimal telling; inductive names the direction of reasoning, from examples to rule.
- **Kilpatrick, Parkhurst, Armstrong, Montessori.** Project method; Dalton Plan; heuristic method; early childhood method.
- **Mastery learning and PSI.** Bloom: class taught together, formative tests and correctives. Keller: each student self-paced through written units with proctors.
- **Criterion-referenced and norm-referenced.** Mastery is judged against a fixed standard (criterion), not against other learners (norm).

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Definitions of a project

- **Kilpatrick:** "a wholehearted purposeful activity proceeding in a social environment".
- **J. A. Stevenson**, as quoted in Indian teacher-education material: "a problematic act carried to completion in its natural setting" (some versions read "its most natural setting"). Note the three ideas: a problem, completion, a natural setting.

Kilpatrick's four types of project

Kilpatrick grouped projects by purpose: to make or do something, to enjoy something aesthetic, to solve a problem, and to acquire a skill or item of knowledge. The labels usually used for the examination are:

1. **Producer (constructive) project:** making something, such as a model, a garden or a magazine.
2. **Consumer (aesthetic) project:** enjoying and appreciating something, such as a poem, a piece of music or a painting.
3. **Problem project:** solving an intellectual question.
4. **Drill (specific learning) project:** mastering a skill or a body of knowledge.

Polya's four steps

The mathematician George Polya, in *How to Solve It*, gave four steps of problem solving: **understand the problem**, **devise a plan**, **carry out the plan**, and **look back** (check the result and reflect on the method). Looking back is where the solver learns something to use in future problems.

Where problem-based learning began

Problem-based learning was introduced in the medical school of **McMaster University**, Canada, in the late 1960s, by **Howard Barrows** and colleagues. Students first met a patient problem and then learnt the science they needed in order to deal with it. Keep apart: the case method with Harvard, PBL with McMaster, and micro-teaching with Stanford.

Variants of the case method

- In the **incident method**, devised by **Paul Pigors**, the group receives only a short statement of an incident and must dig out the remaining facts by questioning the teacher, then define the issue and decide. It trains fact-finding as well as analysis. In an ordinary case study the facts are supplied in full.
- In the **in-basket** (in-tray) exercise, a simulation, each trainee is given a pile of letters, memos and requests such as might lie on a manager's or principal's desk, and must deal with them under time pressure. The decisions are discussed afterwards.

Psychodrama and sociodrama

J. L. Moreno (Jacob Levy Moreno), the founder of psychodrama, distinguished two forms of enactment. **Psychodrama** deals with the private problems of an individual and is used in therapy. **Sociodrama** deals with problems shared by a group or community, such as caste prejudice or dowry. Classroom role play on social issues is sociodrama. In **role reversal** two persons exchange roles, so that each sees the situation from the other's side.

Mathetics and PSI side by side

In a matching item on individualised instruction, remember four lines: Skinner's linear programming, every learner passes through the same small steps; Crowder's branching programming, the learner's response decides the next frame; Gilbert's mathetics, the final step is mastered first and the learner works backwards; Keller's PSI, self-paced units, each passed with a proctor. (Programmed instruction and mathetics are explained in section 4.2.)

Action research

A small project carried out to improve one's own practice often follows the cycle of **action research**, associated with **Kurt Lewin**: **plan, act, observe, reflect**, and then plan again, in a spiral. Its aim is to improve practice in one's own setting, not to produce general laws. Ex post facto research, by contrast, studies effects after the event without any intervention; an experiment manipulates a

variable with a control group and aims at generalisation. Action research is taught more fully under Research Aptitude (Unit II).

CHECK YOURSELF

1. Arrange Kilpatrick's steps of the project method in their correct order:

1. Planning
2. Evaluating
3. Purposing
4. Executing

(A) 3, 1, 2, 4
(B) 1, 3, 4, 2
(C) 3, 1, 4, 2
(D) 3, 4, 1, 2

2. Match List I with List II:

List I (Person)	List II (Associated with)
A. Helen Parkhurst	I. Heuristic method
B. H. E. Armstrong	II. Personalised System of Instruction
C. Fred S. Keller	III. Dalton Plan
D. Benjamin S. Bloom	IV. Mastery learning

(A) A-III, B-I, C-II, D-IV
(B) A-I, B-III, C-II, D-IV
(C) A-III, B-I, C-IV, D-II
(D) A-III, B-II, C-I, D-IV

3. Given below are an Assertion (A) and a Reason (R):

Assertion (A): The project method helps learners see connections among different subjects.

Reason (R): A real-life project draws on knowledge from several subjects at the same time.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. Given below are two statements:

Statement I: In role play, the players learn a written script by heart before the class.

Statement II: Discussion after the enactment is where most of the learning in role play takes place.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

5. Consider the following statements about mastery learning:

1. Formative tests after each unit are used to award final grades.
2. Learners who do not reach the standard receive corrective help.
3. Performance is judged against a fixed standard.
4. It was developed by Benjamin S. Bloom.

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 2, 3 and 4 only
(C) 1 and 4 only
(D) 1, 2, 3 and 4

6. A teacher gives students four sentences in the passive voice, asks them to find what the sentences have in common, and lets them state the rule for forming the passive. The method is:

(A) deductive
(B) inductive
(C) synthetic
(D) lecture-cum-demonstration

7. Which statement about problem-based learning is NOT correct?

(A) The problem is met before the relevant content is studied.
(B) The tutor acts mainly as a facilitator.
(C) The problems are usually open and ill-structured.
(D) Every group must end with a physical product.

8. A law college arranges a mock trial in which students act as judge, lawyers and witnesses, after which the teacher and students analyse the decisions taken. This is best described as:

(A) simulation followed by debriefing
(B) a symposium
(C) programmed instruction
(D) psychodrama

4.5 Off-line and on-line teaching; blended learning and the flipped classroom

Syllabus: Methods of teaching in Institutions of higher learning: Off-line vs. On-line methods

EXAM FOOTPRINT

Asked occasionally; usually a two-statement or single-answer item defining blended or flipped learning.

Core

Off-line and on-line teaching

Off-line teaching (ऑफ़लाइन शिक्षण) is the ordinary face-to-face teaching of a classroom, laboratory or tutorial, where teacher and learners are physically together. **On-line teaching** (ऑनलाइन शिक्षण) uses the internet to connect teacher and learners who may be far apart. On-line teaching can be synchronous or asynchronous, and it can be mixed with face-to-face teaching in blended learning.

Point	Off-line (face-to-face)	On-line
Place	Same room	Anywhere with a connection
Time	Fixed timetable	Live (synchronous) or at the learner's convenience (asynchronous)

Point	Off-line (face-to-face)	On-line
Interaction	Immediate, with gestures and social presence	Through video, chat, forums, e-mail
Practical work	Direct, in laboratory or field	Limited; virtual labs and simulations
Needs	Campus, teachers, timetable	Devices, internet, digital skills, self-discipline
Strength	Social presence, immediate feedback, practical work	Flexibility of time, place and pace; reach to large numbers

Synchronous and asynchronous learning

Synchronous learning (समकालिक अधिगम) happens at the same time for everyone: teacher and learners are online together and interact in real time. Examples: a live video class, a webinar, instant chat.

Asynchronous learning (असमकालिक अधिगम) does not need everyone at the same time. Each learner takes part at a convenient time. Examples: recorded lectures, discussion boards or forums, e-mail, assignments, and a quiz that stays open over a day or a week. Asynchronous learning gives time for reflection before replying.

The test is simple: must everyone be present at the same moment? If yes, synchronous; if no, asynchronous. A quiz that may be taken at any time on Saturday is asynchronous, because learners take it at different times.

Synchronous (same time)	Asynchronous (different times)
Live video class	Recorded lecture
Webinar	Discussion forum or board
Instant chat	E-mail
Live phone or video tutorial	Assignment or quiz open over a period

A **threaded discussion forum** comes closest to a seminar without requiring everyone at once: a paper or prompt is posted, and participants reply to it and to one another over several days. A webinar allows discussion only live. A podcast and a broadcast e-mail are one-way.

Blended learning

Blended learning (मिश्रित अधिगम), also called **hybrid learning**, is a planned mix of face-to-face classroom teaching and online learning within one course, so that each mode does what it does best. It keeps the social presence, immediate feedback and practical work of the classroom and adds the flexibility, variety and self-pacing of online study. Because online activities can be done at any hour, outside the timetable, blended learning also extends learning time beyond the classroom.

Blended learning still needs a campus, devices and, above all, careful planning of what is done in each mode. It does not remove the classroom, and it does not reduce planning.

Distinguish it from three other terms that appear as options:

- **Distance learning** separates teacher and learner for most of the course.
- **Programmed learning** uses self-instructional frames (section 4.2).
- **Mastery learning** requires each unit to be mastered before the next (section 4.4).

The flipped classroom

In the **flipped classroom** (फ़्लिप्ड कक्षा) the usual order is reversed. Students meet new content **before class**, at home, usually through short recorded videos or readings. **Class time** is then used for discussion, clearing doubts, problem-solving and application, with the teacher and classmates at hand. The lecture has not disappeared; it has moved out of the classroom in recorded form.

In terms of Bloom's taxonomy, the flipped classroom sends the lower-order work (remembering and understanding, the first exposure) to home, and keeps the higher-order work (applying, analysing, evaluating and creating) for class, where help is available. That is why a flipped classroom frees class time for higher-order activities. A statement that "in a flipped classroom new content is first presented through the teacher's lecture in class" is false: it describes the conventional classroom.

Flipped and blended. Blended learning is the general idea of combining face-to-face and online learning. The flipped classroom is one model of blended learning. So every flipped classroom is blended, but not every blended course is flipped. Other blended arrangements, such as station rotation or the flex model, are not flipped.

MOOC types in brief

Two kinds of MOOC (massive open online course) are often named in questions on online learning. **cMOOCs** are connectivist: learners build and share knowledge across networks. **xMOOCs** are structured, instructor-led courses of video lectures and quizzes on large platforms. Section 4.6 explains both.

CLASSROOM EXAMPLE

In a B.Com. class of 70 at a college in Fatehabad, the teacher uploads a twelve-minute video on depreciation methods and a short reading on the college's learning portal on Sunday. Students watch it when convenient (asynchronous) and post questions on the course forum. In class on Tuesday, after a three-question entry quiz, groups solve depreciation problems on real company accounts while the teacher moves among them (flipped classroom). On Thursday evening she holds a live online doubt-clearing session (synchronous). The whole course, mixing class and online work, is blended learning.

OFTEN CONFUSED

- **Synchronous and asynchronous.** Same time (live class, webinar, chat) against different times (recorded lecture, forum, e-mail).
- **Blended and flipped.** Blended is the general mix of face-to-face and online; flipped is one blended model in which first exposure to content is online and class is for application.
- **Blended and distance learning.** Blended keeps regular face-to-face classes; distance learning separates teacher and learner for most of the course.
- **Flipped and conventional classroom.** Flipped: content at home, application in class. Conventional: content in class (lecture), application at home (homework).
- **Hybrid and blended.** Two names for the same thing.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Where the flipped classroom came from

The flipped classroom was popularised by **Jonathan Bergmann and Aaron Sams**, two chemistry teachers at a high school in Colorado, United States, who began recording their lectures for students in 2007 and later described the approach in their book *Flip Your Classroom*. A similar "inverted classroom" had already been described by the economists Lage, Platt and Treglia in 2000. Keep apart the pairs often offered as distractors: Hase and Kenyon (heutagogy), Siemens and Downes (connectivism), Joyce and Weil (models of teaching).

When students come unprepared

A common difficulty in a flipped course is that some students arrive without having watched the video. If the teacher re-lectures the whole class each time, the incentive to prepare disappears and the class time is lost. The response most consistent with the model is to use a brief entry quiz or task that holds students accountable, and to go ahead with the planned activity, while the unprepared catch up.

Models of blended learning

The Clayton Christensen Institute describes four main models of blended learning:

- **Rotation model:** students rotate, on a fixed schedule or at the teacher's direction, between online learning and other modes such as small-group teaching or pen-and-paper work. Station rotation, lab rotation, individual rotation and the flipped classroom are sub-types of the rotation model.
- **Flex model:** online learning is the backbone of the course, and teachers give face-to-face support flexibly as needed.
- **A la carte (self-blend) model:** a student takes one or more courses entirely online, in addition to the usual courses on campus.
- **Enriched virtual model:** students do most of their work remotely online and attend required face-to-face sessions from time to time.

Transactional distance

In distance education (दूरस्थ शिक्षा), **Michael G. Moore** proposed the theory of **transactional distance**. The distance that matters, he argued, is psychological and communicative, not geographical. It depends on three variables: **dialogue** (interaction between teacher and learner), **structure** (how rigidly the course is laid out) and **learner autonomy**. Transactional distance is greatest when structure is high and dialogue is low; such a course demands the most autonomy from the learner. More dialogue reduces the distance.

CHECK YOURSELF

1. Consider the following:

1. A live webinar with questions in real time
2. A discussion board on which students post over a week
3. A recorded lecture watched at night
4. Instant chat during an online class

Choose the correct answer from the options given below. The synchronous forms are:

(A) 1 and 2 only
(B) 1 and 4 only
(C) 2 and 3 only
(D) 1, 3 and 4 only

2. Given below are an Assertion (A) and a Reason (R):

Assertion (A): In a flipped classroom, class time is available for problem-solving and discussion.

Reason (R): In a flipped classroom, learners meet new content through videos or readings before class.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

3. Given below are two statements:

Statement I: Every blended course is a flipped classroom.

Statement II: Blended learning removes the need for any face-to-face teaching.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

4. Match List I with List II:

List I (Model)	List II (Description)
A. Rotation	I. Online learning is the backbone; teachers support as needed
B. Flex	II. Students move on a schedule between online and other modes
C. A la carte	III. Mostly remote online work with some required campus sessions
D. Enriched virtual	IV. A student adds a fully online course to campus courses

(A) A-I, B-II, C-IV, D-III
(B) A-II, B-I, C-III, D-IV
(C) A-II, B-I, C-IV, D-III
(D) A-II, B-IV, C-I, D-III

5. In terms of Bloom's taxonomy, the flipped classroom keeps for class time mainly:

(A) remembering facts from the textbook
(B) copying notes from the board
(C) listening to the full lecture
(D) applying, analysing and evaluating

6. According to Moore, transactional distance is least when a course has:
(A) high structure and low dialogue
(B) low dialogue and low autonomy
(C) much dialogue between teacher and learner
(D) a large geographical distance

7. A teacher wants her M.A. students to argue about a critical essay at length over a week, though they live in different districts and have different working hours. The most suitable tool is:
(A) a live webinar
(B) a threaded discussion forum
(C) a podcast
(D) a broadcast e-mail

4.6 MOOCs, SWAYAM, SWAYAM PRABHA and other national digital initiatives

Syllabus: Off-line vs. On-line methods (Swayam, Swayamprabha, MOOCs etc.)

EXAM FOOTPRINT

Asked in most cycles; usually a single-answer item on what a platform such as SWAYAM PRABHA offers, or a match of platforms with coordinating bodies.

Core

This section is mostly facts: names, full forms, who runs what, and a few figures. Learn the full forms exactly, and keep the platforms apart by what each one does. Figures that change are collected in the Fact watch box.

What a MOOC is

A **MOOC** (massive open online course) (विशाल मुक्त ऑनलाइन पाठ्यक्रम) is a course offered through the internet to anyone who wishes to join. Take the name word by word:

- **Massive:** there is no fixed limit on enrolment; thousands of learners anywhere can join the same course. It does not mean a long course, a large fee or large video files.
- **Open:** anyone may enrol, without entry qualification and without being a student of a particular institution or place. Study is usually free; a fee is normally charged only for a certificate.
- **Online:** the course is delivered through the internet.
- **Course:** it is a real course, with instructors, a schedule (usually a few weeks), study material and assessments. “Open” does not mean that there are no instructors, no deadlines or no tests.

MOOCs are video-centred: short recorded lectures are the core, with readings, quizzes and forums around them.

Their best-known weakness is a **low completion rate**. Because enrolment is open and free, very large numbers register, many out of curiosity, and only a small proportion finish. Completion rates are commonly reported in single figures or low double figures (in per cent). Lack of personal contact, the self-discipline needed, and weak incentives are the usual reasons. Openness itself is one cause of low completion.

SWAYAM

SWAYAM stands for **Study Webs of Active-Learning for Young Aspiring Minds**. It is India's national platform for MOOCs, set up by the **Ministry of Education** (then called the Ministry of Human Resource Development, MHRD). It was launched on 9 July 2017 by the President of India at a National Convention on Digital Initiatives, together with the SWAYAM PRABHA channels and the National Academic Depository.

The main facts about SWAYAM:

- It is designed around three principles of education policy: **access, equity and quality**.
- Courses range from school level through undergraduate and postgraduate levels to skill and teacher-training courses.
- Anyone may enrol, of any age, anywhere. There is no age bar, and it is not restricted to IIT students or to any group.
- Learning is **free**. A learner who wants a certificate registers for a **proctored examination** (a supervised examination held at a centre, पर्यवेक्षित परीक्षा), for which a fee is charged, and must pass it.
- Credits earned can be transferred (credit transfer, क्रेडिट हस्तांतरण) to the learner's own university under UGC regulations (see below).
- SWAYAM is one of the components of the Ministry of Education's **PM e-Vidya** programme for multi-mode access to digital education, along with DIKSHA, SWAYAM PRABHA, radio and podcasts.

The four quadrants of a SWAYAM course

Every SWAYAM course is built in four parts, called the **four quadrants**:

1. **e-Tutorial:** video lectures, with animations, simulations and demonstrations where useful.
2. **e-Content:** reading material that can be downloaded or printed: notes, e-books, case studies.
3. **Discussion forum:** an online forum where learners raise doubts and the course team and other learners answer them.
4. **Assessment (self-assessment):** quizzes, assignments and tests.

Printed correspondence lessons sent by post, residential workshops, weekly face-to-face contact classes and compulsory laboratory attendance are not quadrants; they belong to older distance education or to campus teaching. Books differ in the order in which they number the third and fourth quadrants; learn the four parts, not the numbers.

National Coordinators of SWAYAM

SWAYAM courses are prepared through **National Coordinators** (राष्ट्रीय समन्वयक), each responsible for one sector. The Ministry of Education now lists ten:

National Coordinator	Sector
NPTEL (National Programme on Technology Enhanced Learning)	Engineering
UGC (University Grants Commission)	Non-technical postgraduate education

National Coordinator	Sector
CEC (Consortium for Educational Communication)	Non-technical undergraduate education
AICTE (All India Council for Technical Education)	Self-paced and international courses
IGNOU (Indira Gandhi National Open University)	Out-of-school students; diploma and certificate courses
IIM Bangalore (IIMB)	Management studies
NITTTR (National Institute of Technical Teachers Training and Research)	Teacher training
NCERT (National Council of Educational Research and Training)	School education
NIOS (National Institute of Open Schooling)	School education (open schooling)
INI (Institutions of National Importance), coordinated by Banaras Hindu University	Courses from Institutions of National Importance, other than technology

For engineering, the answer is NPTEL; for undergraduate non-technical courses, CEC; for postgraduate non-technical courses, UGC.

SWAYAM PRABHA

SWAYAM PRABHA is a group of **DTH (direct-to-home) television channels** that telecast curriculum-based educational programmes **24 hours a day, 7 days a week**. The signal is carried by the GSAT-15 satellite and uplinked from BISAG-N, Gandhinagar. The channels are free on **DD Free Dish**. New content of at least four hours is shown every day and repeated five more times, so learners can choose a convenient time. Content comes from NPTEL, the IITs, UGC, CEC, IGNOU, NCERT and NIOS, among others.

SWAYAM PRABHA exists for learners who have a television and a dish but **no internet**. It needs no internet connection at the learner's end.

In terms of communication, a television broadcast reaches a large, scattered audience through a mass medium, and viewers cannot reply through the same channel. SWAYAM PRABHA is therefore **one-way mass communication** (एकतरफ़ा जनसंचार). It has no quizzes, no examinations and no credits of its own; interaction comes through its web portal or through SWAYAM courses.

Point	SWAYAM	SWAYAM PRABHA
What it is	Web platform for MOOCs	Group of DTH television channels
Needs internet?	Yes	No (dish and television)
Interaction	Two-way: forums, quizzes, assignments	One-way broadcast
Assessment and credits	Proctored examination, certificate, credit transfer	None of its own
Cost to learner	Free; fee only for certificate examination	Free on DD Free Dish

Credit transfer: the UGC regulations

The UGC has made regulations that let a student's own university count SWAYAM courses towards a degree. Under the **UGC (Credit Framework for Online Learning Courses through SWAYAM) Regulations, 2021**, an institution may allow up to **40 per cent** of the total courses offered in a particular programme in a semester to be taken through SWAYAM, and must give credit for them. The earlier Regulations of 2016 had allowed up to 20 per cent. (See the Fact watch box.) Credit transfer is thus a consequence of this regulatory permission.

NPTEL

NPTEL (National Programme on Technology Enhanced Learning) was started in **2003** by seven Indian Institutes of Technology (Bombay, Delhi, Kanpur, Kharagpur, Madras, Guwahati and Roorkee) and the Indian Institute of Science, Bangalore, with funding from the Ministry of Education (then MHRD). It offers free online courses, chiefly in engineering and science but also in management and the humanities, with optional proctored examinations for certificates. It is the SWAYAM National Coordinator for engineering. It was not started by NCERT or by the IIMs.

Platforms for higher education and research

Initiative	What it does and who runs it
e-PG Pathshala	Curriculum-based, interactive e-content for postgraduate subjects. An initiative under the National Mission on Education through ICT (NME-ICT) of the Ministry of Education, with content developed through the UGC and the portal run by the INFLIBNET Centre
ePathshala	NCERT's portal and mobile app (developed by its Central Institute of Educational Technology, CIET) for school textbooks and other e-resources. Not to be confused with e-PG Pathshala
Shodhganga	Repository of theses and dissertations submitted to Indian universities, maintained by the INFLIBNET Centre
Shodhgangotri	Repository of synopses of approved or ongoing research (research in progress), also run by INFLIBNET
ShodhShuddhi	INFLIBNET's service giving universities access to plagiarism-detection software
National Digital Library of India (NDLI)	A single-window search platform for learning resources from many libraries and institutions, developed at IIT Kharagpur under NME-ICT
Virtual Labs	Web-based, simulation-based laboratory experiments in science and engineering, under NME-ICT, by a consortium coordinated by IIT Delhi
DIKSHA (Digital Infrastructure for Knowledge Sharing)	National platform for school education, led by NCERT under the Ministry of Education; e-content for students and training material for teachers; QR codes in textbooks link to it. Launched 5 September 2017
INFLIBNET (Information and Library Network) Centre, Gandhinagar	An inter-university centre of the UGC that runs Shodhganga, Shodhgangotri, ShodhShuddhi, e-ShodhSindhu, Vidwan and IRINS, and implements One Nation One Subscription
CEC (Consortium for Educational Communication), New Delhi	An inter-university centre of the UGC for electronic media. It coordinates the Educational Multimedia Research Centres (EMRCs) in universities, produces programmes for undergraduate subjects, and is a SWAYAM National Coordinator and a provider of SWAYAM PRABHA channels
One Nation One Subscription (ONOS)	Nationwide access for government-funded institutions to scholarly e-journals of major international publishers through one central subscription, operational from 1 January 2025 and implemented by INFLIBNET

Initiative	What it does and who runs it
NIRF (National Institutional Ranking Framework)	The Ministry of Education's ranking of higher education institutions; not a learning platform

Two quick sorting rules. Web-based services (SWAYAM, e-PG Pathshala, Virtual Labs, NDLI) need the internet at the learner's end; SWAYAM PRABHA through a dish does not. School-level services are DIKSHA and ePathshala; postgraduate e-content is e-PG Pathshala; theses are Shodhganga; synopses are Shodhgangotri.

CLASSROOM EXAMPLE

A first-year M.A. Hindi student in a village near Sirsa has no reliable internet at home. She follows lectures on literature on a SWAYAM PRABHA channel through the family's DD Free Dish. In the college library, with internet, she reads e-PG Pathshala modules on her syllabus and searches Shodhganga for theses on Premchand. Next semester she enrols in a SWAYAM course on research methodology, studies it free, pays the fee for the proctored examination, and her university counts the credits under the UGC regulations.

OFTEN CONFUSED

- **SWAYAM and SWAYAM PRABHA.** SWAYAM: web MOOCs with quizzes, examinations and credits; needs internet. SWAYAM PRABHA: one-way DTH television; no internet; no assessment.
- **e-PG Pathshala and ePathshala.** e-PG Pathshala: postgraduate e-content (UGC, INFLIBNET, NME-ICT). ePathshala: NCERT's school textbooks and resources.
- **Shodhganga and Shodhgangotri.** Shodhganga: submitted theses. Shodhgangotri: synopses of research in progress.
- **Shodhganga and ShodhShuddhi.** Shodhganga stores theses; ShodhShuddhi gives plagiarism-detection software.
- **DIKSHA and SWAYAM.** DIKSHA is mainly for school education and teacher training; SWAYAM offers courses from school to postgraduate level and beyond.
- **NPTEL and NDLI.** NPTEL: online courses by IITs and IISc. NDLI: a digital library search platform (IIT Kharagpur).
- **Virtual Labs and NDLI.** Virtual Labs: simulated experiments. NDLI: a library of learning resources.
- **Massive and open.** Massive: unlimited numbers. Open: no entry requirements, usually free.

FACT WATCH

Fact watch (checked October 2026)

- **SWAYAM launch:** 9 July 2017, by the President of India, at the National Convention on Digital Initiatives (Ministry of Human Resource Development, now Ministry of Education). The same event launched 32 SWAYAM PRABHA channels and the National Academic Depository.
- **SWAYAM National Coordinators: ten** (NPTEL, UGC, AICTE, CEC, IGNOU, IIMB, NITTTR, INI, NCERT, NIOS), as listed in the Ministry of Education's PIB backgrounder on PM e-Vidya of March 2026, with figures as of January 2026. Older books and older SWAYAM pages list **nine**, without INI; Banaras Hindu University was designated National Coordinator for Institutions of National Importance in 2025.
- **SWAYAM PRABHA channels:** the service began in 2017 with **32** channels; later sources gave 34, and then **40**. In 2026 official sources disagree. The PIB backgrounder on PM e-Vidya of 10 March 2026, and a PIB document of June 2026, describe a group of **48** DTH channels. The SWAYAM PRABHA website and a PIB factsheet of August 2026 give **40**. The documents do not explain the difference. If a question names the March 2026 note, its figure is 48; if a question asks only "how many channels", look for 40 or 48 among the options and check the latest official figure before your examination.
- **Credit transfer limit:** up to **40 per cent** of the total courses offered in a programme in a semester, under the UGC (Credit Framework for Online Learning Courses through SWAYAM) Regulations, 2021, notified in the Gazette of India on 25 March 2021. These replaced the Regulations of 2016 (notified 19 July 2016), which allowed up to **20 per cent**. No later amendment to the 40 per cent ceiling was found; a reply in the Lok Sabha in July 2026 still gave 40 per cent.
- **Four quadrants:** e-Tutorial, e-Content, discussion forum, assessment. Revised SWAYAM Guidelines were approved by the SWAYAM Board in 2024 (circulated in May 2024) and describe a broader course framework; the four quadrants remain the description in Ministry of Education material of 2026 and the answer the examination expects.
- **Certificate fee:** charged for the proctored examination; on NPTEL course pages it is usually Rs 1,000 per course, but the amount is set course by course.
- **SWAYAM Plus:** a separate platform for industry-relevant and skill courses, launched in February 2024 and operated by IIT Madras. Do not confuse it with SWAYAM.
- **ShodhShuddhi:** INFLIBNET's plagiarism-detection service began in 2019 with the software Urkund (later called Ouriginal). That contract ended on 30 September 2023, and the service has used DrillBit since 1 November 2023. It is meant for theses, synopses and research articles, through universities.
- **One Nation One Subscription:** approved by the Union Cabinet on 25 November 2024 and operational from 1 January 2025, with INFLIBNET as implementing agency; about 13,000 e-journals of 30 international publishers for government-funded institutions. It builds on the e-ShodhSindhu consortium (formed in December 2015 by merging UGC-INFONET, N-LIST and INDEST-AICTE). Older books name e-ShodhSindhu as the national journal consortium.
- **NDLI:** pilot from 2015; nationally launched on 19 June 2018.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

cMOOCs and xMOOCs

The term MOOC was coined in 2008, usually credited to Dave Cormier (some sources add Bryan Alexander), for the course "Connectivism and Connective Knowledge" (CCK08), run by **George Siemens and Stephen Downes** through the University of Manitoba on the principles of **connectivism** (संयोजनवाद), the theory that learning is the forming of connections across a network of people and resources. These are called **cMOOCs**. In a cMOOC much of the content is found, created, remixed and shared by the learners themselves, and the course acts as a hub of a network rather than as a fixed body of content. Because connectivism treats learning as network-building, learners must take responsibility for the content.

xMOOCs came later. They are structured, instructor-led courses with a fixed syllabus of video lectures, quizzes and examinations, offered on large platforms such as Coursera, edX and SWAYAM. They are content-delivery courses.

So "xMOOCs are based on connectivism" is false, and "cMOOCs follow a fixed syllabus of instructor-made video lectures" is also false: each statement gives one type the feature of the other.

A **SPOC** (small private online course) uses MOOC-style materials, such as video lectures and automatic quizzes, for a limited, selected group, typically the students of one institution. It allows closer interaction and is often used in blended teaching on campus.

Reading a completion rate

A completion rate is a fraction, and its meaning depends on the base (the denominator). Always say which base is used.

WORKED EXAMPLE

A MOOC has 40,000 enrolments. 8,000 learners begin the first week's work. 2,400 complete the course.

Completion rate on enrolments = completers / enrolments = 2,400 / 40,000 = 0.06 = 6 per cent

Completion rate on starters = completers / starters = 2,400 / 8,000 = 0.30 = 30 per cent

Share of enrolled learners who ever started = 8,000 / 40,000 = 0.20 = 20 per cent

The same course can be reported as having a 6 per cent or a 30 per cent completion rate. Neither is wrong; they answer different questions.

USUAL SLIPS

- Dividing by the wrong base: the question says "on enrolments" or "on those who began"; read it twice.
- Forgetting to multiply by 100 when the options are in per cent.
- Dividing the larger number by the smaller (for example 40,000 / 2,400) and getting a figure above 100 per cent.

Parent and host institution

The UGC's credit framework uses two terms. The **parent institution** is the one in which the student is enrolled; it incorporates the SWAYAM credits into the student's record. The **host institution** is the one whose faculty member offers the course on SWAYAM. If a student of University X takes a course taught by a professor of Institute Y, X is the parent and Y the host.

National Academic Depository and Academic Bank of Credits

The **National Academic Depository** (NAD), launched on 9 July 2017 with SWAYAM, is a round-the-clock online store of academic awards, such as degrees, diplomas, certificates and mark-sheets, lodged by institutions and boards. Employers and institutions can verify them online. The UGC implements NAD with **DigiLocker** as its single depository. The **Academic Bank of Credits** (ABC) is different: it records the credits a student earns, so that they can be transferred and counted towards a degree (see section 6.7). NDLI is a library, and Shodhganga holds theses.

Vidwan and IRINS

Two more INFLIBNET services:

- **Vidwan:** a national database of profiles of experts, that is, scientists, researchers and faculty members, used for example to find experts for committees.
- **IRINS** (Indian Research Information Network System): a web-based research information management system that institutions use to collect and show their faculty's publications and research activity.

The National Educational Technology Forum

The National Education Policy 2020 proposes an autonomous body, the **National Educational Technology Forum** (NETF), to provide a platform for the free exchange of ideas on the use of technology to enhance learning, assessment, planning and administration. Do not confuse it with the **National Research Foundation**, which is meant to fund and promote research, or with the **National Testing Agency**, which conducts examinations such as UGC NET.

CHECK YOURSELF

1. Match List I with List II:

List I (Initiative)	List II (Function)
A. Shodhgangotri	I. Plagiarism-detection software for universities
B. ShodhShuddhi	II. Synopses of research in progress
C. Vidwan	III. Single-window search for learning resources
D. NDLI	IV. Database of expert profiles

(A) A-II, B-I, C-III, D-IV
(B) A-II, B-I, C-IV, D-III
(C) A-I, B-II, C-IV, D-III
(D) A-IV, B-I, C-II, D-III

2. Consider the following statements about SWAYAM:

1. Its full form is Study Webs of Active-Learning for Young Aspiring Minds.
2. A learner must pay a fee to watch the video lectures.
3. A certificate requires passing a proctored examination.
4. Its courses are built on four quadrants, one of which is a discussion forum.

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 2 and 4 only
(C) 1, 2, 3 and 4
(D) 1, 3 and 4 only

3. Given below are an Assertion (A) and a Reason (R):

Assertion (A): SWAYAM PRABHA can serve learners who have no internet connection.

Reason (R): SWAYAM PRABHA programmes are telecast on DTH channels received through a dish.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. Given below are two statements:

Statement I: cMOOCs are built on connectivism, with learners creating and sharing content across networks.

Statement II: An xMOOC is a small private course offered only to the students of one college.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

5. On SWAYAM, the National Coordinator for non-technical postgraduate courses is:

(A) NPTEL
(B) CEC
(C) UGC

(D) NIOS

6. A MOOC records 25,000 enrolments, of whom 5,000 begin the course and 1,000 complete it. The completion rate on those who began is:

(A) 4 per cent
(B) 5 per cent
(C) 20 per cent
(D) 25 per cent

7. Under the UGC (Credit Framework for Online Learning Courses through SWAYAM) Regulations, 2021, the ceiling on courses through SWAYAM in a programme in a semester is:

(A) 10 per cent
(B) 20 per cent
(C) 30 per cent
(D) 40 per cent

8. Which of the following is mainly meant for school education?

(A) e-PG Pathshala
(B) DIKSHA
(C) Shodhganga
(D) NPTEL

Answers to “Check yourself”: Chapter 4

Section 4.1

1. (C), because setting a task and guiding learners while they work is the facilitator’s role. Reading out, dictating and demonstrating while students watch keep the teacher as transmitter.

2. (D), because the approach is the underlying view of learning, a method organises whole lessons, and a technique is a small device used within a method.

3. (B), because in role play, the case method and peer tutoring the learners do the main work. The lecture and dictation keep the teacher at the centre, so any option that contains 1 or 4 is wrong.

4. (D), because the teacher’s role changes to designer and guide but does not lose importance, so Statement I is false. A lecture is often the best way to give a large class an overview, so Statement II is true.

5. (A), because both are true and the learner’s own activity is the very reason for better retention.

6. (A), because a common pace belongs to teacher-centred teaching, continuous formative assessment to learner-centred teaching, examples-to-rule to induction and rule-to-examples to deduction.

7. (C), because NEP 2020 asks for experiential, inquiry-driven, discovery-oriented, learner-centred and discussion-based pedagogy. The other options describe what the policy wants to leave behind.

Section 4.2

1. (A), because the lecture is economical and can bring together material not yet in books. It gives little feedback from each learner and does not develop laboratory skills, so statements 2 and 4 are wrong.

2. (A), because both are true and the lack of learner practice is exactly why the lecture cannot build skills.

3. (B), because asking learners to predict, observe and explain keeps them thinking during a demonstration. It does not replace laboratory practice.

4. (C), because small steps means short frames, active responding means answering at every frame, immediate confirmation means knowing at once, and self-pacing means one’s own speed.

5. (C), because remedial frames are the mark of Crowder’s branching. In Skinner’s linear programme the learner writes (constructs) the answer, so Statement II is false.

6. (D), because a lecture has a sender, a message and a channel; what is missing is the return flow from the listeners.

7. (B), because a skill must be shown and then practised by each learner. A demonstration alone (C) shows the skill but gives no practice.

8. (C), because retrogressive chaining, with the last step learnt first, is the mark of Gilbert’s mathetics. Mastery learning (D) concerns reaching a fixed standard in each unit.

Section 4.3

1. (A), because depth on one paper marks a seminar, prepared talks on different aspects a symposium, free conversation under a moderator a panel, and a product made by participants a workshop.

2. (D), because the chairperson introduces, the paper is presented, it is discussed, and the chairperson sums up last, after hearing the discussion.

3. (C), because Osborn's rules aim at quantity and defer judgement until idea generation is over, so Statement II is false.

4. (C), because team teaching does use each teacher's strengths, but it needs more coordination, not less; that is its main difficulty.

5. (B), because the rapporteur records and reports the proceedings. The convener organises, the moderator conducts a panel, and the discussant comments on a paper.

6. (A), because statements 1, 2 and 4 describe cooperative learning correctly. Statement 3 describes a competitive structure (negative interdependence), not cooperation.

7. (C), because thinking alone, then talking with a partner, then sharing with the class is Lyman's Think-Pair-Share.

8. (A), because the Trump Plan gives only about 20 per cent of students' time to small-group discussion, less than either large-group instruction or independent study (usually taught as about 40 per cent each).

Section 4.4

1. (C), because the order is purposing, planning, executing and evaluating (judging the result against the purpose).

2. (A), because Parkhurst devised the Dalton Plan, Armstrong the heuristic method, Keller the Personalised System of Instruction and Bloom mastery learning.

3. (A), because both are true and drawing on several subjects at once is exactly why the project method promotes correlation.

4. (D), because role play is unscripted, so Statement I is false; debriefing is where the experience becomes learning, so Statement II is true.

5. (B), because statements 2, 3 and 4 are correct. The formative tests in mastery learning are not graded; grades come from a later summative test.

6. (B), because the students go from examples to the rule, which is induction.

7. (D), because a physical product is the mark of the project method, not of PBL. The other three statements describe PBL correctly.

8. (A), because a mock trial practises a real task in an artificial setting, and the analysis afterwards is debriefing. Psychodrama deals with an individual's private problems in therapy.

Section 4.5

1. (B), because a live webinar and instant chat need everyone present at the same moment. A discussion board and a recorded lecture are used at different times.

2. (A), because both are true, and moving first exposure out of class is exactly what frees class time.

3. (B), because the flipped classroom is only one model of blended learning, and blended learning keeps face-to-face teaching.

4. (C), because rotation means moving between modes on a schedule; flex has online learning as backbone; a la carte adds a fully online course; enriched virtual is mostly remote.

5. (D), because first exposure (remembering and understanding) happens at home, and the higher-order work is done in class.

6. (C), because more dialogue reduces transactional distance. High structure with low dialogue gives the greatest distance.

7. (B), because a threaded forum allows extended, reasoned exchange without everyone being present at once. A webinar needs everyone together, and a podcast and a broadcast e-mail are one-way.

Section 4.6

1. (B), because Shodhgangotri holds synopses, ShodhShuddhi gives plagiarism-detection software, Vidwan is the expert database and NDLI is the single-window library search.

2. (D), because study on SWAYAM is free; a fee is charged only for the certificate examination. The full form, the proctored examination and the forum quadrant are correct.

3. (A), because both are true and the DTH broadcast is exactly why no internet is needed.

4. (C), because Statement I describes cMOOCs correctly, but Statement II describes a SPOC; an xMOOC is a structured, instructor-led course open to very large numbers.

5. (C), because UGC handles non-technical postgraduate courses. NPTEL handles engineering and CEC non-technical undergraduate courses.

6. (C), because 1,000 / 5,000 = 0.20, that is 20 per cent. On enrolments it would be 1,000 / 25,000 = 4 per cent, option (A), which uses the wrong base.

7. (D), because the 2021 Regulations allow up to 40 per cent. The 20 per cent ceiling, option (B), belonged to the 2016 Regulations.

8. (B), because DIKSHA, led by NCERT, serves school students and teachers. e-PG Pathshala is postgraduate, Shodhganga holds theses and NPTEL is mainly engineering.

Chapter 5
Teaching Support System

A teaching support system is everything a teacher uses, beyond voice and gesture, to help students learn: the chalkboard and the chart, the projector and the language laboratory, the learning management system and the open textbook. The syllabus groups these as traditional, modern and ICT-based. This chapter first explains why aids help and how they are classified, including Edgar Dale's Cone of Experience, and then takes the three groups in turn.

5.1 Teaching aids: purposes, classification and Dale's Cone of Experience

Syllabus: Teaching Support System: Traditional, Modern and ICT based

> **EXAM FOOTPRINT**
> Asked occasionally; usually an item on who proposed the cone of experience or which experience sits at its concrete or abstract end.

Core

A **teaching aid** (शिक्षण सहायक सामग्री) is any material or device that helps the teacher make an idea clear, concrete and memorable. The **teaching support system** (शिक्षण सहायक प्रणाली) is the whole set of such aids and the facilities that make them usable.

Why aids help

- They make abstract ideas **concrete**: a model of the heart, a map of the monsoon.
- They engage more than one **sense**, which holds attention and suits learners with different preferences.
- They show what cannot be brought to class: the very large, the very small, the distant, the dangerous, the past.
- They save time: a prepared diagram is quicker than a long description.
- They add variety and interest, and so support motivation.

But an aid teaches only when it is used well. It is a means, not an end (see 3.4).

Classification by the sense addressed

This is the classification most often asked.

Type	Addresses	Examples
Audio aids (श्रव्य सामग्री)	Hearing only	Radio, tape recorder, audio recording, podcast, language laboratory headphones
Visual aids (दृश्य सामग्री)	Sight only	Chalkboard, chart, map, globe, model, specimen, flash card, flannel board, silent filmstrip, slides
Audio-visual aids (दृश्य-श्रव्य सामग्री)	Both hearing and sight	Sound film, television, video, multimedia presentation with sound

A sound film or television appeals to both the ear and the eye at once. A tape recorder is an audio aid; a silent filmstrip and a flannel board are visual aids. Aids that involve more senses are generally

thought to hold attention better and aid retention, provided they are not overloaded (see Going Further).

Classification as projected and non-projected

- **Projected aids** (प्रक्षेपित सामग्री) need a projector to throw an image on a screen: slides, filmstrips, overhead transparencies, films, opaque projection by an episcope or epidiascope, and computer output through an LCD projector.
- **Non-projected aids** (अप्रक्षेपित सामग्री) are seen directly: chalkboard, charts, maps, models, specimens, flash cards, flannel board, bulletin board.

Be careful with the epidiascope. It can show a page of a book or a leaf, which looks like direct display, but it does so by projecting an enlarged image. It is itself a projector, so it is a projected aid.

Other classifications

- **By dimension**: two-dimensional (charts, maps, photographs, which are flat) and three-dimensional (models, specimens, objects, globes, mock-ups, which have length, breadth and depth). See 5.2.
- **By technology**: traditional, modern (audio-visual and projected equipment) and ICT-based. This is the syllabus's own grouping and the plan of this chapter.

Group (syllabus)	Typical aids	Section
Traditional (पारंपरिक)	Chalkboard, charts, models, specimens, flash cards, flannel board, handouts	5.2
Modern (आधुनिक)	Overhead projector, slide projector, epidiascope, radio, television, language laboratory	5.3
ICT-based (आईसीटी आधारित)	Computer and LCD projector, smart class, multimedia, LMS, webinars, podcasts, wikis, OER	5.4

Edgar Dale's Cone of Experience

Edgar Dale, an American educator, introduced the **Cone of Experience** (अनुभव शंकु) in his book *Audiovisual Methods in Teaching* (first edition 1946). The cone arranges kinds of learning experience from the most **concrete** (मूर्त) at the broad base to the most **abstract** (अमूर्त) at the narrow top. At the base the learner does the thing; at the top the learner meets only words.

In the third edition (1969), the version most often reproduced, the cone has eleven bands. Read them from the bottom (most concrete) to the top (most abstract):

1. Direct, purposeful experiences (the base)
2. Contrived experiences
3. Dramatised experiences
4. Demonstrations
5. Study trips
6. Exhibits
7. Educational television
8. Motion pictures
9. Recordings, radio and still pictures
10. Visual symbols
11. Verbal symbols (the apex)

Band	What it means
Direct, purposeful experiences	First-hand experience with real things, done for a purpose: growing a crop, conducting a survey. In Dale's words, experience that is seen, handled, tasted, touched, felt and smelled.
Contrived experiences	Edited copies of reality: models, mock-ups, simulations, used when the real thing is too large, small, complex or dangerous
Dramatised experiences	Taking part in a re-enactment: role play, a play, a mock parliament
Demonstrations	Watching someone show how something is done
Study trips	Visiting real places to observe: a factory, a museum, a field site
Exhibits	Displays arranged for viewing: museum cases, exhibitions
Educational television	Viewing broadcast or recorded television programmes
Motion pictures	Films: moving images with sound
Recordings, radio and still pictures	Sound alone, or a single fixed image (photograph, slide)
Visual symbols	Charts, graphs, maps, diagrams: abstract signs that still show something to the eye
Verbal symbols	Words, spoken or written, which bear no resemblance to what they stand for

Three positions settle most questions. The **base** is direct, purposeful experiences. The **apex** is verbal symbols, with visual symbols just below it. The band **immediately above contrived experiences** is dramatised experiences, and demonstrations come after that. So, for example, demonstrations, study trips and exhibits run in that order from more concrete to more abstract, and visual symbols are more abstract than motion pictures.

Dale's point was that concrete experience supports abstract understanding. A teacher should give enough concrete experience for words to carry meaning, and the cone helps in choosing aids. It does not say that lower bands are always better.

> **CLASSROOM EXAMPLE**
>
> A teacher of B.A. Economics in a Kurukshetra college is teaching "price determination". She starts with verbal symbols (a definition), and the class is lost. Next year she reverses the order. Students first run a small mock market in class, buying and selling cards (a contrived and dramatised experience). They then watch a short film of a sabzi mandi (motion pictures), study a demand-supply graph (visual symbols), and only then write the definition (verbal symbols). She has moved up Dale's cone from concrete to abstract.

OFTEN CONFUSED

- **Concrete vs abstract in the cone.** Lower means more concrete, higher means more abstract. Direct experience is the base; verbal symbols are the apex.
- **Study trip vs demonstration.** A study trip feels more "real", but in Dale's cone it lies above demonstrations, not below.
- **Visual symbols vs verbal symbols.** Visual symbols (charts, maps, graphs) still show something to the eye and lie just below verbal symbols, which are the most abstract.
- **Visual symbols vs still pictures.** A photograph resembles its object (still pictures band); a graph is an abstract sign (visual symbols band).
- **Projected vs non-projected.** Ask whether a projector throws the image on a screen. The epidiascope, OHP and slide projector are projected; flash cards, charts, models and the flannel board are not.
- **Audio vs audio-visual.** Radio and tape recorder are audio; television and sound film are audio-visual; a silent filmstrip is visual.

FACT WATCH

- **The cone's editions.** Dale introduced the cone in 1946 (first edition of *Audiovisual Methods in Teaching*) with ten bands, including "dramatic participation" and "field trips" and no television. The second edition (1954) renamed dramatic participation as "dramatised experiences" and added television. The third edition (1969) gives the eleven bands above, with "study trips" and "educational television", and relates the cone to Bruner's three modes of learning (checked October 2026).
- **The percentages are not Dale's.** The popular claim that "we remember 10 per cent of what we read, 20 per cent of what we hear, 30 per cent of what we see…" is often printed on or beside Dale's cone. Dale's cone contains no such numbers, and the figures have no known research basis. Subramony, Molenda, Betrus and Thalheimer traced this "mythical retention chart" and its attachment to Dale's cone in a 2014 article in *Educational Technology*. If you see percentages on a cone, the diagram has been altered.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

The cone and Bruner's three modes

In the 1969 edition, Dale related his cone to the three modes of learning proposed by the psychologist Bruner in 1966: **enactive** (learning by doing), **iconic** (learning through observation of images) and **symbolic** (learning through abstractions such as words and symbols). The cone is read in three parts:

Bruner's mode	Bands of Dale's cone	Kind of learning
Enactive (क्रियात्मक)	Direct, contrived and dramatised experiences	Doing
Iconic (प्रतिमात्मक)	Demonstrations, study trips, exhibits, educational television, motion pictures, recordings, radio and still pictures	Observing
Symbolic (प्रतीकात्मक)	Visual symbols and verbal symbols	Symbolising

The word “visual” in “visual symbols” tempts students to choose iconic. But charts, graphs and maps are abstract signs; the iconic band consists of pictures and observed events. The two topmost bands belong to the symbolic mode.

What Dale did and did not claim

Dale called the cone a “visual analogy”, a picture to help thinking, not a ranking of media by effectiveness. He warned that too much reliance on concrete experience may actually obstruct meaningful generalisation, and that realistic experiences can be costly in time and effort. His advice was to balance concrete and abstract experiences. The cone carried no numbers, and Dale reported no experiments on retention. The retention percentages were added by others. Older trainers’ manuals and many websites still print them under Dale’s name; they are wrong.

Integrating an aid with the lesson

An aid becomes support material, not entertainment, only when it is integrated with the lesson. For a film or video:

1. **Preview** it and choose the relevant part.
2. **Prepare** the class: tell them what to look for and give guiding questions.
3. **Show** the relevant part, pausing where useful.
4. **Follow up** with discussion that links what was seen to the objectives.

Screening a film in full without comment, choosing the longest film available, or using a film in place of the planned lesson leaves learning to chance.

Too many aids: cognitive load

Cognitive load theory (संज्ञानात्मक भार सिद्धांत) holds that working memory can handle only a few new elements at once. Every aid makes demands on it. When aids and stimuli multiply, the load exceeds capacity, attention is drawn away from the objective, and learning suffers. So using too many teaching aids in a single lesson can hinder learning, and overload of working memory is the reason. The same idea underlies the design principles for slides in 5.4.

Choosing an aid: objective and audience

Choose the aid by two questions: what must the learners perceive, and how many are they? To show 100 students how the valves of the heart open and close in sequence, the objective concerns movement over time, which only a moving image can show, and the class is large, which calls for projection. An animation shown through a projector is the right choice. A preserved specimen is real but static and too small for a hundred students; a wall chart and flash cards are also static.

CHECK YOURSELF

1. Which of the following is an audio-visual aid?

(A) Tape recorder
(B) Flannel board
(C) Television
(D) Silent filmstrip

2. Arrange the following bands of Dale's Cone of Experience from the most concrete to the most abstract:

1. Motion pictures
2. Dramatised experiences
3. Verbal symbols
4. Study trips
5. Visual symbols

Choose the correct answer from the options given below:

(A) 2, 1, 4, 5, 3
(B) 2, 4, 1, 3, 5
(C) 2, 4, 1, 5, 3
(D) 4, 2, 1, 5, 3

3. Assertion (A): The retention percentages often printed on Dale's cone are reliable findings from Dale's research.

Reason (R): Dale's cone arranges experiences by degree of abstraction.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. Match List I with List II:

List I	List II
A. Contrived experiences	I. Symbolic mode
B. Exhibits	II. Enactive mode
C. Verbal symbols	III. Iconic mode
D. Bruner	IV. Proposed the three modes of learning (1966)

Choose the correct answer from the options given below:

(A) A-II, B-III, C-I, D-IV
(B) A-III, B-II, C-I, D-IV
(C) A-I, B-III, C-II, D-IV
(D) A-II, B-I, C-III, D-IV

5. Which of the following are projected aids?

1. Epidiascope
2. Flash card
3. Overhead transparency
4. Filmstrip

5. Flannel board

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 2, 4 and 5 only
(C) 3, 4 and 5 only
(D) 1, 3 and 4 only

6. Assertion (A): Crowding a lesson with many aids can reduce learning.

Reason (R): Working memory can process only a limited number of new elements at a time.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

7. The band of Dale's cone that lies immediately below demonstrations is:

(A) contrived experiences
(B) dramatised experiences
(C) study trips
(D) exhibits

8. Given below are two statements:

Statement I: The base of Dale's cone consists of direct, purposeful experiences.

Statement II: Dale held that more concrete experience is always better for every learner.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

5.2 Traditional support

Syllabus: Teaching Support System: Traditional

EXAM FOOTPRINT

Rarely asked directly; it supports classification items on teaching aids.

Core

Traditional support (पारंपरिक सहायक सामग्री) means the aids that need no electricity or electronic device: the chalkboard, charts, maps, pictures, models, specimens, flash cards, the flannel board, the bulletin board and printed handouts. They are cheap, always available and fully under the teacher's control.

The chalkboard

The **chalkboard** (श्यामपट्ट) is one of the oldest and most widely used classroom aids. Smart boards, webinar tools and learning apps are ICT-based; the chalkboard is traditional.

Advantages	Limitations
Cheap and always available; needs no power	The teacher's back is turned to the class while writing, so eye contact is lost
Diagrams and arguments can grow step by step before the students' eyes	What is written must be rubbed off and cannot be stored
Writing is slow, which keeps the teacher in step with the students' thinking	Poor handwriting or a crowded board defeats it
Content can be altered at once in response to questions	Chalk dust; glare on a shiny board

The chalkboard's special strength is material that **unfolds during the lesson**: a derivation, a calculation, a grammatical analysis, a diagram built up stage by stage, at the class's pace. Material prepared in advance, such as a colour photograph, a recorded interview or a large table of data, is better handled by projected or digital aids. The usual examination limitation is the loss of eye contact while writing. "Cannot be used without electricity", "can show only matter printed in advance" and "too costly" are all false.

Good chalkboard practice: plan the layout, write large and legibly, keep one part for key terms, stand aside so all can see, and turn to face the class while talking.

Charts, maps, pictures and other two-dimensional aids

Charts, maps, graphs, posters and photographs are **two-dimensional** (द्वि-आयामी): they are flat. A good chart has one main idea, large lettering and an uncrowded layout (see 3.4).

Models, specimens and other three-dimensional aids

Three-dimensional aids (त्रि-आयामी सामग्री) have length, breadth and depth and can be viewed from all sides: models, specimens, real objects, globes and mock-ups.

- A **model** (प्रतिमान or मॉडल) is a recognisable representation of a real thing, enlarged, reduced or simplified. Models are used when the real object is too large (the solar system), too small (a cell), too costly, too dangerous or not accessible.
- A **specimen** (नमूना) is a real object, a sample that represents its class: a rock, a leaf, an insect, a coin. It is not a representation; it is the real thing in sample form.

Flash cards

A **flash card** (फ़्लैश कार्ड) is a small card carrying a word, picture, formula or question that is shown briefly to the class. Flash cards suit quick drill, revision and recall of vocabulary, definitions and number facts. They are simple, non-projected visual aids. They are not suited to the detailed study of long texts, and they do not project or record anything.

The flannel board

A **flannel board** (फलालैन बोर्ड) is a board covered with flannel or a similar rough cloth. Cut-outs backed with rough material stick to it and can be added, moved or removed at will. So a display can be **built up step by step**: the parts of a sentence, the stages of a process, the members of a family tree. This advantage comes from the adhesive cut-outs. The flannel board is also a non-projected visual aid, but that classification does not explain the advantage: a printed chart is non-projected too, and cannot be built up.

Handouts

A **handout** (हैंडआउट) is a printed sheet given to students. A good handout gives the structure of the lecture, key terms, definitions and diagrams. Students can then listen and think instead of copying, and the key material is accurate. It reduces the burden of note-taking. But it supports the lecture

and further reading; it does not make attendance, the textbook or the library unnecessary, and it is not a script for the teacher to read from.

Why traditional aids still matter

Traditional aids have not become irrelevant in ICT-enabled classrooms. They are inexpensive, always available, need no electricity, and stay under the teacher's direct control; a model can be handled as no screen image can. In a rural college with unreliable power they may be the only aids that work every day. Good teaching combines traditional and digital support.

CLASSROOM EXAMPLE

A Botany teacher at a college in Jind brings a basket of real leaves (specimens) to her B.Sc. class, passes them round, and asks students to sort them by venation. She then shows a large model of a leaf in cross-section, since the real tissues are too small to see. On the chalkboard she builds a classification table row by row as students report their findings. At the end she gives a one-page handout with the table and key terms, so that students spend the period observing, not copying.

OFTEN CONFUSED

- **Model vs specimen.** A model represents the real thing (enlarged, reduced or simplified). A specimen is the real thing, a sample of its class.
- **Two-dimensional vs three-dimensional.** Charts, maps and photographs are flat. Models, specimens, globes and mock-ups have depth.
- **Chalkboard vs prepared slide.** The chalkboard suits content that unfolds during the lesson; a slide suits content prepared in advance.
- **Flannel board vs chart.** Both are non-projected visual aids, but only the flannel board lets a display be built up and rearranged.
- **Handout as support vs handout as substitute.** A handout supports listening and reading; it does not replace them.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Kinds of three-dimensional aid

Aid	What it is
Model	A representation of a real object, enlarged, reduced or simplified
Working model	A model with moving parts that shows how the thing works
Cut-away model	A model with part of the outside removed to show the inside, for example of an engine or the earth
Mock-up	A simplified working replica of a device or system, used for training in its operation, for example a cockpit mock-up
Diorama	A three-dimensional scene in miniature, with small figures and objects arranged in front of a painted background: a village, a battle, a habitat
Specimen	A real object or sample that represents its class
Globe	A three-dimensional model of the earth

A mock-up imitates the working of an apparatus for practice; a static miniature scene is a diorama. A cut-away model shows the inside of an object.

Other traditional aids

- **Bulletin board** (सूचना पट्ट): a board for notices, clippings and student work, changed regularly.
- **Wall charts of different kinds**: flow charts (stages of a process), tree charts (branching relationships), tabular charts, pie and bar charts.
- **Puppets** and **real objects** (realia) are also traditional aids, useful for dramatised and direct experience.

Where traditional aids sit on Dale's cone

Models, mock-ups and specimens belong to the lower, concrete part of Dale's cone (contrived and direct experiences). Charts, maps and graphs belong near the top (visual symbols). Pictures belong to the still-pictures band. So "traditional" does not mean "abstract": a specimen is among the most concrete aids a teacher can use.

CHECK YOURSELF

1. Which of the following is a three-dimensional aid?

(A) Wall map
(B) Photograph
(C) Globe
(D) Bar chart

2. Given below are two statements:

Statement I: A specimen is a real object that represents its class.

Statement II: A model is used when the real object is too large, too small or not accessible.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

3. Match List I with List II:

List I	List II
A. Diorama	I. Simplified working replica for training in operation
B. Mock-up	II. Miniature scene before a painted background
C. Cut-away model	III. Real sample of its class
D. Specimen	IV. Shows the inside of an object

Choose the correct answer from the options given below:

(A) A-II, B-I, C-III, D-IV
(B) A-I, B-II, C-IV, D-III
(C) A-II, B-IV, C-I, D-III
(D) A-II, B-I, C-IV, D-III

4. Assertion (A): Traditional aids remain useful in classrooms that also have digital equipment.

Reason (R): Traditional aids are cheap, need no electricity and remain under the teacher's control.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

5. For which purpose is the chalkboard better than a slide prepared in advance?

(A) Showing a photograph of a painting
(B) Working out a problem step by step in response to students' questions
(C) Presenting a large statistical table
(D) Playing an expert's recorded talk

6. Assertion (A): A flannel board allows a diagram to be built up and rearranged during the lesson.

Reason (R): A flannel board does not need a projector.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

7. Which of the following are limitations of the chalkboard?

1. The teacher's back is turned to the class while writing
2. It cannot be used without electricity
3. What is written cannot be stored after it is rubbed off
4. It can display only matter printed in advance

Choose the correct answer from the options given below:

(A) 1 and 2 only
(B) 1 and 3 only
(C) 2 and 4 only
(D) 1, 3 and 4 only

5.3 Modern support

Syllabus: Teaching Support System: Modern

EXAM FOOTPRINT

Rarely asked directly; it supports items classifying aids as traditional, modern or ICT-based.

Core

Modern support (आधुनिक सहायक सामग्री) in the syllabus means the electrical and electronic audio-visual equipment that came before the computer age and is still found in many colleges: projectors of various kinds, radio, television, audio and video recorders, and the language laboratory. Most need electricity. Many have now been replaced by the computer and LCD projector (5.4), but questions on them remain common.

Projectors: what each one projects

The deciding question is always: what kind of material does this projector take, and how does the light reach the screen?

Device	What it projects	Key features
Overhead projector (OHP) (शिरोपरि प्रक्षेपी)	Large transparent sheets (transparencies) laid flat on its glass stage	Bright image in a normally lit room; teacher faces the class and can write or point on the sheet
Slide projector (diascope) (डायस्कोप)	Small mounted transparent slides, by light passing through them	Needs a darkened room
Filmstrip projector	A strip of still frames on film, shown one by one	Visual; usually silent; darkened room
Episcope (एपिस्कोप)	Opaque material (a page, picture, coin, leaf) by reflected light	Needs a darkened room; image less bright

Device	What it projects	Key features
Epidiascope (एपिडायस्कोप)	Both opaque material and transparent slides	Combines episcope and diascope; darkened room
LCD or LED projector	The output of a computer: slides, video, web pages	Works with a computer; the bridge to ICT (see 5.4)

The overhead projector

The **overhead projector** throws an enlarged image of a transparent sheet on to a screen behind the teacher. Its advantages over the slide projector are that the image is bright enough to use in a lit room, so students can take notes, and that the teacher faces the class while writing or pointing on the transparency. It is a projected visual aid. It cannot project opaque objects (that is the work of the episcope or epidiascope), it needs electricity, and it shows no motion or sound. It has now been largely replaced by the LCD projector connected to a computer.

Radio, television and recordings

- **Radio** and the **tape recorder** (and other audio recorders) are audio aids. Educational radio brings expert talks and dramatised programmes to many places at once.
- **Television** and **sound film** are audio-visual aids. Educational television brings demonstrations, visits and experts into the classroom.
- Recordings let a lesson, a speech or a performance be replayed and studied.

The language laboratory

A **language laboratory** (भाषा प्रयोगशाला) is a room with individual audio or computer stations, each with headphones and a microphone. Each learner listens to model speech, records his or her own response, and plays both back to compare, while the teacher monitors and speaks to individuals from a console. It is used primarily for practice in **listening and speaking** skills, the oral and aural skills that ordinary large classes cannot practise. It is an instructional facility, not a store of books, a statistics room or a science laboratory.

> **CLASSROOM EXAMPLE**
>
> An English department in a college at Bhiwani still has an overhead projector. A teacher teaching sentence structure places a transparency of a long sentence on the stage and, facing the class with the lights on, brackets its clauses with a marker as students call them out. Later, in the department's language laboratory, each student records a reading of the same sentence, listens to a model recording, and records again, while the teacher listens in through the console.

OFTEN CONFUSED

- **Episcope vs diascope.** The episcope projects opaque material by reflected light; the diascope (slide projector) projects transparent slides by light passing through them.
- **Epidiascope vs OHP.** The epidiascope projects opaque objects and slides, in a dark room. The OHP projects large transparencies, in a lit room, with the teacher facing the class.
- **Slide projector vs OHP.** Small mounted slides go in the slide projector; large writable sheets go on the OHP.
- **OHP vs LCD projector.** The OHP projects transparencies; the LCD projector shows computer output.
- **Language laboratory vs computer laboratory.** The language laboratory is for listening and speaking practice; a computer laboratory serves general computing.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

The epidiascope in detail

The name tells you what it does. *Epi* (upon) refers to the episcope, which projects opaque material, such as a page of a book, a picture, a coin or a specimen, by light reflected from its surface. *Dia* (through) refers to the diascope, which projects transparent slides by light passing through them. The **epidiascope** combines the two, so it can project opaque objects as well as transparent slides. It cannot project films with a sound track, play audio, or show computer files. Because reflected light is weak, it needs a darkened room. It is a projected visual aid.

Placing modern aids

- **Projected visual**: OHP, slide projector, filmstrip projector, episcope, epidiascope.
- **Audio**: radio, tape recorder, audio player.
- **Audio-visual**: television, sound film projector, video player.
- **Facility**: the language laboratory, a room with equipment, is better classed as an instructional facility (see 3.5).

On Dale’s cone, educational television and motion pictures are separate bands, and recordings, radio and still pictures share a band just above motion pictures (5.1).

CHECK YOURSELF

1. Match List I with List II:

List I	List II
A. Episcope	I. Computer output
B. Slide projector	II. Opaque pictures and objects
C. Overhead projector	III. Small mounted transparent slides
D. LCD projector	IV. Large transparencies on which the teacher can write

Choose the correct answer from the options given below:

(A) A-II, B-III, C-IV, D-I
(B) A-III, B-II, C-IV, D-I
(C) A-II, B-IV, C-III, D-I
(D) A-IV, B-III, C-II, D-I

2. An advantage of the overhead projector is that it:

(A) projects opaque objects such as coins
(B) shows moving pictures with sound
(C) can be used in a lit room while the teacher faces the class
(D) works without electricity

3. Assertion (A): An epidiascope can project a page from a book.

Reason (R): An epidiascope includes an episcope, which projects opaque material by reflected light.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. A language laboratory is primarily meant for:

(A) storing audio-visual equipment
(B) practising listening and speaking
(C) analysing research data
(D) printing study material

5. Given below are two statements:

Statement I: The episcope projects transparent slides by light passing through them.

Statement II: The overhead projector has largely been replaced by the LCD projector.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

6. Which of the following are audio aids?

1. Radio

2. Tape recorder
3. Television
4. Filmstrip

Choose the correct answer from the options given below:

(A) 1 and 2 only
(B) 1, 2 and 3 only
(C) 3 and 4 only
(D) 2 and 4 only

5.4 ICT-based support

Syllabus: Teaching Support System: ICT based

EXAM FOOTPRINT

Asked occasionally; usually a single-answer item on an ICT tool for teaching, overlapping with the ICT unit.

Core

ICT stands for **Information and Communication Technology** (सूचना एवं संचार प्रौद्योगिकी). It covers the digital tools used to create, store, send and receive information: computers, the internet, projectors, mobile devices and software. ICT-based support includes presentations, the smart classroom, multimedia, learning management systems, online meetings, podcasts, wikis and open educational resources. (Not “Integrated”, “Internet” or “Information and Computer” Technology.)

Hardware in the classroom

- An **LCD (or LED) projector** connected to a computer displays slides, videos and web pages on a screen for the whole class. A printer produces paper copies, a scanner converts paper into digital form, and a web camera captures video; none of these projects.
- A **smart classroom** (स्मार्ट कक्षा) is a room equipped with digital teaching tools: a computer, a projector or interactive board, a sound system and an internet connection, so that the teacher can use multimedia and online resources in face-to-face teaching. It is not a class for selected students, not a class without a teacher, and not an online-only class. The teacher remains central; the technology supports the lesson.

Multimedia

Multimedia (बहुमाध्यम) is the integration of several forms of content, namely text, still images, sound, video and animation, usually under computer control and often with interactivity. Because it addresses more than one sense, it can hold attention and suit learners with different preferences. It does not mean using several textbooks, one teacher teaching several subjects, or storing data on several computers.

Learning management systems

A **learning management system** (अधिगम प्रबंधन प्रणाली, LMS) is software for delivering courses online and managing them: uploading material, enrolling students, collecting assignments with deadlines, running quizzes with automatic scoring, holding discussion forums, keeping a grade book, and reporting each learner’s progress. Moodle, Google Classroom, Canvas and Blackboard are familiar examples.

What distinguishes an LMS from a simple file-sharing service is the **tracking and reporting of each learner's progress**. Both can store and share files; only the LMS manages learning. This is what the word "management" refers to.

Moodle stands for **Modular Object-Oriented Dynamic Learning Environment**. It is free and **open-source** (मुक्त स्रोत) software: any institution may download, install and modify it. Blackboard is a commercial product. Google Classroom and Microsoft Teams are proprietary services owned by companies, even when they are offered to institutions without charge. Free of charge is not the same as open source.

An LMS automates routine tasks such as distributing material and scoring objective quizzes. This frees the teacher for designing learning, giving qualitative feedback, guiding discussion and supporting individuals. It does not remove the need for a teacher; those tasks need professional judgement.

Live and recorded: synchronous and asynchronous tools

Synchronous (समकालिक) tools work in real time: everyone takes part at the same moment. **Asynchronous** (असमकालिक) tools let each learner take part at a time of his or her choosing.

Tool	What it is	Mode
Webinar	"Web" + "seminar": a live online session in which a speaker presents to participants at different places, who ask questions by chat or audio	Synchronous
Video conferencing	Live two-way audio and video among several places, so everyone can see, hear and question one another	Synchronous
Webcast	A live or recorded broadcast of audio or video over the internet from one source to many viewers	One-to-many
Podcast	A series of audio (or video) files distributed over the internet for listening on demand	Asynchronous
Screencast	A recording of a computer screen, usually with narration	Asynchronous
Blog	A web journal of dated posts, newest first	Asynchronous
Wiki	A website whose pages any authorised user can create and edit through a browser, with a history of changes	Asynchronous, collaborative
E-mail	Written messages; no sight, sound or immediacy	Asynchronous

So: to let students on three campuses see and question a guest expert at the same time, use **video conferencing**. For students to write and edit a shared set of pages together, use a **wiki**. For audio episodes students download and play whenever they wish, use a **podcast**. A webinar is not a recorded seminar, a website listing seminars, or a printed report.

Open educational resources

Open educational resources (मुक्त शैक्षिक संसाधन, OER) are teaching and learning materials in any medium that are in the public domain or released under an open licence, so that anyone may use, adapt and redistribute them at no cost. Open textbooks, open courseware and openly licensed videos and images are examples. OER are not sold only through bookshops, restricted to enrolled students, or confined to print.

ICT does not guarantee learning

ICT can present content through several media at once; that is true. But it does not automatically ensure better learning. Its effect depends on how the teacher designs and uses it. Poorly used slides or videos can make learners passive. A capability is not a guarantee.

CLASSROOM EXAMPLE

A Commerce teacher at a university college in Sirsa sets up a Moodle course for her M.Com. class. Readings and short screencasts are uploaded each week; a ten-question quiz scores itself; the grade book shows who has not attempted it. Class time, freed from dictating notes, goes to case discussion. When a chartered accountant in Gurugram agrees to speak, the class joins him by video conference and puts questions live. The students then build a shared glossary of taxation terms on a wiki, and she uses openly licensed diagrams in her slides, crediting each source.

OFTEN CONFUSED

- **Webinar vs podcast.** A webinar is live (synchronous); a podcast is listened to on demand (asynchronous).
- **Wiki vs blog.** A wiki is edited jointly by many users; a blog is a journal of posts, usually by one author.
- **Video conferencing vs webcast.** Video conferencing is two-way among several places; a webcast goes one way, from one source to many.
- **LMS vs file sharing.** Both store files; only the LMS tracks and reports each learner's progress.
- **Open source vs free of charge.** Moodle is open source (its code may be modified and shared). Google Classroom may cost nothing but is proprietary.
- **Smart classroom vs online class.** A smart classroom is a face-to-face class with digital tools; an online class meets over the internet.
- **OER vs free-to-read.** OER may be used, adapted and redistributed under an open licence; a free-to-read web page may still forbid copying and adaptation.

FACT WATCH

- **Moodle** is distributed as free, open-source software under version 3 of the GNU General Public License, and its own source code gives the expansion "Modular Object-Oriented Dynamic Learning Environment" (checked October 2026).
- **Creative Commons licences**: the current version is 4.0. Older materials may carry version 2.5 or 3.0 licences, which use the same letters with the same broad meanings (checked October 2026).

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Creative Commons licence elements

Most OER use **Creative Commons** (CC) licences. Each licence is built from four elements:

Code	Name	Condition
BY	Attribution	You must give appropriate credit to the creator, link to the licence and indicate if changes were made
SA	ShareAlike	If you adapt the material, you must distribute your version under the same licence
NC	NonCommercial	You may not use the material for commercial purposes
ND	NoDerivatives	You may not distribute modified versions

So a resource under CC BY-SA may be adapted by a teacher provided credit is given and the adaptation carries the same licence. It contains no bar on commercial use (that would need NC) and no bar on changes (that would need ND). Because the licence already grants permission, no individual written permission is needed.

The SAMR model

The **SAMR model**, proposed by Puentedura, describes four levels of technology use in teaching:

Level	Meaning and example
Substitution	Technology replaces an older tool with no functional change: students type an essay on a word processor instead of writing by hand, and nothing else changes
Augmentation	Technology replaces a tool with functional improvement: the word processor's spell-check and easy revision
Modification	Technology allows the task to be significantly redesigned: essays are shared online and revised after peer comments
Redefinition	Technology allows new tasks that were previously inconceivable: students produce a multimedia documentary for an online audience

The first two levels are said to enhance a task; the last two to transform it.

Designing slides and multimedia: Mayer's principles

Mayer's research on multimedia learning gives principles for designing slides and videos. Two are often tested.

- **Coherence principle**: people learn better when extraneous material is left out. Interesting but irrelevant pictures, sounds or words, however attractive, compete for attention and reduce learning.
- **Modality principle**: people learn better from graphics with spoken narration than from graphics with printed text on screen, because eye and ear then share the load.

So "adding interesting but irrelevant pictures improves learning" is false, and "learners understand a diagram better with on-screen text than with narration" is also false. Both principles follow from the limited capacity of working memory (see 5.1).

Modes of computer-assisted instruction

Computer-assisted instruction (कंप्यूटर सहायक अनुदेशन, CAI) is commonly described in these modes:

- **Tutorial**: teaches new material through dialogue with the learner, presenting short units and asking questions. When it sends a learner to remedial material after a wrong answer, it follows the principle of **branching programming**, associated with Crowder, in which the path depends on the learner's response. A **linear** programme, associated with Skinner, has one fixed path for all (see 4.2).

- **Drill and practice**: rehearses material already taught.
- **Simulation**: models a real system so that learners can experiment with it.
- **Instructional games**: practice within rules and competition.

Self-learning materials in open and distance learning

In open and distance learning, the printed or digital **self-learning material** (स्व-अधिगम सामग्री, SLM) has to do the teacher's work, because the teacher is not present. It is therefore designed to be:

- **self-explanatory**: clear without outside help;
- **self-contained**: needing no other sources to follow;
- **self-directed**: telling the learner what to do and in what order;
- **self-motivating**: arousing and keeping interest;
- **self-evaluating**: letting the learner check progress, through in-text "check your progress" questions with model answers at the end of the unit.

In-text questions with answers serve the self-evaluating feature. Self-contained means that the unit needs no outside sources; it is a different feature.

CHECK YOURSELF

1. Which of the following is an open-source learning management system?

(A) Blackboard
(B) Moodle
(C) Microsoft Teams
(D) Google Classroom

2. Match List I with List II:

List I	List II
A. Webinar	I. Jointly edited web pages with a history of changes
B. Podcast	II. Live online seminar
C. Wiki	III. Web journal of dated posts
D. Blog	IV. Audio episodes for listening on demand

Choose the correct answer from the options given below:

(A) A-II, B-IV, C-III, D-I
(B) A-II, B-IV, C-I, D-III
(C) A-IV, B-II, C-I, D-III
(D) A-II, B-I, C-IV, D-III

3. Assertion (A): A learning management system makes the teacher unnecessary.

Reason (R): A learning management system can score objective quizzes automatically.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. A resource is released under CC BY-NC. A teacher may:

(A) use it in a commercial coaching course without permission
(B) adapt it for her class, giving credit, without commercial use
(C) use it only if she adds no changes at all
(D) use it only after obtaining written permission

5. Students first used a word processor only in place of handwriting; now they post drafts online and revise them after peer comments. In the SAMR model they have moved from:

(A) Augmentation to Redefinition
(B) Substitution to Augmentation
(C) Modification to Redefinition
(D) Substitution to Modification

6. Given below are two statements:

Statement I: Removing decorative but irrelevant images from slides tends to improve learning.

Statement II: An animation is understood better with on-screen text than with spoken narration.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

7. Which of the following are synchronous tools?

1. Video conferencing
2. Podcast
3. Webinar
4. Blog

Choose the correct answer from the options given below:

(A) 1 and 2 only
(B) 2 and 4 only
(C) 1 and 3 only
(D) 1, 3 and 4 only

8. In self-learning material, the feature that the unit can be followed without consulting any other source is called:

(A) self-evaluating
(B) self-motivating
(C) self-directed
(D) self-contained

Answers to “Check yourself”: Chapter 5

Section 5.1

1. (C), because television addresses both sight and hearing. A tape recorder is audio, and a flannel board and silent filmstrip are visual.

2. (C), because the order up the cone is dramatised experiences, study trips, motion pictures, visual symbols, verbal symbols.

3. (D), because the percentages are not Dale’s and have no research basis, so A is false; R correctly describes the cone.

4. (A), because contrived experiences are enactive, exhibits iconic and verbal symbols symbolic, and the three modes are Bruner’s (1966).

5. (D), because the epidiascope, overhead transparency and filmstrip all need projection; flash cards and the flannel board are seen directly.

6. (A), because overload of limited working memory is the mechanism by which too many aids harm learning.

7. (B), because the order from the base is direct, contrived, dramatised, then demonstrations.

8. (C), because the base is direct experience, but Dale warned that too much reliance on concrete experience can obstruct generalisation.

Section 5.2

1. (C), because a globe has length, breadth and depth. Maps, photographs and bar charts are flat.

2. (A), because both statements are correct definitions.

3. (D), because a diorama is a miniature scene, a mock-up a working replica for training, a cut-away model shows the inside, and a specimen is a real sample.

4. (A), because cheapness, independence from power and teacher control are exactly why traditional aids stay useful, so R explains A.

5. (B), because the board suits content that unfolds at the class’s pace and can be altered; prepared images, tables and recordings suit projected or digital aids.

6. (B), because the build-up comes from cut-outs sticking to flannel; not needing a projector is true but explains nothing about building up.

7. (B), because the board needs no electricity and is written on during the lesson, so statements 2 and 4 are false.

Section 5.3

1. (A), because the episcope takes opaque material, the slide projector small slides, the OHP large writable transparencies, and the LCD projector computer output.

2. (C), because the OHP gives a bright image in a lit room with the teacher facing the class. Opaque objects need an episcope, and the OHP needs power and shows no motion.

3. (A), because the episcope part, using reflected light, is what lets the epidiascope project an opaque page, so R explains A.

4. (B), because the language laboratory gives individual practice in listening and speaking.

5. (D), because the episcope projects opaque material by reflected light (Statement I describes the diascope), and the OHP has largely given way to the LCD projector.

6. (A), because radio and tape recorder address hearing only. Television is audio-visual and the filmstrip is visual.

Section 5.4

1. (B), because Moodle is free, open-source software; Blackboard is commercial and Teams and Google Classroom are proprietary.

2. (B), because a webinar is a live seminar, a podcast is on-demand audio, a wiki is jointly edited, and a blog is a journal of posts.

3. (D), because automation of routine tasks is real, but design, feedback, discussion and individual support still need the teacher.

4. (B), because BY requires credit and NC forbids commercial use; there is no ND, so adaptation is allowed, and the licence itself gives permission.

5. (D), because replacing handwriting with no change is Substitution, and redesigning the task through online peer review is Modification.

6. (C), because the coherence principle supports removing extraneous material, while the modality principle favours narration over on-screen text.

7. (C), because video conferencing and webinars are live; podcasts and blogs are used at any time.

8. (D), because self-contained means needing no outside sources; check-your-progress questions serve the self-evaluating feature.

Chapter 6
Evaluation Systems

The last part of Unit I in the syllabus reads: “Evaluation Systems: Elements and Types of evaluation, Evaluation in Choice Based Credit System in Higher education, Computer based testing, Innovations in evaluation systems.” This chapter follows that sentence phrase by phrase.

“Elements of evaluation” is taught in 6.1 (what measurement, assessment and evaluation are, and how they fit together). The tools that carry out evaluation, and what makes them good, follow in 6.2 (qualities of a good test), 6.4 (tools, test items and item analysis) and 6.5 (how to read and calculate test scores).

“Types of evaluation” is taught in 6.3 (formative, summative, diagnostic, placement, norm- and criterion-referenced) and continued in 6.6 (continuous and comprehensive evaluation, and grading).

“Evaluation in Choice Based Credit System in Higher education” is 6.7. “Computer based testing” is 6.8. “Innovations in evaluation systems” is 6.9.

Sections 6.2, 6.4 and 6.5 contain calculations. Each is taught as a method: the steps, a worked example, the usual slips, and practice questions that become harder as you go. Read them with a pencil in hand.

6.1 Elements of evaluation

Syllabus: Evaluation Systems: Elements and Types of evaluation

> **EXAM FOOTPRINT**
>
> Asked occasionally; usually a two-statement item distinguishing measurement, assessment and evaluation.

Core

Four words that look alike: test, measurement, assessment, evaluation

Examiners often give you four words that sound similar and ask which is narrowest, which is broadest, or which one a statement illustrates. Learn them as a ladder.

A **test** (परीक्षण) is an instrument: a set of questions or tasks given to the learner. A question paper is a test. A test is a thing, not a process.

Measurement (मापन) is the process of assigning numbers to a performance according to fixed rules. When the teacher marks the paper and writes 68 out of 100, she has measured. Measurement answers the question “how much?”. It is quantitative (मात्रात्मक). It does not say whether 68 is good or bad.

Testing and **scoring** are simply the ways in which that number is obtained: giving the test, then marking it. They belong to measurement.

Assessment (आकलन) is wider than measurement. It means gathering evidence about learning and making sense of it. The evidence may be numerical (marks) or descriptive (a teacher’s notes on how a student argued in a seminar). Tests, observation, assignments and projects are all ways of assessing.

Evaluation (मूल्यांकन) is the widest term. It uses the evidence of measurement and assessment and adds a **value judgement** (मूल्य-निर्णय): a judgement of worth, made against objectives or standards. “Her 68 per cent shows very good progress this term” is evaluation, because it judges.

Term	What it is	Example
Test	An instrument: a set of tasks	The internal assessment question paper
Measurement	Assigning numbers by rules; answers "how much?"	"Ravi scored 68 out of 100."
Assessment	Gathering and interpreting evidence, numerical and descriptive	Marks plus notes on Ravi's seminar presentation
Evaluation	Judging worth against objectives; includes the others	"Ravi's 68 is a good performance for his class and meets the course objective."

Two orders are asked. From narrowest to broadest: measurement, assessment, evaluation. From instrument to process to judgement: test, measurement, evaluation. In both, evaluation comes last, because it includes what comes before it.

A short formula that many books give is: evaluation = measurement + value judgement. A fuller form, which Indian textbooks attribute to Norman E. Gronlund, is:

> Evaluation = quantitative description (measurement) + qualitative description (non-measurement) + value judgements

The words to notice are "qualitative description". Evaluation does not use numbers alone. An **anecdotal record** of how a student behaved in a group discussion, or a teacher's observation of her laboratory work, is qualitative evidence. Evaluation uses both kinds, and the judgement of worth is what turns description into evaluation. So if a question asks what Gronlund adds to the two descriptions, the answer is value judgements, not statistical analysis or norms (statistics belong to the quantitative description).

Two traps follow from this. First, measurement is never "judgemental" or "qualitative"; it is quantitative. Second, evaluation is never narrower than measurement; it is the broader term, because it contains measurement and qualitative description and judgement. Appraisal is another word for judging worth, so it sits with evaluation, not with measurement.

The evaluation triangle and the steps of evaluation

Evaluation is not an event at the end of a course. It is tied to what the teacher set out to do. This idea comes from Ralph W. Tyler, whose book *Basic Principles of Curriculum and Instruction* (1949) asked four questions: what purposes should education seek, what learning experiences can attain them, how can those experiences be organised, and how can we tell whether the purposes are being attained. Indian books often present the same triad (objectives, learning experiences, evaluation) as Bloom's tripolar view of the educative process; see Section 1.1.

Indian textbooks draw this as the **evaluation triangle** (मूल्यांकन त्रिभुज), with three corners:

- **Objectives** (उद्देश्य): the changes in the learner that the teacher wants to bring about.
- **Learning experiences** (अधिगम अनुभव): the teaching and activities provided to bring those changes about.
- **Evaluation** (मूल्यांकन): finding out how far the changes have actually taken place.

Each corner affects the other two. Objectives decide what experiences are planned and what is evaluated. Evaluation results show whether the experiences worked, and may lead the teacher to revise the objectives or the experiences. The triangle is not "teacher, learner, curriculum" (that is a different idea, about the parties to teaching) and not "input, process, output" (that describes any system).

From the triangle comes the order of the steps of evaluation:

1. Formulating the objectives (in terms of the behaviour expected of the learner).

2. Providing learning experiences suited to those objectives.
3. Applying the tools of evaluation (tests, observation, assignments) to collect evidence.
4. Interpreting the results, and using them to improve teaching.

Objectives always come first, because without them there is nothing to judge against.

Functions of evaluation

The results of evaluation are used for different purposes. Four are commonly listed:

- **Instructional function**: feedback to the teacher on how well the teaching worked and to the learner on what still needs work.
- **Administrative function** (प्रशासनिक कार्य): decisions about promotion, selection, certification and the award of degrees.
- **Guidance function** (निर्देशन कार्य): helping a student make educational and vocational choices, for example which elective courses suit her strengths.
- **Research function**: supplying data for studies of teaching, curricula and learners.

To tell them apart, ask who uses the result and for what. If the result helps a student choose a course or a career, it is guidance. If it improves the teaching of the current course, it is instructional. If it decides promotion, it is administrative.

Why every measurement contains error

Learning cannot be seen directly. We cannot open a student's mind and count what she knows. We give her a sample of tasks and infer her learning from her performance. So educational measurement is **indirect**.

Because it is indirect, it always contains some error. The test samples only some topics; the student may be tired or anxious; two examiners may mark differently. The score the student actually gets is called the **obtained score** (प्राप्त अंक). The score she would get if there were no error at all is the **true score** (सत्य प्राप्तांक). Classical test theory puts it simply:

obtained score = true score + error

No educational measurement is free of error. That is why we need the ideas of reliability and of the standard error of measurement, which are taught in 6.2.

> **CLASSROOM EXAMPLE**
>
> At a government college in Sirsa, Dr Meena gives her BA Part II English class a 25-mark test on *Macbeth*. She marks the scripts: Arjun gets 18 (measurement). She also notes in her register that Arjun spoke well in the class discussion on the witches but could not support his points with quotations (qualitative description). She writes on his script: "Good grasp of the play; now practise using evidence from the text" (value judgement). Together these make evaluation. Because her objective at the start of the unit was "students will support an interpretation with evidence from the text", she can see that this objective has not yet been met for Arjun, and she plans one more class on quoting. Objectives, learning experiences and evaluation have worked as a triangle.

OFTEN CONFUSED

- Measurement vs evaluation: measurement gives a number (“72 out of 100”) and stops; evaluation adds a judgement of worth (“very good progress”). If the statement only reports a quantity, it is measurement.
- Assessment vs evaluation: assessment gathers and interprets evidence; evaluation judges the worth of what the evidence shows. Evaluation includes assessment.
- Test vs measurement: a test is the instrument; measurement is the process of assigning numbers to performance on it.
- Evaluation triangle vs teaching triangle: the evaluation triangle is objectives, learning experiences, evaluation; teacher, learner, curriculum is a different idea.
- Instructional vs guidance function: improving teaching and learning of the present course is instructional; helping a student choose courses or careers is guidance.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

More on the qualitative side

Qualitative tools used in evaluation include anecdotal records, observation schedules, interviews, portfolios and self-reports. They are taught in 6.4. Remember only that a score, a percentile rank or a standard score is quantitative, while an anecdotal record is qualitative.

Measurement error, again

Three main sources of error appear in questions: the sampling of content (the paper happens to ask what the student did or did not revise), the condition of the examinee (illness, anxiety, fatigue), and scoring (different examiners, or the same examiner on different days). Because every obtained score carries error, an obtained score should be read as an estimate. Put together: a test score is an indirect measure from which learning is inferred, and for that very reason it always contains some error. Both halves of that sentence are true.

Evaluating programmes: the CIPP model

So far we have spoken of evaluating students. Programmes, courses and institutions are evaluated too. A well-known model for this is the **CIPP model** of Daniel Stufflebeam, developed in the United States in the 1960s. Its four letters name four kinds of evaluation:

Letter	What is evaluated	Question it answers
Context	Needs, problems and goals of the setting	What needs to be done?
Input	Resources, plans and strategies available	How should it be done?
Process	Implementation of the plan	Is it being done as planned?
Product	Outcomes and results	Did it succeed?

The CIPP model is called decision-oriented, because each part feeds a decision for those who manage the programme. Do not confuse it with invented expansions such as “Content, Instruction, Practice, Performance” or “Curriculum, Input, Pedagogy, Product”. Only Context, Input, Process and Product are correct. Tyler’s approach, by contrast, is objectives-based: it asks how far the stated objectives have been reached.

CHECK YOURSELF

1. Which of the following statements is an example of evaluation, not merely of measurement?

(A) Sunita answered 42 of the 60 items correctly.
(B) Sunita took 55 minutes to complete the paper.
(C) Sunita's score of 70 shows that she has met the unit's objectives well.
(D) Sunita obtained 70 marks out of 100.

2. Arrange the following in order from the narrowest to the broadest in scope:

1. Evaluation
2. Measurement
3. Assessment

Choose the correct answer from the options given below:

(A) 3, 2, 1
(B) 2, 3, 1
(C) 2, 1, 3
(D) 1, 3, 2

3. Given below are two statements:

Statement I: Measurement assigns numbers to performance according to rules and does not judge whether the performance is good or bad.

Statement II: Evaluation may use qualitative information such as anecdotal records along with measurement.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

4. Match List I (function of evaluation) with List II (use of the result):

List I	List II
A. Instructional	I. Deciding which students are promoted to the next semester
B. Administrative	II. Advising a student on a suitable elective
C. Guidance	III. Data for a study of reading habits of first-year students
D. Research	IV. Telling the teacher which topic must be re-taught

Choose the correct answer from the options given below:

(A) A-II, B-I, C-IV, D-III
(B) A-I, B-IV, C-II, D-III
(C) A-IV, B-II, C-I, D-III
(D) A-IV, B-I, C-II, D-III

5. Given below are two statements, one labelled Assertion (A) and the other Reason (R):

Assertion (A): Every score on an educational test contains some error.

Reason (R): Learning is inferred from performance on a sample of tasks and is not observed directly.

In the light of the above statements, choose the correct answer:

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

6. Arrange the steps of the evaluation process in the correct order:

1. Applying tools of evaluation
2. Formulating objectives
3. Interpreting results
4. Providing learning experiences

Choose the correct answer from the options given below:

(A) 2, 1, 4, 3
(B) 4, 2, 1, 3
(C) 2, 4, 1, 3
(D) 2, 4, 3, 1

7. In the CIPP model of programme evaluation, the letter "I" stands for:

(A) Instruction
(B) Implementation
(C) Interpretation
(D) Input

6.2 Qualities of a good test

Syllabus: Evaluation Systems: Elements and Types of evaluation

EXAM FOOTPRINT

Asked occasionally; usually a match of test qualities (validity, reliability, objectivity) with definitions or ways of checking them.

Core

The four qualities

A good test has four main qualities: **validity** (वैधता), **reliability** (विश्वसनीयता), **objectivity** (वस्तुनिष्ठता) and **usability** (व्यावहारिकता), also called practicability. Standardised tests also have **norms** (मानक): tables of how a reference group performed, so that a score can be compared with that group. Learn one sentence for each quality.

Quality	The question it asks	A quick test of the idea
Validity	Does the test measure what it claims to measure?	A reasoning test that mainly rewards fast reading is not valid.
Reliability	Does it give consistent results?	The same students get similar scores if tested again under similar conditions.

Quality	The question it asks	A quick test of the idea
Objectivity	Is the score free of the examiner's personal judgement?	Two examiners give the same mark to the same answer.
Usability	Can it be given, scored and interpreted without undue time, cost or difficulty?	A valid test that costs too much to give is of little use.

Validity is the most important quality. A test that measures the wrong thing is useless for its purpose, however consistent, objective or cheap it may be.

Validity and its types

Validity always means: does the test measure what it is meant to measure? Anything that lets something else affect the scores lowers validity. Suppose a mathematics test is written in long, difficult English. Students weak in English lose marks for a reason that has nothing to do with mathematics. The test is now partly a reading test. Its validity falls, even though its scoring key is perfectly clear (objectivity is unaffected).

There are several kinds of evidence of validity. Learn each with its question.

- **Content validity** (विषयवस्तु वैधता): do the items represent the course content and objectives in proper proportion? It is ensured by preparing a **blueprint** (a table that spreads marks over topics and objectives) before the items are written, and judged by checking the items against the syllabus or blueprint.
- **Criterion-related validity**: do the scores agree with some other measure (a criterion) of the same ability? It has two kinds, which differ only in timing.
- **Concurrent validity** (समवर्ती वैधता): the criterion is available now. A new short reading test is given in the same week as a well-established reading test, and the two sets of scores correlate highly.
- **Predictive validity** (भविष्यसूचक वैधता): the criterion lies in the future. Scores on an entrance or aptitude test are later found to forecast the students' grades in the course. Predictive validity is essential for entrance and aptitude tests.
- **Construct validity** (संरचना वैधता): do the scores reflect the theoretical quality (the construct) that the test claims to measure, such as intelligence, creativity or anxiety? A construct cannot be seen directly, so this evidence is built up through correlations with related measures and through factor analysis.
- **Face validity** (आभासी वैधता): does the test look, on the surface, as if it measures what it should, to candidates and lay observers? This is an impression, not validity in the technical sense. It matters because it affects how seriously candidates take the test.

Remember the timing pair with two words: "now" is concurrent, "later" is predictive.

Reliability and how it is estimated

Reliability is consistency. A reliable test gives similar results when the same students are tested again under similar conditions, or when the test is scored by another examiner, or when another version of the test is used.

Reliability is usually expressed as a **reliability coefficient** (विश्वसनीयता गुणांक). It is found by correlating two sets of scores, and it runs from 0 (no consistency) to 1 (perfect consistency). The nearer to 1, the more reliable the test. It is not a percentage of marks, a percentile or a grade point.

The methods differ in what two sets of scores are correlated:

Method	What is done	What it shows
Test-retest (परीक्षण-पुनर्परीक्षण विधि)	The same test is given twice to the same group, with a gap between the two sittings	Coefficient of stability: steadiness of scores over time
Parallel forms or equivalent forms (समानांतर प्रारूप विधि)	Two equivalent versions of the test are given to the same group	Coefficient of equivalence; if given after an interval, stability and equivalence together
Split-half (अर्ध-विच्छेद विधि)	The test is given once, divided into two halves (usually odd and even items), and the halves are correlated; the Spearman-Brown formula then gives the reliability of the whole test	Internal consistency
Kuder-Richardson (rational equivalence)	The test is given once; formulas KR-20 and KR-21 use the item data of a test scored 1 or 0	Internal consistency
Inter-rater or scorer reliability	Two or more examiners independently mark the same scripts	Agreement between scorers

A sorting rule helps. Two administrations: test-retest (same test twice) and parallel forms (two versions). One administration: split-half and Kuder-Richardson. Scorers: inter-rater.

Longer tests are more reliable. A longer test samples the ability more fully, so the chance errors on single items tend to cancel out. Other things being equal, adding good items raises reliability. Very short tests are usually unreliable. The Spearman-Brown formula expresses this link between length and reliability (see Going Further).

How reliability and validity are related

Learn this pair of sentences: **a test can be reliable without being valid, but it cannot be valid without being reliable.**

Picture a weighing scale that always reads two kilograms too much. It is consistent (reliable), but it gives the wrong weight (not valid). In the same way, a test may give very consistent scores while it measures memory for trivial details instead of the reasoning it claims to measure. It is reliable but not valid.

The reverse, “valid but not reliable”, cannot occur. If a test gives inconsistent scores, they are largely error, and a test made largely of error cannot be measuring its target accurately. So reliability is **necessary but not sufficient** for validity.

Objectivity and usability

Objectivity means that the score does not depend on who marks the paper. Different examiners, or the same examiner at different times, give the same score to the same answer. Multiple-choice items are highly objective, because the key decides the mark. Essay answers are less objective, because marking needs judgement. Objectivity supports reliability, since scorer bias is one source of inconsistent scores.

Usability or practicability covers the practical side: time needed, cost, ease of giving and scoring, clear instructions, and ease of interpreting the results.

CLASSROOM EXAMPLE

The Department of History at a university in Haryana is choosing an entrance test for its MA programme. The committee asks four questions. Do the scores of past entrants forecast their first-semester grades (predictive validity)? Does the paper sample all the BA topics in proportion, as the blueprint says (content validity)? Would a candidate get a similar score if she sat a parallel version a week later (reliability)? Can 600 candidates be tested in two hours and the answers machine-scored (usability and objectivity)? A paper with long essays only would fail the last two questions; a paper of 100 well-made multiple-choice items with a careful blueprint would pass all four.

OFTEN CONFUSED

- Reliability vs validity: reliability is consistency; validity is measuring the right thing. Reliable but not valid is possible; valid but not reliable is not.
- Objectivity vs reliability: objectivity is about the scorer; reliability is about the consistency of scores more generally. Objectivity is one support of reliability.
- Content vs face validity: content validity is judged systematically against a blueprint; face validity is only the surface impression.
- Concurrent vs predictive validity: both compare the test with a criterion; the criterion is available now for concurrent and later for predictive.
- Construct vs content validity: content validity asks whether the course is sampled; construct validity asks whether the theoretical trait is reflected.
- Test-retest vs parallel forms: one test given twice gives stability; two equivalent versions give equivalence.
- Split-half vs Kuder-Richardson: both use one administration and give internal consistency; split-half correlates two halves and needs the Spearman-Brown correction, while KR uses item-level right/wrong data.
- Standard error of measurement vs standard deviation: SEM is the error band around one person's score; SD is the spread of scores across the group.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Method 1: the Spearman-Brown formula (split-half reliability)

When a test is split into halves, the correlation between the halves is the reliability of a half-length test. The full test is twice as long, so it is more reliable. The formula, published in 1910 by Charles Spearman and William Brown in separate papers in the *British Journal of Psychology*, corrects for this.

Steps:

1. Find r, the correlation between the two halves.
2. Full-test reliability = 2r / (1 + r).

WORKED EXAMPLE

Example 1. The odd-item and even-item halves of a test correlate at 0.70. Find the reliability of the whole test.

Full-test reliability = 2 x 0.70 / (1 + 0.70)

= 1.40 / 1.70

= 0.82 (to two decimal places)

The general form is for a test made n times as long: reliability = n r / (1 + (n - 1) r).

Example 2. A test has reliability 0.50. It is made three times as long with similar items. Predict the new reliability.

New reliability = 3 x 0.50 / (1 + 2 x 0.50)

= 1.50 / 2.00

= 0.75

Method 2: the Kuder-Richardson formulas

G. F. Kuder and M. W. Richardson published these formulas in *Psychometrika* in 1937. They use a single administration of a test whose items are scored 1 (right) or 0 (wrong). The symbols are: k = number of items; p = proportion answering an item correctly; q = 1 - p; s^2 = variance of the total scores (the square of their standard deviation); M = mean of the total scores.

KR-20 = [k / (k - 1)] x [1 - (sum of pq over all items) / s^2]

KR-21 = [k / (k - 1)] x [1 - M (k - M) / (k s^2)]

KR-21 is a shortcut. It needs only the mean and the variance, and it assumes that all items are of equal difficulty. When items differ in difficulty, KR-21 comes out lower than KR-20. KR-20 is a special case of Cronbach's alpha, the coefficient used when items are not scored simply right or wrong.

WORKED EXAMPLE

Example 3 (KR-21). A 40-item test has a mean of 24 and a standard deviation of 6, so s^2 = 36.

k / (k - 1) = 40 / 39 = 1.026

M (k - M) / (k s^2) = 24 x 16 / (40 x 36) = 384 / 1440 = 0.267

KR-21 = 1.026 x (1 - 0.267) = 1.026 x 0.733 = 0.75 (to two decimal places)

Example 4 (KR-20). Six students take a 5-item test. Their totals are 5, 4, 3, 2, 1 and 0. The proportions correct on the five items are 5/6, 4/6, 3/6, 2/6 and 1/6.

Mean of totals = 2.5; variance s^2 (dividing by 6) = 2.917

sum of pq = 0.139 + 0.222 + 0.250 + 0.222 + 0.139 = 0.972

KR-20 = (5/4) x (1 - 0.972 / 2.917) = 1.25 x 0.667 = 0.83

KR-21 for the same data = (5/4) x (1 - 2.5 x 2.5 / (5 x 2.917)) = 0.71, lower than KR-20 because the items differ in difficulty

Method 3: the standard error of measurement

Every obtained score contains error (6.1). The **standard error of measurement** (मापन की मानक त्रुटि), SEM, tells how far a person's obtained scores would spread around her true score if she could be tested again and again. It turns the reliability coefficient into a band of marks.

Steps:

1. SEM = SD x square root of (1 - r), where SD is the standard deviation of the test scores and r is the reliability coefficient.
2. Obtained score plus or minus 1 SEM gives a band within which the true score probably lies (about 68 times in 100).
3. Obtained score plus or minus 2 SEM (strictly 1.96 SEM) gives a band with about 95 per cent confidence.

WORKED EXAMPLE

Example 5. A test has SD = 10 and reliability 0.91. Ravi obtains 62.

SEM = 10 x square root of (1 - 0.91) = 10 x square root of 0.09 = 10 x 0.3 = 3

68 per cent band: 62 - 3 to 62 + 3, that is, 59 to 65

95 per cent band: 62 plus or minus 1.96 x 3 = 62 plus or minus 5.88, that is, about 56 to 68

The higher the reliability, the smaller the SEM. If r were 1, SEM would be 0: no error at all.

Strictly speaking, the band is built around the obtained score, so it tells us where the true score probably lies, not that it certainly lies there.

Method 4: the ceiling on validity

A test cannot correlate with any criterion more highly than it correlates with itself. The square root of the reliability coefficient is called the **index of reliability**, and it is the highest validity coefficient the test can reach.

WORKED EXAMPLE

Example 6. A test has reliability 0.81. Its validity can be at most the square root of 0.81 = 0.90. With reliability 0.49, the ceiling is 0.70; with 0.36, it is 0.60.

This is a ceiling only. Actual validity coefficients are usually much lower.

USUAL SLIPS

- Reporting the half-test correlation as the reliability of the whole test. The full test is always more reliable than its halves (0.60 becomes 0.75).
- Writing 2r / (1 + 2r) or 2r / 2 instead of 2r / (1 + r).
- Forgetting the square root in the SEM: with r = 0.91 the error factor is the square root of 0.09, which is 0.3, not 0.09.
- Taking the reliability itself (0.64) as the ceiling on validity; the ceiling is its square root (0.80).
- Confusing SD (spread across a group) with SEM (error around one person’s score).

When split-half goes wrong: speed tests

A **power test** has items of increasing difficulty and enough time for everyone to attempt all of them. A **speed test** has easy items and too little time, so the score depends on how many items the student reaches. On a pure speed test a student who reaches 37 items gets about 19 odd items and 18 even items right. The two halves are almost identical for every student, so the odd-even correlation is close to 1 whatever the real reliability. (A computer simulation of 200 such students, run for this guide, gave an odd-even correlation above 0.99.) Split-half therefore gives a spuriously

high reliability on speed tests. Test-retest or parallel forms should be used for speeded tests. Split-half is suitable for power tests.

Why the group matters: restriction of range

A reliability coefficient is not fixed for a test. It depends on how widely the scores of the group are spread. In classical test theory, reliability is the share of true-score variance in the total variance of scores:

reliability = 1 - (error variance / total variance)

Suppose the error variance is 9 in every group. In a mixed-ability group the total variance is 90, so reliability = 1 - 9/90 = 0.90. In a group of only top students, the scores bunch together and the total variance is 22.5, so reliability = 1 - 9/22.5 = 0.60. The test has not changed; the group has. This is **restriction of range**. A homogeneous group gives a lower coefficient.

Validity threats named by Messick

In a well-known account of validity, Messick (1989) named two main threats to validity:

- **Construct under-representation**: the test is too narrow and leaves out important parts of what it should measure. A research methodology course aims at designing studies, but its examination asks only for definitions.
- **Construct-irrelevant variance**: scores are affected by something outside the construct. A mathematics test in difficult English measures reading ability too; an essay test may reward handwriting.

In short: something essential missing is under-representation; something extra measured is irrelevant variance.

The trade-off in performance assessment

Suppose a teacher-education programme replaces a written test on methods of teaching with an observed classroom lesson judged on a rubric. Observing real teaching gives direct evidence of the skill, so validity rises. But one lesson is a small sample, observers differ, and observation takes much time, so reliability and practicability fall. Rubrics, trained observers and several observations reduce the loss. The general rule: performance tasks are stronger on validity and weaker on reliability; objective tests are the reverse.

CHECK YOURSELF

1. A test is given once, divided into odd and even items, and the two half-scores are correlated at 0.80. By the Spearman-Brown formula, the reliability of the whole test is about:

(A) 0.80
(B) 0.40
(C) 0.89
(D) 0.64

2. Match List I (method) with List II (what it yields):

List I	List II
A. Test-retest	I. Agreement between independent examiners
B. Parallel forms	II. Coefficient of stability
C. Kuder-Richardson	III. Coefficient of equivalence
D. Inter-rater	IV. Internal consistency from one administration

Choose the correct answer from the options given below:

(A) A-III, B-II, C-IV, D-I
(B) A-II, B-III, C-IV, D-I
(C) A-II, B-III, C-I, D-IV
(D) A-IV, B-III, C-II, D-I

3. Given below are two statements, one labelled Assertion (A) and the other Reason (R):

Assertion (A): A test can be highly reliable and still have low validity.

Reason (R): Consistent scores do not by themselves show that a test measures what it is meant to measure.

In the light of the above statements, choose the correct answer:

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. Scores on a college's aptitude test in June are compared with the same students' grades at the end of the first year. The evidence sought is:

(A) concurrent validity
(B) content validity
(C) face validity
(D) predictive validity

5. A test has a standard deviation of 8 and a reliability coefficient of 0.75. A student obtains 50. Within which band does her true score lie, with about 68 per cent confidence?

(A) 48 to 52
(B) 46 to 54
(C) 42 to 58
(D) 44 to 56

6. Consider the following statements:

1. The split-half method needs only one administration of the test.

2. The split-half method is the best choice for a pure speed test.
3. Adding more good items to a test generally raises its reliability.
4. A reliability coefficient tends to fall when the group tested is very homogeneous.

Choose the correct answer from the options given below:

(A) 1, 3 and 4 only
(B) 1 and 2 only
(C) 2, 3 and 4 only
(D) 1, 2, 3 and 4

7. A test has a reliability coefficient of 0.49. The highest validity coefficient it can have is:

(A) 0.24
(B) 0.70
(C) 0.49
(D) 0.98

8. Given below are two statements:

Statement I: A history examination that asks only for dates, though the course aims at historical argument, suffers from construct under-representation.

Statement II: Face validity is established by correlating test scores with a later criterion.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

6.3 Types of evaluation

Syllabus: Evaluation Systems: Elements and Types of evaluation

EXAM FOOTPRINT

Asked in most cycles; usually a scenario asking which type of evaluation a teacher is using, or a match of types with purposes.

Core

Evaluation is classified in two main ways. The first asks **when and why** it is done: before, during or after teaching, and for what purpose. This gives placement, formative, diagnostic and summative evaluation. The second asks **against what** a score is compared: other students, a fixed standard, or the student's own earlier work. This gives norm-referenced, criterion-referenced and ipsative evaluation. Keep the two classifications apart in your mind.

By timing and purpose: four types

Placement evaluation (स्थापन मूल्यांकन) is done **before** instruction. It finds the learner's entry behaviour: what she already knows and can do, and whether she has the prerequisites for the course. Its results are used to decide where she should begin, for example a beginner, intermediate or advanced group in a language course. A placement pre-test answers two questions: does the learner have the prerequisites, and has the learner already achieved the outcomes of the unit?

Learners who have already mastered the unit should skip ahead or be given enrichment or advanced work. Teaching them the same unit again wastes their time.

Formative evaluation (रचनात्मक मूल्यांकन) is done **during** instruction, while teaching and learning are still going on. Its purpose is feedback: to tell the learner how she is progressing and the teacher how well the teaching is working, so that both can improve before the course ends. It is often informal: oral questions, class quizzes, short class tests, observation, and comments on drafts returned for revision. It is not meant for final grades.

Diagnostic evaluation (निदानात्मक मूल्यांकन) looks for the **causes** of learning difficulties that persist in spite of ordinary teaching and feedback. Formative evaluation shows that a student is weak in, say, subtraction; diagnostic evaluation finds out why, using specially designed tests and close observation. Diagnosis is not an end in itself. It is followed by **remedial teaching** (उपचारात्मक शिक्षण) aimed at exactly the cause found.

Summative evaluation (योगात्मक मूल्यांकन) is done at the **end** of a unit, term or course. It sums up how far the objectives have been achieved. Its results are used for grades, promotion and certification. It is usually formal and high-stakes: a final examination, a year-end examination, a degree viva voce.

Type	When	Purpose	Typical tools or uses
Placement	Before instruction	Find entry behaviour; decide starting point and grouping	Pre-test, entry test
Formative	During instruction	Monitor progress; feedback to learner and teacher	Quiz, oral questions, comments on drafts, observation
Diagnostic	During instruction, when difficulties persist	Find causes of persistent difficulties; plan remedy	Specially designed diagnostic test, interview
Summative	End of unit, term or course	Certify achievement; grade; promote	Term-end or year-end examination, final viva

Norman E. Gronlund, as Indian B.Ed. texts quote him, put the formative-diagnostic difference in a picture: formative evaluation gives “first aid” for simple learning problems, while diagnostic evaluation searches for the underlying causes of problems that do not respond to first aid.

Use decides the label

The same instrument can be formative or summative. What decides the label is how the results are used, not what the test looks like.

- A mid-semester unit test given only to find weak areas for re-teaching is **formative**, even though it looks like an examination.
- A term-end examination used to decide promotion is **summative**.
- A mid-term test returned with detailed comments, whose marks also count for part of the final grade, is **both**: the comments are a formative use and the marks a summative use.

So when a question asks “which distinguishes formative from summative evaluation?”, the answer is the use of results to modify instruction still in progress. External setting, whole-course coverage and final grades are typical of summative evaluation but do not define formative evaluation.

Do not confuse diagnostic with prognostic

A **diagnostic test** locates the specific difficulties of a learner and their causes now. A **prognostic test** (पूर्वानुमानात्मक परीक्षण) forecasts how well a person is likely to do in a subject or course in the future;

"prognosis" means forecast. It is used in guidance and selection. An **aptitude test** measures potential for a particular field, and an **intelligence test** measures general mental ability.

By frame of reference: norm, criterion, self

A raw score needs something to be compared with. There are three possible yardsticks.

A **norm-referenced test** (मानक-संदर्भित परीक्षण), NRT, interprets a score by comparing it with the scores of other students in a group. Results are given as ranks, percentiles or standard scores. "Rina scored higher than 85 per cent of her class" is a norm-referenced statement. Competitive examinations for a fixed number of seats work this way: the success of one candidate depends on how others perform.

A **criterion-referenced test** (निकष-संदर्भित परीक्षण), CRT, compares performance with a fixed standard or a defined set of tasks, whatever others do. "Rina solved 17 of the 20 problems", "Rina typed 40 words a minute" and "Rina crossed the pass standard of 50 per cent" are criterion-referenced statements. A driving-licence test is the common example: everyone who meets the standard passes, however many candidates there are.

Ipsative assessment (स्व-संदर्भित आकलन), also called self-referenced assessment, compares a learner with her own earlier performance. A teacher who compares each student's present essay with the essay she wrote two months earlier, and grades the improvement, is using ipsative assessment.

Point	Norm-referenced	Criterion-referenced	Ipsative
Compared with	Other learners	A fixed standard or task domain	The learner's own earlier work
Typical report	Rank, percentile rank	Mastery or non-mastery; percentage of objectives mastered	Progress or gain
Example	Admission of the top 100 in an entrance test	Driving test; licence for all who reach 70 per cent	Improvement since the last assignment
What floats	The cut-off mark floats with rank	The number who pass floats	Nothing is compared with others

Two consequences are asked often. First, certifying that a learner has mastered minimum competencies needs a criterion-referenced test. A norm-referenced test is built to spread scores out for ranking, and a high rank does not show that the competencies have been mastered. Second, a percentile rank is norm-referenced, while the percentage of objectives mastered is criterion-referenced. The two words sound alike but belong to opposite systems.

Many selection examinations combine both. "Qualified candidates must score at least 40 per cent, and must be among the top 6 per cent of those who do" uses a criterion (the fixed 40 per cent, which any number may reach) and then a norm (the top 6 per cent, which depends on rank).

Assessment for, of and as learning

These three phrases are now common. Lorna Earl used all three in her book *Assessment as Learning* (2003).

- **Assessment for learning** (अधिगम के लिए आकलन): the teacher gathers evidence during learning and uses it to adjust teaching. This is formative assessment.
- **Assessment of learning** (अधिगम का आकलन): records and reports what has been achieved at the end. This is summative assessment.

- **Assessment as learning** (अधिगम के रूप में आकलन): the students themselves monitor their work against criteria, judge their own drafts, and set goals. It is self-assessment and self-regulation.

The deciding question between "for" and "as" is: who uses the evidence? If the teacher uses it to adjust teaching, it is for learning. If the students use it to regulate their own learning, it is as learning.

Internal and external evaluation

Internal evaluation (आंतरिक मूल्यांकन) is carried out by the teachers of the institution itself, often continuously through the term. **External evaluation** (बाह्य मूल्यांकन) is carried out by examiners or an agency from outside, usually at the end. External evaluation is usually summative; internal evaluation can serve formative and summative purposes.

CLASSROOM EXAMPLE

In the first week of the BA English (Honours) programme at a college in Hisar, Dr Kaur gives a placement test of grammar. Five students already score above 90 per cent; she gives them an advanced writing project in place of the grammar unit. During the unit she runs a five-minute quiz every Friday (formative). One student, Pooja, keeps writing "he goed" and "she taked" even after feedback. Dr Kaur sets her a short special task with twenty past-tense verbs and finds that Pooja adds "-ed" to every verb, regular or not (diagnostic). She then teaches irregular verbs in groups (remedial). The end-semester examination gives the final grade (summative). The university's rule that a student must get at least 40 per cent to pass is criterion-referenced; the gold medal for the top scorer is norm-referenced.

OFTEN CONFUSED

- Formative vs summative: formative improves learning while it is going on; summative certifies achievement at the end. The use of results decides the label, not the instrument.
- Formative vs diagnostic: formative shows that a weakness exists and gives quick feedback; diagnostic finds the cause of a persistent difficulty and leads to remedial teaching.
- Diagnostic vs prognostic: diagnostic finds causes of present errors; prognostic forecasts future performance.
- Placement vs diagnostic: placement looks at entry behaviour before instruction to decide the starting point; diagnosis searches for causes of difficulties that persist.
- Norm-referenced vs criterion-referenced: comparison with other learners vs comparison with a fixed standard. In NRT the cut-off floats; in CRT the number passing floats.
- Percentile rank vs percentage of objectives mastered: norm-referenced vs criterion-referenced.
- Criterion-referenced vs ipsative: a fixed external standard vs the learner's own earlier work.
- Assessment for learning vs assessment as learning: the teacher uses the evidence vs the students use it themselves.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Who introduced the terms

Michael Scriven introduced the terms formative and summative evaluation in 1967, in his essay "The methodology of evaluation". He was writing about the evaluation of **curricula and programmes**: formative evaluation improves a curriculum while it is being developed; summative evaluation judges the finished curriculum. **Bloom, Hastings and Madaus**, in their *Handbook on Formative and Summative Evaluation of Student Learning* (1971), applied the terms to **student learning**, especially within mastery learning, where short formative tests are given after each learning unit. So the order is: Scriven, curriculum; Bloom and colleagues, student learning. An option that reverses the order is wrong.

Robert Glaser introduced the distinction between norm-referenced and criterion-referenced measurement in a 1963 article in the *American Psychologist* on instructional technology and the measurement of learning outcomes. He argued that new forms of instruction needed tests that showed whether a learner had reached a specified level of performance, not merely her rank. **Ralph W. Tyler** is known for the objectives-based approach to evaluation (6.1). Do not give Tyler or Glaser credit for formative and summative, or Scriven credit for criterion-referenced measurement.

How NRT and CRT differ in construction

Because a norm-referenced test exists to rank people, it needs a wide spread of scores. Its items are chosen to be of middle difficulty and to discriminate well between able and weak students. An item that every examinee answers correctly (or that nobody answers correctly) adds nothing to the differences between people and is dropped from an NRT.

A criterion-referenced test exists to show mastery. Its items are chosen to represent the tasks in the defined domain. If every student answers an item correctly, that may simply show that teaching succeeded, and the item is kept. Results are reported as mastery or non-mastery. After effective teaching, CRT scores are not normally distributed; they bunch at the high end (skewed towards high scores). A normal spread is what an NRT aims at, not a CRT.

How is a CRT cut-off fixed? Often by expert judgement of what a minimally competent candidate can do. In the **Angoff method** (Angoff, 1971), each judge estimates, for each item, the probability that a just-competent candidate will answer it correctly; the estimates are summed and averaged across judges to give the cut-off. A rank-based quota works the opposite way: the proportion who succeed is fixed in advance, and the cut-off mark is whatever the last selected candidate scored.

Three further points:

- A criterion-referenced test can be summative. A final mastery test is both.
- Norm-referencing is a way of interpreting scores, not a property of standardised tests alone. A classroom teacher who ranks her class is interpreting her own test in a norm-referenced way.
- The same scores can be given both interpretations. A score of 70 per cent can be read against a pass mark (criterion) or converted to a percentile rank (norm).

Purpose and user decide the label, again

A term-end examination decides the students' grades, so for them it is summative. If the department later analyses the item-wise results and uses them to redesign the teaching of the course for the next batch, the same results serve a formative purpose for the course. The examination is summative for these students but formative for the course.

The steps of diagnosis, and reading error patterns

Educational diagnosis follows a fixed order:

1. Identify the students who are having difficulty.
2. Locate the specific areas of difficulty.

3. Find the causes of the difficulty.
4. Plan remedial instruction.

The cause cannot be looked for until the area has been located, and remedy cannot be planned until the cause is known. Remedy without diagnosis only repeats the teaching that has already failed.

A diagnostic test contains **many items on each narrow sub-skill**, so that a pattern of errors can be identified reliably for each sub-skill. Diagnostic tests are generally not used for awarding grades; that is a separate feature, not the reason for having many items.

Read the pattern of errors. Careless errors are irregular: different mistakes in different places. A **systematic** error is the same wrong procedure applied every time, and it shows a misconception. A student who writes 64 - 27 = 43 and 52 - 18 = 46 is subtracting the smaller digit from the larger in each column (7 - 4 = 3, 6 - 2 = 4; 8 - 2 = 6, 5 - 1 = 4), whichever number it belongs to. That is a misconception about regrouping (“borrowing”), not carelessness and not low ability. The remedy is to re-teach place value and regrouping, for example with concrete material such as bundles of ten sticks. Pooja’s “goed” and “taked” in the example above are a systematic error of the same kind.

Direct and indirect assessment in outcome-based education

In **outcome-based education**, a course states its course outcomes, and the institution checks how far students attain them. Two kinds of tools are used.

- **Direct assessment** (प्रत्यक्ष आकलन) examines actual student performance: tests, assignments, projects, laboratory work, examinations.
- **Indirect assessment** (अप्रत्यक्ष आकलन) gathers opinions about learning: course-exit surveys of students, alumni surveys, employer surveys. A course-exit survey asking students how well they think they achieved the outcomes is indirect, because it records belief, not performance.

Attainment is usually computed mainly from direct tools, with indirect tools as supporting evidence.

> **WORKED EXAMPLE**
>
> Attainment of a course outcome. Target: at least 60 per cent of students should score 50 per cent or more on the questions linked to the outcome. In a class of 90, 63 students do so.
>
> Attainment = 63 / 90 x 100 = 70 per cent
>
> 70 per cent is above the target of 60 per cent, so the outcome is attained.
>
> Keep three numbers apart: the score level (50 per cent), the target (60 per cent of students) and the result (70 per cent of students). The number of students who reached the level (63) is not itself a percentage.

CHECK YOURSELF

1. Match List I (type of evaluation) with List II (purpose):

List I	List II
A. Placement	I. Locating causes of a persistent difficulty
B. Formative	II. Certifying achievement at the end
C. Diagnostic	III. Deciding the starting point before instruction
D. Summative	IV. Feedback while learning is in progress

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-II, D-I
(B) A-IV, B-III, C-I, D-II
(C) A-III, B-IV, C-I, D-II
(D) A-I, B-IV, C-III, D-II

2. Given below are two statements:

Statement I: A university that admits the top 50 scorers in its test, whatever their marks, is using the test in a criterion-referenced way.

Statement II: A board that certifies every candidate who scores 60 per cent or more is using its test in a criterion-referenced way.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

3. Given below are two statements, one labelled Assertion (A) and the other Reason (R):

Assertion (A): A pre-test given before a unit may show that some learners should skip the unit and take up advanced work.

Reason (R): Placement evaluation checks whether learners have already achieved the intended outcomes of the unit.

In the light of the above statements, choose the correct answer:

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. Consider the following statements:

1. The same test can be formative or summative depending on how its results are used.
2. Assessment as learning means that students monitor and regulate their own learning.
3. Summative evaluation is mainly used for daily feedback.
4. Scores on a criterion-referenced test must be normally distributed.

Choose the correct answer from the options given below:

(A) 1 and 2 only
(B) 1, 2 and 3 only

(C) 2 and 4 only
(D) 1, 3 and 4 only

5. Arrange the following steps of educational diagnosis in order:

1. Finding the causes of the difficulty
2. Planning remedial instruction
3. Identifying the students who have difficulty
4. Locating the specific areas of difficulty

Choose the correct answer from the options given below:

(A) 3, 1, 4, 2
(B) 3, 4, 2, 1
(C) 4, 3, 1, 2
(D) 3, 4, 1, 2

6. The terms formative and summative evaluation were first used by Scriven in connection with:

(A) the selection of teachers
(B) the evaluation of curricula and programmes
(C) mastery learning of students
(D) criterion-referenced measurement

7. A teacher tells a student, “Your score this month is twelve marks better than your score last month.” This is:

(A) norm-referenced interpretation
(B) criterion-referenced interpretation
(C) ipsative interpretation
(D) placement evaluation

8. For a course outcome, the target is that 60 per cent of students score at least half marks on the linked questions. In a class of 75, 42 students do so. Which is correct?

(A) Attainment is 56 per cent; the outcome is not attained
(B) Attainment is 42 per cent; the outcome is not attained
(C) Attainment is 56 per cent; the outcome is attained
(D) Attainment is 60 per cent; the outcome is just attained

6.4 Tools and techniques of evaluation; item analysis

Syllabus: Evaluation Systems: Elements and Types of evaluation

EXAM FOOTPRINT

Rarely asked directly; it supports single-answer items on objective versus essay items and what item analysis measures.

Core

Techniques and tools

A **technique** of evaluation is a way of collecting evidence: testing, observation, self-report, interview, peer rating. A **tool** is the instrument used in that technique: a test, a checklist, a rating scale, a

rubric, an anecdotal record, a cumulative record, a portfolio, a questionnaire. Some tools give **quantitative** evidence (test scores, percentile ranks, standard scores). Others give **qualitative** evidence (anecdotal records, observation notes, portfolios). Evaluation uses both (6.1).

Objective-type and essay-type items

An **objective-type item** (वस्तुनिष्ठ प्रश्न) has one predetermined correct answer, so every examiner gives the same mark. Multiple-choice, true-false, matching and completion items are objective. Many can be answered in a short time, so they sample a wide syllabus, and they are quick and objective to score.

An **essay-type item** (निबंधात्मक प्रश्न) asks for a long written answer that the student selects, organises and expresses in her own words. Only an essay shows whether a student can select, organise, integrate and express ideas, so essays can test higher abilities such as analysis and evaluation. Their two weaknesses are **limited sampling** (a paper has only a few questions, so it covers the syllabus narrowly) and **subjective scoring** (marks vary from examiner to examiner).

Point	Objective-type	Essay-type
Sampling of content	Wide (many items)	Narrow (few questions)
Scoring	Objective, quick	Subjective, slow
What it tests best	Knowledge, understanding, application; with care, higher abilities	Organising and expressing ideas; argument
Guessing	Possible, especially true-false	Little room
Hard part for the teacher	Writing good items	Marking fairly

So objective tests are superior in sampling, objectivity of scoring and speed of scoring. Essay tests are superior in measuring organisation and expression of ideas. Note the common trap: essay tests do not sample the syllabus more widely; they sample it more narrowly.

Objective items are of two kinds:

- **Supply-type** or recall items: the student writes a word, number or phrase. Completion (fill in the blank) and short-answer items.
- **Selection-type** or recognition items: the student chooses from given answers. True-false, matching and multiple-choice items.

The parts of an item

A **multiple-choice item** (बहुविकल्पीय प्रश्न) has three parts. The **stem** poses the problem. The **key** is the correct option. The **distractors** (विकर्षक) are the plausible wrong options, meant to attract those who do not know the answer.

A **matching item** has two columns. The entries of the first column, for which matches are to be found, are the **premises**. The entries of the second column, from which answers are chosen, are the **responses**. Giving more responses than premises reduces guessing by elimination. Stem, key and distractor belong to MCQs; premise and response belong to matching items.

Guessing and the limits of each item type

On a true-false item a blind guess has a 50 per cent chance of being right, because there are only two choices. A four-option MCQ reduces this to 25 per cent; a five-option MCQ to 20 per cent. Supply-type items (essay, short answer, completion), where the student must write the answer, leave little room for guessing.

Item type	Characteristic limitation
True-false	A blind guess succeeds half the time
Matching	Suits only homogeneous material: sets of similar, related facts
Essay	Scoring varies from examiner to examiner
Multiple-choice	Writing several plausible distractors for each item is hard

Rules for writing multiple-choice items

- The stem should present a single, clearly stated problem.
- The distractors should be plausible to a student who does not know the answer.
- The key should be placed at random among the positions, with no regular pattern.
- Options should be similar in length, grammar and detail, so the key does not stand out.
- Options such as “all of the above” are best avoided: a candidate who recognises two correct options can choose it without knowing the rest.

The blueprint and the steps of test construction

A **blueprint** (नील पत्र), also called a **table of specifications**, is a two-way table prepared before the items are written. One side lists the content units; the other lists the objectives (for example knowledge, understanding, application), and often the forms of question. Each cell shows how many marks go to that unit and objective. The blueprint makes the test sample the course in proper proportion, so it supports content validity (6.2). A marking scheme and a scoring key are prepared later, after the items are written.

The steps of constructing an achievement test run in this order:

1. Planning: objectives, content and blueprint.
2. Writing the items.
3. Trying out the test on a sample of students.
4. Item analysis of the try-out responses.
5. Assembling the final form, keeping good items and revising or dropping weak ones.

Item analysis needs response data, so it can never come before the try-out.

Observation tools: checklist, rating scale, rubric

- A **checklist** (जाँच सूची) records only whether a feature is present or absent: yes or no.
- A **rating scale** (निर्धारण मापनी) records the degree of each feature on a numerical or verbal scale (1 to 5; poor to excellent).
- A **rubric** goes further. For each criterion it describes what performance at each level looks like. An **analytic rubric** scores each criterion separately; a **holistic rubric** gives one overall judgement.

The test between a rating scale and a rubric is the descriptor. An instrument with criteria and a five-point scale, but no description of what each point means, is a rating scale.

Records: anecdotal, cumulative, portfolio

- An **anecdotal record** (उपाख्यानात्मक अभिलेख) is a teacher’s brief written description of a single significant incident in a student’s behaviour. It is qualitative.
- A **cumulative record** (संचयी अभिलेख) is a comprehensive, year-by-year account of a student kept by the institution: attendance, health, test scores, activities, teachers’ remarks. It is used for guidance.

- A **portfolio** is a collection of the student's own selected work, usually with her reflections on it.
- A **sociogram** is a chart of the choices that members of a group make about one another (whom they would like to work with), showing who is popular, isolated, or part of a clique.

What item analysis is

Item analysis (पद विश्लेषण) is done after a test has been tried out. For each item it finds:

- the **difficulty index** (कठिनाई सूचकांक), also called the **facility value**: the proportion of examinees who answer the item correctly;
- the **discrimination index** (विभेदन सूचकांक): how well the item separates high scorers from low scorers on the whole test;
- how well each **distractor** is working.

Read the facility value carefully. A facility value of 0.90 means 90 per cent answered correctly: the item is **very easy**. A value of 0.10 means a very difficult item. The name "difficulty index" misleads many students: the higher the value, the easier the item. Items with a facility value near 0.50 can discriminate best; very easy and very hard items discriminate poorly, because nearly everyone (or nearly no one) gets them right.

In a norm-referenced test, an item that every examinee answers correctly contributes nothing to the differences between students, so it is dropped. In a criterion-referenced test the same item may be kept, because it may simply show that teaching succeeded (6.3).

CLASSROOM EXAMPLE

Dr Rao prepares a 50-item objective test in Indian Polity for BA Part I at a college in Rohtak. She first draws a blueprint: five units down the side, knowledge, understanding and application across the top. She writes the items, tries them out on last year's students, and analyses each item. Item 17 has a facility value of 0.95: almost everyone got it right, so she makes it harder. In Item 23 more weak students than strong students chose the key, so she reads it again and finds that the stem is ambiguous. In Item 31 no one chose option (C), so she writes a more plausible distractor. Then she assembles the final test.

OFTEN CONFUSED

- Stem, key, distractor vs premise, response: the first three are parts of an MCQ; the last two are the columns of a matching item.
- Supply-type vs selection-type: the student writes the answer (completion, short answer) vs chooses it (true-false, matching, MCQ).
- Blueprint vs marking scheme: the blueprint is planned before items are written and secures content validity; the marking scheme is prepared for the items once written.
- Checklist vs rating scale vs rubric: present or absent; degree on a scale; degree with a description of each level.
- Anecdotal vs cumulative record vs portfolio: one incident; the institution's year-by-year record; the student's own selection of work.
- Facility value vs discrimination index: how many got it right vs whether the able got it right more often than the weak.
- High difficulty index vs difficult item: a high facility value means an easy item.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Method 1: facility value and discrimination index

Steps:

1. Score all the scripts and arrange them in order of total score.
2. Take the top 27 per cent as the **upper group** and the bottom 27 per cent as the **lower group**. Set the middle aside. T. L. Kelley (1939) showed that upper and lower groups of 27 per cent give the most efficient comparison. For 200 examinees, each group has 0.27 x 200 = 54 scripts; for 300, each has 81. When a question gives the sizes of the groups, simply use them.
3. For each item count RU (number right in the upper group) and RL (number right in the lower group). Let n be the number in one group.
4. Facility value FV = (RU + RL) / (2n), that is, the number right in both groups divided by the total in both groups.
5. Discrimination index DI = (RU - RL) / n, the difference divided by the size of **one** group. Equivalently, DI = proportion right in the upper group minus proportion right in the lower group.

WORKED EXAMPLE

Example 1. Upper and lower groups have 30 students each. An item is answered correctly by 27 of the upper group and 15 of the lower group.

FV = (27 + 15) / (30 + 30) = 42 / 60 = 0.70

DI = (27 - 15) / 30 = 12 / 30 = 0.40

A fairly easy item that discriminates well. Keep it.

Example 2. Same groups. Upper group 12 right, lower group 18 right.

FV = (12 + 18) / 60 = 30 / 60 = 0.50

DI = (12 - 18) / 30 = -6 / 30 = -0.20

A negative index: more low scorers than high scorers got it right. Something is wrong with the item: an ambiguous stem, a wrong key or a misleading option. Revise or drop it.

Example 3. Same groups. Upper group 29 right, lower group 28 right.

FV = 57 / 60 = 0.95; DI = 1 / 30 = 0.03

Very easy and almost no discrimination. In a norm-referenced test it adds little.

How to read the discrimination index. It runs from -1 to +1. A value near zero means high and low scorers did equally well: the item fails to separate able from weak and needs revision. A negative value is a warning sign. A widely quoted rule of thumb, from the work of Ebel, is: 0.40 and above, very good; 0.30 to 0.39, reasonably good but may be improved; 0.20 to 0.29, marginal, usually needs revision; 0.19 and below, poor, to be revised or rejected.

Why items near 0.50 discriminate best. With equal groups, the largest possible DI for an item is twice the smaller of FV and (1 - FV). Python gives these ceilings:

Facility value	Highest possible DI
0.10 or 0.90	0.20
0.30 or 0.70	0.60

Facility value	Highest possible DI
0.50	1.00
1.00	0.00

USUAL SLIPS

- Dividing RU - RL by both groups together (60) instead of one group (30). In Example 1 that gives 12 / 60 = 0.20 instead of 0.40.
- Giving the discrimination index when the facility value is asked, or the reverse.
- Reading a high difficulty index as a hard item. FV 0.90 is very easy.
- Counting those who got the item wrong: 30 right out of 100 gives FV 0.30, not 0.70.
- Taking 25 per cent when the question says 27 per cent: 27 per cent of 200 is 54, not 50.

Method 2: distractor analysis

A good distractor attracts more low scorers than high scorers, because it draws those who do not know the answer. Two warning signs:

- A distractor chosen by **more high scorers than low scorers** may be defensible as a correct answer, or the item may be wrongly keyed. Review the item.
- A distractor chosen by **more examinees than the key** is over-attractive; check it for ambiguity. It is certainly functioning, but perhaps too well.

A distractor that almost nobody chooses is **non-functional**: it does no work. A common rule, following Haladyna and Downing (1993), counts a distractor as non-functional if fewer than 5 per cent of examinees choose it.

WORKED EXAMPLE

Example 4. Forty students (upper 20, lower 20) answer a four-option item whose key is A.

Option	Upper group	Lower group	Total	Per cent of 40
A (key)	15	6	21	52.5
B	0	1	1	2.5
C	1	10	11	27.5
D	4	3	7	17.5

FV = 21 / 40 = 0.525; DI = (15 - 6) / 20 = 0.45

5 per cent of 40 = 2 students. B, chosen by 1 student (2.5 per cent), is non-functional; rewrite it.

C works well: 10 low scorers, 1 high scorer.

D draws more high scorers (4) than low scorers (3). Check whether D could be defended as correct.

Method 3: correction for guessing and the chance score

On objective tests a student may guess. The usual **correction for guessing** (formula scoring) assumes that every wrong answer came from a blind guess:

Corrected score S = R - W / (n - 1)

Here R = number right, W = number wrong, n = number of options per item. **Omitted items are not counted**: they are neither rewarded nor penalised.

The **chance score** is what a candidate would be expected to get by blind guessing on every item:

Chance score = number of items x 1 / (number of options)

WORKED EXAMPLE

Example 5. Five-option items. A student has 50 right, 20 wrong and leaves 10 blank.

S = 50 - 20 / (5 - 1) = 50 - 5 = 45

Example 6. True-false items (n = 2). A student has 60 right and 20 wrong.

S = 60 - 20 / 1 = 40

Example 7. Chance scores. 100 five-option items: 100 x 1/5 = 20. 40 true-false items: 40 x 1/2 = 20. 60 four-option items: 60 x 1/4 = 15.

A score close to the chance score shows no real knowledge.

USUAL SLIPS

- Subtracting all the wrong answers (R - W) on a four-option test. The divisor is n - 1 = 3.
- Dividing by n instead of n - 1.
- Counting omitted items as wrong.
- Using the wrong number of options for the chance score: 10 would be the chance score for 50 five-option items, not four-option items (12.5).

Two flaws that help the test-wise

When the key is noticeably longer and more carefully qualified than the distractors, test-wise students learn to choose the longest option. The item then rewards test-wiseness instead of knowledge, which lowers its validity, and weak but test-wise students succeed, which lowers its discrimination. The cure is options of similar length, grammar and detail. “All of the above” is a similar giveaway.

Interpretive exercises and cloze tests

An **interpretive exercise** gives introductory material (a passage, a table, a graph, a map) followed by several objective items that ask the examinee to interpret, infer and judge. Because the facts are supplied, the items test thinking, not recall. The reading comprehension and data interpretation sets of Paper 1 are interpretive exercises. A **cloze test** also uses a passage, but deletes words at regular intervals; the student fills the gaps. It measures reading comprehension, not a set of interpretive questions.

CHECK YOURSELF

1. On an item, 24 of the 40 students in the upper group and 8 of the 40 in the lower group answer correctly. The facility value and discrimination index are, respectively:

(A) 0.40 and 0.20
(B) 0.40 and 0.40
(C) 0.80 and 0.40
(D) 0.32 and 0.16

2. Match List I (term) with List II (description):

List I	List II
A. Premise	I. Plausible wrong option of an MCQ
B. Distractor	II. Entry in the first column of a matching item
C. Stem	III. The correct option
D. Key	IV. Part of an MCQ that poses the problem

Choose the correct answer from the options given below:

(A) A-II, B-III, C-IV, D-I
(B) A-I, B-II, C-IV, D-III
(C) A-IV, B-I, C-II, D-III
(D) A-II, B-I, C-IV, D-III

3. Given below are two statements, one labelled Assertion (A) and the other Reason (R):

Assertion (A): Essay tests are superior to objective tests in measuring the ability to organise and express ideas.

Reason (R): Essay tests sample the content of a course more widely than objective tests.

In the light of the above statements, choose the correct answer:

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. On a test of four-option items, a student has 36 right, 15 wrong and 9 omitted. The score corrected for guessing is:

(A) 21
(B) 31
(C) 33
(D) 36

5. Arrange the steps of constructing an achievement test in order:

1. Trying out the test
2. Preparing the blueprint
3. Assembling the final form
4. Writing the items
5. Item analysis

Choose the correct answer from the options given below:

(A) 2, 4, 1, 5, 3

(B) 4, 2, 1, 5, 3
(C) 2, 4, 5, 1, 3
(D) 2, 1, 4, 5, 3

6. Consider the following statements:

1. A rating scale without a description of each level is not a rubric.
2. A cumulative record is a student's own selection of her best work.
3. An anecdotal record describes a single significant incident.
4. A checklist records only whether a feature is present or absent.

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 2 and 4 only
(C) 1, 3 and 4 only
(D) 1, 2, 3 and 4

7. An item analysis is to be done for 400 examinees using the upper and lower 27 per cent. How many scripts are there in each group?

(A) 108
(B) 27
(C) 100
(D) 216

8. Item analysis shows that option (B) of an item, which is not the key, was chosen by 12 high scorers and 3 low scorers. The most likely conclusion is that:

(A) option (B) is a well-functioning distractor
(B) the item is too difficult for the group
(C) option (B) may be defensible or the item may be wrongly keyed
(D) option (B) is non-functional and should be removed

6.5 Reading test scores: a method chapter

Syllabus: Evaluation Systems: Elements and Types of evaluation

EXAM FOOTPRINT

Rarely asked directly; it supports Data Interpretation and Mathematical Reasoning items on averages and spread.

Core

A raw score means little by itself

A **raw score** (अपरिष्कृत प्राप्तांक) is the number of marks a student actually gets. A raw score of 35 tells us almost nothing. Out of 40 or out of 100? Was the paper easy or hard? Did the others score 20 or 38? Was the pass mark 33? A raw score needs a **frame of reference** before it can be interpreted: either a **norm** (how others performed) or a **criterion** (a fixed standard). These are the two frames you met in 6.3. This section teaches the numbers used with the norm frame: averages, spread, percentile ranks, standard scores and correlation.

Method 1: mean, median and mode

These are measures of **central tendency** (केंद्रीय प्रवृत्ति): each gives one number that represents the general level of a group.

- The **mean** (माध्य) is the arithmetic average: add all the scores and divide by how many there are.
- The **median** (माध्यिका) is the middle score when the scores are arranged in order. With an even number of scores, it is the average of the two middle scores.
- The **mode** (बहुलक) is the score that occurs most often.

WORKED EXAMPLE

Example 1. Ten students score: 12, 15, 15, 18, 20, 22, 24, 25, 28, 31.

Mean = sum / number = 210 / 10 = 21

Median: the scores are already in order; the two middle ones (5th and 6th) are 20 and 22; median = (20 + 22) / 2 = 21

Mode = 15 (it occurs twice)

Example 2. Seven students score: 34, 41, 28, 50, 45, 39, 47.

First arrange in order: 28, 34, 39, 41, 45, 47, 50

Median = the 4th score = 41

Example 3. Why the median is sometimes better. Seven scores: 40, 42, 45, 47, 48, 50, 95.

Mean = 367 / 7 = 52.4; median = 47

One very high score (95) pulls the mean up; the median is not affected. When a few extreme scores are present, the median represents the group better.

Method 2: the standard deviation

Two classes can have the same mean and still be very different. One may be made of students who are all about average; another may mix very weak and very strong students. The **standard deviation** (मानक विचलन), SD, measures this spread: roughly, how far scores lie from the mean on average. A larger SD means a more varied group. (The range, the highest score minus the lowest, is a rougher measure of spread.)

Steps:

1. Find the mean.
2. Subtract the mean from each score to get its deviation.
3. Square each deviation.
4. Add the squares and divide by the number of scores. This is the **variance**.
5. Take the square root of the variance. This is the SD.

WORKED EXAMPLE

Example 4. The ten scores of Example 1 (mean 21).

Deviations: -9, -6, -6, -3, -1, 1, 3, 4, 7, 10

Squares: 81, 36, 36, 9, 1, 1, 9, 16, 49, 100; sum = 338

Variance = 338 / 10 = 33.8

SD = square root of 33.8 = 5.81

Some books divide by (N - 1) instead of N when the group is a sample from a larger population; that gives 338 / 9, and SD = 6.13. Examination questions at this level usually give the SD or use the simple form.

How to read it. Two classes have the same mean of 60 on an entry test. In Class A the SD is 5; in Class B it is 15. Class A is fairly homogeneous: most students are near 60. Class B contains both very weak and very strong students. A lesson pitched at the average will miss many of them, so **Class B needs more differentiated instruction**. Equal means hide unequal spread.

Method 3: percentile and percentile rank

A **percentile rank** (शततमक कोटि), PR, is the percentage of the group whose scores fall below a given score. A **percentile** (शततमक) is the reverse: the score below which a given percentage of the group falls. If Rina's percentile rank is 80, then 80 per cent of the group scored below her; her score is the 80th percentile of that group.

WORKED EXAMPLE

Example 5. In a class of 40, 30 students score below Arjun.

PR = 30 / 40 x 100 = 75

In a group of 60, 18 score below Meena: PR = 18 / 60 x 100 = 30.

A percentile rank is a **norm-referenced** measure: it tells the position of a student among others. It is not the percentage of marks. Percentage of marks compares a score with the maximum possible; percentile rank compares it with other examinees. On a hard paper a student with 45 per cent marks may stand at the 80th percentile. "PR 80" does not mean that the student answered 80 per cent of the items correctly, nor that exactly 80 students scored lower.

CLASSROOM EXAMPLE

In the internal assessment of MA Hindi at a university in Kurukshetra, Section A and Section B both average 18 out of 30. Dr Yadav finds that the SD is 2 in Section A and 6 in Section B. In Section B she forms small groups: some students work on basic reading of the prescribed poems, others on comparative criticism. Section A, much more uniform, she teaches mostly as a whole class. She also tells the students their percentile ranks within the class, and explains that "PR 70" means 70 per cent of the class scored below them, not that they got 70 per cent marks.

OFTEN CONFUSED

- Percentile rank vs percentage of marks: position among others vs share of the maximum marks.
- Percentile vs percentile rank: a score vs a percentage of the group below a score.
- Mean vs median: the arithmetic average, pulled by extreme scores, vs the middle score, not affected by them.
- Mean vs standard deviation: general level vs spread.
- Standard deviation vs standard error of measurement: spread across the group vs error around one person's score (6.2).
- Correlation vs causation: a correlation shows that two things go together, not that one causes the other.
- Correlation coefficient vs shared variance: r = 0.60 means shared variance $r^2 = 0.36$, not 60 per cent.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Scales of measurement

S. S. Stevens, in a 1946 paper in *Science* ("On the theory of scales of measurement"), described four levels of measurement (मापन के स्तर). Each level has all the properties of the one before it, plus one more.

Scale	What it allows	Example
Nominal	Labels categories only; no order	Roll numbers, gender, stream (Arts, Science)
Ordinal	Order, but unequal or unknown differences	Ranks: first, second, third in class
Interval	Order and equal units, but no true zero	Temperature in degrees Celsius
Ratio	Equal units and a true (absolute) zero	Height, weight, time taken

Ranks are ordinal. The gap between first and second need not equal the gap between second and third, so ranks are not interval.

Test scores are not on a ratio scale. A ratio scale needs a true zero that means complete absence of the quality. A test samples only part of a subject, so a score of zero does not mean zero knowledge, and a score of 80 does not show twice the knowledge of a score of 40. Scores do show order (60 is more than 40); what they lack is an absolute zero.

Method 4: standard scores, z and T

A **standard score** (मानक प्राप्तांक) re-expresses a raw score as its distance from the mean in units of the SD. It lets us compare scores from different tests.

$$z = (X - \text{mean}) / SD$$

$$T = 50 + 10z$$

A z-score has a mean of 0 and an SD of 1. Scores above the mean have positive z, scores below have negative z. A T-score has a mean of 50 and an SD of 10, which removes the minus signs and most decimals.

WORKED EXAMPLE

Example 6. A test has mean 60 and SD 8.

X = 72: z = (72 - 60) / 8 = 12 / 8 = 1.5; T = 50 + 10 x 1.5 = 65

X = 52: z = (52 - 60) / 8 = -8 / 8 = -1; T = 50 + 10 x (-1) = 40

X = 60: z = 0; T = 50

Example 7. Which is the better performance? Kiran scores 72 in English (class mean 66, SD 4) and 80 in History (class mean 70, SD 10).

English: z = (72 - 66) / 4 = 1.5

History: z = (80 - 70) / 10 = 1.0

Her raw mark is higher in History, but relative to her classmates she did better in English.

Standard scores are norm-referenced: they say where a student stands in the group.

Combining marks from different examiners: scaling

Suppose Examiner A's marks have mean 70 and SD 5, and Examiner B's marks, for a comparable group, have mean 50 and SD 15. If raw marks are simply added, the examiner whose marks vary more decides the order of merit. Converting each set to standard scores first puts both on a common scale. This is the idea behind **scaling** or **moderation**.

WORKED EXAMPLE

Example 8. Two students, P and Q.

P: A gives 75, B gives 50. Raw total = 125. z(A) = (75 - 70) / 5 = 1.0; z(B) = (50 - 50) / 15 = 0. Sum of z = 1.0

Q: A gives 65, B gives 65. Raw total = 130. z(A) = (65 - 70) / 5 = -1.0; z(B) = (65 - 50) / 15 = 1.0. Sum of z = 0

Raw totals put Q above P. Standard scores put P above Q. The raw ranking was decided by Examiner B's wider spread.

The normal curve

Many sets of scores, when there are enough of them, are close to the bell-shaped **normal distribution** (प्रसामान्य वितरण): symmetrical, with mean, median and mode at the centre. The areas under the curve, computed in Python for this guide, are:

Region	Share of cases
Between the mean and 1 SD on one side	34.13 per cent
Within 1 SD on either side of the mean	68.27 per cent (about 68)
Within 2 SD on either side	95.45 per cent (about 95)
Within 3 SD on either side	99.73 per cent (about 99.7)
Below +1 SD (z = 1)	84.13 per cent
Beyond +2 SD on one side	2.28 per cent

WORKED EXAMPLE

Example 9. Suppose IQ scores in a population are normally distributed with mean 100 and SD 15.

IQ 85 to 115 is -1 SD to +1 SD: about 68 per cent of people.

IQ 70 to 130 is -2 SD to +2 SD: about 95 per cent.

IQ 115 is z = +1, so about 84 per cent of people lie below it, a percentile rank of about 84.

Method 5: correlation and shared variance

Correlation (सहसंबंध) shows how far two sets of scores go together. The correlation coefficient r runs from -1 to +1. A positive r means high scores on one go with high scores on the other; a negative r means high goes with low; r near 0 means no linear relation. Reliability and criterion validity coefficients (6.2) are correlations.

Steps for the product-moment correlation coefficient:

1. Find the mean of X and the mean of Y.
2. Find each deviation from its mean: dx and dy.
3. Multiply each pair: dx x dy, and add the products.
4. Square the dx values and add them; square the dy values and add them.
5. r = (sum of dx dy) / square root of [(sum of dx^2) x (sum of dy^2)].

WORKED EXAMPLE

Example 10. Five students: hours in the library per week (X) and marks out of 10 (Y).

Student	X	Y	dx	dy	dx dy
1	1	2	-2	-2	4
2	2	4	-1	0	0
3	3	3	0	-1	0
4	4	6	1	2	2
5	5	5	2	1	2

Mean X = 3, mean Y = 4

Sum of dx dy = 8; sum of dx^2 = 10; sum of dy^2 = 10

r = 8 / square root of (10 x 10) = 8 / 10 = 0.80

Shared variance = r^2 = 0.64, or 64 per cent

Two cautions. First, **correlation is not causation**. A correlation of +0.60 between attendance and marks shows that the two go together, moderately strongly. It does not show that attendance causes the marks: motivated students may both attend more and study more. Second, r is not a percentage. The share of variation in one variable that goes with variation in the other is the **shared variance**, r^2. For r = 0.60 it is 0.36 (36 per cent); for r = 0.50 it is 0.25; for r = -0.60 it is again 0.36.

USUAL SLIPS

- Dividing the raw score by the SD without first subtracting the mean: (65 / 10) is not a z-score.
- Forgetting the minus sign for scores below the mean.
- Writing T = 10 + 50z instead of T = 50 + 10z.
- Reading a percentile rank as a percentage of marks.
- Calling r = 0.60 “60 per cent shared”, or “60 per cent caused by”. Shared variance is r^2 = 0.36.
- Taking the 34 per cent (mean to 1 SD on one side) as the share within 1 SD on both sides (68 per cent).
- Not arranging scores in order before finding the median.

CHECK YOURSELF

1. Five students score 4, 7, 7, 9 and 13. The mean, median and mode are, respectively:

(A) 7, 7, 8
(B) 8, 7, 7
(C) 8, 9, 7
(D) 7, 8, 7

2. A test has mean 50 and SD 4. A student who scores 58 has a z-score and a T-score of, respectively:

(A) 2.0 and 70
(B) 8.0 and 58
(C) 2.0 and 52
(D) 14.5 and 195

3. Given below are two statements, one labelled Assertion (A) and the other Reason (R):

Assertion (A): A raw score of 42 cannot be interpreted until we know how others performed or what standard was set.

Reason (R): Raw scores must always be converted to T-scores before they are reported.

In the light of the above statements, choose the correct answer:

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

4. Match List I (example) with List II (scale of measurement):

List I	List II
A. Hostel room numbers	I. Ratio
B. Merit list positions	II. Interval
C. Temperature in degrees Celsius	III. Nominal
D. Time taken to finish a race	IV. Ordinal

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-II, D-I
(B) A-IV, B-III, C-II, D-I
(C) A-III, B-II, C-IV, D-I
(D) A-III, B-IV, C-I, D-II

5. In a normal distribution, approximately what percentage of cases lies between 2 SD below and 2 SD above the mean?

(A) 34 per cent
(B) 68 per cent
(C) 99.7 per cent
(D) 95 per cent

6. The correlation between scores on a reading test and a writing test is 0.70. Which statement is correct?

(A) Reading ability causes 70 per cent of writing ability.

(B) About 49 per cent of the variation in one is shared with the other.
(C) 70 per cent of the students are good at both.
(D) The two abilities are unrelated.

7. Two classes have the same mean score. Class X has an SD of 3 and Class Y an SD of 12. Given below are two statements:

Statement I: Class Y is more heterogeneous than Class X.

Statement II: Because the means are equal, the two classes need the same teaching plan.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

8. Examiner A's marks have mean 60 and SD 4; Examiner B's marks have mean 60 and SD 16, for comparable groups. If the raw marks of the two examiners are added to make a merit list:

(A) both examiners will have equal influence on the order of merit
(B) Examiner A will have greater influence because his marks vary less
(C) Examiner B will have greater influence because his marks vary more
(D) neither examiner will have any influence because the means are equal

6.6 Continuous and Comprehensive Evaluation and grading

Syllabus: Evaluation Systems: Elements and Types of evaluation

EXAM FOOTPRINT

Asked occasionally; usually a single-answer or two-statement item on continuous and comprehensive evaluation or grading.

Core

What CCE means

Continuous and Comprehensive Evaluation (सतत एवं व्यापक मूल्यांकन), usually shortened to CCE, is a school-level scheme of evaluation (मूल्यांकन). Its name has two parts, and the examination often asks what each part means.

- **Continuous** means that assessment (आकलन) goes on regularly through the whole year, during teaching, and not only in one examination at the end. Class work, assignments, quizzes, projects and observation all count. This is formative evaluation (रचनात्मक मूल्यांकन) built into daily teaching, together with periodic summative evaluation (योगात्मक मूल्यांकन).
- **Comprehensive** means that assessment covers the whole child: both the **scholastic (शैक्षिक)** areas and the **co-scholastic (सह-शैक्षिक)** areas.

Scholastic areas are the curricular subjects, such as languages, mathematics, science and social science, and the intellectual abilities developed through them. Co-scholastic areas are everything else that education is meant to develop: life skills (thinking, social and emotional skills), attitudes, values, health and physical education, and literary, artistic and cultural activities such as music, art, sports and games.

Keep the two words apart. ‘Continuous’ is about time (regular, all through the year). ‘Comprehensive’ is about coverage (scholastic plus co-scholastic). Neither word is about written versus oral tests, internal versus external examiners, or school versus college. Those are tempting wrong options.

Where CCE comes from

The Right of Children to Free and Compulsory Education Act, 2009 (the RTE Act) made this kind of evaluation a duty for elementary schools. Section 29(2)(h) of the Act says that the curriculum and evaluation procedure shall take into consideration “comprehensive and continuous evaluation of child’s understanding of knowledge and his or her ability to apply the same”. Notice that the Act puts ‘comprehensive’ first; the scheme is still commonly called CCE.

The Central Board of Secondary Education (CBSE) introduced grading in Classes IX and X from the 2009–10 session, with CCE beginning in Class IX that year and then covering Class X. Under it, marks were replaced by grades, the year was divided into formative and summative assessments, and co-scholastic areas were graded on the report card. CBSE later ended the scheme for Class X and brought back a compulsory board examination from the 2017–18 session.

Why grades instead of marks

Grading (श्रेणीकरण) means reporting performance in a few broad bands, each named by a letter grade, instead of exact marks. A student does not get 67 or 68; the student gets, say, B2.

Why do this? Because marks look more precise than they are. Every examination has **measurement error (मापन त्रुटि)**: the same answer could get 66 from one examiner and 69 from another, or on another day. A difference of one or two marks between two students therefore tells us almost nothing. Fine marks invite students, parents and schools to treat these meaningless differences as real.

Grades put students into broad bands. This brings three benefits that the examination likes to ask about.

- It reduces over-fine distinctions and misclassification. Two students with 67 and 68 are in the same band, as they should be.
- It reduces unhealthy competition for every single mark.
- It reduces stress and fear of examinations.

Grades do not remove the need for examinations. Some assessment is still needed to decide the grade. Nor do grades raise the pass percentage by themselves.

Absolute and relative grading

There are two ways of deciding which student gets which grade. This pair is asked very often.

In **absolute grading (निरपेक्ष श्रेणीकरण)** the grade boundaries are fixed in advance, in marks or per cent. For example, 80 per cent or more is A, 70 to 79 is B, and so on. Every student who crosses a boundary gets that grade, however many such students there are. If the whole class does well, the whole class can get A. The student is judged against a fixed standard. This is the logic of **criterion-referenced (निकष-संदर्भित)** interpretation.

In **relative grading (सापेक्ष श्रेणीकरण)** the grade depends on the student’s position in the group. For example, the top 10 per cent get A, the next 20 per cent get B, and so on. The proportions are fixed; the mark needed for each grade floats with the performance of the group. This is the logic of **norm-referenced (मानक-संदर्भित)** interpretation. Relative grading is also called **grading on the curve**, because the proportions are often taken from the normal (bell-shaped) curve.

Point	Absolute grading	Relative grading
Fixed in advance	Mark boundaries for each grade	Proportion of students in each grade

Point	Absolute grading	Relative grading
Grade depends on	The student's own marks against the standard	The student's rank in the group
Can everyone get A?	Yes, if everyone reaches the standard	No; only the fixed top proportion
Logic	Criterion-referenced	Norm-referenced
Example	A for 80 per cent and above	A for the top 10 per cent

The chief objection to relative grading appears clearly in a small class of uniformly able students. Relative grading must spread the class over the whole range of grades. So small differences in marks are stretched, and some good students must receive low grades however well they have done. Absolute grading avoids this, because it uses a fixed standard. The opposite risk, everyone getting the top grade, belongs to lenient absolute grading, never to relative grading.

In an analogy question, keep the order: relative grading is to absolute grading as norm-referenced interpretation is to criterion-referenced interpretation. The reversed pair is a common trap. Norm-referenced and criterion-referenced tests are taught fully in Section 6.3.

CLASSROOM EXAMPLE

At a government college in Sirsa, a B.A. class of 40 takes an internal test in English. Dr Kaur announces in advance: 80 and above is A, 65 to 79 is B, 50 to 64 is C. Twelve students score above 80 and all twelve get A. This is absolute grading. In a parallel section, Dr Yadav decides that the top four students (10 per cent) will get A, whatever their marks. One student with 83 gets B, because four classmates scored higher. This is relative grading, or grading on the curve. The same mark has led to different grades because the rule is different.

OFTEN CONFUSED

- Continuous and comprehensive. Continuous = regular, through the year (time). Comprehensive = scholastic plus co-scholastic (coverage).
- Scholastic and co-scholastic. Scholastic = curricular subjects and the intellectual abilities they develop. Co-scholastic = life skills, attitudes, values, health and physical education, cultural activities.
- Absolute and relative grading. Absolute = fixed mark boundaries, anyone can reach any grade (criterion-referenced). Relative = fixed proportions, grade depends on rank (norm-referenced).
- Grading on the curve and criterion-referenced grading. On the curve = proportions fixed in advance, so it is norm-referenced, not criterion-referenced.
- Direct and indirect grading. Direct = grade given straight from qualitative judgement. Indirect = marks first, then converted to grades.

FACT WATCH

- RTE Act 2009, section 29(2)(h): evaluation procedure to provide for "comprehensive and continuous evaluation" (as of October 2026).
- CBSE introduced grading in Classes IX and X, with CCE, from 2009–10 (circular of September 2009) and ended the scheme for Class X, restoring a compulsory board examination, from 2017–18. Older books still describe CCE as the current CBSE scheme.
- The National Education Policy 2020 continues the idea in schools through regular, formative, competency-based assessment and a holistic progress card (see Section 6.9).

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Direct and indirect grading

Grades can be reached in two ways. In **direct grading (प्रत्यक्ष श्रेणीकरण)** the examiner does not award marks at all. The examiner reads the answer, judges its quality as a whole, and writes the grade straight away. Direct grading is quick and suits qualitative work, but it depends heavily on the examiner's judgement.

In **indirect grading (अप्रत्यक्ष श्रेणीकरण)** marks are awarded first and then converted into grades. The conversion can follow fixed mark ranges (that is absolute grading) or the student's position in the group (that is relative grading). So absolute and relative grading are both forms of indirect grading.

The CBSE nine-point scale under CCE

Students sometimes meet the CBSE scale in questions. Under CCE, scholastic areas in Classes IX and X were graded on a nine-point scale: A1 (91–100, grade point 10), A2 (81–90, 9), B1 (71–80, 8), B2 (61–70, 7), C1 (51–60, 6), C2 (41–50, 5), D (33–40, 4), E1 (21–32) and E2 (20 and below). E1 and E2 meant the student needed to improve. CBSE allowed an indicative percentage to be read off by multiplying the grade point average by 9.5.

WORKED EXAMPLE

A Class X student under CCE had a cumulative grade point average of 8.4. What was the indicative percentage?

Indicative percentage = grade point average x 9.5

= 8.4 x 9.5 = 79.8 per cent

This multiplier belongs to CBSE's school scheme only. It is not a rule of the university Choice Based Credit System, which is the subject of Section 6.7.

Limits of CCE in practice

Teachers often found CCE heavy in record-keeping. Continuous assessment can turn into continuous testing, and co-scholastic grading can become a formality if teachers lack time and training. These are criticisms of how the scheme was carried out, not of its two principles.

CHECK YOURSELF

1. In Continuous and Comprehensive Evaluation, the word ‘continuous’ refers mainly to:

(A) assessment of both scholastic and co-scholastic areas
(B) regular assessment spread through the whole year
(C) assessment by both internal and external examiners
(D) assessment through both written and oral tests

2. Which of the following belong to the co-scholastic areas of CCE?

1. Life skills
2. Mathematics
3. Values and attitudes
4. Health and physical education
5. Social science

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 2 and 5 only
(C) 1, 3 and 4 only
(D) 1, 3, 4 and 5 only

3. Given below are two statements:

Statement I: In absolute grading, the number of students who can receive the top grade is fixed before the examination.

Statement II: In relative grading, a student’s grade depends on his or her position in the group.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

4. Assertion (A): Reporting results in broad grades rather than exact marks can reduce misclassification of students.

Reason (R): A difference of one or two marks between students often lies within the error of measurement.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

5. Match List I with List II.

List I	**List II**
A. Grade A for all who score 75 per cent or more	I. Direct grading
B. Grade A for the top 15 per cent of the class	II. Indirect grading
C. Examiner writes a grade without awarding marks	III. Absolute grading

List I	List II
D. Marks awarded first and then converted into grades	IV. Relative grading

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II
(B) A-IV, B-III, C-I, D-II
(C) A-III, B-IV, C-II, D-I
(D) A-I, B-II, C-III, D-IV

6. A teacher of a small honours class in which all eight students are very able decides to give grades by a fixed distribution. The likely result is that:

(A) every student will receive the top grade
(B) the grades will not depend on any test
(C) some capable students will receive low grades
(D) the class average cannot be calculated

6.7 Evaluation in the Choice Based Credit System

Syllabus: Evaluation in Choice Based Credit System in Higher education

EXAM FOOTPRINT

Asked fairly often; usually a single-answer or two-statement item on credits, grade points and the choice-based credit system.

Core

What the Choice Based Credit System is

The **Choice Based Credit System (विकल्प आधारित क्रेडिट प्रणाली)**, or CBCS, is the pattern of courses, credits and grades that the University Grants Commission (UGC) asked Indian universities to adopt through its guidelines of 2015. The abbreviation is often tested: it is Choice Based Credit System, not ‘Course Based’, ‘Curriculum’ or ‘Scheme’.

The guidelines came from the UGC, the body for university education. Do not confuse it with NCERT (school curriculum), NCTE (teacher education) or NAAC (assessment and accreditation of institutions).

The system has three ideas in its name.

- **Choice.** The student does not take a fixed package. Some courses are compulsory, but the student chooses others from a pool. The UGC describes this as a ‘cafeteria’ approach: students can take courses of their choice, learn at their own pace, take additional courses and follow an interdisciplinary path.
- **Credit.** Every course carries a stated number of credits, which shows how much study it involves.
- **Grades.** Performance in each course is reported as a letter grade with a grade point, and these are combined into averages for the semester and for the whole programme.

Because every university states its courses in credits on a common pattern, credits earned in one institution can be counted by another. This is **credit transfer (क्रेडिट हस्तांतरण)**, and it allows students to move between institutions. Today this is done through the Academic Bank of Credits, described below. Credit transfer means credits, not marks, fees or teachers, moving from one institution to another.

Under CBCS, performance is not reported in raw marks alone. It is reported through letter grades and grade points attached to credits, summarised as the SGPA for each semester and the CGPA over all semesters completed so far.

What a credit is

A **credit (क्रेडिट)** is the unit by which course work is measured. The 2015 guidelines define it by the weekly teaching time: one credit is equivalent to one hour of teaching (lecture or tutorial) or two hours of practical work or field work per week, through a semester.

So a credit tells us the weight of a course. A 4-credit course carries twice the weight of a 2-credit course when averages are calculated. A credit is not a fee, a rank or an attendance figure.

To find the credits of a course, count lecture and tutorial hours one for one, and practical hours two for one.

> **WORKED EXAMPLE**
>
> A course has 3 hours of lectures and 2 hours of practical work every week through the semester. How many credits does it carry?
>
> Lecture: 3 hours x 1 credit per hour = 3 credits
>
> Practical: 2 hours / 2 hours per credit = 1 credit
>
> Total = 3 + 1 = 4 credits

The newer UGC Curriculum and Credit Framework for Undergraduate Programmes (CCFUP, December 2022) says the same thing per semester rather than per week. In a semester of 15 weeks, one credit of theory means 15 hours of lecture or tutorial, and one credit of practical or field work means 30 hours. The National Credit Framework (NCrF) goes further and counts all ‘notional learning hours’, including self-study: 30 hours of learning make one credit, and a year of 1,200 hours makes 40 credits. Weekly or per semester, the answer to the worked example is the same: 4 credits.

Kinds of courses under CBCS (2015)

The 2015 guidelines place courses in three groups. Many questions are built on this table.

Course type	What it is
Core course (मूल पाठ्यक्रम)	Compulsory for every student of the programme, as a core requirement of the discipline.
Elective course (वैकल्पिक पाठ्यक्रम)	Chosen from a pool. Three kinds: **Discipline Specific Elective (DSE):** offered by the main discipline. **Generic Elective (GE):** chosen generally from an unrelated discipline, to gain wider exposure. **Dissertation/project:** a course for special or advanced knowledge that the student studies on his or her own with advisory support from a teacher.

Course type	What it is
Ability Enhancement Course (क्षमता संवर्धन पाठ्यक्रम)	Two kinds: **Ability Enhancement Compulsory Course (AECC):** Environmental Science, and English or MIL (Modern Indian Language) Communication, taken by all. **Skill Enhancement Course (SEC) (कौशल संवर्धन पाठ्यक्रम):** value-based or skill-based courses chosen from a pool.

Three traps follow from this table.

- An elective from an unrelated discipline is a Generic Elective. An elective from one's own discipline is a Discipline Specific Elective.
- The two compulsory ability enhancement courses are Environmental Science and English/MIL Communication. A short skill-based course chosen from a pool, such as 'Technical Writing', is a Skill Enhancement Course, not an AECC. Both are 'enhancement' courses, but only the AECCs are fixed for all students.
- In the 2015 classification a dissertation or project is an elective course carrying credits. Many programmes make a project compulsory in practice, but in the UGC table it sits under electives, not core.

Letter grades and grade points

The UGC recommends a ten-point grading system. Learn this table exactly.

Letter grade	Descriptor	Grade point
O	Outstanding	10
A+	Excellent	9
A	Very Good	8
B+	Good	7
B	Above Average	6
C	Average	5
P	Pass	4
F	Fail	0
Ab	Absent	0

Points to fix in memory:

- O stands for Outstanding, not 'Ordinary' or 'Optional'.
- Plain A is Very Good with 8 points. Excellent with 9 points is A+.
- B is Above Average (6) and C is Average (5). The order of words can trick you: 'above average' is higher than 'average'.
- P (Pass) is the lowest passing grade and carries 4 points, not 5.
- F and Ab both carry 0. A student with F has failed and must reappear in the examination.
- There is no D grade on the UGC scale. Some older books and notes print one; do not use it in a UGC question.

The UGC leaves universities free to decide the marks or grade needed to pass a course and the CGPA needed for a degree, taking into account the norms of professional councils. So individual universities may differ on pass marks, but the letters and points above are the UGC scale.

SGPA: the method

The **Semester Grade Point Average (सेमेस्टर ग्रेड पॉइंट औसत)**, or SGPA, measures performance in one semester. It is a credit-weighted average.

1. For each course, multiply its credits by the grade point earned. The result is the course's **credit points**. (Credits are multiplied, not added, divided or subtracted.)
2. Add the credit points of all the courses of the semester.
3. Add the credits of all the courses of the semester.
4. Divide step 2 by step 3. Round to two decimal places, as the UGC guidelines require.

In symbols, SGPA = sum of (Ci x Gi) / sum of Ci, where Ci is the number of credits of the ith course and Gi is the grade point scored in it. The denominator is the total credits, not the number of courses.

WORKED EXAMPLE

This is the UGC's own illustration. Six courses: Course 1 (3 credits, A), Course 2 (4 credits, B+), Course 3 (3 credits, B), Course 4 (3 credits, O), Course 5 (3 credits, C), Course 6 (4 credits, B).

Credit points: 3 x 8 = 24; 4 x 7 = 28; 3 x 6 = 18; 3 x 10 = 30; 3 x 5 = 15; 4 x 6 = 24

Total credit points = 24 + 28 + 18 + 30 + 15 + 24 = 139

Total credits = 3 + 4 + 3 + 3 + 3 + 4 = 20

SGPA = 139 / 20 = 6.95

WORKED EXAMPLE

A student earns A+ (9) in a 4-credit course, B (6) in a 4-credit course, O (10) in a 3-credit course, P (4) in a 2-credit course and B+ (7) in a 3-credit course.

Credit points: 4 x 9 = 36; 4 x 6 = 24; 3 x 10 = 30; 2 x 4 = 8; 3 x 7 = 21

Total credit points = 119; total credits = 16

SGPA = 119 / 16 = 7.4375, reported as 7.44

The simple average of the grade points, (9 + 6 + 10 + 4 + 7) / 5 = 7.2, is wrong: it gives the 2-credit course the same weight as the 4-credit ones.

When all courses carry equal credits, the credits cancel out and the SGPA is simply the mean of the grade points. For four equal-credit courses with O, A+, B+ and P, the SGPA is (10 + 9 + 7 + 4) / 4 = 30 / 4 = 7.5.

CGPA: the method

The **Cumulative Grade Point Average (संचयी ग्रेड पॉइंट औसत)**, or CGPA, measures performance over all the semesters completed so far. It is calculated in the same way, over all courses of all semesters: total credit points divided by total credits. If you are given each semester's SGPA and credits, first turn each SGPA back into credit points by multiplying by that semester's credits.

WORKED EXAMPLE

The UGC's illustration of CGPA. Six semesters carry 20, 22, 25, 26, 26 and 25 credits, with SGPAs of 6.9, 7.8, 5.6, 6.0, 6.3 and 8.0.

Credit points: 20 x 6.9 = 138; 22 x 7.8 = 171.6; 25 x 5.6 = 140; 26 x 6.0 = 156; 26 x 6.3 = 163.8; 25 x 8.0 = 200

Total credit points = 969.4; total credits = 144

CGPA = 969.4 / 144 = 6.7319…, reported as 6.73

WORKED EXAMPLE

Three semesters carry 22, 24 and 20 credits, with SGPAs of 6.8, 7.5 and 8.2.

Credit points: 22 x 6.8 = 149.6; 24 x 7.5 = 180; 20 x 8.2 = 164

Total = 493.6 over 66 credits

CGPA = 493.6 / 66 = 7.4787…, reported as 7.48

The simple average of the three SGPAs is 7.5, which is wrong because the semesters carry unequal credits.

Both SGPA and CGPA appear on the transcript. SGPA belongs to one semester; CGPA to all semesters taken together, never to one course or only the final semester.

USUAL SLIPS

- Dividing by the number of courses instead of the total credits.
- Adding credits and grade points instead of multiplying them.
- Averaging SGPAs (or a CGPA and a new SGPA) without weighting them by credits.
- Using wrong grade points: P is 4 (not 5), A is 8 (not 9), B+ is 7, B is 6.
- Inventing a D grade. The UGC scale has none.
- Leaving a failed course out of the denominator when the question says to apply the UGC formula (see Going Further).

Multiple entry and exit, and the Academic Bank of Credits

The National Education Policy 2020 and the UGC's CCFUP (2022) built a four-year undergraduate programme with **multiple entry and exit (बहु-प्रवेश एवं बहु-निकास)**. A student may leave after any year with an award and come back later.

Leaves after successfully completing	Award
One year (two semesters)	Undergraduate certificate
Two years (four semesters)	Undergraduate diploma
Three years (six semesters)	Bachelor's degree
Four years (eight semesters)	Bachelor's degree with honours, or honours with research

The usual trap is to swap the first two: one year gives a certificate, two years give a diploma.

Credits make this possible. The **Academic Bank of Credits (एकेडमिक बैंक ऑफ क्रेडिट्स)**, or ABC, set up under the UGC (Establishment and Operation of Academic Bank of Credits in Higher Education) Regulations, 2021, is a digital store of credits. Credits earned by a student in recognised institutions

are deposited in the student's account, can be transferred, and are later redeemed towards a certificate, diploma or degree. It works like a bank account for credits, which is why it supports multiple entry and exit.

Keep it apart from the **National Academic Depository (राष्ट्रीय शैक्षणिक निक्षेपागार)**, or NAD. The NAD is a digital store of academic awards: degrees, diplomas, certificates and mark-sheets, lodged by institutions, which students and employers can retrieve and verify online through DigiLocker. In one line: ABC stores credits; NAD stores awards.

CLASSROOM EXAMPLE

Ritu studies B.A. at a college affiliated to a state university in Haryana. In her first semester she takes four core papers of 4 credits each and the compulsory course in Environmental Science of 2 credits; she also chooses a generic elective in Economics. Her transcript shows A in Indian Writing in English (4 credits), B+ in Phonetics (4 credits), and so on, with an SGPA at the foot. At the end of the second year family needs force her to leave. Because her credits are lodged in her ABC account, she receives an undergraduate diploma and can return two years later to complete the degree. Her diploma itself is lodged in the NAD.

OFTEN CONFUSED

- SGPA and CGPA. SGPA = one semester. CGPA = all semesters completed so far.
- Credits and credit points. Credits = weight of the course. Credit points = credits x grade point.
- Discipline Specific Elective and Generic Elective. DSE = from the main discipline. GE = from an unrelated discipline.
- AECC and SEC. AECC = Environmental Science and English/MIL Communication, compulsory for all. SEC = skill- or value-based, chosen from a pool.
- A and A+. A = Very Good, 8. A+ = Excellent, 9.
- B and C. B = Above Average, 6. C = Average, 5.
- P and F. P = Pass, 4 (lowest passing grade). F = Fail, 0 (as is Ab).
- Certificate and diploma. Certificate after one year; diploma after two.
- ABC and NAD. ABC stores credits; NAD stores awards (degrees, mark-sheets).
- CBCS 2015 and CCFUP 2022. CBCS: core, DSE, GE, AECC, SEC. CCFUP: major, minor, multidisciplinary, AEC, SEC, VAC, internship.

FACT WATCH

- UGC CBCS guidelines (2015): ten-point scale O, A+, A, B+, B, C, P, F, Ab with points 10, 9, 8, 7, 6, 5, 4, 0, 0; SGPA and CGPA rounded to two decimal places (checked October 2026). Some older books print a D grade; it is not on the UGC scale.
- Credit: per week in the 2015 guidelines (one hour of lecture or tutorial, or two hours of practical, per week); per semester in CCFUP, December 2022 (15 hours theory or 30 hours practical per credit) (checked October 2026). NCrF counts 30 notional learning hours per credit.
- Undergraduate course categories: the 2015 CBCS names (core, DSE, GE, AECC, SEC) are still the ones asked in CBCS questions; the CCFUP of December 2022 uses major, minor, multidisciplinary courses, AEC, SEC, value-added courses and internship (as of October 2026).
- Academic Bank of Credits: UGC Regulations, 2021 (as of October 2026).

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Failed and absent courses in the SGPA

The UGC formula divides by "the sum of the number of credits of all the courses undergone by a student". Read as written, a course with F or Ab stays in the denominator with its full credits and adds nothing (0 points) to the numerator. So a failure pulls the SGPA down. The UGC's printed illustration does not itself contain an F row, and universities differ in how they record a course that is later cleared, but when a question says to apply the CBCS formula, keep the failed course in.

WORKED EXAMPLE

A student fails a 4-credit course (F, 0), gets A (8) in a 4-credit course, B+ (7) in a 4-credit course and O (10) in a 2-credit course.

Credit points: 4 x 0 = 0; 4 x 8 = 32; 4 x 7 = 28; 2 x 10 = 20; total = 80

Credits of all courses taken = 4 + 4 + 4 + 2 = 14

SGPA = 80 / 14 = 5.714..., reported as 5.71

Leaving out the failed course gives 80 / 10 = 8.0, which is wrong under the UGC formula.

When a failed course is later cleared and the new grade replaces F, only the numerator changes. Each grade point gained adds (credits of that course x points gained) to the total.

WORKED EXAMPLE

Six courses of 4 credits each; grade points 9, 7, 7, 6, 8 and 0 (F). The student later clears the failed course with P (4).

Before: 4 x (9 + 7 + 7 + 6 + 8 + 0) = 148; 148 / 24 = 6.1666..., reported as 6.17

After: 4 x (9 + 7 + 7 + 6 + 8 + 4) = 164; 164 / 24 = 6.8333..., reported as 6.83

Rise = 16 extra credit points / 24 credits = 0.666..., about 0.67

Divide by the total credits of all six courses, not by the number of courses or the credits of one course.

Target-grade problems

Examiners like to turn the formula round: what must a student score next to reach a target? The method is the same every time.

1. Credit points already earned = credits so far x present CGPA.
2. Credit points needed = total credits after the next semester x target CGPA.
3. Points still required = step 2 minus step 1.
4. Required SGPA = step 3 / credits of the next semester.

WORKED EXAMPLE

A student has a CGPA of 6.4 over 60 credits. The next semester carries 24 credits. What SGPA will raise the CGPA to 6.8?

Points earned = 60 x 6.4 = 384

Points needed = (60 + 24) x 6.8 = 84 x 6.8 = 571.2

Still required = 571.2 - 384 = 187.2

Required SGPA = 187.2 / 24 = 7.8

The tempting answer 7.2 (since 6.8 is midway between 6.4 and 7.2) treats the two parts as equal; the earlier 60 credits outweigh the new 24.

WORKED EXAMPLE

Five courses of equal credit. Four grades are already known: A (8), B+ (7), B (6) and A+ (9). What grade in the fifth course gives an SGPA of 7.4?

With equal credits, SGPA = mean of grade points

Total needed = 5 x 7.4 = 37

Known total = 8 + 7 + 6 + 9 = 30

Fifth course needs 37 - 30 = 7 points, which is B+

The same reasoning tells you how far one good semester moves a CGPA. A strong semester added to many earlier credits moves the CGPA only a little, because the earlier credits weigh more.

Grade points and scales of measurement

Letter grades only rank performance, so strictly they form an **ordinal scale (क्रमसूचक मापनी)**. Adding and averaging are meaningful only when the steps between values are equal, which is the property of an **interval scale (अंतराल मापनी)**. Calculating an SGPA therefore treats the grade points as if they formed at least an interval scale, with equal steps from P to C to B and so on. This is a convention that the whole system rests on. A nominal scale only names categories, and a ratio scale needs a true zero; neither describes what averaging assumes. Scales of measurement are treated in Section 6.5.

From CBCS 2015 to CCFUP 2022

The CCFUP of December 2022 keeps credits, grades, SGPA and CGPA but renames and reshapes the course types for the four-year programme.

CCFUP 2022 category	What it covers
Major	Courses in the main discipline, studied in depth (the place of the old core and DSE)
Minor	Courses in another discipline or a vocational area

CCFUP 2022 category	What it covers
Multidisciplinary courses (MDC)	Introductory courses in broad areas outside the major
Ability Enhancement Courses (AEC)	Language and communication courses
Skill Enhancement Courses (SEC)	Practical and job-related skills
Value-Added Courses (VAC)	Courses such as environmental education, understanding India, and health and wellness
Internship and research project	Summer internship; a research project or dissertation in the fourth year

So Environmental Science, which was an AECC in 2015, falls among the value-added courses in 2022, while language communication stays as the AEC. Answer a question by the framework it names. If it says ‘CBCS’ or ‘UGC guidelines of 2015’, use core, DSE, GE, AECC and SEC.

The National Higher Education Qualifications Framework

The UGC’s **National Higher Education Qualifications Framework (राष्ट्रीय उच्चतर शिक्षा अर्हता रूपरेखा)**, or NHEQF, places higher education qualifications on levels 4.5 to 8, each described by learning outcomes. Levels 1 to 4 belong to school education.

Level	Qualification
4.5	Undergraduate certificate (one year)
5	Undergraduate diploma (two years)
5.5	Bachelor’s degree (three years)
6	Bachelor’s degree (honours or honours with research, four years); postgraduate diploma
6.5	Master’s degree (two years after a three-year bachelor’s, or one year after a four-year bachelor’s)
7	Master’s degree such as M.E. or M.Tech. (two years after a four-year bachelor’s)
8	Doctoral degree

The doctorate is at level 8, the top of the higher education framework. Not every master’s degree is at 6.5: the general master’s is at 6.5, and a two-year master’s after a four-year bachelor’s, such as M.Tech., is at level 7.

CHECK YOURSELF

1. Match List I (letter grade) with List II (grade point) on the UGC ten-point scale.

List I	List II
A. A+	I. 4
B. B	II. 9
C. P	III. 6
D. B+	IV. 7

Choose the correct answer from the options given below:

(A) A-II, B-IV, C-I, D-III
(B) A-II, B-III, C-I, D-IV
(C) A-IV, B-III, C-II, D-I
(D) A-II, B-I, C-III, D-IV

2. In a semester a student obtains O in a 4-credit course, B in a 3-credit course, B+ in a 3-credit course and C in a 2-credit course. The SGPA is:

(A) 7.00
(B) 7.25
(C) 7.42
(D) 7.67

3. Semester I carries 18 credits with an SGPA of 7.5 and Semester II carries 22 credits with an SGPA of 6.5. The CGPA after two semesters is:

(A) 6.95
(B) 7.00
(C) 7.05
(D) 6.90

4. A student's CGPA is 7.2 over 90 credits. The next semester carries 30 credits. The SGPA needed in that semester to bring the CGPA to 7.5 is:

(A) 7.8
(B) 8.1
(C) 8.4
(D) 9.0

5. A course has 3 lecture hours, 1 tutorial hour and 2 practical hours every week through the semester. Under the CBCS definition of a credit, it carries:

(A) 4 credits
(B) 5 credits
(C) 6 credits
(D) 3 credits

6. Given below are two statements:

Statement I: Under the UGC CBCS guidelines of 2015, Environmental Science is an Ability Enhancement Compulsory Course.

Statement II: A Generic Elective is an elective offered by the student's own main discipline.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

7. Assertion (A): Dividing the total credit points of a semester by the number of courses gives a wrong SGPA when courses carry unequal credits.

Reason (R): The SGPA is a credit-weighted average, so its denominator is the total credits.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

8. Which of the following statements are correct?

1. The Academic Bank of Credits stores credits earned by a student and allows them to be redeemed for an award.
2. The National Academic Depository stores degrees and mark-sheets in digital form.
3. A student who leaves a four-year undergraduate programme after one year receives an undergraduate diploma.
4. In the NHEQF the doctoral degree is placed at level 8.

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 2, 3 and 4 only
(C) 1 and 3 only
(D) 1, 2 and 4 only

6.8 Computer-based testing

Syllabus: Computer based testing

EXAM FOOTPRINT

Rarely asked directly; it supports items on innovations in evaluation and online assessment.

Core

What a computer-based test is

In a **computer-based test (कंप्यूटर आधारित परीक्षण)**, or CBT, the questions appear on a computer screen and the candidate answers by clicking with a mouse or using the keyboard. The test is taken at a test centre on computers set up by the testing body. Responses are stored electronically and scored by the computer. CBT stands for Computer Based Test, not ‘Centre Based’ or ‘Common Board’.

The UGC NET is conducted by the **National Testing Agency (NTA)**, which the Government of India set up in November 2017 as an autonomous testing body registered as a society. The NTA conducted the UGC NET for the first time in December 2018, and it did so as a computer-based test. Before that the examination was conducted by the CBSE with paper and OMR sheets.

How the screen works

A CBT screen has features that an OMR (optical mark recognition) answer sheet cannot offer.

- A **question palette** shows the number of every question, coloured to show which are answered, not answered, not visited or marked for review.
- An answer can be changed, cleared or **marked for review** any number of times until the test is submitted. On an OMR sheet a darkened circle cannot be altered.
- An on-screen **timer** shows the time left, and the test closes itself at the end.

The candidate does not choose the difficulty of the paper, cannot see a scoring key, and still needs a centre with a steady power supply.

NTA and similar bodies put **mock tests** online before the examination. A mock test lets candidates practise the interface: moving between questions, marking and changing answers, using the palette and watching the timer. Its value is familiarity, which lowers anxiety and prevents marks being lost through handling errors. It does not show the actual questions, and it does not raise anyone's score by itself.

Advantages and limitations

Advantages	Limitations
Responses recorded electronically, so scoring is quick and free from human marking error	Needs test centres, computers, power and network; technical failure can disrupt a session
Results declared faster	Candidates differ in familiarity with computers (the digital divide)
Questions and options can be shuffled from a question bank, so neighbours do not see the same item in the same place; copying becomes difficult	Identity checks and invigilation are still needed
Every action is logged: time spent on each item, answer changes	Long reading on screen can tire some candidates
Tests can run in many sessions on many days	Different sessions need different papers, which must then be equated

In a country like India the chief limitation is the **digital divide (डिजिटल विभाजन)**. A candidate from a rural or poor background who has seldom used a computer may lose marks for reasons that have nothing to do with the subject. That is a threat to fairness. Objective scoring, speed of results and shuffling of items are advantages, not limitations.

Remember also what CBT cannot do. It changes how items are delivered and scored, not what they measure. A poorly written item stays poor on a screen, so the validity (वैधता) of the items must still be established by the test makers.

Normalisation across shifts

A large CBT is held in several shifts, often over several days, with a different question paper in each shift. The papers cannot be made exactly equal in difficulty. So raw scores from different shifts cannot be compared directly. They are converted by a statistical procedure called **normalisation (सामान्यीकरण)**, usually into **percentile scores (शततमक अंक)**: a candidate's percentile shows the percentage of candidates in the same shift who scored at or below that candidate. (Section 6.5 defines percentile rank as the percentage scoring below a score; both conventions are in use, so read the definition given in the question.)

Normalisation is the answer when a question asks how scores from different shifts are made comparable. Randomisation only shuffles questions within a paper. Negative marking only penalises guessing. Revaluation only rechecks scoring. None of them equates two different papers.

Adaptive testing

In a **computerised adaptive test (कंप्यूटरीकृत अनुकूली परीक्षण)**, or CAT, the next question depends on the candidate's answers so far. The program estimates the candidate's ability after each response. A correct answer brings a harder item, a wrong answer an easier one. Each candidate therefore meets items suited to his or her level, and a precise score is reached with fewer items than a fixed test needs.

Kind of CBT	How the items are chosen
Linear fixed-form	Every candidate in a session gets the same set of items, perhaps in shuffled order
Linear-on-the-fly (LOFT)	A different form is assembled for each candidate before the test starts, but it does not change during the test
Computerised adaptive	Each next item is chosen during the test according to the responses so far

LOFT and CAT both give candidates different items, so they are easily confused. The deciding point is adaptation during the test: only CAT adapts.

CLASSROOM EXAMPLE

A student from a village near Fatehabad prepares for the UGC NET. She has used a smartphone but never a desktop computer. Her teacher at the college makes her take the NTA mock test three times in the computer laboratory. The first time she loses five minutes finding her way between the buttons and the question palette. By the third attempt she moves through the palette with ease. On the day of the examination none of her time goes on the interface. The mock test has not given her any questions in advance; it has removed a handicap that had nothing to do with her subject.

OFTEN CONFUSED

- CBT and OMR testing. CBT = answers on screen, changeable until submission. OMR = darkened circles on paper, which cannot be altered.
- Normalisation and randomisation. Normalisation = making scores from different papers comparable. Randomisation = shuffling the order of items or options.
- Adaptive testing and linear-on-the-fly testing. CAT adapts each item to the responses during the test. LOFT builds a different form for each candidate before the test, without adapting.
- Mock test and actual test. Mock = practice with the interface, different questions. Actual = the real paper.
- Item response theory and classical test theory. IRT = item parameters independent of the group tested (needed for CAT). CTT = difficulty and discrimination indices that change with the group.

FACT WATCH

- National Testing Agency: set up by the Government of India in November 2017 (as of October 2026).
- First UGC NET conducted by the NTA: December 2018, in computer-based mode. Older books describe the NET as a paper-and-pencil test conducted by the CBSE.
- The mode of the examination is fixed for each cycle by the NTA's information bulletin. It has been computer-based in normal practice since December 2018, but check the mode announced in the bulletin for your own cycle.

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Why adaptive tests need item response theory

In an adaptive test, two candidates answer different items. To compare them, the computer must know how difficult and how discriminating each item is on a single scale that does not depend on who happened to take it. **Item response theory (एकांश अनुक्रिया सिद्धांत)**, or IRT, provides such item parameters (difficulty, discrimination and, in some models, guessing). The items of the bank are calibrated in advance on large samples, and the candidate's ability is estimated on the same scale.

Classical test theory, the basis of the difficulty index and discrimination index taught in Section 6.4, gives values that change with the group tested. An item that looks easy in a strong group looks hard in a weak one. Such sample-dependent values cannot place candidates who took different items on one scale. That is why CAT depends on IRT, not on classical statistics alone. Bloom's taxonomy and Flanders' categories have nothing to do with calibration.

What percentiles across shifts assume

When percentiles are worked out within each shift, the 90th percentile in Shift 1 is treated as equal to the 90th percentile in Shift 2. That is fair only if the candidates in each shift are of similar ability on average. The testing agency secures this by allotting candidates to shifts at random, from a large pool. The papers of different shifts are not identical; if they were, no normalisation would be needed. The assumption is about the candidates, not the papers.

What the log of responses offers

Because a computer records every click, a testing agency can study how long candidates spend on each item and how often they change answers. Very short response times on hard items can signal guessing; patterns of identical answers can signal malpractice. Item statistics from the log also help to improve the question bank. This is a gain in information, not a guarantee of validity.

CHECK YOURSELF

1. In a computer-based test, which of the following can a candidate do that an OMR sheet does not allow?

(A) Choose an easier version of the question paper
(B) Change a marked answer before final submission
(C) See the answer key after each question
(D) Take the test without any electricity

2. Given below are two statements:

Statement I: Computer-based testing removes the effect of unequal familiarity with computers among candidates.

Statement II: Computer-based testing makes it possible to record the time a candidate spends on each item.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

3. Match List I with List II.

List I	List II
A. Normalisation	I. Shuffling the order of items and options
B. Randomisation	II. Making scores of different shift papers comparable
C. Adaptive testing	III. Practice with the test interface
D. Mock test	IV. Next item chosen by the responses so far

Choose the correct answer from the options given below:

(A) A-II, B-I, C-III, D-IV
(B) A-I, B-II, C-IV, D-III
(C) A-II, B-I, C-IV, D-III
(D) A-IV, B-I, C-II, D-III

4. Assertion (A): A computerised adaptive test can reach a precise estimate of ability with fewer items than a fixed test.

Reason (R): In an adaptive test every candidate answers the same items in the same order.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

5. A test in which a different form is assembled for each candidate before the test begins, but does not change during the test, is called:

(A) linear-on-the-fly testing
(B) computerised adaptive testing
(C) linear fixed-form testing

(D) open-book testing

6. Which of the following are advantages of computer-based testing?

1. Quick scoring and declaration of results
2. Shuffling of questions to make copying difficult
3. Freedom from the need for test centres
4. A record of time taken on each item

Choose the correct answer from the options given below:

(A) 1, 2 and 3 only
(B) 2, 3 and 4 only
(C) 1, 3 and 4 only
(D) 1, 2 and 4 only

6.9 Innovations in evaluation

Syllabus: Innovations in evaluation systems

EXAM FOOTPRINT

Asked occasionally; usually a multiple-select item on newer assessment practices such as portfolios, rubrics, open-book and peer assessment.

Core

Why innovations are needed

A single written examination at the end of a course tests mainly what a student can recall in three hours. It says little about what the student can do, how the student has grown, or how the student thinks about his or her own work. The innovations in this section try to fill these gaps. Most of them are tools of formative evaluation (रचनात्मक मूल्यांकन); some also change summative examinations.

The National Education Policy 2020 states the direction plainly. In its words (paragraph 4.34), assessment should shift from one that is “summative and primarily tests rote memorization skills” to one that is “more regular and formative, is more competency-based, promotes learning and development”, and “tests higher-order skills, such as analysis, critical thinking, and conceptual clarity”. It also proposes a holistic, multidimensional progress card. The shift runs from summative rote testing towards regular, formative, competency-based assessment, never the other way.

Rubrics

A **rubric (रूब्रिक, मूल्यांकन मानदंड-तालिका)** is a scoring guide. It states the criteria on which a piece of work will be judged and describes what performance looks like at each level, for example excellent, good, fair and poor. A rubric makes marking transparent and consistent, and when it is shared in advance it tells students what is expected. A rubric is not a question bank, a timetable or a marks register.

A rubric has three parts.

- **Criteria**: the aspects of performance to be judged, such as content, organisation, language and referencing.
- **Scale**: the levels of performance, such as 4 (excellent) to 1 (poor).

- **Descriptors**: the wording in each cell that says what that level of that criterion looks like. Descriptors are what make two markers give the same score.

There are two kinds of rubric.

- An **analytic rubric** scores each criterion separately and describes levels for each. It gives a profile of strengths and weaknesses as well as a total. It is slower but diagnostic.
- A **holistic rubric** describes levels of overall quality. The marker chooses the level that best fits the work as a whole and gives a single score. It is quick but gives little diagnostic feedback.

Do not confuse either with a **checklist**, which only records whether a feature is present or absent, or an **anecdotal record**, which is a narrative note of an observed incident. Both are taught in Section 6.4.

Portfolios

A **portfolio (पोर्टफोलियो)** is a purposeful collection of a student's work gathered over a period: drafts, assignments, projects, reflections. Because it covers time, it shows growth and process as well as finished products. Two features separate a portfolio from a mere folder of papers: the work is **selected** against stated criteria, and the student adds a **reflection** explaining why each piece was chosen and what it shows. Collecting everything without selection and reflection is only storage.

An **e-portfolio** is a portfolio kept in electronic form: documents, presentations, audio and video, with the student's reflections and the teacher's comments. It can be shared online and updated over years. It is not an e-book, e-journal or e-library.

Open-book and on-demand examinations

In an **open-book examination (खुली पुस्तक परीक्षा)** candidates may consult books and notes while answering. Since facts can be looked up, questions of recall lose their value. A good open-book paper asks candidates to apply, interpret, analyse and judge, so that the books serve as tools and not as answer keys. A paper that can be answered by copying passages tests only speed in finding pages. Open-book does not mean choosing one's own date, examiner or questions.

In an **on-demand examination (मांग पर परीक्षा)** the learner takes the examination when he or she feels ready, instead of on a date fixed for everyone. A unique paper is generated for each sitting from a large **question bank (प्रश्न बैंक)** according to a blueprint, so that papers are of equal standard. The system suits open and distance learning; the National Institute of Open Schooling (NIOS) runs an On-Demand Examination. Its defining feature is freedom of timing. Adaptive testing (Section 6.8) also draws on an item bank, but its defining feature is that difficulty changes during the test.

Self-assessment and peer assessment

In **self-assessment (स्व-आकलन)** students judge their own work against clear criteria. The main gain is awareness of what they have mastered and what they have not, so that they can plan their next steps. This awareness of one's own learning is **metacognition (अधिसंज्ञान)**. Self-assessment supplements the teacher's feedback (प्रतिपुष्टि); it is not a way of avoiding it, ranking oneself against classmates, or getting marks without study. It is mainly reflection and need not produce a grade at all.

In **peer assessment (सहपाठी आकलन)** students judge one another's work against given criteria, usually with a rubric. It can be confined to comments that help a classmate improve a draft, without any marks. Its main benefit to the student who assesses is a clearer grasp of what good work looks like: applying the criteria to someone else's work makes the standard explicit in one's own mind and improves one's own later work. A lighter marking load is a benefit to the teacher, not to the learner.

Peer marks can be distorted by friendship, rivalry and vague standards. Four remedies improve their reliability (विश्वसनीयता): a clear rubric that gives every marker the same criteria; anonymous marking;

several markers per script, averaged; and training with sample scripts. Letting friends mark each other, leaving criteria to each marker or using one peer per script all make matters worse.

Performance-based and authentic assessment

In **performance-based assessment (निष्पादन आधारित आकलन)** students show a skill by doing something: an experiment, a presentation, a field report, a moot court argument. It reveals skills that paper-and-pencil tests cannot show. Its weakness is on the scoring side: it takes longer to mark and depends on the judgement of the assessor, which is why rubrics are used. A multiple-choice test is quicker and more objective to score; a performance task is not.

Authentic assessment (प्रामाणिक आकलन) goes one step further. The task reproduces real-life or work conditions: a real product, a real audience and often a real deadline, such as a news bulletin for the campus radio, a case analysis for a local firm, or a report for a panchayat. It shows what students can do with what they know.

Term	Deciding feature
Authentic assessment	A task that reproduces real-world conditions
Portfolio assessment	Work collected and selected over time, with reflection
Ipsative assessment	The learner compared with his or her own earlier performance
Objective assessment	Fixed-answer items such as multiple-choice

Remote proctoring

In online examinations **remote proctoring (दूरस्थ निरीक्षण)** lets candidates take a test from their own location while software, a human invigilator, or both, watch them through the webcam, microphone and screen. It verifies identity and flags suspicious behaviour, doing the work of an invigilator in an examination hall. It has nothing to do with sending papers by courier or evaluating scripts at a distant centre.

> **CLASSROOM EXAMPLE**
>
> In the M.A. English programme at a state university, Dr Malik replaces one sessional test with a portfolio. Each student keeps drafts of three essays, the comments of two classmates on each draft (given with a four-criterion analytic rubric), and a final one-page reflection on how the writing changed. The students mark each other's drafts anonymously and only with comments. At the end, Dr Malik grades the portfolio with the same rubric, and a colleague second-marks a sample to check consistency. The students learn the criteria by using them, and the final grade rests on selected work and reflection, not on everything a student happened to write.

OFTEN CONFUSED

- Analytic and holistic rubric. Analytic = separate score for each criterion, with a profile. Holistic = one overall score.
- Rubric and checklist. Rubric = levels described for each criterion. Checklist = present or absent only.
- Criteria and descriptors. Criteria = aspects judged. Descriptors = what each level of each aspect looks like.
- Portfolio and folder. Portfolio = selected work with reflection. Folder = everything, unselected.
- Open-book and on-demand examination. Open-book = books allowed during the test. On-demand = learner chooses when to take the test.
- On-demand and adaptive testing. On-demand = choice of timing. Adaptive = difficulty changes during the test.
- Self- and peer assessment. Self = judging one's own work. Peer = judging classmates' work.
- Authentic and ipsative assessment. Authentic = real-world task. Ipsative = comparison with one's own earlier performance.
- Positive and negative washback. Positive = a test that encourages desirable learning. Negative = a test that narrows teaching to what is tested.
- Washback and practice effect. Washback = effect of a test on teaching and learning. Practice effect = score gain from having taken a test before.

FACT WATCH

- National Education Policy 2020, paragraph 4.34: assessment to shift from summative, rote-memory testing to regular, formative, competency-based assessment testing higher-order skills (as of October 2026).
- UGC document: Evaluation Reforms in Higher Educational Institutions (2019) (as of October 2026).

Going Further

This layer is for Sets 11 to 20 of the booklet. Read it after you have finished Set 10.

Kinds of portfolio

Kind	What it holds
Working portfolio	Work in progress, including drafts
Showcase (display) portfolio	The pieces the student selects as his or her best
Assessment portfolio	Work chosen to show achievement of stated objectives, judged against criteria

Portfolios are strong in authenticity and in showing growth. Their weakness is in measurement: different assessors can judge the same portfolio very differently, so inter-rater reliability is low. The usual remedies are clear rubrics, training of raters, double marking and moderation meetings. Authenticity is the strength of the portfolio, not its weakness; negative marking and longer time limits have nothing to do with the problem.

Weighted rubric scores

When the criteria of an analytic rubric carry different weights, multiply each criterion score by its weight and add.

> **WORKED EXAMPLE**
>
> A rubric has three criteria weighted 50, 25 and 25 per cent, each scored on a four-point scale. A student gets 2, 4 and 4.
>
> Weighted score = 0.50 x 2 + 0.25 x 4 + 0.25 x 4 = 1.0 + 1.0 + 1.0 = 3.0 out of 4
>
> As a percentage: 3.0 / 4 x 100 = 75 per cent
>
> The unweighted mean, (2 + 4 + 4) / 3 = 3.33, or 83.3 per cent, ignores the heavier first criterion.

Fair marks for group work

Giving every member of a group the same mark lets those who did little benefit from the work of others (free riding). A fairer procedure combines a mark for the group product with peer ratings of each member's contribution, made against agreed criteria. One common way is to multiply the group mark by a contribution factor: the member's average peer rating divided by the group's average rating.

> **WORKED EXAMPLE**
>
> A group project earns 70. Members' average peer ratings (out of 5) are Aman 4.5, Bela 4.0, Chetan 3.5 and Dev 2.0. The group average is 3.5.
>
> Aman: 70 x 4.5 / 3.5 = 90
>
> Bela: 70 x 4.0 / 3.5 = 80
>
> Chetan: 70 x 3.5 / 3.5 = 70
>
> Dev: 70 x 2.0 / 3.5 = 40

Washback

Washback (परीक्षा का पश्च प्रभाव), also called backwash, is the effect of a test on teaching and learning. It is **negative** when a test narrows the curriculum, for example when a high-stakes multiple-choice test makes teachers drill answering technique and drop extended writing. It is **positive** when the test encourages desirable learning, for example when an examination with open-ended problems makes teachers teach problem-solving. Alderson and Wall examined the idea in their paper 'Does washback exist?' (1993). Messick counted such consequences of testing as part of a full account of validity (consequential validity). Washback is not the practice effect (score gain from retaking a test) or the halo effect (one impression colouring other judgements).

Comment-only feedback

In a study published in 1988, the Israeli psychologist Ruth Butler gave school pupils one of three kinds of feedback on their work: comments only, grades only, or grades with comments. Those who received comments only improved most and stayed most interested. Those who received grades with comments did no better than those given grades alone: the grade drew attention away from the advice. Paul Black and Dylan Wiliam, whose 'Inside the Black Box' (1998) made the case for formative assessment, drew on Butler's work, and her findings are widely cited in support of comment-only marking of formative work. Two kinds of information seem better than one, but here they are not.

360-degree feedback

360-degree feedback collects judgements about a person from all around: from superiors, colleagues, those served, and the person himself or herself. In higher education it is used mainly to appraise teachers, combining feedback from students, peers and the head of department with the teacher's self-appraisal. Its strength is a rounder picture than any single source gives; its risk is that unguided raters repeat each other's biases, so criteria are needed here too.

Question banks

A **question bank** is a large, classified store of items, each tagged by unit, objective, cognitive level and, after try-out, difficulty and discrimination. Papers can then be drawn from it according to a blueprint, so that different papers are comparable. Question banks make on-demand examinations, shuffled computer-based tests and adaptive tests possible. Their risk is exposure: an item seen too often becomes known.

Outcome-based assessment and the UGC's reform documents

In **outcome-based education (परिणाम आधारित शिक्षा)** the programme first states its learning outcomes (अधिगम परिणाम): what a graduate should know and be able to do. Teaching and assessment are then aligned to those outcomes, and each question or task is checked against the outcome it is meant to measure. The UGC's Learning Outcomes-based Curriculum Framework (LOCF) and the NHEQF (Section 6.7) both describe qualifications in this way.

The UGC's document Evaluation Reforms in Higher Educational Institutions (2019) applied these ideas to examinations in colleges and universities. It argued for continuous internal assessment alongside the end-semester examination, assessment linked to stated learning outcomes instead of rote memory, and wider use of technology in examinations. Read it as the higher education counterpart of the school-level shift in NEP 2020.

CHECK YOURSELF

1. Match List I with List II.

List I	List II
A. Holistic rubric	I. Records only whether a feature is present or absent
B. Analytic rubric	II. One overall score for the whole work
C. Checklist	III. Narrative note of an observed incident
D. Anecdotal record	IV. Separate score for each criterion

Choose the correct answer from the options given below:

(A) A-IV, B-II, C-I, D-III
(B) A-II, B-IV, C-I, D-III
(C) A-II, B-IV, C-III, D-I
(D) A-I, B-IV, C-II, D-III

2. Assertion (A): Recall questions are of little value in an open-book examination.

Reason (R): In an open-book examination candidates may consult books and notes while answering.

(A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

3. Given below are two statements:

Statement I: A showcase portfolio contains every piece of work a student has produced during the course.

Statement II: The chief measurement weakness of portfolio assessment is low agreement among raters.

In the light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

4. A rubric has three criteria weighted 30, 30 and 40 per cent, each scored on a four-point scale. A student scores 4, 3 and 2. The weighted score as a percentage of the maximum is:

(A) 72.5 per cent
(B) 75 per cent
(C) 80 per cent
(D) 70 per cent

5. Which of the following improve the reliability of peer assessment?

1. A clear rubric shared with all markers
2. Anonymous scripts
3. Averaging the marks of several peer markers
4. Allowing each marker to choose his or her own criteria

Choose the correct answer from the options given below:

(A) 1 and 4 only
(B) 2, 3 and 4 only
(C) 1, 2 and 3 only
(D) 1, 2, 3 and 4

6. After a university entrance test began to consist only of short factual multiple-choice items, coaching classes in a town stopped setting essays and drilled elimination tricks. This is an example of:

(A) positive washback
(B) the practice effect
(C) the halo effect
(D) negative washback

7. A learner in an open schooling programme takes a mathematics paper on a date of her own choosing; the paper is generated for her from a large bank of items according to a blueprint. This is:

(A) adaptive testing
(B) on-demand examination
(C) open-book examination
(D) remote proctoring

8. Research on formative feedback, such as Ruth Butler's, suggests that pupils' later performance improves most when they receive:

(A) comments without grades
(B) grades together with comments
(C) grades without comments
(D) their rank in the class

Answers to “Check yourself”: Chapter 6

Section 6.1

1. (C), because only (C) passes a judgement of worth against objectives. (A), (B) and (D) report quantities (items, time, marks), which is measurement.

2. (B), because measurement only assigns numbers, assessment gathers and interprets evidence, and evaluation judges worth and includes both.

3. (A), because measurement stops at the number, and evaluation combines quantitative and qualitative description with a value judgement.

4. (D), because re-teaching is instructional, promotion is administrative, choice of elective is guidance and data for a study is research. (C) wrongly swaps promotion and guidance.

5. (A), because both are true and the indirect, sampled nature of measurement is exactly why error is always present.

6. (C), because objectives come first, then learning experiences, then tools are applied, then results are interpreted. (D) interprets before collecting evidence.

7. (D), because CIPP stands for Context, Input, Process and Product. Implementation is what the Process part studies, but it is not the name of a letter.

Section 6.2

1. (C), because 2 x 0.80 / (1 + 0.80) = 1.60 / 1.80 = 0.89. Choosing 0.80 reports the half-test correlation, which is always lower than the full-test reliability.

2. (B), because test-retest gives stability, parallel forms give equivalence, Kuder-Richardson gives internal consistency from one sitting, and inter-rater gives agreement between examiners.

3. (A), because both are true, and the fact that reliability does not guarantee validity is exactly why a consistent test can still measure the wrong thing.

4. (D), because the criterion (first-year grades) comes later. If both measures had been taken at the same time, it would be concurrent validity.

5. (B), because SEM = 8 x square root of 0.25 = 8 x 0.5 = 4, and 50 plus or minus 4 gives 46 to 54. (D) uses 8 x 0.75 = 6, which forgets the square root.

6. (A), because statement 2 is wrong: on a pure speed test the odd and even halves are nearly identical, so split-half gives a spuriously high estimate.

7. (B), because the ceiling is the square root of the reliability, which is 0.70. (C) repeats the reliability itself.

8. (C), because the narrow examination leaves out an essential part of the construct, while face validity is only a surface impression; correlation with a later criterion is predictive validity.

Section 6.3

1. (C), because placement decides the starting point, formative gives feedback during learning, diagnostic finds causes, and summative certifies at the end. (A) wrongly gives the causes to summative.

2. (D), because admitting a fixed number by rank is norm-referenced; certifying all who reach a fixed standard is criterion-referenced.

3. (A), because both are true, and it is precisely because placement evaluation checks for outcomes already achieved that it can send some learners ahead.

4. (A), because statement 3 is wrong (daily feedback is formative) and statement 4 is wrong (CRT scores are often bunched at the high end after good teaching).

5. (D), because diagnosis goes: who has difficulty, where, why, and then remedy. (A) looks for causes before locating the area.

6. (B), because Scriven wrote about curriculum and programme evaluation in 1967; Bloom and colleagues later applied the terms to student learning and mastery learning.

7. (C), because the student is compared with her own earlier performance, not with others or with a fixed standard.

8. (A), because 42 / 75 = 0.56, or 56 per cent of students, which is below the target of 60 per cent. (B) treats the number of students as a percentage.

Section 6.4

1. (B), because FV = (24 + 8) / 80 = 0.40 and DI = (24 - 8) / 40 = 16 / 40 = 0.40. (A) divides the difference by 80, both groups together.

2. (D), because a premise is in the first column of a matching item, a distractor is a plausible wrong option, the stem poses the problem and the key is the correct option.

3. (C), because the Assertion is true but the Reason is false: essay tests have few questions and sample the course narrowly.

4. (B), because S = 36 - 15 / 3 = 36 - 5 = 31; the 9 omitted items are not counted. (A) subtracts all 15 wrong answers.

5. (A), because planning (blueprint) comes first, then writing, try-out, item analysis and the final form. (C) puts item analysis before the try-out, which gives it no data.

6. (C), because statement 2 describes a portfolio; a cumulative record is the institution's year-by-year record of the student.

7. (A), because 0.27 x 400 = 108. (D) is both groups together.

8. (C), because a working distractor attracts more low scorers than high scorers; when able students prefer it, the item needs review.

Section 6.5

1. (B), because the mean is 40 / 5 = 8, the middle score is 7 and 7 occurs twice. (A) puts the mean and mode in the wrong places.

2. (A), because z = (58 - 50) / 4 = 2.0 and T = 50 + 10 x 2 = 70. (D) divides the raw score by the SD without subtracting the mean.

3. (C), because a raw score needs a norm or a criterion to be meaningful, but there is no rule that it must be turned into a T-score; it may equally be read against a standard.

4. (A), because room numbers only label, merit positions show order, Celsius has equal units but no true zero, and time has a true zero.

5. (D), because about 95 per cent of cases lie within 2 SD either way. 68 per cent is within 1 SD.

6. (B), because shared variance is $r^2 = 0.49$. Correlation does not show cause, and r is not a percentage of students.

7. (C), because the larger SD shows a more varied class, which needs more differentiated teaching; equal means hide unequal spread.

8. (C), because when raw marks are added, the set with the larger spread decides the ranking. Converting both to standard scores first removes this unfairness.

Section 6.6

1. (B), because ‘continuous’ is about time: assessment goes on regularly through the year. Option (A) describes ‘comprehensive’, the other half of the name.

2. (C), because life skills, values and attitudes, and health and physical education are co-scholastic. Mathematics and social science are curricular subjects, so they are scholastic.

3. (D), because in absolute grading it is the mark boundaries that are fixed, not the number of students; any number can get the top grade. Statement II correctly describes relative grading.

4. (A), because grades are less prone to misclassification precisely because small mark differences are mostly measurement error; R explains A.

5. (A), because a fixed mark boundary is absolute grading, a fixed top proportion is relative grading, a grade without marks is direct grading, and marks converted into grades is indirect grading.

6. (C), because a fixed distribution must spread even a uniformly able class over the grades, so some good students are pushed down. Everyone getting the top grade is the risk of lenient absolute grading, not of relative grading.

Section 6.7

1. (B), because A+ is 9, B (Above Average) is 6, P is 4 and B+ is 7.

2. (C), because credit points are 4 x 10 + 3 x 6 + 3 x 7 + 2 x 5 = 40 + 18 + 21 + 10 = 89 over 12 credits, and 89 / 12 = 7.4166…, reported as 7.42. The simple mean of the grade points, 7.00, ignores the credits.

3. (A), because 18 x 7.5 + 22 x 6.5 = 135 + 143 = 278 over 40 credits gives 6.95. The simple average 7.00 forgets that Semester II carries more credits.

4. (C), because points earned are 90 x 7.2 = 648, points needed are 120 x 7.5 = 900, and the 252 still required over 30 credits give 8.4. The answer 7.8 wrongly treats the two parts as equal.

5. (B), because lectures and tutorials count one credit per weekly hour (3 + 1 = 4) and two practical hours make one credit, so 4 + 1 = 5. Counting each practical hour as a credit gives the wrong answer 6.

6. (C), because Environmental Science is one of the two AECCs, but a Generic Elective is taken from an unrelated discipline; one offered by the student’s own discipline is a Discipline Specific Elective.

7. (A), because the SGPA weights each course by its credits, so the correct denominator is total credits; R explains why A is true.

8. (D), because statements 1, 2 and 4 are correct. Exit after one year earns a certificate; the diploma comes after two years.

Section 6.8

1. (B), because a CBT answer can be changed, cleared or marked for review until submission, while a darkened OMR circle cannot be altered. A CBT candidate cannot choose the difficulty of the paper.

2. (D), because unequal familiarity with computers is a limitation of CBT, not something it removes; but the computer does log time spent on each item.

3. (C), because normalisation equates shift papers, randomisation shuffles items, adaptive testing chooses each item by earlier responses, and a mock test is practice with the interface.

4. (C), because A is true but R is false: in an adaptive test candidates meet different items chosen by their own responses.

5. (A), because a linear-on-the-fly test builds a separate form for each candidate in advance but does not adapt during the test; adaptation is the mark of CAT.

6. (D), because quick scoring, shuffling and time logs are advantages; CBT still needs test centres.

Section 6.9

1. (B), because a holistic rubric gives one overall score, an analytic rubric scores each criterion, a checklist records presence or absence, and an anecdotal record narrates an incident.

2. (A), because recall loses value exactly because the facts can be looked up; R explains A.

3. (D), because a showcase portfolio holds only the pieces the student selects as best, while low inter-rater agreement is indeed the chief weakness of portfolios.

4. (A), because 0.3 x 4 + 0.3 x 3 + 0.4 x 2 = 1.2 + 0.9 + 0.8 = 2.9 out of 4, which is 72.5 per cent. The unweighted mean, 3 out of 4 or 75 per cent, ignores the weights.

5. (C), because a common rubric, anonymity and averaging several markers all reduce bias and error; letting each marker choose criteria makes marks less consistent.

6. (D), because the test has narrowed teaching to what it tests, which is negative washback. The practice effect is a score gain from taking a test again.

7. (B), because freedom to choose the date, with a paper drawn from a bank by blueprint, defines on-demand examination; adaptive testing changes difficulty during the test.

8. (A), because comment-only feedback produced the most improvement, and adding a grade cancelled the benefit of the comments.

Quick-Reference Tables

These tables gather, in one place, the names, orders, full forms, dated facts and formulae taught in the chapters; the last column gives the section to reread.

Thinkers and what each is linked with

The year is given only where the guide gives one. Where the guide says that a link is the one "Indian textbooks" make, the table keeps that caution.

Thinker	Linked with	Year	See
Allen, Dwight W. (and colleagues)	Micro-teaching, developed at Stanford University	1963 (the year usually given)	1.8
Amidon, Edmund (with Hunter)	Teaching as "an interactive process, primarily involving classroom talk, between teacher and pupils"		1.1
Anderson, Lorin and Krathwohl, David	Revised cognitive taxonomy: Remember to Create; knowledge dimension	2001	1.4
Armstrong, H. E. (Henry Edward)	Heuristic method in science teaching		4.4
Arnett, Jeffrey Jensen	Emerging adulthood (about 18 to 25)	2000	2.1
Ausubel, David	Meaningful reception learning; advance organiser; subsumption; "what the learner already knows"	1968 (*Educational Psychology: A Cognitive View*)	2.2, 3.3
Bandura, Albert	Observational learning (Bobo doll); attention, retention, reproduction, motivation; social cognitive theory; self-efficacy	1977 (self-efficacy)	2.3, 2.6, 3.2
Bigge, Morris L.	Three levels of teaching; "thoughtless" to "thoughtful"; *Learning Theories for Teachers*; with Hunt, *Psychological Foundations of Education* (1962)	1964	1.5
Binet, Alfred (with Théodore Simon)	First workable intelligence scale; mental age	1905	2.8
Bloom, Benjamin S.	*Taxonomy of Educational Objectives*, *Handbook I* (cognitive, with others); mastery learning; with Hastings and Madaus, formative and summative applied to student learning (1971)	1956	1.4, 4.4, 6.3
Bourdieu, Pierre	Cultural capital: embodied, objectified, institutionalised ("The Forms of Capital")	1986	3.3
Bruner, Jerome	Discovery learning; enactive, iconic, symbolic modes; spiral curriculum (*The Process of Education*, 1960); concept attainment model; scaffolding (with Wood and Ross, 1976)	1960; 1966 (*Toward a Theory of Instruction*)	1.9, 2.2
Cattell, Raymond	Fluid and crystallised intelligence		2.8
Chickering and Gamson	Seven principles for good practice in undergraduate education	1987	1.7
Crowder	Branching programming (intrinsic programming)		4.2

Thinker	Linked with	Year	See
Dale, Edgar	Cone of Experience (*Audiovisual Methods in Teaching*)	1946 (first edition); 1969 (third edition)	5.1
Das, J. P., Naglieri, Jack and Kirby, John	PASS theory of intelligence, drawing on A. R. Luria		2.8
Dave, R. H.	Psychomotor taxonomy: five levels, Imitation to Naturalisation	usually dated 1970	1.4
Deci, Edward and Ryan, Richard	Self-determination theory: autonomy, competence, relatedness		2.6
Dewey, John	Reflective thinking in five steps (*How We Think*); learning by doing; progressive education; basis of project and problem-solving methods	1910 (*How We Think*)	1.5, 2.3, 4.4
Dweck, Carol	Fixed (entity) and growth (incremental) mindsets		2.6
Elkind, David	Adolescent egocentrism: imaginary audience and personal fable	1967	2.1
Erikson, Erik H.	Eight psychosocial stages; identity crisis (*Childhood and Society*)	1950	2.1
Flanders	Interaction Analysis Categories (ten categories, a tally every three seconds); rule of two-thirds	late 1950s (University of Minnesota)	1.8
Fleming, Neil	VARK questionnaire	1987	2.4
Freire, Paulo	Critique of banking education; dialogue (*Pedagogy of the Oppressed*)		2.7
Gage, N. L.	Teaching as “a form of interpersonal influence aimed at changing the behaviour potential of another person”; *The Scientific Basis of the Art of Teaching*	1978 (the book)	1.1
Gagné, Robert M.	*The Conditions of Learning*; eight types of learning; nine events of instruction; five learned capabilities	1965	1.9, 2.3
Gardner, Howard	Multiple intelligences (*Frames of Mind*): seven, later eight with naturalistic	1983	2.8
Gilligan, Carol	Ethic of care, against Kohlberg’s ethic of justice (*In a Different Voice*)	1982	2.5
Glaser, Robert	Basic Teaching Model (1962); norm-referenced and criterion-referenced measurement (1963)	1962; 1963	1.9, 6.3
Goleman, Daniel	Popularised emotional intelligence (*Emotional Intelligence*)	1995	2.6
Gordon, William J. J.	Synectics (personal family)	1961	1.9
Green, Thomas F.	Map (topology) of activities related to teaching, as Indian textbooks credit it (*The Activities of Teaching*)	1971 (the book)	1.1
Gronlund, Norman E.	General objectives clarified by specific learning outcomes; evaluation formula that Indian textbooks attribute to him; “first aid” picture of formative evaluation, as Indian B.Ed. texts quote him		1.3, 6.1, 6.3

Thinker	Linked with	Year	See
Guilford, J. P.	Structure-of-intellect model; convergent and divergent production		2.8
Hall, G. Stanley	Storm and stress (*Adolescence*)	1904	2.1
Harrow, A. J.	Psychomotor taxonomy: six levels, Reflex movements to Non-discursive communication	1972	1.4
Havighurst, Robert J.	Developmental tasks (*Developmental Tasks and Education*)	1948 (first edition)	2.1
Herbart, Johann Friedrich	Memory level; apperception and apperceptive mass; four stages, later the five formal steps	1776–1841	1.5, 1.6
Herzberg, Frederick	Two-factor theory: hygiene factors and motivators		2.6
Hunt, Maurice P.	Reflective level (the name the examination and most Indian textbooks use); with Lawrence E. Metcalf, *Teaching High School Social Studies*	1955	1.5
Jackson, Philip W.	Pre-active and interactive teaching (*Life in Classrooms*); the three phases are linked with him	1968	1.2
Joyce, Bruce and Weil, Marsha	*Models of Teaching*; four families of models; focus, syntax, social system, principles of reaction, support system	1972	1.9
Keller, Fred S.	Personalised System of Instruction (Keller Plan); “Good-bye, Teacher…”	1968	4.4
Kilpatrick, William Heard	Project method: “a wholehearted purposeful activity proceeding in a social environment”	1918	4.4
Knowles, Malcolm	Andragogy (popularised it); six assumptions about adult learners; self-directed learning; learning contract		2.7
Kohlberg, Lawrence	Three levels and six stages of moral reasoning; Heinz dilemma		2.5
Köhler, Wolfgang	Insight (the chimpanzee Sultan); Gestalt psychology		2.3
Kolb, David A.	Experiential learning cycle; diverging, assimilating, converging, accommodating styles (*Experiential Learning*)	1984	2.4
Kounin, Jacob	Withitness, overlapping, momentum, smoothness (*Discipline and Group Management in Classrooms*)	1970	1.7
Krathwohl, Bloom and Masia	Affective taxonomy, *Handbook II*: Receiving to Characterisation	1964	1.4
Lewin, Kurt	Leadership styles (with Lippitt and White, 1939); field theory, B = f(P, E); action research	1939 (leadership studies)	1.7, 2.3, 4.4
Mager, Robert F.	Three-part objective: performance, condition, criterion (*Preparing Objectives for Programmed Instruction*)	1962	1.3
Marcia, James	Identity statuses: diffusion, foreclosure, moratorium, achievement	1966	2.1
Maslow, Abraham	Hierarchy of needs; deficiency and growth needs		2.6
McClelland, David	Need for achievement (n-Ach); needs for achievement, affiliation and power		2.6

Thinker	Linked with	Year	See
Mezirow, Jack	Transformative learning; disorienting dilemma; perspective transformation		2.7
Mishra and Koehler	TPACK (Technological Pedagogical Content Knowledge)	2006	1.8
Morrison, Henry Clinton	Understanding level; unit plan and mastery formula (*The Practice of Teaching in the Secondary School*); teaching as contact of "a more mature personality and a less mature one"	1926 (the book); 1934 (the definition)	1.1, 1.5
Osborn, Alex F.	Brainstorming (*Applied Imagination*)		4.3
Parkhurst, Helen	Dalton Plan		4.4
Passi, B. K.	Indian model of micro-teaching (with NCERT)		1.8
Pavlov, Ivan	Classical conditioning		2.3
Piaget, Jean	Four stages; schema, assimilation, accommodation, equilibration; cognitive constructivism; heteronomous and autonomous morality		2.2, 2.5
Puentedura, Ruben	SAMR model of technology use		1.8, 5.4
Rogers, Carl	Non-directive teaching (personal family); teacher as facilitator		1.9, 2.6
Rosenthal and Jacobson	Pygmalion effect (*Pygmalion in the Classroom*, "Oak School")	1968	3.2
Salovey, Peter and Mayer, John	Introduced the term emotional intelligence	1990	2.6
Scriven, Michael	Introduced formative and summative evaluation, for curricula and programmes ("The methodology of evaluation")	1967	6.3
Shulman, Lee	Pedagogical content knowledge (PCK)	1986	1.8
Siemens, George and Downes, Stephen	Connectivism; the first cMOOC, CCK08	2008 (CCK08)	2.3, 4.6
Simpson, E. J.	Psychomotor taxonomy: seven levels, Perception to Origination	1972	1.4
Skinner, B. F.	Operant conditioning; schedules of reinforcement; programmed instruction; linear programming		2.3, 4.2
Slavin, Robert	STAD; group goals with individual accountability		4.3
Smith, B. O.	Teaching as "a system of actions intended to induce learning"		1.1
Spearman, Charles	Two-factor theory (g and s); Spearman-Brown formula (with William Brown, in separate papers)	1910 (Spearman-Brown)	2.8, 6.2
Sternberg, Robert	Triarchic theory: analytical, creative, practical		2.8
Suchman	Inquiry training model		1.9
Terman, Lewis (and William Stern)	Stern divided mental age by chronological age; Terman multiplied by 100 in the Stanford-Binet (IQ)		2.8
Thorndike, E. L. (Edward Lee)	Trial and error; connectionism; laws of readiness, exercise and effect; named the halo effect (1920)	1920 (halo effect)	2.3, 3.6
Thurstone, L. L.	Seven primary mental abilities		2.8

Thinker	Linked with	Year	See
Tolman, Edward C.	Latent learning; cognitive map		2.3
Tomlinson, Carol Ann	Differentiated instruction: content, process, product, learning environment		2.9
Tyler, Ralph W.	*Basic Principles of Curriculum and Instruction*; objectives-based evaluation; basis of the evaluation triangle	1949	1.3, 6.1
Vroom, Victor	Expectancy theory: expectancy, instrumentality, valence		2.6
Vygotsky, Lev	Zone of proximal development; social constructivism; private and inner speech (*Mind in Society*)	1978 (in English)	2.2
Wechsler, David	Deviation IQ (mean 100, SD 15)		2.8
Yerkes and Dodson	Inverted-U law of arousal and performance	1908	2.6

Levels, stages and sequences to learn in order

Each order runs from first to last, or from lowest to highest, as the guide teaches it.

Sequence	Order	See
Teaching continuum (least to most use of understanding)	Conditioning, Training, Instruction, Teaching	1.1
Phases of teaching (Jackson)	Pre-active, Interactive, Post-active (no “reactive” phase)	1.2
Functions of teaching	Diagnostic, Prescriptive, Evaluative	1.2
Aims to objectives (broad to narrow)	Aim, Goal, Instructional objective, Learning outcome	1.3
Bloom’s cognitive domain, 1956 (lowest to highest)	Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation	1.4
Revised cognitive domain, 2001 (lowest to highest)	Remember, Understand, Apply, Analyse, Evaluate, Create	1.4
Affective domain, Krathwohl, Bloom and Masia, 1964 (lowest to highest)	Receiving, Responding, Valuing, Organisation, Characterisation by a value or value complex	1.4
Psychomotor domain, Dave (5 levels)	Imitation, Manipulation, Precision, Articulation, Naturalisation	1.4
Psychomotor domain, Simpson (7 levels)	Perception, Set, Guided response, Mechanism, Complex overt response, Adaptation, Origination	1.4
Psychomotor domain, Harrow (6 levels)	Reflex movements, Basic fundamental movements, Perceptual abilities, Physical abilities, Skilled movements, Non-discursive communication	1.4
Levels of teaching (lowest to highest)	Memory (Herbart), Understanding (Morrison), Reflective (Hunt)	1.5
Herbart’s own four stages	Clearness, Association, System, Method	1.5
Herbartian five formal steps	Preparation (ending with statement of aim), Presentation, Comparison and association, Generalisation, Application	1.5
Morrison’s mastery formula	Pre-test, teach, test the result, adapt procedure, teach and test again to the point of actual learning	1.5
Morrison’s five steps (E-P-A-O-R)	Exploration, Presentation, Assimilation, Organisation, Recitation	1.5

Sequence	Order	See
Reflective-level teaching	Recognising and defining the problem, Formulating hypotheses, Testing the hypotheses, Drawing conclusions	1.5
Dewey's five steps of reflective thinking (*How We Think*)	Felt difficulty, Location and definition, Suggestion of a possible solution, Development by reasoning of its bearings, Further observation and experiment	1.5, 4.4
Micro-teaching cycle	Plan, Teach, Feedback, Re-plan, Re-teach, Re-feedback	1.8
Indian model of micro-teaching, minutes	Teach 6, Feedback 6, Re-plan 12, Re-teach 6, Re-feedback 6; total 36 (initial planning not counted)	1.8
Phases of a micro-teaching programme	Knowledge acquisition, Skill acquisition, Transfer	1.8
Glaser's Basic Teaching Model	Instructional objectives, Entering behaviour, Instructional procedures, Performance assessment	1.9
Concept attainment model	Presentation of data and identification of the concept, Testing attainment of the concept, Analysis of thinking strategies	1.9
Gagné's nine events of instruction	Gaining attention, Informing learners of the objective, Stimulating recall of prior learning, Presenting the stimulus, Providing learning guidance, Eliciting performance, Providing feedback, Assessing performance, Enhancing retention and transfer	1.9, 2.3
Erikson's stages around the college years	4 Industry versus inferiority (school age), 5 Identity versus role confusion (adolescence), 6 Intimacy versus isolation (young adulthood), 7 Generativity versus stagnation (middle adulthood)	2.1
Piaget's stages	Sensorimotor (birth to 2), Pre-operational (2 to 7), Concrete operational (7 to 11), Formal operational (11 or 12 onwards)	2.2
Vygotsky's development of speech	Social speech, Private speech, Inner speech	2.2
Bruner's modes of representation	Enactive, Iconic, Symbolic	2.2
Multi-store model of memory (Atkinson and Shiffrin)	Sensory register, Short-term (working) memory, Long-term memory	2.2
Gagné's eight types of learning (simplest to most complex)	Signal learning, Stimulus-response learning, Chaining, Verbal association, Discrimination learning, Concept learning, Rule (principle) learning, Problem solving	2.3
Bandura's processes of observational learning	Attention, Retention, Reproduction, Motivation	2.3
Kolb's experiential learning cycle	Concrete experience, Reflective observation, Abstract conceptualisation, Active experimentation	2.4
SQ3R (Robinson)	Survey, Question, Read, Recite, Review	2.4
Zimmerman's self-regulation cycle	Forethought, Performance, Self-reflection	2.4
Kohlberg's moral reasoning	Pre-conventional (1 punishment and obedience; 2 self-interest), Conventional (3 approval of others; 4 law and order), Post-conventional (5 social contract; 6 universal ethical principles)	2.5
Maslow's hierarchy (base to top)	Physiological, Safety, Love and belonging, Esteem, Self-actualisation	2.6

Sequence	Order	See
Learner autonomy (increasing)	Pedagogy, Andragogy, Heutagogy	2.7
Mezirow's transformative learning	Disorienting dilemma, Critical reflection, Perspective transformation	2.7
Seminar session	Chairperson introduces theme and speaker, Paper presented, Discussion (often a discussant first), Chairperson sums up	4.3
Kilpatrick's project method	Purposing, Planning, Executing, Evaluating	4.4
Problem-solving method	Recognising and defining the problem, Collecting data, Forming hypotheses, Testing hypotheses, Drawing a conclusion	4.4
Polya's problem solving	Understand the problem, Devise a plan, Carry out the plan, Look back	4.4
Action research (Lewin)	Plan, Act, Observe, Reflect, then plan again	4.4
Dale's Cone of Experience, 1969 (bottom, most concrete, to top, most abstract)	Direct purposeful experiences, Contrived experiences, Dramatised experiences, Demonstrations, Study trips, Exhibits, Educational television, Motion pictures, Recordings radio and still pictures, Visual symbols, Verbal symbols	5.1
Using a film in class	Preview, Prepare the class, Show, Follow up	5.1
SAMR (Puentedura)	Substitution, Augmentation, Modification, Redefinition	5.4
Narrow to broad	Measurement, Assessment, Evaluation (and Test, Measurement, Evaluation: instrument, process, judgement)	6.1
Steps of evaluation	Formulating objectives, Providing learning experiences, Applying tools of evaluation, Interpreting results	6.1
CIPP model (Stufflebeam)	Context, Input, Process, Product	6.1
Constructing an achievement test	Planning (blueprint), Writing items, Try-out, Item analysis, Assembling the final form	6.4
Steps of diagnosis	Identify students in difficulty, Locate the areas, Find the causes, Plan remedial instruction	6.3
Scales of measurement (Stevens)	Nominal, Ordinal, Interval, Ratio	6.5
Multiple exit in the four-year undergraduate programme	One year: certificate; two years: diploma; three years: bachelor's degree; four years: bachelor's (honours or honours with research)	6.7
NHEQF levels	4.5 certificate, 5 diploma, 5.5 bachelor's (three years), 6 bachelor's honours (four years) and postgraduate diploma, 6.5 master's, 7 master's such as M.Tech. after a four-year bachelor's, 8 doctorate	6.7

Acronyms and full forms

Acronym	Full form	See
ABC	Academic Bank of Credits	4.6, 6.7
ABCD	Audience, Behaviour, Condition, Degree (format for objectives)	1.3
AEC	Ability Enhancement Course (CCFUP 2022)	6.7
AECC	Ability Enhancement Compulsory Course (CBCS 2015)	6.7
AICTE	All India Council for Technical Education	4.6

Acronym	Full form	See
AQAR	Annual Quality Assurance Report	3.7
ARCS	Attention, Relevance, Confidence, Satisfaction (Keller)	2.6
ATI	Aptitude-treatment interaction	3.3
CA / MA	Chronological age / Mental age	2.8
CAI	Computer-assisted instruction	5.4
CAS	Cognitive Assessment System	2.8
CAT	Computerised adaptive test	6.8
CBCS	Choice Based Credit System	6.7
CBSE	Central Board of Secondary Education	6.6, 6.8
CBT	Computer Based Test	6.8
CC (BY, SA, NC, ND)	Creative Commons (Attribution, ShareAlike, NonCommercial, NoDerivatives)	5.4
CCE	Continuous and Comprehensive Evaluation	6.6
CCFUP	Curriculum and Credit Framework for Undergraduate Programmes	6.7
CEC	Consortium for Educational Communication	4.6
CGPA	Cumulative Grade Point Average	6.7
CIET	Central Institute of Educational Technology (NCERT)	4.6
CIPP	Context, Input, Process, Product	6.1
cMOOC / xMOOC	Connectivist MOOC / structured, instructor-led MOOC	4.6
COs, POs, PSOs, PEOs	Course outcomes, Programme outcomes, Programme-specific outcomes, Programme educational objectives	1.3
CPD	Continuous professional development	1.10
CRT / NRT	Criterion-referenced test / Norm-referenced test	6.3
DIKSHA	Digital Infrastructure for Knowledge Sharing	4.6
DSE / GE	Discipline Specific Elective / Generic Elective	6.7
DTH	Direct-to-home (television)	4.6
ERG	Existence, Relatedness, Growth (Alderfer)	2.6
FIAC (FIACS)	Flanders' Interaction Analysis Categories (FIACS when the category system is meant)	1.8
ICT	Information and Communication Technology	5.4
IGNOU	Indira Gandhi National Open University	4.6
IIMB	IIM Bangalore	4.6
INFLIBNET	Information and Library Network (Centre, Gandhinagar)	4.6
INI	Institutions of National Importance	4.6
IQ	Intelligence quotient	2.8
IQAC	Internal Quality Assurance Cell	3.7
IRINS	Indian Research Information Network System	4.6
IRT	Item response theory	6.8
KR-20, KR-21	Kuder-Richardson formulas 20 and 21	6.2

Acronym	Full form	See
LMS	Learning management system	5.4
LOCF	Learning Outcomes-based Curriculum Framework	6.9
LOFT	Linear-on-the-fly testing	6.8
MDC	Multidisciplinary courses (CCFUP 2022)	6.7
MHRD	Ministry of Human Resource Development (now Ministry of Education)	4.6
MIL	Modern Indian Language	6.7
MKO	More knowledgeable other	2.2
MOOC	Massive open online course	4.6
Moodle	Modular Object-Oriented Dynamic Learning Environment	5.4
n-Ach	Need for achievement (McClelland)	2.6
NAAC	National Assessment and Accreditation Council	3.7
NAD	National Academic Depository	4.6, 6.7
NCERT	National Council of Educational Research and Training	4.6
NCrF	National Credit Framework	6.7
NDLI	National Digital Library of India	4.6
NEP 2020	National Education Policy 2020	4.1, 6.9
NETF	National Educational Technology Forum	4.6
NHEQF	National Higher Education Qualifications Framework	6.7
NIOS	National Institute of Open Schooling	4.6, 6.9
NIRF	National Institutional Ranking Framework	4.6
NITTTR	National Institute of Technical Teachers Training and Research	4.6
NME-ICT	National Mission on Education through ICT	4.6
NPTEL	National Programme on Technology Enhanced Learning	4.6
NTA	National Testing Agency	6.8
OBE	Outcome-based education	1.3
OER	Open educational resources	5.4
OMR	Optical mark recognition	6.8
ONOS	One Nation One Subscription	4.6
PASS	Planning, Attention-arousal, Simultaneous processing, Successive processing	2.8
PBL	Problem-based learning	4.4
PCK	Pedagogical content knowledge	1.8
PR	Percentile rank	6.5
PSI	Personalised System of Instruction (Keller Plan)	4.4
RCEM	Regional College of Education, Mysore (approach to objectives)	1.3
RPwD Act	Rights of Persons with Disabilities Act, 2016	2.9
RTE Act	Right of Children to Free and Compulsory Education Act, 2009	2.9, 6.6
SAMR	Substitution, Augmentation, Modification, Redefinition	5.4

Acronym	Full form	See
SD	Standard deviation	6.5
SEC	Skill Enhancement Course	6.7
SEM	Standard error of measurement	6.2
SGPA	Semester Grade Point Average	6.7
SLM	Self-learning material	5.4
SPOC	Small private online course	4.6
SQ3R	Survey, Question, Read, Recite, Review	2.4
STAD	Student Teams-Achievement Divisions	4.3
SWAYAM	Study Webs of Active-Learning for Young Aspiring Minds	4.6
TPACK	Technological Pedagogical Content Knowledge	1.8
UCS, UCR, CS, CR	Unconditioned stimulus, Unconditioned response, Conditioned stimulus, Conditioned response	2.3
UDL	Universal Design for Learning	2.9
UGC	University Grants Commission	4.6, 6.7
VAC	Value-Added Courses (CCFUP 2022)	6.7
VARK	Visual, Aural, Read/Write, Kinaesthetic	2.4
ZPD	Zone of proximal development	2.2

Dates and official facts

These are the positions printed in the guide, checked in October 2026. Facts that can change are listed again, with where to check them, in “Facts to Recheck before Your Examination”.

Fact	Position as given in the guide	See
Bloom’s taxonomy in current use	Revised taxonomy of Anderson and Krathwohl (2001), Create at the top; cognitive handbook 1956, affective handbook 1964	1.4
NEP 2020: teachers’ professional development	At least 50 hours of continuous professional development a year for every teacher; school principals and school complex leaders 50 hours or more	1.10
NEP 2020: school teacher qualification	By 2030 the minimum degree qualification for school teaching will be a four-year integrated B.Ed. (a target date)	1.10
UGC Code of Professional Ethics	The one in the UGC Regulations of 2018; older books may quote an earlier version	1.10
RPwD Act, 2016	21 specified disabilities; benchmark disability not less than 40 per cent; not less than 5 per cent of seats in government and government-aided higher education, with 5 years’ upper age relaxation (Section 32); not less than 4 per cent of vacancies in government employment (Section 34); assent December 2016, in force April 2017. The 1995 Act: 7 disabilities and 3 per cent	2.9
RTE Act, 2009	Section 29(2)(h): “comprehensive and continuous evaluation”; free and compulsory education for children aged 6 to 14	2.9, 6.6

Fact	Position as given in the guide	See
Hawthorne effect	Named after studies of 1924 to 1932 at the Western Electric Company's Hawthorne Works; John Henry effect named by Gary Saretsky in 1972; halo effect named by Edward Thorndike in 1920	3.6
IQAC and NAAC	NAAC asks every accredited institution to set up an IQAC, which submits an AQAR each year; NAAC's framework has been under revision, so do not rely on older grading scales	3.7
SWAYAM launch	9 July 2017, by the President of India, at the National Convention on Digital Initiatives, with 32 SWAYAM PRABHA channels and the National Academic Depository	4.6
SWAYAM National Coordinators	Ten (NPTEL, UGC, AICTE, CEC, IGNOU, IIMB, NITTTR, INI, NCERT, NIOS), per the PIB backgrounder of March 2026; older sources list nine, without INI; Banaras Hindu University designated for INI in 2025	4.6
SWAYAM PRABHA channels	Began with 32 (2017); later 34, then 40. In 2026 official sources disagree: 48 in the PIB backgrounder of 10 March 2026 and a PIB document of June 2026; 40 on the SWAYAM PRABHA website and in a PIB factsheet of August 2026	4.6
SWAYAM PRABHA delivery	24 hours a day, 7 days a week; GSAT-15; uplinked from BISAG-N, Gandhinagar; free on DD Free Dish; at least four hours of new content daily, repeated five more times	4.6
SWAYAM credit transfer	Up to 40 per cent of the total courses in a programme in a semester (UGC Regulations, 2021, Gazette 25 March 2021); the 2016 Regulations (19 July 2016) allowed 20 per cent; a Lok Sabha reply of July 2026 still gave 40 per cent	4.6
SWAYAM four quadrants	e-Tutorial, e-Content, Discussion forum, Assessment; revised SWAYAM Guidelines approved in 2024 (circulated May 2024)	4.6
SWAYAM certificate fee	Charged for the proctored examination; usually Rs 1,000 per course on NPTEL pages, set course by course	4.6
SWAYAM Plus	Separate platform for industry-relevant and skill courses, launched February 2024, operated by IIT Madras	4.6
NPTEL	Started 2003 by seven IITs and IISc Bangalore; National Coordinator for engineering	4.6
DIKSHA	Launched 5 September 2017; led by NCERT	4.6
ShodhShuddhi	Began 2019 with Urkund (later Ouriginal) until 30 September 2023; DrillBit since 1 November 2023	4.6
One Nation One Subscription	Approved by the Union Cabinet 25 November 2024; operational from 1 January 2025; INFLIBNET the implementing agency; about 13,000 e-journals of 30 publishers; builds on e-ShodhSindhu (December 2015)	4.6
NDLI	Pilot from 2015; nationally launched 19 June 2018; developed at IIT Kharagpur	4.6
Dale's cone editions	1946: ten bands; 1954: "dramatised experiences" and television; 1969: eleven bands, related to Bruner's modes; the retention percentages are not Dale's (traced by Subramony and others, 2014)	5.1

Fact	Position as given in the guide	See
Moodle and Creative Commons	Moodle: free, open-source, GNU General Public License version 3; Creative Commons current version 4.0	5.4
CBSE and CCE	Grading with CCE in Classes IX and X from 2009–10; board examination for Class X restored from 2017–18	6.6
UGC CBCS grading (2015)	O 10, A+ 9, A 8, B+ 7, B 6, C 5, P 4, F 0, Ab 0; no D grade; SGPA and CGPA to two decimal places	6.7
Definition of a credit	2015: one hour of lecture or tutorial, or two hours of practical, per week; CCFUP (December 2022): 15 hours theory or 30 hours practical per semester; NCrF: 30 notional learning hours, 1,200 hours a year = 40 credits	6.7
Course categories	CBCS 2015: core, DSE, GE, AECC, SEC; CCFUP 2022: major, minor, MDC, AEC, SEC, VAC, internship	6.7
Academic Bank of Credits	UGC Regulations, 2021	6.7
NHEQF	Higher education qualifications on levels 4.5 to 8; doctorate at level 8	6.7
National Testing Agency	Set up November 2017; first conducted UGC NET in 2018, as a computer-based test; the mode is fixed by the NTA information bulletin	6.8
NEP 2020, paragraph 4.34	Assessment to shift from summative, rote-memory testing to regular, formative, competency-based assessment of higher-order skills	6.9
UGC Evaluation Reforms	Evaluation Reforms in Higher Educational Institutions (2019)	6.9

Formulae

Work every formula as the guide's worked examples do, with the steps written out.

Formula	Meaning of symbols	See
Higher-order share = (items at higher-order levels / total items) x 100	Higher-order usually Analyse, Evaluate, Create, unless the question says otherwise	1.4
One micro-teaching cycle = 6 + 6 + 12 + 6 + 6 = 36 minutes; complete cycles = session minutes / 36 (whole number only)	Teach, feedback, re-plan, re-teach, re-feedback (Indian model)	1.8
Teacher talk % = (categories 1 to 7) / total x 100; Student talk % = (8 + 9) / total x 100; Silence % = 10 / total x 100	Flanders tallies; total includes all ten categories	1.8
I/D ratio = (tallies in 1 to 4) / (tallies in 5 to 7)	Indirect over direct; below 1 means more direct	1.8
Time observed (minutes) = total tallies x 3 / 60 = total / 20	One tally every 3 seconds	1.8
Rule of two-thirds: teacher talk share of class time = 2/3 x 2/3 = 4/9 (about 44 per cent); direct teacher talk = 8/27 (about 30 per cent)	Flanders	1.8
B = f(P, E)	Behaviour is a function of the person and the psychological environment (Lewin)	2.3
Motivation = expectancy of success x value of the goal	Expectancy-value theory	2.6

Formula	Meaning of symbols	See
Motivational force = expectancy x instrumentality x valence	Vroom; multiplied, so one zero makes the force zero	2.6
IQ = (MA / CA) x 100; MA = IQ x CA / 100; CA = MA x 100 / IQ	MA mental age, CA chronological age, in the same units (months if needed)	2.8
Deviation IQ = 100 + 15 x z	z = (person's score - mean of age group) / SD of age group	2.8
Completion rate = completers / base	Base is enrolments or starters; say which	4.6
Evaluation = measurement + value judgement; fuller form that Indian textbooks attribute to Gronlund: quantitative description + qualitative description + value judgements	Measurement is the quantitative description	6.1
Obtained score = true score + error	Classical test theory	6.1
Full-test reliability = 2r / (1 + r)	Spearman-Brown; r = correlation between the two halves	6.2
Reliability of a test n times as long = n r / (1 + (n - 1) r)	r = present reliability	6.2
KR-20 = [k / (k - 1)] x [1 - (sum of pq) / s^2]	k items; p proportion right on an item; q = 1 - p; s^2 variance of total scores	6.2
KR-21 = [k / (k - 1)] x [1 - M (k - M) / (k s^2)]	M mean of total scores; assumes equal item difficulty	6.2
SEM = SD x square root of (1 - r)	r reliability; obtained score plus or minus 1 SEM about 68 per cent, plus or minus 1.96 SEM about 95 per cent	6.2
Highest possible validity = square root of reliability	The index of reliability	6.2
Reliability = 1 - (error variance / total variance)	Explains restriction of range	6.2
Attainment of an outcome = students reaching the level / students in class x 100	Compare with the target	6.3
Facility value FV = (RU + RL) / (2n)	RU, RL right in upper and lower groups; n size of one group; groups usually 27 per cent each	6.4
Discrimination index DI = (RU - RL) / n	Divide by one group only; runs from -1 to +1	6.4
Highest possible DI = 2 x the smaller of FV and (1 - FV)	Equal groups	6.4
Corrected score S = R - W / (n - 1)	R right, W wrong, n options per item; omitted items not counted	6.4
Chance score = number of items x 1 / number of options	Expected score by blind guessing	6.4
Mean = sum of scores / number of scores	Median is the middle score after ordering; mode the most frequent	6.5
Variance = sum of squared deviations from the mean / N; SD = square root of variance	Some books divide by N - 1 for a sample	6.5
PR = number scoring below / number in group x 100	Percentile rank as defined in 6.5 (6.8 notes the "at or below" convention used in normalisation)	6.5, 6.8
z = (X - mean) / SD; T = 50 + 10z	X raw score	6.5

Formula	Meaning of symbols	See
r = (sum of dx dy) / square root of [(sum of dx^2) x (sum of dy^2)]	dx, dy deviations from the means of X and Y	6.5
Shared variance = r^2	r = 0.60 gives 0.36	6.5
Normal curve	Mean to 1 SD: 34.13 per cent; within 1 SD: 68.27; within 2 SD: 95.45; within 3 SD: 99.73	6.5
Indicative percentage = grade point average x 9.5	CBSE school CCE scheme only; not a CBCS rule	6.6
Credits per week = lecture and tutorial hours + practical hours / 2	2015 UGC definition	6.7
Credit points = credits x grade point	For one course	6.7
SGPA = sum of (Ci x Gi) / sum of Ci	Ci credits of the ith course, Gi grade point; F and Ab stay in the denominator	6.7
CGPA = sum of (semester credits x SGPA) / total credits	Weight each semester by its credits	6.7
Required SGPA = [(credits so far + next credits) x target CGPA - credits so far x present CGPA] / next credits	Target-grade problems	6.7
Weighted rubric score = sum of (weight x criterion score)	Weights as fractions	6.9
Member’s mark = group mark x member’s average peer rating / group’s average rating	Group-work contribution factor	6.9

Confusable Pairs at a Glance

Every “Often confused” box in the guide is collected here, in the order of the sections. Cover the middle column and test yourself: can you state the deciding difference for each pair?

Pair	The deciding difference	Section
Training and teaching	Training builds one skill to a standard by practice; teaching also seeks understanding and judgement	1.1
Instruction and teaching	Instruction passes on organised knowledge and can be done by a book or machine; teaching involves a person and aims at understanding; all teaching includes instruction, not the reverse	1.1
Indoctrination and teaching	Decided by manner and aim, not subject: indoctrination wants belief without question; teaching gives reasons and welcomes objections	1.1
Conditioning and training	Conditioning links stimulus and response with no understanding (lowest); training builds a skill by practice	1.1
Gage and B. O. Smith definitions	Gage: interpersonal influence on behaviour potential; Smith: system of actions intended to induce learning	1.1
Morrison and Amidon definitions	Morrison: more mature and less mature personality; Amidon: interactive process of classroom talk	1.1
Teacher-learner-content and objectives-learning experiences-evaluation	First is the triad of teaching (three focal points); second is the tripolar view of the educative process	1.1
Task sense and achievement sense of teaching	Task: trying, need not succeed; achievement: implies learning took place	1.1
Pre-active and post-active phase	Pre-active plans before the lesson (fixing objectives); post-active evaluates after it (selecting tests, changing strategies)	1.2
Sizing up the class and evaluating outcomes	Sizing up is interactive (facing the class); evaluating outcomes is post-active	1.2
Teacher as independent variable and method as independent variable	In analysis of teaching the teacher is independent; in an experiment on methods the method is independent	1.2
Intervening and moderator variable	Intervening carries the effect on the causal path; moderator changes the strength or direction of the relation	1.2
Diagnostic and prescriptive function	Diagnostic finds entry behaviour and difficulties; prescriptive chooses content and methods to remedy them	1.2
Aim and objective	Aim is broad, long-term, not directly testable; objective is narrow and testable at the end of a lesson or unit	1.3
Objective and teacher activity	Objective states what the learner will do; “the teacher will explain” is not an objective	1.3
Condition and criterion (Mager)	Condition is the circumstance (“given a map”); criterion is the standard (“90 per cent accuracy”)	1.3
Mager and Bloom	Mager: how to write one objective; Bloom: how to classify objectives into levels	1.3
Mager and Gronlund	Mager: single objective with performance, condition, criterion; Gronlund: general objective clarified by specific learning outcomes	1.3

Pair	The deciding difference	Section
Programme outcomes and programme educational objectives	POs: at graduation; PEOs: a few years into the career	1.3
Original top and revised top of Bloom's taxonomy	1956: Evaluation highest, Synthesis fifth; 2001: Create highest, Evaluate fifth	1.4
Analyse and Evaluate	Analyse breaks into parts and relates them; Evaluate judges against criteria	1.4
Comprehension (translation) and Application	Changing words into a graph is translation (Comprehension); Application uses a rule in a new situation	1.4
Evaluate (cognitive) and Valuing (affective)	Evaluate: justifying a choice against criteria; Valuing: attaching worth and commitment	1.4
Valuing and Organisation	Valuing concerns one value; Organisation builds several values into a system	1.4
Organisation and Characterisation	Organisation builds the value system; Characterisation is when it governs conduct consistently	1.4
Receiving and Responding	Mere attention is Receiving; active reaction (answering, complying, volunteering) is Responding	1.4
Imitation and Receiving	Imitation is Dave's first psychomotor level; Receiving is the first affective level	1.4
Simpson and Dave	Simpson: seven levels, Perception to Origination; Dave: five levels, Imitation to Naturalisation	1.4
Complexity, internalisation, coordination	Organising principles of cognitive, affective and psychomotor domains respectively	1.4
Herbart's steps and Morrison's steps	Herbart: preparation, presentation, comparison/association, generalisation, application; Morrison: exploration, presentation, assimilation, organisation, recitation; only presentation is common	1.5
Preparation and exploration	Preparation (Herbart) recalls related knowledge and states the aim; exploration (Morrison) finds what learners already know of the unit	1.5
Application and recitation	Application is Herbart's last step; recitation is Morrison's last step	1.5
Assimilation and organisation	Assimilation is supervised study ending in a mastery test; organisation is an unaided written outline	1.5
Understanding level and reflective level	Understanding: teacher explains, learner grasps and applies in familiar cases; reflective: learner meets an unfamiliar problem and reaches and tests conclusions	1.5
Hunt and Herbart	Hunt: reflective level; Herbart: memory level	1.5
Extrinsic and intrinsic motivation (levels)	Extrinsic (marks, praise) marks the memory level; intrinsic (felt problem) marks the reflective level	1.5
Objective tests and essay tests (levels)	Memory: recall and recognition items; understanding: both; reflective: mainly essay type	1.5
Concrete to abstract vs particular to general	First is about kind of experience (things to ideas); second is about scope (cases to a rule); separate maxims	1.6
Particular to general vs empirical to rational	Several cases lead to one rule vs one observed fact given its reasoned explanation	1.6

Pair	The deciding difference	Section
Whole to part vs analysis to synthesis	Present the whole first vs break into elements and then rebuild	1.6
Psychological order vs logical order	Follows the learner's interests and readiness vs follows the subject's expert structure; teach psychological first	1.6
Inductive vs deductive method	Examples to rule, slow, suits discovery vs rule to examples, quick, suits practice; combine induction then deduction	1.6
Analytic vs synthetic method	Unknown (conclusion) back to known (data), shows why vs known to unknown, short finished proof	1.6
Maxim "known to unknown" vs analytic method "unknown to known"	Linking new knowledge to old vs direction of a proof from conclusion back to data	1.6
Deduction vs analysis	Applying a general rule to cases vs working backwards from a conclusion	1.6
Reflective practice vs subject mastery	Examining one's own teaching to improve it vs knowing the subject	1.7
Democratic vs laissez-faire teaching	Guides, reasons, invites participation and still corrects errors vs gives up direction	1.7
Authoritarian vs democratic groups (Lewin, Lippitt and White)	Worked mainly while the leader watched vs kept working in the leader's absence with good morale	1.7
Classroom management vs punishment	Creates conditions for engaged learning time and prevents problems vs reacts after problems	1.7
Input vs outcome of teaching	Lectures, notes, coverage vs learning achieved; outcome judges teaching	1.7
Withitness vs overlapping (Kounin)	Showing awareness of the whole room vs attending to two matters at once	1.7
Presage vs context vs process vs product variables	What the teacher brings vs the setting vs classroom behaviour vs pupil outcomes	1.7
Teacher immediacy vs clarity vs credibility vs efficacy	Reducing psychological distance vs intelligible explanation vs perceived competence and trust vs teacher's belief in own capability	1.7
Set induction vs closure	Opens the lesson, links to previous knowledge vs ends it and consolidates	1.8
Set induction vs stimulus variation	Attention at the opening only vs attention throughout by changes in teacher behaviour	1.8
Reinforcement vs probing	Praises or accepts a response vs asks follow-up questions to deepen it	1.8
Probing vs illustrating with examples	Works on a student's answer vs clarifies the idea for the whole class	1.8
Prompting vs redirection (probing)	Hints after no or wrong answer vs same question put to other students	1.8
Convergent vs divergent question	One correct answer (even if computed) vs many acceptable answers, usually higher-order	1.8
Micro-teaching vs simulated teaching	Scaled-down cycle for one skill with feedback vs role-play among trainees with rotated roles	1.8

Pair	The deciding difference	Section
Micro-teaching vs link practice	One skill at a time vs several skills combined in longer lessons with larger groups	1.8
Stanford model vs Indian model of micro-teaching	Real pupils and videotape vs peers as pupils, supervisor and peer feedback, no costly equipment; same cycle	1.8
Knowledge acquisition vs skill acquisition phase	Reading, discussion, model lesson vs plan-teach-feedback-re-plan-re-teach	1.8
Indirect vs direct influence (Flanders)	Categories 1-4 (feelings, praise, students' ideas, questions) vs 5-7 (lecture, directions, criticism)	1.8
Student response (8) vs student initiation (9)	Answers the teacher vs talk the student starts	1.8
I/D ratio vs its inverse	Indirect over direct (120/240 = 0.5) vs direct over indirect (2.0)	1.8
PCK vs TPACK	Content plus pedagogy (Shulman 1986) vs adds technology (Mishra and Koehler 2006)	1.8
TPACK vs SAMR	Teacher knowledge framework vs levels of technology use	1.8
Social system vs support system (models)	Roles and relationships of people vs materials and facilities	1.9
Syntax vs focus vs principles of reaction	Sequence of phases vs central aim vs how the teacher responds	1.9
Concept attainment vs advance organiser	Bruner; examples and non-examples vs Ausubel; general framework before detail	1.9
Inquiry training vs group investigation	Suchman, information-processing, individual questioning vs Thelen, social, democratic group process	1.9
Inquiry training vs synectics	Questioning a puzzling event vs creativity through analogy (Gordon, personal family)	1.9
Glaser's basic teaching model vs Gagné's nine events	Four components of a teaching system with feedback loops vs nine events of a lesson	1.9
Academic freedom vs freedom from rules	Inquiry, expression, choice of method vs not ignoring curriculum or evaluation norms	1.10
Academic freedom vs plagiarism	Right to publish one's own ideas vs no right to others' work	1.10
Confidentiality vs accountability	Keeping personal information private vs answering for one's work	1.10
Marks of a profession vs conditions of employment	Specialised knowledge, training, service motive, code vs salary, post, facilities	1.10
Gift vs ghost authorship	Non-contributor put in vs real contributor left out	1.10
NEP 2020 CPD vs four-year integrated B.Ed.	In-service growth (50 hours a year) vs pre-service qualification (by 2030); separate provisions	1.10
Identity vs role confusion / intimacy vs isolation	Identity is Erikson's adolescent crisis; intimacy belongs to young adulthood	2.1
Industry vs inferiority / identity vs role confusion	Industry is school age (about 6-12); identity is adolescence	2.1
Hall / Havighurst / Erikson	Hall: storm and stress; Havighurst: developmental tasks; Erikson: psychosocial stages	2.1
Moratorium / diffusion	Both lack commitment; moratorium explores, diffusion does not	2.1

Pair	The deciding difference	Section
Foreclosure / achievement	Both committed; foreclosure without exploration, achievement after exploration	2.1
Imaginary audience / personal fable	Being watched and judged vs being unique and invulnerable (both Elkind)	2.1
Arnett / Levinson	Emerging adulthood (18-25) vs seasons of adult life	2.1
Assimilation / accommodation	Fit experience into old schema vs change the schema	2.2
Equilibration / conservation	Balancing process behind development vs a concrete-operational achievement	2.2
Centration / conservation	Focusing on one feature (pre-operational) is why conservation fails	2.2
Concrete operational / formal operational	Logic on real objects vs abstract, hypothetical reasoning	2.2
Zone of proximal development / level of potential development	The zone is the gap; potential level is its upper edge	2.2
Cognitive constructivism / social constructivism	Piaget (individual) vs Vygotsky (interaction and culture)	2.2
More knowledgeable other / generalised other	Helper in the ZPD vs G. H. Mead's term for society's attitudes	2.2
Scaffolding / advance organiser	Help during the task, withdrawn vs framework given before the lesson	2.2
Enactive / iconic	Action vs images	2.2
Bruner / Ausubel	Discovery learning vs meaningful reception learning	2.2
Egocentric speech / private speech	Piaget's reading (immaturity) vs Vygotsky's (tool guiding thought)	2.2
Comparative / expository organiser	Relates to familiar material vs framework for wholly new material	2.2
Progressive differentiation / integrative reconciliation	General to detailed vs relating and reconciling ideas	2.2
Maintenance / elaborative rehearsal	Plain repetition vs linking to existing knowledge	2.2
Serial position / von Restorff effect	Position decides recall vs distinctiveness decides recall	2.2
Proactive / retroactive interference	Old disturbs new vs new disturbs old	2.2
Classical / operant conditioning	Involuntary response by pairing vs voluntary behaviour by consequences	2.3
Negative reinforcement / punishment	Removes unpleasant, behaviour rises vs behaviour falls	2.3
Negative punishment / negative reinforcement	Removes pleasant to reduce vs removes unpleasant to increase	2.3
Extinction / punishment	Reinforcer withheld vs unpleasant applied or pleasant removed	2.3
Stimulus generalisation / discrimination	Similar stimuli evoke response vs only the original does	2.3

Pair	The deciding difference	Section
Stimulus generalisation / response generalisation	Many stimuli, one response vs one stimulus, many responses	2.3
Law of readiness / law of effect	Preparedness before acting vs consequences after acting	2.3
Trial and error / insight	Gradual (Thorndike) vs sudden (Köhler)	2.3
Insight / latent learning	Köhler's chimpanzees vs Tolman's rats	2.3
Ratio / interval schedules	Number of responses vs time	2.3
Direct / vicarious reinforcement	Received oneself vs seen given to a model	2.3
Bruner / Gagné	Discovery learning vs conditions and hierarchy of learning	2.3
Shaping / chaining	Refining one response by approximations vs linking separate responses	2.3
Connectivism / constructivism	Learning in networks and devices vs learner's own constructions	2.3
VARK / Bruner's modes	Learning preferences vs ways knowledge is represented	2.4
Diverger / assimilator	Both reflective observation; concrete experience vs abstract conceptualisation	2.4
Converger / accommodator	Both active experimentation; abstract conceptualisation vs concrete experience	2.4
Field-independent / field-dependent	Analytical, self-directed vs global, socially oriented	2.4
Impulsive / reflective tempo	Fast and inaccurate vs slow and accurate (Kagan)	2.4
Reflector / theorist	Watches and gathers data vs builds logical theory	2.4
Pragmatist / activist	Tests ideas in practice vs plunges into new experience	2.4
Pre-conventional / conventional	Punishment and self-interest vs approval and law and order	2.5
Stage 4 / stage 5 (Kohlberg)	Upholds rules as such vs rules as changeable social contract	2.5
Kohlberg's levels / Piaget's moralities	Pre-/conventional/post-conventional vs heteronomous/autonomous	2.5
Moral reasoning / identity	Kohlberg vs Erikson and Marcia	2.5
intrinsic / extrinsic motivation	reward inside the activity (interest, satisfaction) or outside it (marks, scholarship, fear)	2.6
deficiency / growth needs (Maslow)	physiological, safety, belonging, esteem stop motivating once met; self-actualisation keeps motivating; esteem is still a deficiency need	2.6
introduced / popularised emotional intelligence	Salovey and Mayer introduced the term; Goleman popularised it (1995)	2.6
emotional intelligence / Gardner's personal intelligences	Goleman's EI is a separate theory from Gardner's interpersonal and intrapersonal intelligences	2.6
self-efficacy / self-esteem	belief in capability for a task, or general feeling of worth	2.6
Yerkes-Dodson: moderate / maximum arousal	performance peaks at moderate arousal (inverted U); optimum lower for hard tasks	2.6
Weiner: effort / ability	both internal; effort unstable, ability stable	2.6
learned helplessness / test anxiety	stops trying altogether after uncontrollable failure, or fear interfering during a test	2.6

Pair	The deciding difference	Section
fixed (entity) / growth (incremental) mindset	ability unchangeable, or grows with effort; “incremental” = growth	2.6
mastery / performance goal	aim to understand and improve, or to look able compared with others	2.6
Alderfer / McClelland three-need sets	existence, relatedness, growth; or achievement, affiliation, power	2.6
Deci and Ryan / Alderfer “relatedness”	only Deci and Ryan have autonomy and competence	2.6
ARCS confidence / satisfaction	expectation of success before and during; good feeling after achievement	2.6
overjustification / Hawthorne / Pygmalion / Zeigarnik	reward kills interest; being observed; teacher expectation; unfinished tasks remembered	2.6
Vroom: multiply / add	terms are multiplied; any zero gives zero	2.6
pedagogy / andragogy	teaching children (dependent, subject-centred, external) or helping adults learn (self-directing, problem-centred, internal)	2.7
readiness / orientation (Knowles)	when adults become ready (role demands), or how content is organised (around problems)	2.7
self-directed / incidental / programmed learning	planned by learner; unplanned; path fixed by programme	2.7
learning contract / portfolio	plan drawn up in advance, or evidence of work already done	2.7
andragogy / heutagogy	self-directed within a teacher-designed course, or learner decides what and how (self-determined)	2.7
Mezirow / Kolb	transforming a frame of reference after a disorienting dilemma, or general cycle of learning from experience	2.7
situational / institutional / dispositional barriers	learner’s life circumstances; provider’s practices; learner’s attitudes	2.7
inter- / intra-individual differences	between persons, or within one person’s abilities	2.8
aptitude / achievement	potential to learn with training (future), or what has been learnt	2.8
aptitude / attitude (and Hindi अभिक्षमता / अभिवृत्ति)	potential to learn, or settled favourable/unfavourable tendency	2.8
Spearman / Thurstone	g plus s, or seven primary mental abilities	2.8
Thurstone / Guilford	both many abilities; three-dimensional operations-contents-products is Guilford’s	2.8
fluid / crystallised intelligence	reasoning on new problems (declines), or stored knowledge (maintained or grows)	2.8
interpersonal / intrapersonal intelligence	understanding others, or understanding oneself	2.8
Sternberg creative / practical	dealing with novelty, or adapting to everyday settings (“street-smart”)	2.8
convergent / divergent production	one right answer, or many answers (creativity)	2.8
ratio / deviation IQ	MA ÷ CA × 100, or standing among same-age peers (mean 100, SD 15)	2.8

Pair	The deciding difference	Section
Flynn / Matthew effect	rise in average IQ across generations, or widening gaps between individuals	2.8
enrichment / acceleration	deeper, wider work at same level, or faster movement through curriculum	2.9
slow learner / underachiever	low ability with matching attainment, or adequate ability with low attainment	2.9
slow learner / learning disability	generally below-average ability, or specific difficulty despite average or higher intelligence	2.9
dyslexia / dysgraphia / dyscalculia / dyspraxia	reading; handwriting; number; movement	2.9
alt text / captions	for learners who cannot see, or who cannot hear	2.9
integration / inclusion	learner fits unchanged system, or system changes for learner	2.9
RPwD 5 per cent / 4 per cent	seats in government and aided higher education, or vacancies in government employment	2.9
RPwD Act 2016 / RTE Act 2009	rights of persons with disabilities, or free and compulsory education for children	2.9
differentiated instruction / UDL	teacher varies content, process, product for learners present, or variety designed in from the start for all	2.9
Support material vs instructional facilities	Material is made or chosen for a particular content (chart, worksheet, slides, e-journal); facilities are lasting infrastructure the institution provides for all teaching (laboratory, library, projector, network)	3.1, 3.4, 3.5
Learning environment vs instructional facilities	Environment is the condition of the surroundings (light, noise, seating, climate); facilities are the equipment and spaces themselves	3.1, 3.5
Necessary vs sufficient condition	Necessary: cannot succeed without it; sufficient: enough on its own. Subject mastery and access to facilities are necessary, not sufficient	3.1, 3.5
Pygmalion effect vs Golem effect	Same self-fulfilling mechanism; high expectations raise performance (Pygmalion), low expectations lower it (Golem)	3.2
Pygmalion effect vs Hawthorne effect	Pygmalion works through the teacher's expectations; Hawthorne through the subjects' awareness of being observed	3.2, 3.6
Pygmalion effect vs halo effect	Pygmalion changes actual performance; halo changes only the rater's judgement	3.2, 3.6
Teacher efficacy vs teacher autonomy	Efficacy is belief in one's capacity to bring about learning; autonomy is freedom to take professional decisions	3.2
Maturation vs learning	Maturation comes from growth with age, independent of training; learning comes from experience and practice	3.3
Aptitude vs prior knowledge	Aptitude is potential to learn; prior knowledge is what has already been learned	3.3
Cognitive vs affective learner factors	Intelligence, aptitude, prior knowledge are cognitive; interest, attitude, motivation, anxiety are affective	3.3
Cultural capital vs aptitude	Cultural capital is familiarity with educated culture gained at home; aptitude is the potential to learn	3.3
Suitability vs accuracy of support material	Suitability concerns the learners' level; accuracy concerns the truth of the content; separate criteria, neither explains the other	3.4

Pair	The deciding difference	Section
Physical environment vs psychological climate	Light, air, noise, seating are physical; relationships, warmth, fear are psychological	3.6
Psychological safety vs laissez-faire climate	Safety keeps guidance and high expectations; laissez-faire withdraws guidance	3.6
Hawthorne effect vs John Henry effect	Hawthorne: the studied group improves because it is observed; John Henry: the control group works harder to compete	3.6
Halo effect vs leniency error vs central tendency error	Halo: one trait colours other ratings of the same person; leniency: everyone rated high; central tendency: everyone rated near the middle	3.6
Board of Studies vs Academic Council	Board of Studies frames the syllabus of one subject; Academic Council approves courses, syllabi and regulations for the university	3.7
Academic Council vs IQAC	Academic Council approves academic matters; IQAC plans, guides and monitors quality and prepares the AQAR	3.7
Institutional climate vs classroom climate	Institutional climate (leadership, recognition, staff relations) shapes teacher morale; classroom climate shapes student participation	3.7
Approach / method	Approach is the underlying view of learning; method is the organised way of teaching a lesson	4.1
Method / technique	Method organises a whole lesson; technique is a small device inside it (questioning, buzz group)	4.1
Facilitator / transmitter	Facilitator guides active learners (learner-centred); transmitter delivers content to listeners (teacher-centred)	4.1
Inductive / deductive	Inductive: examples to rule, slower, discovery; deductive: rule to examples, quicker, practice	4.1
Lecture / demonstration	Lecture: learners hear; demonstration: learners also see a process performed; neither gives hands-on practice	4.2
Demonstration / laboratory practice	Demonstration: teacher performs; practice: each learner performs; skills come from practice	4.2
Economy / participation (lecture)	Lecture is economical (many learners per teacher) but not participatory	4.2
Immediate confirmation / self-pacing	Knowing at once whether a response is right / going at one's own speed	4.2
Linear / branching programming	Linear (Skinner): small frames, constructed responses, same path; branching (Crowder): larger frames, multiple choice, remedial frames, path depends on response	4.2
Mathetics / linear programming	Mathetics (Gilbert): backward chaining, last step first; linear: forward small steps	4.2
Seminar / symposium	Seminar: one paper discussed in depth; symposium: several speakers on different aspects in turn	4.3
Symposium / panel discussion	Symposium: prepared talks, no conversation among speakers; panel: informal conversation under a moderator	4.3
Seminar / workshop	Seminar: paper and discussion; workshop: practical work and a product	4.3
Seminar / conference	Conference is larger and broader, with plenary and parallel sessions; both include papers	4.3

Pair	The deciding difference	Section
Chairperson / moderator / rapporteur / convener	Chairperson conducts seminar or symposium; moderator conducts panel; rapporteur records and reports; convener organises	4.3
Tutorial / lecture	Tutorial: small group, remedial, supplements; lecture: large group, presentation	4.3
Team teaching / peer tutoring	Several teachers share one course / students teach students	4.3
Buzz group / fishbowl	Buzz: many small groups talk at once briefly; fishbowl: inner circle talks, outer circle observes	4.3
Forum / panel	Forum: the audience talks from the floor; panel: experts on the platform talk	4.3
Positive / negative interdependence	Cooperation: each succeeds only if others succeed; competition: one succeeds only if others fail	4.3
Jigsaw / STAD	Jigsaw: members teach one another their parts; STAD: team scores from individual improvement points	4.3
Project / case method	Project: real task in the field producing something; case: analysis of a situation already written up	4.4
Project / problem-solving	Problem-solving is chiefly mental and ends in a solution; project is real-life activity ending in a product	4.4
Project method / PBL	PBL: ill-structured problem first, facilitator, no product required; product is the mark of a project	4.4
Role play / simulation	Role play: acting a person's part for empathy; simulation: practising a task in a reality-like setting for skills and decisions	4.4
Role play / dramatisation	Role play is unscripted; dramatisation uses a learnt script	4.4
Heuristic / inductive	Heuristic: learner discovers by own experiment, minimal telling; inductive: reasoning from examples to rule	4.4
Kilpatrick / Parkhurst / Armstrong / Montessori	Project method / Dalton Plan / heuristic method / early childhood method	4.4
Mastery learning / PSI	Bloom: class taught together with formative tests and correctives; Keller: self-paced written units with proctors	4.4
Psychodrama / sociodrama	Psychodrama: individual's private problem, therapy; sociodrama: problem shared by a group	4.4
Case study / incident method	Case: full facts given; incident (Pigors): brief incident, facts obtained by questioning	4.4
Induction / deduction (logic)	Induction: instances to rule, conclusion probable; deduction: rule to case, certain if premises true	4.4
Action research / ex post facto	Action research: plan-act-observe-reflect to improve own practice; ex post facto: studies effects after the event without intervention	4.4
Synchronous / asynchronous	Same time (live class, webinar, chat) / different times (recorded lecture, forum, e-mail)	4.5
Blended / flipped	Blended: general mix of face-to-face and online; flipped: one blended model with content online first, class for application	4.5
Blended / distance learning	Blended keeps regular face-to-face classes; distance learning separates teacher and learner for most of the course	4.5

Pair	The deciding difference	Section
Flipped / conventional classroom	Flipped: content at home, application in class; conventional: content in class, homework at home	4.5
Rotation / flex model	Rotation: move between modes on a schedule; flex: online learning is the backbone with flexible support	4.5
SWAYAM / SWAYAM PRABHA	SWAYAM: web MOOCs with quizzes, exams, credits, needs internet; SWAYAM PRABHA: one-way DTH TV, no internet, no assessment	4.6
e-PG Pathshala / ePathshala	e-PG Pathshala: postgraduate e-content (UGC, INFLIBNET, NME-ICT); ePathshala: NCERT school textbooks	4.6
Shodhganga / Shodhgangotri	Submitted theses / synopses of research in progress	4.6
Shodhganga / ShodhShuddhi	Thesis repository / plagiarism-detection software access	4.6
DIKSHA / SWAYAM	DIKSHA: school education and teacher training; SWAYAM: courses school to PG and beyond	4.6
NPTEL / NDLI	Online courses by IITs and IISc / digital library search platform (IIT Kharagpur)	4.6
Massive / open (MOOC)	Massive: unlimited enrolment; open: no entry requirements, usually free	4.6
cMOOC / xMOOC / SPOC	cMOOC: connectivist, learner networks (Siemens, Downes); xMOOC: structured instructor-led; SPOC: small private course for one group	4.6
NAD / ABC	NAD stores degrees and mark-sheets for verification (DigiLocker); ABC records credits earned	4.6
Parent / host institution	Parent: where the student is enrolled and credits are recorded; host: whose faculty offers the SWAYAM course	4.6
NETF / NRF / NTA	NETF: forum on educational technology; NRF: funds research; NTA: conducts tests	4.6
Study trip vs demonstration (Dale's cone)	Demonstrations lie below study trips; a study trip is more abstract in Dale's order	5.1
Visual symbols vs verbal symbols	Visual symbols (charts, maps, graphs) lie just below verbal symbols (words), which form the apex	5.1
Visual symbols vs still pictures	A photograph resembles its object (still pictures band, iconic); a graph is an abstract sign (visual symbols, symbolic)	5.1
Projected vs non-projected aids	Projected aids need a projector to throw an image on a screen (epidiascope, OHP, slides, filmstrip); non-projected are seen directly (chalkboard, charts, flash cards, flannel board, models)	5.1
Audio vs audio-visual aids	Radio and tape recorder address hearing only; television and sound film address both; a silent filmstrip is visual	5.1
Model vs specimen	A model represents the real thing (enlarged, reduced or simplified); a specimen is the real thing in sample form	5.2
Diorama vs mock-up vs cut-away model	Diorama: miniature scene before a painted background; mock-up: simplified working replica for training; cut-away: shows the inside	5.2
Chalkboard vs prepared slide	Chalkboard suits content that unfolds during the lesson; slide suits content prepared in advance	5.2
Flannel board vs chart	Both non-projected visual aids; only the flannel board lets a display be built up and rearranged	5.2

Pair	The deciding difference	Section
Episcope vs diascope	Episcope projects opaque material by reflected light; diascope (slide projector) projects transparent slides	5.3
Epidiascope vs OHP	Epidiascope: opaque objects and slides, darkened room; OHP: large transparencies, lit room, teacher faces class	5.3
Webinar vs podcast	Webinar is live (synchronous); podcast is on demand (asynchronous)	5.4
Wiki vs blog	Wiki is jointly edited with change history; blog is a journal of dated posts	5.4
Video conferencing vs webcast	Video conferencing is two-way among several places; webcast is one-way from one source to many	5.4
LMS vs file-sharing service	Both store files; only an LMS tracks and reports each learner's progress	5.4
Open source vs free of charge	Open source code may be modified and shared (Moodle); a free service may still be proprietary (Google Classroom)	5.4
CC SA vs ND vs NC	SA: adaptations under the same licence; ND: no adaptations distributed; NC: no commercial use	5.4
SAMR Substitution vs Augmentation	Substitution: no functional change; Augmentation: functional improvement (e.g. spell-check)	5.4
Self-contained vs self-evaluating (SLM)	Self-contained: needs no outside sources; self-evaluating: in-text questions with answers let the learner check progress	5.4
Measurement vs evaluation	Measurement gives a number and stops; evaluation adds a judgement of worth	6.1
Assessment vs evaluation	Assessment gathers and interprets evidence; evaluation judges worth and includes assessment	6.1
Test vs measurement	Test is the instrument; measurement is the process of assigning numbers	6.1
Evaluation triangle vs teaching triangle	Objectives, learning experiences, evaluation vs teacher, learner, curriculum	6.1
Instructional vs guidance function	Improving the present course vs helping choose courses or careers	6.1
Reliability vs validity	Consistency vs measuring the right thing; reliable-not-valid possible, valid-not-reliable impossible	6.2
Objectivity vs reliability	Freedom from scorer judgement vs consistency of scores generally	6.2
Content vs face validity	Systematic check against blueprint vs surface impression	6.2
Concurrent vs predictive validity	Criterion available now vs criterion obtained later	6.2
Construct vs content validity	Theoretical trait reflected vs course adequately sampled	6.2
Test-retest vs parallel forms	Same test twice (stability) vs two equivalent versions (equivalence)	6.2
Split-half vs Kuder-Richardson	Halves correlated and Spearman-Brown corrected vs item-level 1/0 data	6.2
SEM vs standard deviation	Error band around one person's score vs spread across the group	6.2
Construct under-representation vs construct-irrelevant variance	Something essential missing vs something extra measured	6.2
Formative vs summative	Use during learning to improve vs use at end to certify; use decides, not instrument	6.3

Pair	The deciding difference	Section
Formative vs diagnostic	Shows a weakness and gives quick feedback vs finds cause of persistent difficulty	6.3
Diagnostic vs prognostic	Causes of present errors vs forecast of future performance	6.3
Placement vs diagnostic	Entry behaviour before instruction vs causes of persisting difficulty	6.3
Norm-referenced vs criterion-referenced	Comparison with others (cut-off floats) vs fixed standard (number passing floats)	6.3
Percentile rank vs percentage of objectives mastered	Norm-referenced vs criterion-referenced	6.3
Criterion-referenced vs ipsative	Fixed external standard vs learner's own earlier work	6.3
Assessment for learning vs assessment as learning	Teacher uses the evidence vs students use it to regulate themselves	6.3
Scriven vs Bloom (formative/summative)	Scriven 1967, curriculum; Bloom, Hastings, Madaus 1971, student learning	6.3
Direct vs indirect assessment	Examines performance vs gathers opinions (surveys)	6.3
Stem/key/distractor vs premise/response	Parts of an MCQ vs columns of a matching item	6.4
Supply-type vs selection-type items	Student writes the answer vs chooses it	6.4
Blueprint vs marking scheme	Planned before item writing for content validity vs prepared for written items	6.4
Checklist vs rating scale vs rubric	Present/absent vs degree vs degree with descriptors per level	6.4
Anecdotal vs cumulative record vs portfolio	Single incident vs institution's year-by-year record vs student's selected work	6.4
Facility value vs discrimination index	How many got it right vs whether able did better than weak	6.4
High difficulty index vs difficult item	High facility value means an easy item	6.4
Percentile rank vs percentage of marks	Position among examinees vs share of maximum marks	6.5
Percentile vs percentile rank	A score vs the percentage of the group below a score	6.5
Mean vs median	Average pulled by extremes vs middle score unaffected	6.5
Correlation vs causation	Going together vs one producing the other	6.5
r vs shared variance	r = 0.60 gives shared variance r squared = 0.36	6.5
Ordinal vs interval scale	Order only (ranks) vs equal units	6.5
Interval vs ratio scale	No true zero vs true zero; test scores lack a true zero	6.5
Continuous / comprehensive (CCE)	Continuous = regular, through the year (time); comprehensive = scholastic plus co-scholastic (coverage)	6.6
Scholastic / co-scholastic	Scholastic = curricular subjects and intellectual abilities; co-scholastic = life skills, attitudes, values, health and physical education, cultural activities	6.6

Pair	The deciding difference	Section
Absolute / relative grading	Absolute = fixed mark boundaries, anyone can reach any grade (criterion-referenced); relative = fixed proportions, grade by rank (norm-referenced)	6.6
Grading on the curve / criterion-referenced grading	On the curve fixes proportions in advance, so it is norm-referenced	6.6
Direct / indirect grading	Direct = grade straight from qualitative judgement; indirect = marks first, then converted	6.6
SGPA / CGPA	SGPA = one semester; CGPA = all semesters completed so far	6.7
Credits / credit points	Credits = weight of course; credit points = credits x grade point	6.7
Discipline Specific Elective / Generic Elective	DSE = from the main discipline; GE = from an unrelated discipline	6.7
AECC / SEC	AECC = Environmental Science and English/MIL Communication, compulsory; SEC = skill- or value-based, chosen from a pool	6.7
A / A+ (UGC scale)	A = Very Good, 8; A+ = Excellent, 9	6.7
B / C (UGC scale)	B = Above Average, 6; C = Average, 5	6.7
P / F (UGC scale)	P = Pass, 4 (lowest pass); F = Fail, 0 (Ab also 0)	6.7
UG certificate / UG diploma	Certificate after one year; diploma after two years	6.7
Academic Bank of Credits / National Academic Depository	ABC stores credits; NAD stores awards (degrees, mark-sheets)	6.7
CBCS 2015 / CCFUP 2022 course types	CBCS: core, DSE, GE, AECC, SEC; CCFUP: major, minor, MDC, AEC, SEC, VAC, internship	6.7
CBT / OMR test	CBT answers changeable until submission; OMR circles cannot be altered	6.8
Normalisation / randomisation	Normalisation equates scores of different shift papers; randomisation shuffles item order	6.8
Adaptive testing / linear-on-the-fly testing	CAT adapts each item during the test; LOFT builds a unique form before the test without adapting	6.8
Item response theory / classical test theory	IRT item parameters are sample-independent (needed for CAT); CTT indices change with the group	6.8
Mock test / actual test	Mock = practice with the interface, different questions	6.8
Analytic / holistic rubric	Analytic = score per criterion, profile; holistic = one overall score	6.9
Rubric / checklist	Rubric describes levels per criterion; checklist records present/absent only	6.9
Criteria / descriptors	Criteria = aspects judged; descriptors = what each level of each aspect looks like	6.9
Portfolio / folder	Portfolio = selected work with reflection; folder = everything unselected	6.9
Open-book / on-demand examination	Open-book = books allowed during the test; on-demand = learner chooses when to sit	6.9
On-demand / adaptive testing	On-demand = choice of timing; adaptive = difficulty changes during the test	6.9
Self- / peer assessment	Self = own work; peer = classmates' work	6.9

Pair	The deciding difference	Section
Authentic / ipsative assessment	Authentic = real-world task; ipsative = comparison with own earlier performance	6.9
Positive / negative washback	Positive = test encourages desirable learning; negative = test narrows teaching	6.9
Washback / practice effect	Washback = effect of a test on teaching and learning; practice effect = score gain from retaking	6.9

Revision Sheets

One sheet per chapter, for the last week. The most-asked points come first; the number in brackets is the section that teaches each point in full. If a point is not clear, go back to that section. The last line of each sheet lists the commonest confusions.

Revision Sheet 1: Teaching: Concept, Objectives, Levels, Characteristics and Basic Requirements

- **Levels**, lowest to highest: memory (Johann Friedrich **Herbart**), understanding (Henry Clinton **Morrison**), reflective (Maurice P. **Hunt**; also Morris L. Bigge). If Hunt is an option, choose Hunt. Each level includes the one below; they differ in the amount of thinking demanded. (1.5)
- Herbart's steps: preparation, presentation, comparison/association, generalisation, application. Morrison's: exploration, presentation, assimilation, organisation, recitation. (1.5)
- Memory: teacher dominant, extrinsic motivation, objective tests. Understanding: essay and objective tests. Reflective: felt problem, teacher a democratic guide, intrinsic motivation, mainly essay tests. (1.5)
- **Bloom 1956**: Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation. **2001** (Anderson and Krathwohl): Remember, Understand, Apply, Analyse, Evaluate, Create; Create is now highest. Knowledge dimension: factual, conceptual, procedural, metacognitive. (1.4)
- Affective (1964): Receiving, Responding, Valuing, Organisation, Characterisation. Psychomotor: Simpson, Perception to Origination (7); Dave, Imitation to Naturalisation (5); Harrow, Reflex movements to Non-discursive communication (6). (1.4)
- Teaching is purposeful and triadic (teacher, learner, subject matter). Continuum: conditioning, training, instruction, teaching. Indoctrination differs by manner and aim, not subject. (1.1)
- Definitions: Morrison, mature and less mature personality; Gage, interpersonal influence on behaviour potential; B. O. Smith, system of actions intended to induce learning; Amidon, interactive process, classroom talk. (1.1)
- Phases: pre-active (fixing objectives), interactive (sizing up the class, diagnosing), post-active (selecting testing devices, evaluating). Teacher independent, learner dependent, content and methods intervening variables. (1.2)
- An objective states what the learner will do. Mager: performance, condition, criterion; ABCD adds Audience. (1.3)
- **Micro-teaching**: Stanford, Dwight W. Allen and colleagues, usually dated 1963; 5 to 10 pupils, 5 to 10 minutes, one skill. Cycle: plan, teach, feedback, re-plan, re-teach, re-feedback. Indian model: 6, 6, 12, 6, 6 = 36 minutes. (1.8)
- **Flanders**: ten categories; 1 to 4 indirect (4 is asking questions), 5 to 7 direct, 8 and 9 student talk, 10 silence; a tally every 3 seconds; I/D = indirect / direct. (1.8)
- Set induction opens the lesson; stimulus variation works throughout; probing deepens a partial answer. (1.8)
- Effective teaching is judged by learning: interactive, feedback-based, flexible, democratic. Lewin, Lippitt and White: democratic groups worked even when the leader was away. (1.7)
- Maxims run from the easier end: known to unknown, simple to complex, concrete to abstract, particular to general, whole to part. Inductive: particular to general; deductive: general to particular. (1.6)
- Joyce and Weil: focus, syntax, social system, principles of reaction, support system. Concept attainment (Bruner), advance organiser (Ausubel), inquiry training (Suchman). (1.9)
- NEP 2020: at least 50 hours of continuous professional development a year. (1.10)

Traps: Herbart (memory) against Hunt (reflective); never “Herbert”. Evaluation tops 1956, Create tops 2001. Sizing up the class is interactive, not post-active. Flanders category 4 is indirect. Bruner with concept attainment, Ausubel with advance organisers.

Revision Sheet 2: Learner’s Characteristics

- **Learning theories**: Pavlov, classical conditioning; Thorndike, trial and error, laws of readiness, exercise and effect; Skinner, operant conditioning; Köhler, insight; Tolman, latent learning; Bandura, observational learning (attention, retention, reproduction, motivation). (2.3)
- Reinforcement increases behaviour; punishment decreases it. Negative reinforcement removes something unpleasant. Ratio schedules count responses; interval schedules count time. (2.3)
- Gagné’s hierarchy ends: discrimination, concept, rule, problem solving. Dewey: learning by doing. (2.3)
- **Piaget**: sensorimotor (0 to 2), pre-operational (2 to 7), concrete operational (7 to 11), formal operational (11 or 12 onwards). Conservation marks the concrete stage. Order fixed, ages approximate. (2.2)
- **Vygotsky**: the ZPD is the gap between what the learner does alone and with help. Scaffolding is temporary help (term from Wood, Bruner and Ross, 1976). (2.2)
- **Bruner**: discovery; enactive, iconic, symbolic; spiral curriculum. **Ausubel**: meaningful reception learning; advance organiser given before the lesson. (2.2)
- Miller: seven plus or minus two; chunking; distributed practice beats massed. (2.2)
- **Adolescence**: Hall, storm and stress; Erikson, identity versus role confusion; Havighurst, developmental tasks; Marcia, diffusion, foreclosure, moratorium, achievement; Arnett, emerging adulthood (18 to 25). (2.1)
- Maslow, from the base: physiological, safety, love and belonging, esteem, self-actualisation; the lower four are deficiency needs. Intrinsic reward lies inside the activity. (2.6)
- Yerkes-Dodson: inverted U, moderate arousal best. Emotional intelligence: introduced by Salovey and Mayer, popularised by Goleman. (2.6)
- **Andragogy** (Knowles): adults are self-directing, use experience, are ready when life roles demand, problem-centred, internally motivated. (2.7)
- IQ = mental age / chronological age x 100. Spearman, g and s; Thurstone, primary mental abilities; Guilford, convergent and divergent; Cattell, fluid and crystallised; Gardner, multiple intelligences (eight, with naturalistic); Sternberg, triarchic. (2.8)
- VARK (Fleming): Visual, Aural, Read/Write, Kinaesthetic. Kolb’s cycle: concrete experience, reflective observation, abstract conceptualisation, active experimentation. (2.4)
- Kohlberg: pre-conventional, conventional (stage 4, law and order), post-conventional. (2.5)
- RPwD Act 2016: 21 disabilities; 5 per cent of higher education seats, 4 per cent of government vacancies. Inclusion changes the institution; integration does not. (2.9)

Traps: Negative reinforcement is not punishment. The ZPD is the gap, not the potential level. Identity (adolescence) against intimacy (young adulthood). Readiness (life roles) against orientation (problem-centred). Interpersonal (others) against intrapersonal (oneself).

Revision Sheet 3: Factors Affecting Teaching

- Six groups: teacher, learner, support material, instructional facilities, learning environment, institution. Place a factor by asking where it lives and who controls it. (3.1)
- Subject mastery is necessary but not sufficient; no single factor “alone ensures” good teaching. (3.1)

- Teacher factors: subject mastery, communication, questioning, warmth and enthusiasm, expectations. (3.2)
- Pygmalion effect (Rosenthal and Jacobson, 1968): high expectations raise performance, a self-fulfilling prophecy. Golem effect: low expectations lower it. (3.2)
- Learner factors: prior knowledge (entry behaviour), intelligence, interest (affective), anxiety (emotional), health, maturation and readiness. (3.3)
- Support material is made for a lesson (chart, slides); relevance to the objective comes first, then suitability, accuracy, visibility. (3.4)
- Facilities are lasting infrastructure (laboratory, library, projector, network); without a laboratory, psychomotor objectives suffer. Facilities enable but do not cause good teaching. (3.5)
- Environment: physical (light, ventilation, seating, noise) and psychological (respect, freedom from fear). Noise distracts attention; it does not change intelligence. U-shaped seating suits discussion. (3.6)
- Hawthorne effect: people improve because they know they are observed. John Henry effect: the control group competes. Halo effect: one trait colours other ratings. (3.6)
- Institution factors: class size, timetable, workload, evaluation policy, leadership, climate (teacher morale). The Board of Studies frames the syllabus; the Academic Council approves it. (3.7)

Traps: Support material against facility. Physical against psychological environment. Hawthorne (students' awareness) against Pygmalion (teacher's expectations). Interest is affective, not cognitive.

Revision Sheet 4: Methods of Teaching in Institutions of Higher Learning

- Teacher-centred (lecture, demonstration): teacher transmits, learners passive. Learner-centred (project, problem-solving, discussion, case, role play): teacher facilitates; based on constructivism. No single best method. (4.1)
- Approach is widest, technique smallest. (4.1)
- **SWAYAM**: Study Webs of Active-Learning for Young Aspiring Minds; launched 9 July 2017; free, fee only for the proctored examination. Quadrants: e-Tutorial, e-Content, discussion forum, assessment. Credit transfer up to 40 per cent (2021 Regulations; 20 per cent in 2016). (4.6)
- Coordinators: NPTEL engineering; CEC undergraduate non-technical; UGC postgraduate non-technical; IIM Bangalore management; NCERT and NIOS school. (4.6)
- **SWAYAM PRABHA**: DTH channels, free on DD Free Dish, no internet, one-way, no credits. Began with 32 channels; 2026 sources give 40 or 48, so check before the examination. (4.6)
- e-PG Pathshala (postgraduate) against ePathshala (NCERT, school); Shodhganga (theses) against Shodhgangotri (synopses). (4.6)
- MOOC: massive, open, online, course; low completion. cMOOCs connectivist, xMOOCs instructor-led. (4.6)
- Seminar: one paper in depth, chairperson. Symposium: several talks on different aspects, no conversation. Panel: experts converse, moderator. Workshop: hands-on product. (4.3)
- Brainstorming (Osborn): defer judgement, aim at quantity. Team teaching: several teachers; peer tutoring: students teach students. (4.3)
- Lecture: strength economy; weak in feedback and skills. Linear programming (Skinner): small steps, same path. Branching (Crowder): multiple-choice, remedial frames. (4.2)
- Project method: Kilpatrick, after Dewey; purposing, planning, executing, evaluating. Dalton Plan: Parkhurst. Heuristic: Armstrong. PBL: McMaster, problem first. (4.4)
- Mastery learning: Bloom. Keller Plan: self-paced units with proctors. (4.4)

- Synchronous (live) against asynchronous (forum, recording). Flipped classroom: content at home, application in class; it is one model of blended learning. (4.5)

Traps: Seminar, symposium and panel. Linear against branching. SWAYAM against SWAYAM PRABHA. Shodhganga against Shodhgangotri. Blended against flipped.

Revision Sheet 5: Teaching Support System

- **Dale's Cone** (*Audiovisual Methods in Teaching*, 1946; eleven bands in 1969): base, direct purposeful experiences; then contrived, dramatised, demonstrations, study trips, exhibits, educational television, motion pictures, recordings and still pictures, visual symbols; apex, verbal symbols. (5.1)
- Study trips lie above demonstrations. The retention percentages often printed on the cone are not Dale's. (5.1)
- Bruner's modes on the cone: enactive (lowest three bands), iconic (middle), symbolic (visual and verbal symbols). (5.1)
- Audio (radio, tape recorder); visual (chart, model, silent filmstrip); audio-visual (television, sound film). (5.1)
- Projected (OHP, slides, epidiascope) against non-projected (chalkboard, chart, model, flash card). Too many aids overload working memory. (5.1)
- Chalkboard: suits material that unfolds in the lesson; loses eye contact. A model represents; a specimen is real. (5.2)
- OHP: transparencies, lit room, teacher faces class. Episcope: opaque material. Epidiascope: opaque and transparent. (5.3)
- Language laboratory: listening and speaking practice. (5.3)
- An LMS tracks each learner's progress. Moodle is open source; Google Classroom is proprietary. (5.4)
- Webinar and video conferencing are synchronous; podcast and wiki are asynchronous. OER may be used, adapted and redistributed under an open licence. (5.4)

Traps: The epidiascope is projected. Visual symbols are symbolic, not iconic. Free is not open source. Webinar (live) against podcast (on demand).

Revision Sheet 6: Evaluation Systems

- Placement before, formative during, diagnostic for causes of persistent difficulty, summative at the end. The use of results decides the label. (6.3)
- Norm-referenced: compared with others (percentile rank). Criterion-referenced: fixed standard (driving test). Ipsative: own earlier work. (6.3)
- Scriven (1967): formative and summative. Glaser (1963): norm- and criterion-referenced. (6.3)
- Test (instrument), measurement (numbers), assessment (evidence), evaluation (adds a value judgement; widest). (6.1)
- Evaluation triangle: objectives, learning experiences, evaluation. Obtained score = true score + error. (6.1)
- Validity matters most; reliability, objectivity, usability follow. Concurrent validity: criterion now; predictive: criterion later. Content validity: blueprint. (6.2)
- Test-retest gives stability; parallel forms, equivalence; split-half and Kuder-Richardson, internal consistency. A test can be reliable without being valid, not valid without being reliable. (6.2)
- Objective items sample widely; essay items sample narrowly and are scored subjectively. MCQ: stem, key, distractors. (6.4)

- Facility value = (RU + RL) / 2n; high means easy. Discrimination = (RU - RL) / n. Correction for guessing: R - W / (n - 1). (6.4)
- Percentile rank: percentage below, not percentage of marks. z = (X - mean) / SD; T = 50 + 10z. About 68 and 95 per cent lie within 1 and 2 SD. (6.5)
- CCE: continuous (time), comprehensive (scholastic plus co-scholastic). Absolute grading is criterion-referenced; relative grading is norm-referenced. (6.6)
- CBCS (UGC, 2015): one credit = one hour of lecture or two of practical a week. Core, DSE, GE, AECC, SEC. (6.7)
- O 10, A+ 9, A 8, B+ 7, B 6, C 5, P 4, F 0, Ab 0; no D. SGPA = sum of (credits x grade points) / total credits. (6.7)
- One year certificate, two years diploma. ABC stores credits; NAD stores awards. (6.7)
- NTA, set up November 2017, conducts UGC NET as a computer-based test. Shifts are compared by normalisation. Adaptive tests change items during the test. (6.8)
- Rubric: criteria, scale, descriptors. Portfolio: selection and reflection. On-demand: choice of timing. (6.9)

Traps: A high difficulty index means an easy item. Percentile rank is not percentage. Relative grading is norm-referenced. Formative against diagnostic. P is 4, not 5.

Bilingual Glossary

English term, the Hindi term used in this guide, and a one-line meaning. Hindi terms follow NTA's Hindi syllabus where it uses a settled term, and otherwise the term standard in Hindi-medium textbooks.

English	हिंदी	Meaning
ability enhancement course	क्षमता संवर्धन पाठ्यक्रम	AECC and SEC under CBCS 2015
absolute grading	निरपेक्ष श्रेणीकरण	Grades by fixed mark boundaries
abstract	अमूर्त	Removed from direct perception
Academic Bank of Credits	एकेडमिक बैंक ऑफ क्रेडिट्स	Digital store of a student's credits
Academic Council	विद्या परिषद	University's chief academic body
academic freedom	शैक्षणिक स्वतंत्रता	freedom to inquire, teach, research and publish within obligations
acceleration	त्वरण	faster progress through the curriculum
accommodation	समायोजन	changing a schema to fit new experience
achievement	उपलब्धि	what has already been learnt
achievement motivation	उपलब्धि अभिप्रेरणा	desire to succeed against a standard of excellence (McClelland)
action zone	क्रिया क्षेत्र	Front and centre seats where most interaction occurs
administrative function	प्रशासनिक कार्य	use of evaluation for promotion, selection, certification
adolescence	किशोरावस्था	period between childhood and adulthood
adolescent	किशोर	a person between childhood and adulthood
adolescent egocentrism	किशोर अहंकेंद्रिता	Elkind's adolescent self-centredness
advance organiser	अग्रिम संगठक	general introduction given before new material (Ausubel)
advance organiser model	अग्रिम संगठक प्रतिमान	general framework given before detailed material (Ausubel)
affective	भावात्मक	Concerning feelings, interests, attitudes
affective domain	भावात्मक पक्ष	area of feelings, attitudes and values
aim	लक्ष्य	broad, long-term purpose of education
analytic method	विश्लेषण विधि	unknown (conclusion) back to known (data)
andragogy	प्रौढ़ शिक्षाशास्त्र	art and science of helping adults learn (Knowles)
anecdotal record	उपाख्यानात्मक अभिलेख	brief description of a single significant incident
anxiety	दुश्चिंता	worry and tension about what may happen
apperception	अवबोधन	absorbing a new idea into existing ideas (Herbart)
apperceptive mass	पूर्व-ज्ञान पुंज	body of existing ideas into which new ideas are absorbed (Herbart)
approach	उपागम	the broad view of teaching and learning behind a teacher's work
aptitude	अभिक्षमता	potential to learn a skill with training

English	हिंदी	Meaning
aptitude-treatment interaction	अभिक्षमता-उपचार अंतःक्रिया	Effect of a method depends on learner characteristics
assessment	आकलन	gathering and interpreting evidence about learning
assessment as learning	अधिगम के रूप में आकलन	students monitor and regulate their own learning
assessment for learning	अधिगम के लिए आकलन	teacher uses evidence during learning to adjust teaching
assessment of learning	अधिगम का आकलन	records achievement at the end
assimilation	आत्मसात्करण	fitting new experience into an existing schema
asynchronous	असमकालिक	At each learner’s own time
asynchronous learning	असमकालिक अधिगम	learning in which each takes part at a convenient time
attitude	अभिवृत्ति	settled favourable or unfavourable tendency to respond
attribution	गुणारोपण	the cause a person gives for success or failure (Weiner)
audio aids	श्रव्य सामग्री	Aids addressing hearing
audio-visual aids	दृश्य-श्रव्य सामग्री	Aids addressing both sight and hearing
authentic assessment	प्रामाणिक आकलन	Real-world task assessment
behavioural objective	व्यवहारगत उद्देश्य	objective stated as observable learner behaviour
blended learning	मिश्रित अधिगम	planned mix of face-to-face and online learning
blueprint	नील पत्र	two-way table of content x objectives before item writing
Board of Studies	अध्ययन मंडल	Body that frames a subject’s syllabus
brainstorming	मस्तिष्क उद्वेलन	group production of many ideas with judgement deferred
branching programming	शाखीय अभिक्रमण	Crowder’s style: response decides next frame
bulletin board	सूचना पट्ट	Board for notices and displays
case method	व्यष्टि अध्ययन विधि	analysis of a written real or realistic situation
central tendency	केंद्रीय प्रवृत्ति	one number representing the group’s general level
chalkboard	श्यामपट्ट	Board for writing with chalk
checklist	जाँच सूची	records presence or absence of features
Choice Based Credit System	विकल्प आधारित क्रेडिट प्रणाली	UGC 2015 system of credits, choice and grades
chronological age	कालानुक्रमिक आयु	actual age since birth
chunking	खंडीकरण	grouping items into meaningful units
class size	कक्षा का आकार	Number of students in a class
classical conditioning	शास्त्रीय अनुबंधन	learning by pairing stimuli (Pavlov)
classroom climate	कक्षा का वातावरण	Socio-emotional atmosphere of a class
classroom management	कक्षा प्रबंधन	organising space, time, routines and relationships for engaged learning
closure	समापन	consolidating the lesson at its end
co-scholastic area	सह-शैक्षिक क्षेत्र	Life skills, attitudes, values, health, cultural activities
code of conduct	आचार संहिता	accepted rules of right conduct for members

English	हिंदी	Meaning
cognitive	संज्ञानात्मक	to do with knowing and thinking
cognitive domain	संज्ञानात्मक पक्ष	area of knowing and thinking
cognitive load theory	संज्ञानात्मक भार सिद्धांत	Working memory’s limited capacity governs learning design
cognitive style	संज्ञानात्मक शैली	typical way of perceiving and thinking
communication	संप्रेषण	sending a message from sender to receiver through a channel
communication skill	संप्रेषण कौशल	Ability to make ideas reach learners
computer based testing	कंप्यूटर आधारित परीक्षण	Test taken and scored on computer
computer-assisted instruction	कंप्यूटर सहायक अनुदेशन	Teaching through computer programs
computerised adaptive test	कंप्यूटरीकृत अनुकूली परीक्षण	Test whose next item depends on earlier responses
concept attainment model	संप्रत्यय उपलब्धि प्रतिमान	concepts formed from examples and non-examples (Bruner)
concrete	मूर्त	Directly perceived
concurrent validity	समवर्ती वैधता	agreement with a criterion measured at the same time
conditioning	अनुबंधन	building a stimulus-response link by pairing or reinforcement, without understanding
Cone of Experience	अनुभव शंकु	Dale’s arrangement of experiences from concrete to abstract
conference	सम्मेलन	large meeting on a broad theme with paper sessions
confidentiality	गोपनीयता	keeping personal or examination information private
connectivism	संयोजनवाद	theory that learning is forming connections across networks
conservation	संरक्षण	quantity stays same despite change in appearance
construct validity	संरचना वैधता	scores reflect a theoretical trait
constructivism	रचनावाद	learners build their own knowledge
content validity	विषयवस्तु वैधता	items represent course content and objectives in proportion
Continuous and Comprehensive Evaluation	सतत एवं व्यापक मूल्यांकन	Regular, whole-year assessment of scholastic and co-scholastic areas
continuous professional development	सतत व्यावसायिक विकास	lifelong in-service learning of teachers (CPD)
conventional	परंपरागत	Kohlberg level guided by approval and law
convergent production	अभिसारी चिंतन	thinking towards one correct answer (Guilford)
cooperative learning	सहकारी अधिगम	small groups working to a shared goal
core course	मूल पाठ्यक्रम	Compulsory course of the programme
correlation	सहसंबंध	degree to which two sets of scores go together
credit	क्रेडिट	Unit measuring course work by teaching hours
credit transfer	क्रेडिट हस्तांतरण	counting credits earned elsewhere towards one’s degree

English	हिंदी	Meaning
criterion-referenced	निकष-संदर्भित	Judged against a fixed standard
criterion-referenced test	निकष-संदर्भित परीक्षण	interprets a score against a fixed standard
critical reflection	आलोचनात्मक चिंतन	examining one's own assumptions
crystallised intelligence	क्रिस्टलीकृत बुद्धि	stored knowledge and skills (Cattell)
cultural capital	सांस्कृतिक पूँजी	Familiarity with educated culture passed on by family (Bourdieu)
Cumulative Grade Point Average	संचयी ग्रेड पॉइंट औसत	Credit-weighted average over all semesters
cumulative record	संचयी अभिलेख	institution's year-by-year comprehensive record
Dalton Plan	डाल्टन योजना	Parkhurst's plan of individual monthly assignments
debate	वाद-विवाद	two sides argue for and against a motion
deductive method	निगमन विधि	rule first, applied to examples
democratic teaching	लोकतांत्रिक शिक्षण	respectful teaching that invites questions and participation
demonstration method	प्रदर्शन विधि	teacher shows step by step how something is done
dependent variable	आश्रित चर	the learner (in analysis of teaching); the measured outcome (in an experiment)
developmental tasks	विकासात्मक कार्य	tasks society expects at each life stage (Havighurst)
diagnostic and remedial	निदानात्मक एवं उपचारात्मक	finding difficulties and helping overcome them
diagnostic evaluation	निदानात्मक मूल्यांकन	finding causes of persistent difficulties
diagnostic function	निदानात्मक कार्य	finding entry behaviour and difficulties
diascope	डायस्कोप	Slide projector
differentiated instruction	विभेदित अनुदेशन	varying content, process and product to suit learners
difficulty index / facility value	कठिनाई सूचकांक	proportion answering an item correctly
digital divide	डिजिटल विभाजन	Unequal access to and familiarity with computers
direct assessment	प्रत्यक्ष आकलन	examines actual performance
direct grading	प्रत्यक्ष श्रेणीकरण	Grade given without marks
discovery learning	खोज अधिगम	learning by finding out oneself (Bruner)
discrimination index	विभेदन सूचकांक	upper-group minus lower-group proportion correct
distance education	दूरस्थ शिक्षा	education with teacher and learner mostly separated
distractor	विकर्षक	plausible wrong option
divergent production	अपसारी चिंतन	thinking that produces many answers; creativity (Guilford)
effective teaching	प्रभावी शिक्षण	teaching judged by the learning it produces
egocentrism	अहंकेंद्रिता	difficulty seeing another's point of view
elective course	वैकल्पिक पाठ्यक्रम	Course chosen from a pool
emerging adulthood	उभरती वयस्कता	Arnett's stage, about 18 to 25
emotion	संवेग	a feeling such as fear, joy or anger that affects behaviour

English	हिंदी	Meaning
emotional intelligence	संवेगात्मक बुद्धि	ability to recognise and manage one's own and others' emotions
enactive	क्रियात्मक	Bruner's mode: learning by doing
enrichment	संवर्धन	deeper and wider work at the same level
entry behaviour	प्रवेश व्यवहार	what the learner already knows and can do before instruction
environment	वातावरण	what a person experiences: family, culture, schooling, health
epidiascope	एपिडायस्कोप	Projects opaque material and slides
episcope	एपिस्कोप	Projects opaque material by reflected light
equilibration	संतुलनीकरण	balancing assimilation and accommodation
essay-type item	निबंधात्मक प्रश्न	item needing a long organised written answer
evaluation	मूल्यांकन	judging worth against objectives; includes measurement and assessment
evaluation triangle	मूल्यांकन त्रिभुज	objectives, learning experiences and evaluation as three linked corners
evaluative function	मूल्यांकनात्मक कार्य	checking whether objectives have been achieved
experiential learning	अनुभवात्मक अधिगम	learning from experience and reflection (Kolb)
external evaluation	बाह्य मूल्यांकन	by outside examiners or agency
extinction	विलोपन	fading of a response when not reinforced
extrinsic motivation	बाह्य अभिप्रेरणा	motivation from marks, praise, rewards
face validity	आभासी वैधता	surface appearance of relevance
facilitator	सुगमकर्ता	teacher who arranges conditions and guides learning
factors affecting teaching	शिक्षण को प्रभावित करने वाले कारक	Influences that make teaching more or less successful
feedback	प्रतिपुष्टि	information returned to the sender about the effect of a message or action
field-dependent	क्षेत्र-आश्रित	perceives globally, relies on outside cues
field-independent	क्षेत्र-स्वतंत्र	able to separate item from background
flannel board	फलालैन बोर्ड	Cloth board on which backed cut-outs stick
flash card	फ़्लैश कार्ड	Card shown briefly for drill
flipped classroom	फ़्लिप्ड कक्षा	content studied before class; class used for application
fluid intelligence	तरल बुद्धि	reasoning in new situations (Cattell)
focus (of a model)	केंद्र-बिंदु	central aim of a model
formative evaluation	रचनात्मक मूल्यांकन	frequent evaluation that guides learning
from concrete to abstract	मूर्त से अमूर्त की ओर	things learners can perceive before ideas and definitions
from empirical to rational	अनुभवजन्य से तर्कसंगत की ओर	observed fact before reasoned explanation
from indefinite to definite	अनिश्चित से निश्चित की ओर	vague ideas made clear and exact
from known to unknown	ज्ञात से अज्ञात की ओर	start from what learners know and move to the new

English	हिंदी	Meaning
from particular to general	विशिष्ट से सामान्य की ओर	several cases before the rule; basis of induction
from psychological to logical	मनोवैज्ञानिक से तार्किक की ओर	learner's interests before subject structure
from simple to complex	सरल से जटिल की ओर	easy elements before difficult ones
from whole to part	पूर्ण से अंश की ओर	complete picture before details (Gestalt)
Gestalt	गेस्टाल्ट	whole, form; school of Wertheimer, Köhler, Koffka
gifted learner	प्रतिभाशाली शिक्षार्थी	learner far above age group in ability or performance
Golem effect	गोलेम प्रभाव	Low teacher expectations lower performance
grading	श्रेणीकरण	Reporting performance in broad lettered bands
guidance function	निर्देशन कार्य	use of evaluation to help educational and vocational choices
halo effect	प्रभामंडल प्रभाव	One favourable trait colours other ratings
handout	हैंडआउट	Printed sheet given to students
Hawthorne effect	हॉथोर्न प्रभाव	Improvement due to awareness of being studied
hearing impairment	श्रवण बाधिता	partial or total loss of hearing
heredity	वंशानुक्रम	what a person inherits from parents
heuristic method	अन्वेषण विधि	learner finds out as a discoverer
heutagogy	स्व-निर्धारित अधिगम	self-determined learning (Hase and Kenyon)
hierarchy of needs	आवश्यकता पदानुक्रम	Maslow's ordered levels of needs
higher-order thinking	उच्च-स्तरीय चिंतन	analysing, evaluating and creating
hypothesis	परिकल्पना	tentative solution to be tested
iconic	प्रतिमात्मक	Bruner's mode: learning through images
ICT-based	आईसीटी आधारित	Using digital technology
identity statuses	पहचान स्थितियाँ	Marcia's four outcomes of exploration and commitment
identity versus role confusion	पहचान बनाम भूमिका भ्रम	Erikson's adolescent crisis
imaginary audience	काल्पनिक दर्शक	feeling of being always watched
impartiality	निष्पक्षता	just and equal treatment of all students
inclusive education	समावेशी शिक्षा	all learners, with and without disability, learn together with needed supports
independent variable	स्वतंत्र चर	the teacher (in analysis of teaching); the manipulated factor (in an experiment)
indirect assessment	अप्रत्यक्ष आकलन	gathers opinions about learning (surveys)
indirect grading	अप्रत्यक्ष श्रेणीकरण	Marks first, then converted to grades
individual differences	वैयक्तिक भिन्नताएँ	ways in which learners differ from one another
indoctrination	मतारोपण	implanting beliefs to be held beyond question
inductive method	आगमन विधि	examples first, rule drawn from them
Information and Communication Technology	सूचना एवं संचार प्रौद्योगिकी	Digital tools to create, store, send, receive information
inquiry training model	पृच्छा प्रशिक्षण प्रतिमान	inquiry into a puzzling event by questioning (Suchman)

English	हिंदी	Meaning
insight	अंतर्दृष्टि	sudden grasp of relations in a whole situation
institution	संस्था	The college or university with its policies and leadership
institution-related factors	संस्थागत कारक	Policies, structures and leadership of the institution
institutional climate	संस्थागत वातावरण	General atmosphere of leadership and staff relations
instruction	अनुदेशन	systematic passing on of organised knowledge; possible without a teacher
instructional facilities	अनुदेशात्मक सुविधाएँ	Infrastructure the institution provides for all teaching
instructional objective	शिक्षण उद्देश्य	precise, testable statement of what the learner will do after instruction
intelligence	बुद्धि	general capacity to learn, reason and adapt
intelligence quotient	बुद्धि-लब्धि	IQ: MA/CA x 100, or deviation score
interaction analysis	अंतःक्रिया विश्लेषण	systematic coding of classroom talk (Flanders)
interactive phase	अंतःक्रियात्मक अवस्था	classroom stage of teacher-learner interaction
interest	रुचि	liking for an activity
internal evaluation	आंतरिक मूल्यांकन	by the institution's own teachers
Internal Quality Assurance Cell	आंतरिक गुणवत्ता आश्वासन प्रकोष्ठ	Institution's body for planning and monitoring quality (NAAC)
internalisation	आंतरीकरण	degree to which a value becomes one's own
interval scale	अंतराल मापनी	Scale with equal units, no true zero
intervening variable	मध्यवर्ती चर	content, methods, strategies that carry the teacher's influence
intrinsic motivation	आंतरिक अभिप्रेरणा	motivation from the task or problem itself
ipsative assessment	स्व-संदर्भित आकलन	compares a learner with her own earlier performance
item analysis	पद विश्लेषण	finding difficulty, discrimination and distractor functioning of items
item response theory	एकांश अनुक्रिया सिद्धांत	Theory giving sample-independent item parameters
knowledge dimension	ज्ञान आयाम	factual, conceptual, procedural and metacognitive knowledge
laissez-faire	हस्तक्षेप-रहित	Climate in which the teacher withdraws guidance
language laboratory	भाषा प्रयोगशाला	Stations for listening and speaking practice
late adolescence	उत्तर किशोरावस्था	the last part of adolescence, about 17 to 19 years
latent learning	अव्यक्त अधिगम	learning without reinforcement, shown later (Tolman)
law of effect	प्रभाव का नियम	Thorndike: satisfying results strengthen a response
law of exercise	अभ्यास का नियम	Thorndike: use strengthens, disuse weakens
law of readiness	तत्परता का नियम	Thorndike: readiness makes action satisfying
leadership style	नेतृत्व शैली	authoritarian, democratic or laissez-faire way of leading a group
learned helplessness	अधिगत असहायता	giving up because one believes actions make no difference (Seligman)

English	हिंदी	Meaning
learner	शिक्षार्थी	the person who learns
learner-centred method	शिक्षार्थी-केंद्रित विधि	method in which learners are the main actors
learning	अधिगम	relatively permanent change in behaviour from experience or practice
learning contract	अधिगम अनुबंध	written plan of objectives, resources, evidence and criteria agreed in advance
learning environment	अधिगम वातावरण	Physical and psychological surroundings of learning
learning experiences	अधिगम अनुभव	teaching and activities provided to bring about the objectives
learning management system	अधिगम प्रबंधन प्रणाली	Software to deliver and manage online courses
learning outcome	अधिगम परिणाम	what the learner can demonstrate as a result of learning
learning style	अधिगम शैली	preferred way of taking in information
lecture method	व्याख्यान विधि	teacher presents organised content orally to listeners
levels of teaching	शिक्षण के स्तर	memory, understanding and reflective levels
levels/scales of measurement	मापन के स्तर	nominal, ordinal, interval, ratio
linear programming	रेखीय अभिक्रमण	Skinner’s style: same small steps for every learner
locus of control	नियंत्रण-स्थल	where a person places control of outcomes: self or outside (Rotter)
massive open online course (MOOC)	विशाल मुक्त ऑनलाइन पाठ्यक्रम	online course open to unlimited numbers without entry requirements
mastery learning	निपुणता अधिगम	every learner must reach a fixed standard in each unit
maturation	परिपक्वता	Unfolding of capacities with age, independent of training
maxim of teaching	शिक्षण सूत्र	working rule for the order in which material is presented
mean	माध्य	arithmetic average
meaningful learning	सार्थक अधिगम	new material related to existing knowledge
measurement	मापन	assigning numbers to performance by rules; answers ‘how much’
measurement error	मापन त्रुटि	Chance variation in a score not due to real ability
median	माध्यिका	middle score in order
memory level	स्मृति स्तर	teaching to fix and reproduce facts
mental age	मानसिक आयु	age level of a person’s test performance
metacognition	अधिसंज्ञान	thinking about one’s own thinking
metacognitive	अधिसंज्ञानात्मक	knowledge of thinking and of one’s own thinking
method of teaching	शिक्षण विधि	organised step-by-step way of teaching a lesson or unit
micro-teaching	सूक्ष्म शिक्षण	scaled-down practice lesson for one skill, with feedback
mnemonic devices	स्मृति सहायक युक्तियाँ	memory aids such as acronyms and rhymes
mode	बहुलक	most frequent score
model	प्रतिमान / मॉडल	Representation of a real thing

English	हिंदी	Meaning
model of teaching	शिक्षण प्रतिमान	complete planned pattern of teaching (Joyce and Weil)
modern	आधुनिक	Electrical audio-visual equipment
modern support	आधुनिक सहायक सामग्री	Electrical audio-visual equipment
moral development	नैतिक विकास	growth of reasoning about right and wrong
motivation	अभिप्रेरणा	arousing and sustaining the learner's will to learn
multimedia	बहुमाध्यम	Integration of text, image, sound, video, animation
multiple entry and exit	बहु-प्रवेश एवं बहु-निकास	Leaving and rejoining a programme with awards
multiple intelligences	बहु-बुद्धि	Gardner's theory of several independent intelligences
multiple-choice item	बहुविकल्पीय प्रश्न	stem with options: key and distractors
National Academic Depository	राष्ट्रीय शैक्षणिक निक्षेपागार	Digital store of degrees and mark-sheets
National Coordinator	राष्ट्रीय समन्वयक	body responsible for SWAYAM courses in one sector
National Higher Education Qualifications Framework	राष्ट्रीय उच्चतर शिक्षा अर्हता रूपरेखा	UGC framework of levels 4.5-8
noise	शोर	Physical disturbance; barrier to communication
non-projected aids	अप्रक्षेपित सामग्री	Aids seen directly
non-verbal communication	अशाब्दिक संप्रेषण	communication by face, gesture, eye contact, movement
norm-referenced	मानक-संदर्भित	Judged against the group
norm-referenced test	मानक-संदर्भित परीक्षण	interprets a score against other learners
normal distribution	प्रसामान्य वितरण	symmetrical bell-shaped distribution
normalisation	सामान्यीकरण	Making scores of different papers comparable
norms	मानक	tables of a reference group's performance for comparing scores
objective	उद्देश्य	what the learner should achieve
objective-type item	वस्तुनिष्ठ प्रश्न	item with one predetermined correct answer
objectivity	वस्तुनिष्ठता	freedom of scores from the scorer's judgement
observational learning	अवलोकनात्मक अधिगम	learning by watching a model (Bandura)
obtained score	प्राप्त अंक	the score actually obtained on a test
off-line teaching	ऑफ़लाइन शिक्षण	face-to-face teaching with teacher and learners together
on-demand examination	मांग पर परीक्षा	Learner chooses the date; paper from a bank
on-line teaching	ऑनलाइन शिक्षण	teaching through the internet
one-way mass communication	एकतरफ़ा जनसंचार	broadcast to a large audience with no return channel
open educational resources	मुक्त शैक्षिक संसाधन	Openly licensed teaching materials
open source	मुक्त स्रोत	Software whose code may be used, modified and shared
open-book examination	खुली पुस्तक परीक्षा	Books allowed during the test
operant conditioning	क्रियाप्रसूत अनुबंधन	behaviour shaped by consequences (Skinner)
ordinal scale	क्रमसूचक मापनी	Scale giving order only
outcome-based education	परिणाम आधारित शिक्षा	planning that starts from intended outcomes

English	हिंदी	Meaning
overhead projector	शिरोपरि प्रक्षेपी	Projects large transparencies in a lit room
panel discussion	पैनल चर्चा	experts converse informally under a moderator
parallel forms method	समानांतर प्रारूप विधि	two equivalent versions; equivalence
pedagogical content knowledge	शैक्षणिक विषयवस्तु ज्ञान	knowing how to make a particular topic understandable to learners
pedagogy	शिक्षाशास्त्र	art and science of teaching children
peer assessment	सहपाठी आकलन	Students judging classmates' work
peer group	समवयस्क समूह	group of people of the same age and status
peer tutoring	सहपाठी शिक्षण	students teaching fellow students
percentile	शततमक	score below which a given percentage falls
percentile rank	शततमक कोटि	percentage of the group scoring below a given score
percentile score	शततमक अंक	Percentage of candidates scoring at or below a score
performance-based assessment	निष्पादन आधारित आकलन	Assessment by doing a task
personal fable	व्यक्तिगत कल्पित कथा	belief in being unique and invulnerable
Personalised System of Instruction	वैयक्तिकृत अनुदेशन प्रणाली	Keller's self-paced, unit-mastery plan with proctors
personality	व्यक्तित्व	Characteristic pattern of a person's behaviour and traits
phases of teaching	शिक्षण की अवस्थाएँ	pre-active, interactive and post-active stages of teaching
physical environment	भौतिक वातावरण	Light, air, noise, space, furniture, seating
placement evaluation	स्थापन मूल्यांकन	before instruction; entry behaviour and starting point
plagiarism	साहित्यिक चोरी	presenting another's work as one's own
portfolio	पोर्टफोलियो	Purposeful collection of work with reflection
post-active phase	उत्तर-क्रियात्मक अवस्था	evaluation and reflection stage after teaching
post-conventional	उत्तर-परंपरागत	Kohlberg level guided by principles
pre-active phase	पूर्व-क्रियात्मक अवस्था	planning stage before teaching
pre-conventional	पूर्व-परंपरागत	Kohlberg level guided by punishment and reward
predictive validity	भविष्यसूचक वैधता	scores forecast a later criterion
prescriptive function	उपचारात्मक कार्य	choosing content and methods to suit and remedy the diagnosis
principles of reaction	प्रतिक्रिया के सिद्धांत	how the teacher responds to learners
prior knowledge	पूर्व ज्ञान	What the learner already knows
probing questioning	खोजपूर्ण प्रश्न	follow-up questions that deepen a student's answer
problem-based learning	समस्या आधारित अधिगम	learning driven by an ill-structured problem met first
problem-solving method	समस्या समाधान विधि	learners solve a problem by steps of scientific inquiry
proctored examination	पर्यवेक्षित परीक्षा	examination held under supervision at a centre
profession	व्यवसाय	occupation with specialised knowledge, training, service motive, code
professional ethics	व्यावसायिक नैतिकता	standards of right conduct expected in a profession

English	हिंदी	Meaning
prognostic test	पूर्वानुमानात्मक परीक्षण	forecasts future performance
programmed instruction	अभिक्रमित अनुदेशन	self-instruction through a sequence of small frames with immediate confirmation
project method	परियोजना विधि	learners carry out a purposeful real-life task to completion
projected aids	प्रक्षेपित सामग्री	Aids whose image a projector throws on a screen
psychological environment	मनोवैज्ञानिक वातावरण	Relationships, warmth, freedom from fear
psychological safety	मनोवैज्ञानिक सुरक्षा	Feeling that one will not be humiliated for a question or mistake
psychomotor	मनोगत्यात्मक	Concerning physical and manipulative skill
psychomotor domain	मनोगत्यात्मक पक्ष	area of physical skills needing mind-muscle coordination
psychosocial	मनोसामाजिक	relating mind and society (Erikson’s stages)
punishment	दंड	consequence that decreases behaviour
Pygmalion effect	पिग्मेलियन प्रभाव	High teacher expectations raise performance
quantitative	मात्रात्मक	expressed in numbers
question bank	प्रश्न बैंक	Classified store of items
questioning skill	प्रश्न कौशल	Skill of framing and distributing questions
rating scale	निर्धारण मापनी	records degree on a numerical or verbal scale
raw score	अपरिष्कृत प्राप्तांक	marks actually obtained, before conversion
readiness	तत्परता	Maturity and preparation sufficient for a task
reasonable accommodation	उचित सुविधा	necessary changes without undue burden so persons with disabilities enjoy rights equally
reflective level	चिंतनशील स्तर	problem-centred teaching for critical, independent thinking
reflective practice	चिंतनशील अभ्यास	examining one’s own teaching to improve it
reinforcement	पुनर्बलन	encouraging desired responses by praise and approval
relative grading	सापेक्ष श्रेणीकरण	Grades by position in the group
reliability	विश्वसनीयता	consistency of scores
reliability coefficient	विश्वसनीयता गुणांक	correlation expressing reliability, 0 to 1
remedial teaching	उपचारात्मक शिक्षण	teaching that corrects learning deficiencies
remote proctoring	दूरस्थ निरीक्षण	Online invigilation by webcam and screen
reversibility	उत्क्रमणीयता	mentally undoing an action
role play	भूमिका निर्वाह	learners act out roles unscripted, followed by discussion
rote learning	रटंत अधिगम	learning by repetition without meaning
rubric	रूब्रिक (मूल्यांकन मानदंड-तालिका)	Scoring guide with criteria, levels, descriptors
scaffolding	आलंबन, पाड़	temporary support withdrawn as learner gains competence
schema	स्कीमा, मानसिक संरचना	organised pattern of thought

English	हिंदी	Meaning
scholastic area	शैक्षिक क्षेत्र	Curricular subjects and intellectual abilities
self-actualisation	आत्मसिद्धि	realising one's full potential; top of Maslow's hierarchy
self-assessment	स्व-आकलन	Judging one's own work against criteria
self-directed learning	स्व-निर्देशित अधिगम	learning in which the learner diagnoses needs, sets goals, finds resources, evaluates
self-efficacy	आत्म-प्रभावकारिता	belief in one's capability to succeed at a task (Bandura)
self-fulfilling prophecy	स्वतः पूर्ण होने वाली भविष्यवाणी	Expectation that causes itself to come true
self-learning material	स्व-अधिगम सामग्री	Distance-learning material designed for study without a teacher
self-regulated learner	स्व-नियमित शिक्षार्थी	learner who plans, monitors and evaluates own learning
Semester Grade Point Average	सेमेस्टर ग्रेड पॉइंट औसत	Credit-weighted average for one semester
seminar	संगोष्ठी	paper presented and discussed in depth
set induction	प्रस्तावना कौशल	skill of introducing a lesson and linking it to prior knowledge
simulated teaching	अनुरूपित शिक्षण	role-play practice among trainees (teacher, pupil, observer)
simulation	अनुरूपण	practice in an artificial situation resembling reality
skill enhancement course	कौशल संवर्धन पाठ्यक्रम	Skill- or value-based course chosen from a pool
slow learner	मंद अधिगमकर्ता	learner with somewhat below-average ability
smart classroom	स्मार्ट कक्षा	Classroom with digital teaching tools
social characteristics	सामाजिक विशेषताएँ	ways a learner relates to others
social constructivism	सामाजिक रचनावाद	knowledge built through social interaction (Vygotsky)
social system	सामाजिक प्रणाली	roles and relationships in a model
specific learning disability	विशिष्ट अधिगम अक्षमता	difficulty in one area despite average or higher intelligence
specimen	नमूना	Real sample representing its class
split-half method	अर्ध-विच्छेद विधि	one administration split into halves; internal consistency
standard deviation	मानक विचलन	spread of scores around the mean
standard error of measurement	मापन की मानक त्रुटि	error band around an individual's obtained score
standard score	मानक प्राप्तांक	score expressed in SD units from the mean (z, T)
stimulus generalisation	उद्दीपक सामान्यीकरण	similar stimuli evoke the conditioned response
stimulus variation	उद्दीपन परिवर्तन	changes in teacher behaviour to hold attention
storm and stress	तूफ़ान और तनाव	Hall's description of adolescence as turbulent
strategy	शिक्षण व्यूह रचना	overall plan for reaching an objective, often combining methods
study habits	अध्ययन आदतें	regular ways of studying
subject mastery	विषय पर अधिकार	Deep and accurate command of content
summative evaluation	योगात्मक मूल्यांकन	evaluation at the end for grading

English	हिंदी	Meaning
support material	सहायक सामग्री	Aids and resources used to teach a particular content
support system	सहायक प्रणाली	materials and facilities a model needs
symbolic	प्रतीकात्मक	Bruner’s mode: learning through symbols
symposium	परिसंवाद	several prepared talks on aspects of one theme
synchronous	समकालिक	In real time, all at once
synchronous learning	समकालिक अधिगम	learning in which all take part at the same time
syntax (of a model)	संरचना	sequence of phases of a model
synthetic method	संश्लेषण विधि	known data forward to unknown conclusion
taxonomy of educational objectives	शैक्षिक उद्देश्यों का वर्गीकरण	ordered classification of educational objectives
teacher	शिक्षक	Person who teaches
teacher efficacy	शिक्षक प्रभावकारिता	Teacher’s belief in own capacity to bring about learning
teacher-centred method	शिक्षक-केंद्रित विधि	method in which the teacher controls content and pace
teaching	शिक्षण	purposeful activity intended to bring about learning in another person
teaching aid	शिक्षण सहायक सामग्री	Material that helps make ideas clear
Teaching Aptitude	शिक्षण अभिवृत्ति	title of Unit I
teaching skill	शिक्षण कौशल	small definable set of teacher behaviours that aid learning
teaching support system	शिक्षण सहायक प्रणाली	The whole set of aids and facilities
teaching to the test	परीक्षा-केंद्रित शिक्षण	Narrowing teaching to likely exam questions
team teaching	दल शिक्षण	several teachers jointly plan, teach and evaluate
technique	प्रविधि	small device used within a method
test	परीक्षण	an instrument: a set of questions or tasks given to the learner
test-retest method	परीक्षण-पुनर्परीक्षण विधि	same test given twice; stability
three-dimensional	त्रि-आयामी	Having length, breadth and depth
three-dimensional aids	त्रि-आयामी सामग्री	Aids having length, breadth and depth
traditional	पारंपरिक	Not needing electricity or electronics
traditional support	पारंपरिक सहायक सामग्री	Aids that need no electricity
training	प्रशिक्षण	building a specific skill by practice to a set standard
transformative learning	रूपांतरणकारी अधिगम	learning that changes a frame of reference (Mezirow)
trial and error	प्रयास और त्रुटि	learning by trying and dropping wrong responses (Thorndike)
true score	सत्य प्राप्तांक	the score free of measurement error
tutorial	अनुशिक्षण	tutor with a small group giving remedial and personal help
two-dimensional	द्वि-आयामी	Flat (length and breadth)
understanding level	बोध स्तर	teaching to see relationships and apply principles

English	हिंदी	Meaning
usability / practicability	व्यावहारिकता	ease of giving, scoring and interpreting a test in time and cost
validity	वैधता	measuring what the test claims to measure
value judgement	मूल्य-निर्णय	a judgement of worth or desirability
variable	चर	anything that can vary
vicarious reinforcement	प्रतिनिधिक पुनर्बलन	reinforcement seen given to a model
visual aids	दृश्य सामग्री	Aids addressing sight
visual impairment	दृष्टि बाधिता	partial or total loss of sight
washback	परीक्षा का पश्च प्रभाव	Effect of a test on teaching and learning
working memory	कार्यकारी स्मृति	short-term memory in use now
workload	कार्यभार	Teaching and other duties assigned
workshop	कार्यशाला	hands-on working session producing something
young adult	युवा वयस्क	person in the twenties, after adolescence
zone of proximal development	समीपस्थ विकास का क्षेत्र	gap between unaided and assisted performance

Booklet Index

This index links every question in the Unit I MCQ booklet to the section of this guide that teaches its answer. Find the set, then the question number; the figure beside it is the section to reread. For example, “23 · 6.7” means that Question 23 of that set is taught in Section 6.7.

Read the whole section, not only the line you think you need. Examiners build distractors from the neighbouring ideas.

Set 1 (Very Easy)

1 · 6.6	**2** · 4.4	**3** · 1.5	**4** · 4.3	**5** · 1.8	**6** · 1.4	**7** · 6.3	**8** · 2.2	**9** · 6.2	**10** · 2.7
11 · 4.2	**12** · 1.4	**13** · 1.7	**14** · 6.3	**15** · 3.7	**16** · 2.4	**17** · 4.3	**18** · 2.8	**19** · 1.1	**20** · 3.6
21 · 5.1	**22** · 1.4	**23** · 3.3	**24** · 1.4	**25** · 2.8	**26** · 1.8	**27** · 2.1	**28** · 4.5	**29** · 6.3	**30** · 2.6
31 · 1.1	**32** · 1.5	**33** · 4.3	**34** · 4.6	**35** · 6.1	**36** · 1.2	**37** · 6.9	**38** · 6.7	**39** · 5.2	**40** · 2.2
41 · 4.6	**42** · 2.3	**43** · 3.2	**44** · 1.5	**45** · 2.3	**46** · 4.1	**47** · 1.4	**48** · 1.5	**49** · 6.9	**50** · 1.2

Set 2 (Very Easy)

1 · 1.4	**2** · 2.2	**3** · 6.2	**4** · 4.6	**5** · 2.9	**6** · 6.1	**7** · 3.2	**8** · 1.5	**9** · 4.5	**10** · 4.3
11 · 1.3	**12** · 1.8	**13** · 1.3	**14** · 1.7	**15** · 1.5	**16** · 2.2	**17** · 6.3	**18** · 2.6	**19** · 4.6	**20** · 6.4
21 · 2.8	**22** · 6.9	**23** · 6.3	**24** · 6.8	**25** · 4.4	**26** · 2.2	**27** · 1.4	**28** · 2.3	**29** · 6.2	**30** · 2.8
31 · 1.6	**32** · 6.7	**33** · 1.1	**34** · 4.2	**35** · 2.7	**36** · 1.5	**37** · 4.3	**38** · 3.7	**39** · 1.4	**40** · 5.4
41 · 6.3	**42** · 4.4	**43** · 6.7	**44** · 2.1	**45** · 1.5	**46** · 1.3	**47** · 1.8	**48** · 3.6	**49** · 5.3	**50** · 4.2

Set 3 (Very Easy)

1 · 6.5	**2** · 1.5	**3** · 4.6	**4** · 2.8	**5** · 5.4	**6** · 1.5	**7** · 6.1	**8** · 1.5	**9** · 1.7	**10** · 1.2
11 · 1.4	**12** · 3.3	**13** · 4.3	**14** · 1.3	**15** · 1.1	**16** · 4.4	**17** · 6.3	**18** · 1.5	**19** · 2.2	**20** · 6.7
21 · 2.6	**22** · 3.2	**23** · 1.4	**24** · 2.9	**25** · 5.1	**26** · 3.6	**27** · 6.9	**28** · 2.3	**29** · 2.7	**30** · 6.8
31 · 6.4	**32** · 5.4	**33** · 2.5	**34** · 1.4	**35** · 4.5	**36** · 6.2	**37** · 4.2	**38** · 1.6	**39** · 1.4	**40** · 2.9
41 · 1.1	**42** · 2.2	**43** · 6.2	**44** · 4.3	**45** · 3.7	**46** · 4.1	**47** · 1.8	**48** · 2.2	**49** · 3.3	**50** · 4.4

Set 4 (Easy)

1 · 1.4	**2** · 1.8	**3** · 1.4	**4** · 6.4	**5** · 2.2	**6** · 5.2	**7** · 4.6	**8** · 1.8	**9** · 6.8	**10** · 1.5
11 · 6.1	**12** · 4.5	**13** · 1.5	**14** · 6.9	**15** · 2.6	**16** · 2.2	**17** · 4.1	**18** · 2.9	**19** · 4.4	**20** · 5.4
21 · 4.3	**22** · 4.6	**23** · 6.7	**24** · 1.5	**25** · 2.6	**26** · 4.4	**27** · 3.7	**28** · 4.2	**29** · 1.7	**30** · 2.8
31 · 6.6	**32** · 1.6	**33** · 2.7	**34** · 6.2	**35** · 1.4	**36** · 2.2	**37** · 1.5	**38** · 1.2	**39** · 1.8	**40** · 2.3
41 · 1.1	**42** · 6.3	**43** · 4.3	**44** · 6.2	**45** · 2.1	**46** · 1.1	**47** · 6.3	**48** · 6.7	**49** · 1.3	**50** · 3.6

Set 5 (Easy)

1 · 5.4	**2** · 6.3	**3** · 1.4	**4** · 6.6	**5** · 1.6	**6** · 6.8	**7** · 2.8	**8** · 6.9	**9** · 2.1	**10** · 4.2
11 · 1.8	**12** · 6.7	**13** · 3.3	**14** · 2.6	**15** · 1.4	**16** · 3.5	**17** · 6.2	**18** · 1.5	**19** · 1.3	**20** · 3.6
21 · 6.3	**22** · 4.5	**23** · 1.5	**24** · 2.7	**25** · 4.6	**26** · 1.4	**27** · 1.1	**28** · 1.1	**29** · 1.5	**30** · 1.10
31 · 2.2	**32** · 6.4	**33** · 4.4	**34** · 1.2	**35** · 1.7	**36** · 2.2	**37** · 2.9	**38** · 6.2	**39** · 1.4	**40** · 4.4
41 · 1.5	**42** · 4.3	**43** · 2.3	**44** · 2.8	**45** · 5.2	**46** · 4.5	**47** · 3.2	**48** · 4.2	**49** · 2.2	**50** · 4.3

Set 6 (Easy)

1 · 4.4	**2** · 1.5	**3** · 6.4	**4** · 2.6	**5** · 2.2	**6** · 5.1	**7** · 1.3	**8** · 6.3	**9** · 1.5	**10** · 6.9
11 · 4.6	**12** · 1.4	**13** · 2.8	**14** · 1.4	**15** · 2.7	**16** · 4.4	**17** · 6.7	**18** · 2.9	**19** · 1.8	**20** · 6.4
21 · 1.9	**22** · 4.4	**23** · 1.5	**24** · 4.2	**25** · 6.2	**26** · 6.7	**27** · 3.7	**28** · 2.3	**29** · 6.8	**30** · 5.4
31 · 4.6	**32** · 2.1	**33** · 6.4	**34** · 2.3	**35** · 1.1	**36** · 3.2	**37** · 1.4	**38** · 4.1	**39** · 2.9	**40** · 1.6
41 · 6.2	**42** · 1.5	**43** · 2.9	**44** · 5.4	**45** · 1.8	**46** · 1.6	**47** · 4.3	**48** · 1.7	**49** · 1.1	**50** · 2.2

Set 7 (Moderate)

1 · 1.4	**2** · 6.4	**3** · 4.5	**4** · 2.3	**5** · 4.2	**6** · 1.4	**7** · 6.6	**8** · 4.4	**9** · 2.6	**10** · 1.8
11 · 2.7	**12** · 3.3	**13** · 2.2	**14** · 1.4	**15** · 6.8	**16** · 1.5	**17** · 1.6	**18** · 2.4	**19** · 6.4	**20** · 1.5
21 · 1.8	**22** · 2.2	**23** · 1.4	**24** · 1.2	**25** · 2.9	**26** · 4.3	**27** · 4.6	**28** · 6.3	**29** · 2.3	**30** · 1.7
31 · 1.4	**32** · 5.4	**33** · 3.4	**34** · 4.3	**35** · 6.9	**36** · 2.1	**37** · 1.7	**38** · 6.3	**39** · 4.4	**40** · 1.5
41 · 5.3	**42** · 1.2	**43** · 4.2	**44** · 1.8	**45** · 6.7	**46** · 6.3	**47** · 1.5	**48** · 2.8	**49** · 4.6	**50** · 1.1

Set 8 (Moderate)

1 · 4.5	**2** · 5.2	**3** · 6.4	**4** · 3.3	**5** · 1.5	**6** · 1.8	**7** · 1.5	**8** · 6.7	**9** · 1.1	**10** · 2.8
11 · 6.9	**12** · 2.7	**13** · 6.8	**14** · 2.9	**15** · 4.4	**16** · 6.1	**17** · 6.7	**18** · 3.5	**19** · 6.3	**20** · 4.5
21 · 2.2	**22** · 2.2	**23** · 4.4	**24** · 3.7	**25** · 6.2	**26** · 1.5	**27** · 4.2	**28** · 6.2	**29** · 1.9	**30** · 2.5
31 · 1.4	**32** · 4.1	**33** · 1.7	**34** · 2.6	**35** · 1.6	**36** · 4.6	**37** · 2.4	**38** · 2.3	**39** · 6.3	**40** · 1.4
41 · 5.4	**42** · 1.5	**43** · 1.3	**44** · 1.2	**45** · 4.4	**46** · 6.6	**47** · 1.4	**48** · 2.3	**49** · 4.3	**50** · 1.10

Set 9 (Moderate)

1 · 5.3	**2** · 1.1	**3** · 5.4	**4** · 1.5	**5** · 5.2	**6** · 1.3	**7** · 4.6	**8** · 4.4	**9** · 4.3	**10** · 1.9
11 · 2.2	**12** · 1.8	**13** · 2.3	**14** · 1.5	**15** · 1.4	**16** · 4.6	**17** · 3.2	**18** · 4.6	**19** · 6.4	**20** · 6.4
21 · 1.7	**22** · 2.2	**23** · 2.9	**24** · 6.9	**25** · 2.8	**26** · 4.3	**27** · 2.7	**28** · 6.8	**29** · 1.8	**30** · 6.3
31 · 2.1	**32** · 1.4	**33** · 6.3	**34** · 6.7	**35** · 1.5	**36** · 1.2	**37** · 5.1	**38** · 6.1	**39** · 1.4	**40** · 2.6
41 · 1.8	**42** · 2.4	**43** · 2.3	**44** · 4.2	**45** · 4.3	**46** · 1.8	**47** · 1.4	**48** · 4.4	**49** · 1.6	**50** · 1.5

Set 10 (Moderate)

1 · 1.5	**2** · 4.1	**3** · 4.3	**4** · 2.2	**5** · 2.6	**6** · 6.4	**7** · 1.8	**8** · 5.1	**9** · 1.1	**10** · 5.4
11 · 1.2	**12** · 6.3	**13** · 4.2	**14** · 2.4	**15** · 2.8	**16** · 6.1	**17** · 2.3	**18** · 1.4	**19** · 6.9	**20** · 4.5
21 · 4.2	**22** · 4.4	**23** · 6.7	**24** · 1.6	**25** · 1.4	**26** · 2.6	**27** · 1.5	**28** · 1.9	**29** · 2.9	**30** · 1.4
31 · 6.7	**32** · 3.7	**33** · 6.4	**34** · 2.1	**35** · 2.3	**36** · 1.5	**37** · 6.9	**38** · 1.2	**39** · 4.4	**40** · 6.6
41 · 4.6	**42** · 1.10	**43** · 6.3	**44** · 3.4	**45** · 1.4	**46** · 1.5	**47** · 1.7	**48** · 2.7	**49** · 4.5	**50** · 6.2

Set 11 (Tough)

1 · 3.6	**2** · 2.8	**3** · 1.5	**4** · 6.3	**5** · 1.3	**6** · 1.5	**7** · 2.9	**8** · 6.7	**9** · 1.5	**10** · 4.3
11 · 4.6	**12** · 1.8	**13** · 4.3	**14** · 6.8	**15** · 1.4	**16** · 1.9	**17** · 2.4	**18** · 5.2	**19** · 1.8	**20** · 1.4
21 · 4.4	**22** · 2.6	**23** · 1.4	**24** · 6.4	**25** · 1.1	**26** · 1.8	**27** · 4.1	**28** · 4.5	**29** · 6.2	**30** · 4.4
31 · 1.6	**32** · 1.8	**33** · 2.3	**34** · 4.6	**35** · 2.2	**36** · 2.2	**37** · 1.3	**38** · 3.7	**39** · 6.5	**40** · 6.9
41 · 4.3	**42** · 6.4	**43** · 2.3	**44** · 3.5	**45** · 6.3	**46** · 2.2	**47** · 3.2	**48** · 1.1	**49** · 2.7	**50** · 1.5

Set 12 (Tough)

1 · 4.6	**2** · 1.7	**3** · 6.9	**4** · 4.4	**5** · 2.3	**6** · 2.9	**7** · 1.2	**8** · 4.3	**9** · 2.3	**10** · 6.3
11 · 1.6	**12** · 2.7	**13** · 1.4	**14** · 6.4	**15** · 1.5	**16** · 2.3	**17** · 1.4	**18** · 2.1	**19** · 4.4	**20** · 6.2
21 · 1.4	**22** · 1.5	**23** · 6.9	**24** · 6.3	**25** · 6.1	**26** · 4.6	**27** · 1.4	**28** · 2.4	**29** · 1.1	**30** · 1.5
31 · 1.10	**32** · 6.3	**33** · 6.7	**34** · 1.1	**35** · 5.1	**36** · 2.6	**37** · 4.5	**38** · 4.2	**39** · 5.1	**40** · 2.2
41 · 1.8	**42** · 2.6	**43** · 6.7	**44** · 5.4	**45** · 1.8	**46** · 1.2	**47** · 4.3	**48** · 1.5	**49** · 2.8	**50** · 6.6

Set 13 (Tough)

1 · 3.6	**2** · 4.6	**3** · 1.6	**4** · 2.9	**5** · 2.2	**6** · 6.4	**7** · 3.5	**8** · 2.6	**9** · 1.8	**10** · 4.3
11 · 1.3	**12** · 6.5	**13** · 1.8	**14** · 6.3	**15** · 2.4	**16** · 4.4	**17** · 2.3	**18** · 4.4	**19** · 5.3	**20** · 1.5
21 · 3.3	**22** · 6.2	**23** · 1.5	**24** · 6.2	**25** · 6.9	**26** · 5.4	**27** · 2.2	**28** · 1.4	**29** · 3.7	**30** · 1.5
31 · 1.9	**32** · 1.4	**33** · 4.1	**34** · 4.3	**35** · 2.8	**36** · 5.2	**37** · 1.1	**38** · 1.8	**39** · 6.7	**40** · 1.9
41 · 1.6	**42** · 2.2	**43** · 1.4	**44** · 6.5	**45** · 2.5	**46** · 4.6	**47** · 1.1	**48** · 2.7	**49** · 6.9	**50** · 1.5

Set 14 (Tough)

1 · 5.1	**2** · 6.7	**3** · 1.5	**4** · 1.1	**5** · 2.4	**6** · 4.3	**7** · 5.4	**8** · 6.3	**9** · 1.5	**10** · 6.1
11 · 1.10	**12** · 2.3	**13** · 2.7	**14** · 4.5	**15** · 1.7	**16** · 1.2	**17** · 6.3	**18** · 1.4	**19** · 2.3	**20** · 4.6
21 · 3.4	**22** · 1.5	**23** · 2.1	**24** · 6.5	**25** · 2.8	**26** · 2.2	**27** · 1.3	**28** · 2.6	**29** · 4.6	**30** · 4.2
31 · 2.3	**32** · 1.8	**33** · 1.3	**34** · 6.9	**35** · 6.4	**36** · 1.7	**37** · 6.7	**38** · 4.4	**39** · 1.5	**40** · 4.2
41 · 6.8	**42** · 6.1	**43** · 4.3	**44** · 6.4	**45** · 3.6	**46** · 1.1	**47** · 1.4	**48** · 2.7	**49** · 1.2	**50** · 4.3

Set 15 (Critical)

1 · 1.2	**2** · 6.2	**3** · 4.4	**4** · 1.5	**5** · 5.1	**6** · 2.9	**7** · 4.2	**8** · 6.3	**9** · 2.2	**10** · 1.8
11 · 1.7	**12** · 2.7	**13** · 6.4	**14** · 1.4	**15** · 4.4	**16** · 1.5	**17** · 1.3	**18** · 1.1	**19** · 6.9	**20** · 2.8
21 · 1.5	**22** · 5.1	**23** · 1.5	**24** · 4.6	**25** · 6.3	**26** · 1.8	**27** · 2.2	**28** · 4.6	**29** · 2.3	**30** · 1.2
31 · 2.1	**32** · 4.2	**33** · 2.6	**34** · 6.1	**35** · 1.4	**36** · 3.6	**37** · 1.4	**38** · 1.1	**39** · 4.3	**40** · 3.1
41 · 4.5	**42** · 2.7	**43** · 6.7	**44** · 6.3	**45** · 2.3	**46** · 1.6	**47** · 1.8	**48** · 2.8	**49** · 6.8	**50** · 1.4

Set 16 (Critical)

1 · 1.9	**2** · 1.8	**3** · 4.5	**4** · 1.4	**5** · 4.3	**6** · 1.5	**7** · 2.3	**8** · 6.2	**9** · 3.1	**10** · 6.3
11 · 4.4	**12** · 2.1	**13** · 1.6	**14** · 5.3	**15** · 2.6	**16** · 6.9	**17** · 1.4	**18** · 3.3	**19** · 1.5	**20** · 1.8
21 · 6.3	**22** · 2.6	**23** · 2.8	**24** · 6.7	**25** · 1.5	**26** · 1.2	**27** · 4.6	**28** · 2.3	**29** · 1.4	**30** · 1.5
31 · 2.7	**32** · 3.5	**33** · 6.3	**34** · 4.2	**35** · 6.2	**36** · 4.2	**37** · 1.8	**38** · 1.1	**39** · 1.6	**40** · 6.1
41 · 1.4	**42** · 6.6	**43** · 5.2	**44** · 2.9	**45** · 4.6	**46** · 2.3	**47** · 4.6	**48** · 2.2	**49** · 6.7	**50** · 2.2

Set 17 (Critical)

1 · 1.4	**2** · 4.5	**3** · 6.4	**4** · 6.2	**5** · 3.4	**6** · 6.3	**7** · 5.1	**8** · 2.5	**9** · 3.6	**10** · 1.5
11 · 1.10	**12** · 6.4	**13** · 1.5	**14** · 4.1	**15** · 1.5	**16** · 6.4	**17** · 2.3	**18** · 6.4	**19** · 6.5	**20** · 2.6
21 · 1.6	**22** · 4.3	**23** · 2.2	**24** · 1.4	**25** · 1.4	**26** · 1.8	**27** · 1.7	**28** · 2.7	**29** · 4.4	**30** · 1.1
31 · 4.4	**32** · 2.9	**33** · 4.6	**34** · 4.3	**35** · 3.5	**36** · 1.2	**37** · 2.7	**38** · 4.6	**39** · 6.3	**40** · 6.7
41 · 2.6	**42** · 1.1	**43** · 6.2	**44** · 2.4	**45** · 1.5	**46** · 1.3	**47** · 2.2	**48** · 3.6	**49** · 5.4	**50** · 4.3

Set 18 (Advanced)

1 · 2.7	**2** · 4.6	**3** · 6.3	**4** · 2.2	**5** · 2.2	**6** · 2.2	**7** · 5.4	**8** · 2.3	**9** · 5.4	**10** · 2.2
11 · 6.2	**12** · 1.2	**13** · 1.4	**14** · 1.9	**15** · 6.4	**16** · 2.6	**17** · 4.4	**18** · 6.5	**19** · 1.5	**20** · 4.2
21 · 5.2	**22** · 1.2	**23** · 6.7	**24** · 1.8	**25** · 6.7	**26** · 2.8	**27** · 1.4	**28** · 4.5	**29** · 4.3	**30** · 1.1
31 · 1.5	**32** · 4.6	**33** · 4.4	**34** · 2.2	**35** · 6.5	**36** · 1.10	**37** · 6.3	**38** · 6.5	**39** · 4.3	**40** · 2.4
41 · 6.4	**42** · 2.2	**43** · 5.1	**44** · 1.8	**45** · 6.9	**46** · 1.4	**47** · 6.4	**48** · 6.3	**49** · 6.8	**50** · 3.7

Set 19 (Advanced)

1 · 6.7	**2** · 4.6	**3** · 1.5	**4** · 2.9	**5** · 6.5	**6** · 2.3	**7** · 1.4	**8** · 2.2	**9** · 1.3	**10** · 6.7
11 · 1.1	**12** · 6.9	**13** · 4.3	**14** · 1.5	**15** · 2.6	**16** · 1.4	**17** · 4.4	**18** · 3.7	**19** · 6.2	**20** · 4.3
21 · 1.9	**22** · 4.3	**23** · 4.6	**24** · 3.3	**25** · 6.3	**26** · 3.6	**27** · 1.5	**28** · 4.4	**29** · 2.2	**30** · 1.3
31 · 1.4	**32** · 1.7	**33** · 2.2	**34** · 6.3	**35** · 6.2	**36** · 6.7	**37** · 5.4	**38** · 2.9	**39** · 4.5	**40** · 2.1
41 · 2.3	**42** · 1.8	**43** · 4.1	**44** · 5.1	**45** · 4.6	**46** · 1.7	**47** · 1.4	**48** · 2.8	**49** · 1.6	**50** · 1.10

Set 20 (Advanced)

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Use this list the other way round: when you finish a section, these are the booklet questions it prepares you for.

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Facts to Recheck before Your Examination

The facts below can change when a ministry, the UGC or the NTA issues a new notice; every one was checked in October 2026, and each line gives the position printed in this guide and the official place to check it.

- **SWAYAM PRABHA channels** (4.6). Began with 32 channels in 2017. In 2026 official sources disagreed: two PIB documents (March and June 2026) gave 48, while the SWAYAM PRABHA website and a PIB factsheet of August 2026 gave 40. Check the SWAYAM PRABHA website and the latest PIB release.
- **SWAYAM National Coordinators** (4.6). Ten, including INI (Institutions of National Importance, coordinated by Banaras Hindu University); older sources list nine. Check the SWAYAM website and the Ministry of Education's website.
- **UGC credit-transfer limit for SWAYAM** (4.6). Up to 40 per cent of the total courses in a programme in a semester, under the UGC Regulations of 2021; the 2016 Regulations allowed 20 per cent. Check UGC's website.
- **Definition of a credit** (6.7). CCFUP (December 2022): 15 hours of theory or 30 hours of practical per semester for one credit. NCrF: 30 notional learning hours for one credit, 1,200 hours a year for 40 credits. Check UGC's website.
- **NHEQF levels** (6.7). Higher education qualifications on levels 4.5 to 8, with the doctorate at level 8. Check UGC's website.
- **NAAC's accreditation framework** (3.7). Under revision; do not rely on the grading scale in older books. Check NAAC's website.
- **One Nation One Subscription** (4.6). Operational from 1 January 2025, implemented by INFLIBNET; about 13,000 e-journals of 30 publishers. Check INFLIBNET's website.
- **Mode of the examination** (6.8). Computer-based in normal practice since the NTA took over the UGC NET in 2018, but fixed afresh each time. Check the NTA information bulletin.
- **NEP 2020 targets with dates** (1.10). At least 50 hours of continuous professional development a year for every teacher; by 2030 a four-year integrated B.Ed. as the minimum degree qualification for school teaching. Check the Ministry of Education's website.
- **RPwD Act, 2016** (2.9). 21 specified disabilities; not less than 5 per cent of seats in government and government-aided higher education, with 5 years' upper age relaxation; not less than 4 per cent of vacancies in government employment. Check the Government of India's published text of the Act.

If an official source now says something different, follow the official source.